



UNIVERSITY OF SAN CARLOS
SCHOOL OF ENGINEERING

PHYSICALLY CONSTRAINED MACHINE
LEARNING SURROGATES OF ACTIVATED
SLUDGE SYSTEMS FOR PROCESS OPTIMIZATION
DISSERTATION

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Cebu City, Philippines, 2026
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Declaration

I hereby declare that the work presented in the dissertation entitled “Physically Constrained Machine Learning Surrogates of Activated Sludge Systems for Process Optimization” is original and the result of my own research. Formulations and ideas taken from other sources are cited as such.

Cebu City, Philippines, 2026

Egberto F. Selerio Jr.

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Abstract

Activated sludge process models can support wastewater treatment analysis and optimization, but repeated mechanistic simulations can be computationally demanding. Machine learning surrogates offer a faster alternative, yet prediction accuracy alone does not make a surrogate suitable for engineering decisions. A model may reproduce overall water-quality indicators while misrepresenting the individual components that produce them; statistically fitted predictions may violate mass conservation or return negative concentrations; an apparently interpretable regression may become difficult to explain after component coupling and physical correction; and a surrogate that predicts a connected treatment plant well on average may still lead an optimizer to a poor operating decision. This dissertation addresses these linked problems by developing a unified framework that connects complete component prediction, physical enforcement, model interpretation, and independent verification of operating decisions.

The study begins with mechanistic steady-state activated sludge simulations that provide reference component states over defined influent and operating conditions. Several statistical learning methods are compared using the complete reactor response rather than only aggregate treatment indicators, while changes in data availability and predictions beyond the sampled domain are also examined. Their raw predictions are then projected onto a physically admissible state so that the returned concentrations satisfy the specified mass-conservation and non-negativity requirements. The study next develops invariant-constrained second-order regression (ICSOR), an interpretable surrogate that represents operating effects, influent loading, nonlinear relationships, interactions, and learned coupling among treatment components. Because physical enforcement can alter the response produced by the regression, interpretation follows the complete prediction process from the fitted equation through component coupling and projection. Finally, the framework is extended from a single reactor to a connected activated sludge plant containing mixing, multiple biological reactors, secondary clarification, recycle streams, and stored solids. The resulting connected surrogate predicts the plant as a linked system, reconciles its predicted streams and inventories through joint projection, and is used to search for operating settings. These settings are then independently evaluated using the original mechanistic model and compared with decisions obtained through direct mechanistic optimization.

The results show that no single learning method is best under all data and operating

conditions. Flexible learners become especially effective when more simulation data are available, whereas the structured regression provides useful predictive performance when data are more limited while retaining a response that can be directly inspected. Physical projection consistently enforces the specified balances and concentration bounds, but it does not necessarily improve every prediction; instead, it can redistribute error among treatment components. This demonstrates that physical consistency and predictive accuracy are related but distinct properties. At the plant scale, joint projection generally improves agreement with the mechanistic reference and provides a coherent representation of the connected treatment system. The surrogate also enables faster initial exploration of operating alternatives. However, independent mechanistic evaluation shows that the faster surrogate searches do not necessarily produce decisions as good as those obtained through direct mechanistic optimization. Thus, computational speed, physical validity, predictive accuracy, interpretability, and decision quality cannot be treated as interchangeable evidence.

The main contribution of this work is therefore not simply a faster activated sludge model, but a transparent and physically checked workflow for using surrogate models in process analysis and optimization. The framework allows rapid generation and screening of candidate operating conditions while preserving component-level information, enforcing declared physical requirements, exposing the structure of an interpretable prediction, and retaining the mechanistic model as the final reference for important operating decisions. In doing so, the study provides a clearer basis for determining what a surrogate predicts well, what physical properties it guarantees, how its response can be interpreted, and whether the decisions made with it remain acceptable when evaluated with the original process model. The findings establish this framework for simulated steady-state activated sludge systems and provide a basis for future extensions to uncertainty-aware prediction, dynamic operation, adaptive sampling, and validation with measured plant data.

Keywords. activated sludge process, machine learning, statistical surrogate, physics enforcement, wastewater treatment, mass conservation, non-negativity, interpretable regression, process optimization.

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Principal Symbols

Symbol	Meaning
F, P, K	Numbers of component states, reaction processes, and independent stoichiometric invariants
ν	Stoichiometric matrix with processes in rows and components in columns
ρ	Vector of mechanistic process rates
A	Full-row-rank stoichiometric invariant operator
V_0, Z	Orthonormal bases of the stoichiometric right null space and the null space of A
c_{in}, c_{out}	Influent state and mechanistic reference effluent state
$c_{raw}, \tilde{c}$	Raw component prediction and projected component prediction
I_{comp}	Fixed component-to-reported-composite map
D_i	Hydraulic dilution rate for the stated reactor and flow basis
ω, e_{SO}	Net dissolved-oxygen transfer rate and dissolved-oxygen coordinate selector
u	Standalone-reactor operating input vector
ϕ, B, Γ, R	Second-order feature vector, driver coefficients, learned component coupling, and coupled-system matrix
$\hat{C}, \Phi$	Auxiliary fitted component states and training feature matrix
N	Number of observations in the stated training-size design
n_r, L	Numbers of biological reactor stages and clarifier layers
ϑ	Connected-plant control vector
q_P, q_C, q_E, q_U	Flow ratios for the reactor train, clarifier feed, overflow, and underflow relative to fresh flow
m, c_i, c_E, c_U	Mixer, reactor-stage, overflow, and underflow concentrations
g_E, g_U	Flow-weighted clarifier outlet component responses
χ, M_{cl}	Joint plant response and clarifier solids inventory
D_χ	Positive scaling matrix for the plant response
J	Dimensionless engineering objective

Symbols specific to a formulation are defined with their equations. A numerical

coordinate is interpreted using the unit of its corresponding component. The plant projection uses u locally for its standardized response projection.

Chapter 1

Introduction and Research Purpose

1.1 Background and rationale

1.1.1 Wastewater treatment as an engineering need

Wastewater from communities and industries carries organic matter, nutrients, and suspended particles that must be removed or transformed before the water is discharged or reused. Treatment requirements depend on the composition of the incoming water, called influent, while the quality achieved is assessed in the outgoing water, called effluent. The varied sources and environmental effects of pollutants described by Madhav et al. (2020), together with the different health consequences examined by Lin et al. (2022), show why this assessment requires more than a single measure of water quality. Engineers need to understand which materials remain in the effluent and how treatment changes their amounts and forms.

In the Philippines, these treatment needs are closely tied to the conditions under which infrastructure is financed, managed, and operated. Tabios III (2020) discusses the physical setting and governance of Philippine water resources, while Damalerio et al. (2022) identify technical, financial, social, and institutional barriers to wastewater management in developing countries and relate them to Philippine technology selection. This context motivates methods that help engineers compare treatment alternatives transparently. Faster calculations can support the technical assessment used in planning and operation, although infrastructure and governance barriers also require responses beyond process modeling.

A useful treatment assessment must therefore consider the resources available to the community or operator alongside the required water quality. The guide edited by Ulrich et al. (2009) connects decentralized treatment requirements with local implementation conditions, and Jafarinejad (2020) considers energy efficiency, cost, reliability, and sustainability within the treatment-design problem. Together, these perspectives place engineering value in the balance between treatment performance and resource use. Prediction accuracy contributes to that assessment, but cannot establish the value of a

treatment choice on its own.

Biological wastewater treatment uses microorganisms to transform organic matter and nutrients. In activated sludge treatment, microorganisms grow as suspended biomass in tanks containing wastewater. A secondary clarifier separates much of this biomass from the treated water by settling. Part of the settled sludge returns to the biological tanks, and another part is withdrawn to control solids accumulation (Henze et al., 2008; Takács et al., 1991).

Within the biological tanks, microorganisms use organic substrate for energy or growth, either oxidizing it or incorporating it into biomass. Nutrients also change form as treatment proceeds. Nitrogen moves among ammonium, nitrite, nitrate, biomass, and nitrogen gas through pathways that include nitrification, which oxidizes ammonium to nitrite and nitrate, and denitrification, which converts oxidized nitrogen to nitrogen gas under suitable conditions. Phosphorus moves among dissolved, stored, and particulate forms. The nitrogen pathways described by Duan et al. (2022) and the connections between nitrogen and phosphorus removal explained by Wentzel et al. (1992) show why these transformations must be considered together. Tracking only one material quantity would leave part of the treatment response unexplained.

The activated sludge model (ASM) family represents these transformations through component balances and process rates (Henze et al., 1987, 1999, 2000). Each component is a material category, such as dissolved ammonium, soluble organic substrate, or a biomass group, and may combine chemical species that share a process role. Their concentrations at a specified location form the component state. Process rates describe how quickly biological or chemical transformations consume or produce these components, allowing the model to follow the changing composition of the wastewater (Henze et al., 2000).

Engineers often report water quality through quantities that combine selected components. Chemical oxygen demand (COD) expresses the oxygen-demand equivalents associated with oxidizable material, while total nitrogen (TN) and total phosphorus (TP) summarize their respective nutrients on the adopted reporting basis. Total suspended solids (TSS) measures suspended particulate material (Henze et al., 2000, 2008). These combined quantities, called composites in this dissertation, provide useful summaries but do not identify a unique component state. Two effluents with the same TN, for example, can contain different proportions of ammonium and nitrate. A model that reproduces the total may therefore still misrepresent the treatment condition that produces it.

Operating choices influence both this composition and the movement of material through the plant. Hydraulic retention time (HRT), calculated as reactor volume divided by the flow rate specified for that reactor, describes the nominal liquid residence time. Aeration supplies oxygen for biological reactions, recycle returns liquid or settled sludge to an earlier treatment location, and wasting withdraws sludge to control the

solids retained in the plant (Henze et al., 2008). Because these choices affect several components and locations together, comparing candidate settings requires a view of the full treatment response as well as the resulting effluent quality.

1.1.2 From process modeling to repeated operating calculations

Mechanistic models provide this view by calculating component concentrations from equations for biological conversion, chemical reactions, liquid flow, and solids settling. Given influent concentrations and operating settings, a steady-state calculation describes conditions at which the modeled inventories, or amounts of material held within treatment units, no longer change with time. The steady-state analysis established by Marais and Ekama (1976) and the process knowledge and limitations reviewed by Hauduc et al. (2013) provide the basis for this approach. Its predictions remain conditional on the adopted component definitions, parameter values, and system boundary, which specifies the units and material streams included in the balance.

By calculating responses at conditions that have not been directly observed, simulation allows engineers to examine alternatives before choosing a design or operating setting. Model validation for an activated sludge process is addressed by Calise et al. (2020), while Rivas et al. (2008) connect model-based calculations with treatment-plant design. Using a model for such decisions requires more than obtaining a numerical response. Each candidate setting must be evaluated, its treatment response interpreted, and its feasibility assessed against the intended engineering requirements.

These comparisons can be made through repeated simulation and engineering judgment or through numerical optimization, which searches for settings that improve a stated objective while satisfying specified limits. For activated sludge operation, that objective can combine effluent quality and resource use, requiring repeated treatment calculations as the search proceeds. A statistical surrogate reduces the cost of these evaluations by approximating the responses of a more detailed model (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019). Here, it predicts component concentrations from influent conditions and operating settings. Whether evaluated with a surrogate or a mechanistic model, a feasible setting satisfies the imposed limits but may not be the best available choice. Nonlinear reactions and recycle relationships can produce several locally favorable settings, so the termination of a numerical search does not establish the best setting across the entire allowed domain (Buzzi-Ferraris & Manenti, 2013; de Araujo et al., 2013).

Aeration illustrates the engineering trade-off behind these repeated calculations. Changing oxygen supply can alter substrate removal and nitrification while also changing resource demand. The operating sensitivities examined by de Araujo et al. (2013) and the aeration optimization and control strategies reviewed by Gu et al. (2023) place these

effects within a broader treatment assessment. A preferred aeration setting consequently depends on the joint treatment response and the declared engineering priorities, rather than on the aeration variable or a single effluent quantity alone.

To approximate that response, statistical surrogates are trained on paired influent and operating inputs and their corresponding mechanistic outputs (Fang et al., 2011; Selerio, 2026). Training fits the model parameters to these pairs, with each sample in this research representing an accepted steady-state simulation rather than a measurement from an operating plant. The use of surrogates in wastewater resource-recovery optimization by Durkin et al. (2024) and the biological wastewater applications of machine learning reviewed by Sundui et al. (2021) motivate this route to faster calculations. For activated sludge analysis, however, computational savings must be accompanied by checks on the physical meaning of the predicted component state.

1.1.3 Why accurate prediction is insufficient

A prediction can be close to a mechanistic reference while violating a physical requirement. Consider a hypothetical effluent ammonium concentration of 0.2 g N m^{-3} and a prediction of -0.1 g N m^{-3} . Its absolute error is only 0.3 g N m^{-3} , but a negative ammonium concentration is physically invalid. Larger substrate or biomass concentrations can dominate an average error score and conceal this problem. A small average error can also conceal a violation of mass conservation distributed across several components (Hansen et al., 2024; Kircher & Votsmeier, 2025).

Physical assessment must therefore check mass conservation and non-negativity separately on the same predicted effluent. Mass conservation requires a consistent account of the material entering, leaving, accumulating, and reacting within the stated system boundary. The reaction accounting comes from stoichiometry, which specifies the relative amounts of components consumed and produced. A weighted combination of components that these reactions leave unchanged is a stoichiometric invariant (Henze et al., 2000; Sturm & Wexler, 2020, 2022). Such invariants provide a basis for balance checks, but their use in a reactor also requires the relevant boundary inputs and outputs to be included (Takács & Vanrolleghem, 2006).

Non-negativity imposes a complementary requirement that every concentration be zero or greater. These two requirements must be enforced together because correcting one defect can leave or create another. Replacing a negative ammonium prediction with zero can disturb the nitrogen balance, while imposing a balance alone can still leave individual concentrations negative (Harder et al., 2022; Kircher & Votsmeier, 2025).

Projection provides a way to make this joint adjustment after a fitted model has produced its component prediction. It moves the raw prediction into a feasible set of concentration states that satisfy specified constraints, using a criterion that measures the change from the original prediction. The resulting state is called the projected prediction. In this research, the constraints include mass conservation and non-negativity, following

their joint enforcement in reactive-chemistry and activated sludge surrogates (Kircher & Votsmeier, 2025; Selerio, 2026). The projection developed by Kircher and Votsmeier (2025) for mechanistic kinetics surrogates can be applied to different fitted models. Its guarantee is tied to the constraints imposed, so an adjusted ammonium, substrate, or biomass concentration need not reproduce every reaction rate or have a smaller prediction error (Selerio, 2026; Sturm & Silva, 2025).

Predictive agreement requires its own assessment against reference responses excluded from the fit being evaluated. Internal validation supports model selection, while a held-out assessment set evaluates the selected procedure (Hastie et al., 2009; Kaufman et al., 2012). Because the reference responses in this research come from activated sludge simulations, these checks establish agreement with the adopted mechanistic model. Agreement with a measured plant remains a separate question that requires field evidence (Machado et al., 2014; Newhart et al., 2019).

A credible assessment must also reflect the information available when a prediction is requested. Influent concentrations and selected aeration settings are known before the treatment response is calculated, whereas the resulting effluent ammonium concentration is still unknown and cannot be supplied as an ordinary predictive input. Allowing unavailable outputs or reserved assessment information to influence fitting or model selection introduces information leakage (Kaufman et al., 2012; Newhart et al., 2019). Keeping this information separate protects the meaning of the assessment, although extrapolation beyond the sampled input range still needs to be examined in its own right.

Interpretability adds another requirement by asking whether the dependence of a prediction on its inputs can be examined. A structured regression makes fitted terms for operation, influent composition, curvature, and component coupling explicit (Selerio, 2026), supporting inspection of the relationships represented by the model (Brunton et al., 2016; Rudin, 2019). Curvature allows a response to change at different rates as an input changes. Interaction terms allow an aeration response to depend on influent loading, or the material supplied through incoming concentrations and flow. Component coupling represents fitted dependence among outputs such as substrate and biomass concentrations. Together, these terms describe the statistical response, although they need not uniquely identify the biological mechanisms responsible for it.

This interpretation must continue through physical enforcement because projection can change both a concentration and its sensitivity to an operating input (Selerio, 2026). Understanding the returned effluent therefore requires examining how the raw regression response is altered by projection, as well as the relationships expressed by the fitted equation.

Even a well-assessed predictor introduces a further question when it is used for optimization. A search may select an aeration or recycle setting where predicted effluent quality is favorable but the surrogate is inaccurate, despite a small average held-out

error. The methods for checking and managing model approximations developed by Alexandrov et al. (1998) and the comparison with reference process calculations by Pedrozo et al. (2025) support independent mechanistic evaluation at the selected controls. Mass conservation and non-negativity establish necessary properties of the component prediction, while the treatment response and resource demand calculated at the chosen setting determine the quality of the operating decision.

1.1.4 Rationale for the research

These concerns motivate a framework that makes fast treatment predictions understandable, physically checked, and usable as complete component states. Methods for statistical approximation, projection, and process optimization are available in the literature (Bhosekar & Ierapetritou, 2018; Durkin et al., 2024; Kircher & Votsmeier, 2025), but their engineering use requires a clear account of what each method establishes. Prediction error measures agreement with a reference, mass conservation and non-negativity test specified physical requirements, interpretation examines the basis of the returned response, and decision assessment evaluates the selected operating choice with the reference model. Connecting these methods allows their evidence to be considered together without treating the different properties as interchangeable.

The need for this connection becomes more demanding as the analysis moves from a single reactor to a plant. Reactor projection can impose mass conservation and non-negativity on a complete component prediction (Selerio, 2026). In a connected plant, the returned response must also reconcile mixing, reactor, clarifier, recycle, and inventory relationships (Alex et al., 2008; Jeppsson & Diehl, 1996). A state that is physically acceptable within one unit can still disagree with the material passed to another, making consistency across the unit connections an additional requirement.

The research follows this progression through three linked capabilities built on a common mechanistic foundation. It first examines statistical approximation and projection of the complete reactor component response. It then develops an interpretable second-order regression and a defined procedure for projecting its predictions. Finally, it extends the analysis to a connected plant surrogate used to examine operating choices and independently evaluate selected controls. This progression connects component prediction and physical constraints with interpretation and the engineering decisions made from the resulting response.

The intended benefit is a more transparent basis for treatment analysis. Researchers can distinguish the errors of the fitted predictor from the changes introduced by projection, while engineers can inspect the complete returned state and check a selected operating setting with the original process model. Projection establishes mass conservation and non-negativity under its stated assumptions (Kircher & Votsmeier, 2025; Selerio, 2026). Assessing accuracy, computational cost, and decision quality alongside that guarantee shows what the surrogate can support and where further evaluation is

needed.

1.2 Statement of the problem

The central problem is how to use machine learning surrogates for activated sludge analysis when accurate approximation alone does not establish a usable engineering response. A complete framework must address four connected limitations concerning the prediction target, physical requirements, interpretation, and plant decisions.

1.2.1 Uncertain agreement with the complete component response

Effluent TN can be predicted accurately even when the underlying ammonium, nitrite, nitrate, and biomass concentrations are poorly represented. Errors can compensate within the total, and discrepancies in small concentrations can be hidden by an aggregate score (Henze et al., 2000; Selerio, 2026). Agreement also depends on how many reactor cases are available for training and whether the influent and operating inputs lie within the sampled ranges (Li et al., 2024; Viering & Loog, 2023). A shared assessment of the complete component response is therefore needed to distinguish the effects of learner choice, data availability, and input domain while keeping the prediction target fixed.

1.2.2 Mass conservation and non-negativity are not established by training alone

Ordinary regression fits predictions by minimizing error, without requiring every new activated sludge effluent to satisfy mass conservation and concentration bounds. Physical penalties can add a fitting cost for a balance residual, which is the numerical discrepancy in a balance equation. A finite penalty may reduce that discrepancy without eliminating violations at every new influent and operating setting (Hansen et al., 2024; Raissi et al., 2019). Projection followed by independent numerical checks offers a way to enforce and assess the two physical requirements on the component state actually returned by the model (Kircher & Votsmeier, 2025; Selerio, 2026).

1.2.3 A visible regression structure can still be difficult to interpret

An explicit regression equation makes its fitted coefficients visible, but their meaning depends on their units and their role in the complete calculation. A coefficient is the fitted weight of an input or input product, so a term multiplying aeration and influent substrate describes a different relationship from one multiplying aeration alone. Learned

component coupling and projection can then alter the ammonium or biomass response produced by those terms (Selerio, 2026). Interpretation must follow these stages to explain the final prediction, and the fitting and validation that support it must use only information available for the intended prediction task (Kaufman et al., 2012).

1.2.4 Unit predictions do not establish the quality of plant decisions

At plant scale, separately predicted unit responses can disagree about the material passed between them. A predicted clarifier underflow may imply a return-sludge contribution that differs from the one represented at the mixer, even though consistent connections are required by the component-transport and plant models (Alex et al., 2008; Jeppsson & Diehl, 1996). Joint projection can reconcile specified plant balances, but satisfaction of these relations leaves full agreement with the mechanistic equations outside the projection guarantee (Kircher & Votsmeier, 2025; Selerio, 2026). An operating search can consequently select controls affected by remaining approximation errors. Independent mechanistic evaluation of the selected aeration, recycle, and wasting settings, under the same treatment and resource priorities, is needed to assess the decision itself. This follows the distinction between approximation quality and optimization behavior developed in surrogate optimization research (Alexandrov et al., 1998; Pedrozo et al., 2025).

1.3 Research questions and objectives

The general objective is to develop a unified framework that connects activated sludge surrogate prediction, projection, interpretation, and operating assessment. Within this framework, mass conservation and non-negativity are explicit requirements, while predictive error and decision quality remain separate assessment endpoints. Four research questions guide its development.

The first question concerns how accurately different statistical learners approximate the complete reactor component response as training data availability changes and inputs move beyond the sampled domain. The corresponding objective is to establish a benchmark, or defined simulation task for model comparison, with shared component targets, data partitions, and evaluation scales. This provides a consistent basis for examining how learning performance changes across those conditions.

The second question concerns what projection changes when it is applied to the same raw predictions. Its objective is a paired assessment of mass conservation, non-negativity, prediction error, displacement, and computational cost before and after projection. Keeping the fitted learner fixed allows the effects of physical enforcement to be distinguished from differences between prediction models.

The third question concerns how an explicit second-order regression with learned component coupling can provide interpretable activated sludge predictions that satisfy mass conservation and non-negativity. Second-order terms represent individual inputs, their squares, and pairwise products, forming a fitted driver that is evaluated before component coupling. The objective is to formulate and examine this model, its fitting problem, and its projection procedure, tracing interpretation from the driver through the raw component prediction to the final projected effluent. Chapter 5 develops the detailed formulations.

The fourth question concerns how joint prediction and projection can support operating choices for a connected activated sludge plant. Its objective is to impose the declared unit connections, formulate common engineering priorities, and independently evaluate the controls selected by an operating search. Comparing surrogate-assisted search with direct mechanistic optimization on the original reference basis connects the predicted plant response with the quality of the resulting decisions.

1.4 Significance of the study

1.4.1 Scientific significance

The scientific value of the study lies in connecting predicted activated sludge responses with the component accounting of the mechanistic model. Reaction stoichiometry defines relative changes among components, while reactor and plant boundaries identify the material streams included in mass conservation. Non-negativity applies to every modeled concentration, including dissolved nutrients and biomass. Making these requirements explicit allows the physical meaning of a predicted treatment state to be examined alongside its agreement with the reference response.

This connection also makes the limits of a physical guarantee clearer. A state can conserve mass and contain no negative concentrations while still failing to satisfy the full nonlinear kinetic equations. Similarly, a reported nutrient total can differ from the inventory used in mass conservation. Keeping these distinctions explicit supports a more precise interpretation of scientific machine learning in biological wastewater treatment (Henze et al., 2000; Takács & Vanrolleghem, 2006).

1.4.2 Methodological significance

The study provides a comparison design that distinguishes predictive performance from the effects of projection. Within each assessment, models receive common influent and operating inputs and predict the same component concentrations. Preprocessing transforms the data into fitting coordinates, while tuning uses internal validation to choose hyperparameters that control model structure or fitting, rather than the coefficients estimated during training. Both steps use only the relevant training information

to avoid information leakage (Kaufman et al., 2012). Paired raw and projected effluents then reveal the changes attributable to physical enforcement. The reactor and plant assessments retain their own validation designs, allowing them to answer related questions without implying a single model ranking across the studies.

Assessment combines errors in individual components and reported water quality quantities with mass conservation residuals, negative concentrations, projection displacement, and computational effort. Displacement measures how much the concentration vector changes during enforcement, revealing whether a prediction needs substantial adjustment to satisfy the imposed balances. Reporting these measures together makes visible the limitations of an average effluent error score and distinguishes predictive agreement from physical compliance, as illustrated by component-level projection and activated sludge assessments (Selerio, 2026; Sturm & Silva, 2025).

The structured regression adds a model whose fitted relationships can be inspected directly. Named terms and learned component coupling support coefficient analysis and sensitivity calculations, where sensitivity measures the local change in a predicted concentration as an input such as aeration changes. Examining projection as a further stage of the effluent calculation carries this interpretation through to the returned response. The resulting analysis explains the fitted treatment relationships without assigning causal biological meanings to the coefficients (Brunton et al., 2016; Selerio, 2026).

1.4.3 Engineering and practical significance

For engineers comparing operating conditions, a useful plant prediction must maintain a consistent account of material across connected treatment units (Alex et al., 2008; Henze et al., 2008). The connected surrogate predicts mixer, reactor, clarifier-outlet, and inventory responses together, and joint projection reconciles specified relations that separate unit predictions can leave inconsistent. Numerical checks then identify mass conservation and non-negativity violations before the response is used in an operating search, providing a clearer basis for evaluating candidate settings.

Independent mechanistic evaluation extends this assessment to the decisions actually selected, following the use of reference-model checks in surrogate optimization (Alexandrov et al., 1998; Pedrozo et al., 2025). It calculates treatment responses at the chosen controls, checks the original engineering requirements, and distinguishes available decisions from failed or ineligible cases. This makes it possible to assess whether a faster search supports the intended operating choice. Claims of lower operating cost or better field treatment still require evaluation of those outcomes.

This transparent and computationally manageable assessment can also support technical capacity in wastewater planning. Philippine and other resource-limited settings need decisions that connect treatment requirements with the resources and conditions available for implementation (Damalerio et al., 2022; Tabios III, 2020; Ulrich

et al., 2009). The framework contributes a way to examine those technical choices, while practical application continues to depend on plant calibration, reliable measurements, operating expertise, and institutional capacity.

1.4.4 Significance for further research

The formulations provide a defined starting point for research on uncertainty, dynamic activated sludge operation, and other biological treatment configurations. Their value lies in keeping physical constraints, statistical agreement, interpretation, and decision evidence distinct as the application changes. Each extension must define its own components and boundary streams. A dynamic treatment model, for example, must account for changing reactor and clarifier inventories (Alex et al., 2008; Takács et al., 1991), so using the same learning method does not by itself carry the steady-state physical guarantees into dynamic operation.

1.5 Contributions of the research

The first contribution is a common reactor assessment that retains the complete component prediction target while examining changes in data availability and input domain. Reporting component and composite scores gives each its appropriate role, making discrepancies in the underlying state visible even when a reported total is favorable.

The second contribution is a paired assessment of projection for mass conservation and non-negativity across different learners. Following the projection principles of Kircher and Votsmeier (2025), it applies common physical requirements while evaluating mathematical feasibility, displacement, prediction error, and computational cost separately. The contribution lies in organizing these methods and measures for activated sludge analysis, with the underlying projection method drawn from the existing literature.

The third contribution is an interpretable second-order regression that combines explicit feature groups, learned component coupling, and checked projection. The feature groups make the activated sludge regression available for structural inspection (Selerio, 2026), while the fitting formulation distinguishes soft physical penalties from the projection that imposes mass conservation and non-negativity on the returned state. Tracing the complete calculation connects the fitted terms with the response ultimately used for analysis.

The fourth contribution extends this reasoning to a connected plant through joint projection and independent operating verification. The formulation links unit responses through the declared mixing, reaction-inventory, clarification, and stored-solids relations. Evaluating selected controls with the original mechanistic equations and a common

objective then connects the physical meaning of the surrogate state with the quality of the operating decision made from it.

Table 1.1 links each research problem to its methodological contribution and the evidence used to assess it. These assessment measures describe how the contributions are examined, with empirical outcomes reported in the corresponding results chapters.

Table 1.1: Research problems, methodological contributions, and assessment evidence.

Research problem	Contribution	Assessment evidence
Complete response is hidden by a few reported totals	Common component-level reactor benchmark	Held-out component and composite errors, learning curves, and beyond-domain assessment
Training does not establish mass conservation and non-negativity	Common projection with paired raw and projected assessment	Equality residuals, concentration checks, error differences, displacement, and cost
Model terms do not directly explain the final response	Explicit second-order regression and component coupling with checked projection	Named coefficient groups, full response sensitivities, and stage-specific diagnostics
Consistent predictions do not establish useful operating decisions	Joint plant projection and original-model verification of selected controls	Unit and inventory checks, common reference objectives, engineering eligibility, and failure accounting

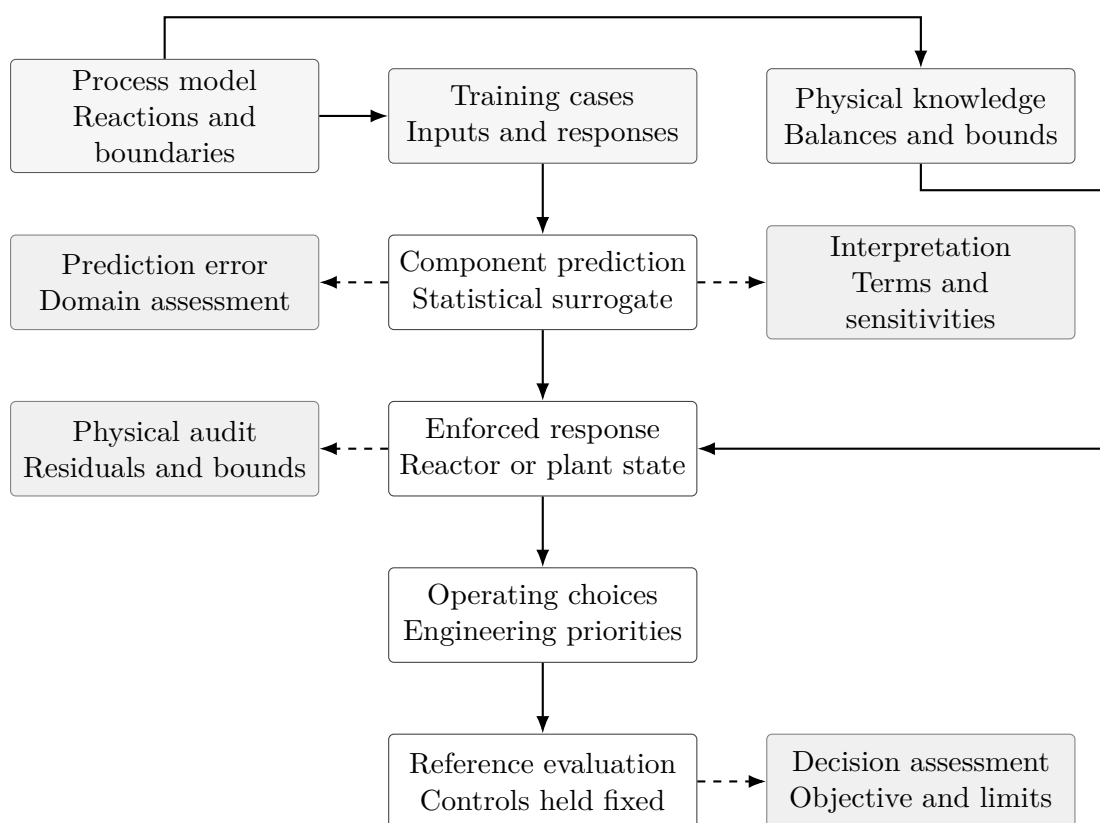
Together, these mathematical and methodological contributions provide a coherent way to examine the stated limitations. Their value is assessed through the evidence outlined above, with comparative performance established by the empirical results rather than presumed from the model formulation.

1.6 Research framework

The framework begins with a mechanistic activated sludge model evaluated over specified influent concentrations and operating settings, which together define the input domain. Each accepted simulation pairs these known inputs with steady-state component concentrations to provide the data for statistical fitting. Reaction stoichiometry and boundary flows supply the mass conservation relations (Henze et al., 2000; Sturm & Wexler, 2022), while non-negativity sets the concentration requirement. Projection then adjusts the raw effluent prediction to satisfy both, following the joint enforcement approaches of Kircher and Votsmeier (2025) and Selerio (2026).

The same logic connects prediction with interpretation and operating analysis. A structured predictor exposes fitted terms and sensitivities, and the connected plant formulation adds relations among treatment units and stored solids. An operating search uses the resulting plant response to select controls, which are independently evaluated with the mechanistic model. Each stage provides evidence for a different aspect of the framework, so a successful physical check cannot substitute for a prediction assessment or a reference evaluation of the selected decision.

Figure 1.1 shows how process knowledge and training data lead to component prediction, projection, operating choice, and reference evaluation. The accompanying assessment boxes identify the evidence needed to examine prediction error, physical compliance, interpretation, and decision quality throughout this sequence.



Solid arrows connect modeling steps. Dashed arrows identify assessment.

Figure 1.1: Conceptual research logic linking process knowledge, component prediction, projection, interpretation, and independent assessment of operating choices.

The mechanistic component concentrations serve as supervised targets paired with known influent and operating inputs. Predicted ammonium, substrate, biomass, and other concentrations are compared with these targets to assess error, while separate checks assess mass conservation, non-negativity, and the additional plant constraints. Structural inspection examines the fitted treatment relationships, and decision assessment evaluates the mechanistic response at the selected controls. Both the reactor and

connected plant follow this logic, with their predictors and physical boundaries specified separately.

1.7 Scope and limitations

The study evaluates simulated steady-state responses of a standalone activated sludge reactor and a connected plant with biological reactors, secondary clarification, and recycle streams. Reference responses come from mechanistic equations with fixed process definitions and parameter values. The reactor and plant assessments each use a separately specified operating domain, so their findings are interpreted within the conditions examined.

The represented state contains organic, nitrogen, phosphorus, oxygen, alkalinity, biomass, and solids coordinates on the adopted model basis. Mass conservation is imposed within that basis and the stated system boundary, and non-negativity applies to the represented concentrations. These requirements define the extent of the physical guarantee. They do not establish complete elemental accounting for unrepresented species, full kinetic feasibility of every projected state, or dynamic stability beyond the specified checks (Selerio, 2026; Takács & Vanrolleghem, 2006).

Predictive agreement in this study is agreement with the selected mechanistic reference under simulated conditions. Establishing performance at an operating treatment plant would require field evidence addressing calibration uncertainty, measurement error, microbial adaptation, and unrepresented reactions (Machado et al., 2014; Newhart et al., 2019). Reliable calibration and process understanding therefore remain essential for any practical extension of the framework (Meijer, 2004; Petersen et al., 2003).

Exploratory simulation informed the design of the connected plant assessment, so its reserved holdout cases provide descriptive evidence conditional on those design choices. This is termed post-selection assessment. Although fitting and tuning exclude the holdout responses, that separation does not make the assessment an independent external confirmation. Mechanistic evaluation supplies a reference check at the selected controls, while the statistical limitation associated with the assessment design remains.

Attached-growth treatment, spatial transport, resource-recovery networks, and dynamic control inform the background and possible extensions of the work, but are outside the evaluated case studies. The connected operating problem is confined to the declared plant topology and local numerical assessment. Its findings consequently do not certify a global optimum or establish a new network configuration.

Numerical examples and conceptual figures are explicitly hypothetical and explain the mathematical operations. Empirical treatment, prediction, optimization, and timing findings are presented separately in the reactor projection, structured-regression, and connected-plant assessments.

Chapter 2

Review of Related Literature

Activated sludge surrogate modeling draws on several bodies of research that address different parts of the same engineering problem. Mechanistic models define the treatment processes, component transformations, and material balances that a surrogate must approximate. Data-driven studies show how treatment quantities can be estimated from measurements or simulations. Physical enforcement methods constrain predictions to satisfy specified balances and concentration bounds. Interpretable modeling provides ways to examine the fitted relationships, while optimization studies determine whether those predictions can support useful operating decisions. This chapter reviews these contributions as a connected body of work and identifies the questions that remain when they are brought together for complete-component activated sludge analysis.

2.1 Mechanistic foundations of activated sludge modeling

2.1.1 Development of component and nutrient models

Activated sludge modeling developed from the need to relate operating conditions to organic matter removal, oxygen demand, biomass growth, and sludge production. Early formulations established the basic connection between biological transformation and material accounting. Marais and Ekama (1976) formulated steady-state activated sludge analysis around substrate utilization and biomass, while Dold et al. (1981) developed a more general model that linked several biological processes within a common treatment calculation. Together, these contributions established the basis for calculating treatment responses from material balances and biological rates rather than treating effluent quality as an isolated empirical outcome.

This framework was expanded as activated sludge models became more component-resolved. Henze et al. (1987) presented the single-sludge framework associated with Activated Sludge Model No. 1 (ASM1), which distinguishes substrate, biomass, and

nitrogen components so that carbon oxidation, nitrification, and denitrification can be calculated together. Later models addressed processes that this formulation did not represent explicitly. Henze et al. (1999) extended phosphorus-removal modeling to include the denitrifying activity of phosphorus-accumulating organisms, while Gujer et al. (1999) introduced Activated Sludge Model No. 3 (ASM3), which represents organic substrate storage and uses endogenous respiration to describe biomass decay.

These developments addressed different biological modeling needs. The phosphorus-removal extension connects nitrate use with phosphate behavior, while the storage-based formulation separates substrate uptake from its later use by biomass. The collected formulations in Henze et al. (2000) provide a common reference for these component definitions and their associated reaction terms. For surrogate modeling, their importance lies in defining what must be predicted. Effluent COD or TN can summarize treatment quality, but the underlying component concentrations describe the material forms that produce those totals. A surrogate intended to preserve process accounting therefore requires more than agreement with a small set of aggregate indicators.

The comparison of seven activated sludge models by Hauduc et al. (2013) further shows why the mechanistic reference must be specified before statistical models are compared. Their review examined how the models represent hydrolysis, fermentation, microbial growth and decay, and biological storage. The differences were not merely mathematical; they reflected different assumptions about treatment processes and available process knowledge. Model selection therefore depends on the treatment system and the question being studied. For a surrogate comparison, this means that all candidate predictors must approximate the same selected process model. Otherwise, differences in prediction may reflect different biological assumptions rather than differences among statistical learners.

2.1.2 Biological dependence and calibration uncertainty

The mechanistic literature also shows why operating conditions and influent composition cannot be interpreted independently. Nutrient-removal processes depend on the material entering the system and on how operating choices alter the conditions under which microorganisms compete and react. Wentzel et al. (1992) examined interactions among nitrification, denitrification, and excess phosphorus removal. Guerrero et al. (2011) studied competition for organic substrate in a pilot system designed for simultaneous nitrogen and phosphorus removal and found that the supplied carbon source affected the ability to maintain phosphorus removal when nitrate entered the initially anaerobic zone. Their results illustrate a specific biological dependence among substrate composition, recycle conditions, and nutrient removal.

For statistical modeling, these findings support retaining both operating conditions and component-resolved influent information. A model based only on total organic loading can overlook differences among carbon sources that perform different biological

roles. Likewise, a model that treats aeration or recycle independently of loading can miss the conditions under which those operating changes influence nutrient removal. This provides a process-based reason to examine interaction terms in a treatment surrogate. It does not, however, imply that a fitted interaction coefficient identifies a microbial mechanism by itself. The statistical relationship and the biological mechanism remain distinct forms of evidence.

Calibration research addresses a related problem: even a mechanistically detailed model requires evidence that its parameters and internal quantities represent the treatment system of interest. Petersen et al. (2003) reviewed experimental designs for estimating activated sludge parameters, while Vanrolleghem et al. (1999) examined the use of respirometry, which measures oxygen consumption, to estimate model parameters and component combinations. These methods show that the information supplied by an experiment must correspond to the parameter or internal quantity being estimated.

The full-scale calibration study of Machado et al. (2014) makes this issue more explicit. They calibrated a phosphorus-removal model using data from a treatment plant in Manresa, Spain, assessed parameter confidence intervals, and examined how influent and operating variables affected the model fit. They also varied these inputs while retaining default kinetic and stoichiometric parameters to evaluate output sensitivity. Their analysis therefore considered uncertainty in both process inputs and biological parameters. For surrogate modeling, this distinction is important because simulated training targets inherit the assumptions of the mechanistic model that generates them. Agreement with those targets establishes how well the surrogate approximates that reference. Agreement with an operating treatment plant requires the additional evidence supplied by calibration and measurement.

The mechanistic literature therefore establishes the component definitions, reaction relationships, and calibration practices needed to give a surrogate prediction physical meaning. The unresolved question for this dissertation is not how to replace that mechanistic foundation, but how to approximate its defined treatment responses efficiently while retaining their material accounting. The surrogate is evaluated against an adopted mechanistic reference; calibration of that reference to a particular operating plant remains a separate task.

2.1.3 Clarification and connected plant balances

Biological reactors alone do not determine the final treatment response. The material leaving the reactor is subsequently separated, recycled, wasted, or discharged, so clarification and solids storage become part of the plant-level material account. Takács et al. (1991) developed a clarification and thickening model based on solids fluxes and balances across layers of a one-dimensional settler. The model calculates solids profiles and overflow and underflow concentrations under steady and changing conditions. Härtel and Pöpel (1992) likewise modeled secondary clarification with

sludge thickening. These studies provide mechanistic descriptions of how suspended material is retained, concentrated, recycled, and discharged.

The difficulty of connecting biological and clarification models was examined directly by Jeppsson and Diehl (1996). They studied how biological components propagate through a secondary clarifier and identified problems that arise because the biological reactor and settler models represent particulate material differently. Their evaluation of component-transport algorithms focused particularly on the composition of sludge returned to the reactor. This result is important for surrogate modeling because it shows that predicting aggregate solids alone is not sufficient when the recycled stream must retain a biologically meaningful component composition.

Alex et al. (2008) addressed the broader need for comparable whole-plant assessment through Benchmark Simulation Model No. 1. The benchmark connects five biological compartments, a secondary settler, internal recycle, return sludge, and wasting, and specifies influent conditions, control settings, and assessment criteria. Its contribution is not only a plant configuration but a common setting in which alternative operating strategies can be compared. It makes explicit that biological conversion, solids separation, recirculation, and wasting form one connected treatment response.

Together, these studies define the relationships that a connected surrogate must preserve. A clarifier prediction affects both the discharged effluent and the recycled biomass returned to the biological tanks. Stored solids contribute to the inventory used in solids-retention calculations. Shared streams carry material from one unit to another and must therefore agree across unit boundaries. A single-reactor surrogate captures only part of this system. Extending physical enforcement to the plant level requires consistent component flows at these connections together with an explicit account of retained solids.

2.2 Data-driven prediction of wastewater treatment responses

2.2.1 Soft sensors and reported treatment quantities

Data-driven wastewater treatment research developed partly because operating plants collect large sets of measurements that are difficult to interpret jointly and because some important treatment quantities are costly, delayed, or difficult to measure directly. Newhart et al. (2019) reviewed methods for fault detection, variable prediction, and advanced process control and identified missing observations, outliers, correlated variables, and changing process behavior as recurring features of plant data. Sundui et al. (2021) reviewed machine learning applications across biological wastewater treatment. Together, these reviews establish a range of useful prediction tasks and emphasize that the choice of method depends on both the available information and the intended

application.

Soft sensors are one example. A soft sensor estimates a quantity that is difficult or slow to measure from other observations that are easier to obtain. Ching et al. (2022) developed a soft sensor for five-day biochemical oxygen demand using extreme gradient boosting (XGBoost). They evaluated influent and effluent estimates at two treatment plants and used supporting measurements selected from variables such as COD, solids, nutrients, and simpler sensor readings. The model covered a wide range of oxygen-demand values and could accommodate missing supporting observations. Its contribution is therefore practical and specific: it provides a faster estimate of an otherwise slow water-quality measurement.

A similar distinction appears in effluent nutrient prediction. Y. Wang et al. (2025) compared a mechanistic activated sludge model with six machine learning models for effluent TN prediction using one year of high-resolution municipal plant data. The mechanistic model reproduced the direction of TN variation, while the learned models achieved greater predictive accuracy in the assessed dataset. Feature-attribution analysis was also used to identify influential inputs. This study shows that statistical models can improve a defined effluent-prediction task while still allowing the fitted response to be inspected.

Other data-driven studies address operational quantities beyond effluent quality. Ekinci et al. (2023) predicted waste-sludge production using 208 daily observations from an advanced biological treatment plant in Turkey. Their inputs included flow, polymer dosage, and the amounts of solids and pollutants removed. They compared learning methods and reported improved prediction after feature selection. This adds a solids-management target to the quality indicators considered elsewhere. At the same time, its use of measured removal quantities illustrates how the available inputs depend on the prediction task. A model used before a proposed operating condition has been evaluated would require inputs available at that earlier decision stage.

The common achievement of these studies is reliable estimation of selected treatment quantities from available plant information. Such predictions can support monitoring, diagnosis, and particular operating analyses. The requirement in this dissertation is broader. When the goal is to check mass conservation, non-negativity, and the distribution of material among biological forms, the complete modeled component state becomes the relevant prediction target. Accuracy for TN, oxygen demand, or sludge production supplies useful evidence for those particular quantities, but it does not establish the accuracy or physical consistency of every modeled component.

2.2.2 Component estimation and simulation-based surrogates

Component-estimation studies move closer to the information used by mechanistic activated sludge models. P. Wang et al. (2024) used a backpropagation neural network to estimate five COD-related component quantities from measured influent COD. They first

characterized those components in an operating plant and then evaluated the neural estimates against the measurements. Their reported prediction accuracy exceeded 80% for the five assessed components. The study addresses a practical difficulty in mechanistic modeling: repeated experimental fractionation of influent COD components is demanding, yet those fractions are needed as model inputs.

That contribution differs from predicting the effluent component state produced by treatment. Influent fractionation estimates how incoming organic material is distributed among model categories. An effluent surrogate instead estimates how a defined influent changes under specified biological and operating conditions. The two tasks can complement one another, but they have different targets and therefore require different evidence. The component estimates of P. Wang et al. (2024) improve characterization of the material entering the model; they do not by themselves establish the complete treatment mapping investigated in this dissertation.

Simulation-based surrogates address that mapping more directly. Fang et al. (2011) combined an extended activated sludge model with a support vector surrogate and a genetic optimization search for a full-scale nutrient-removal system. Mechanistic simulations supplied the input–output relationship learned by the surrogate. Their optimization indicated that anoxic volume and internal recycle could be reduced while retaining the specified discharge quality. The study is an early example of a workflow in which mechanistic simulation generates the training data, statistical learning accelerates response evaluation, and optimization uses the learned approximation to select operating conditions.

He et al. (2023) used a similar simulation-to-surrogate logic for greenhouse-gas emissions from papermaking wastewater treatment. They fitted a Kriging surrogate to steady-state process simulations and reported a coefficient of determination above 0.96. The surrogate then supported analysis and optimization of controllable treatment variables. This work addresses the computational cost of repeatedly evaluating an environmental objective. Its prediction target, however, is the emissions measure rather than the complete component state required for the material-balance assessment of this dissertation.

A closer precedent is provided by Selerio (2026), who addressed complete single-reactor component prediction using invariant-constrained second-order regression. The study used 10,000 simulated steady-state cases with 20 effluent components and compared the structured regression with nine statistical predictors. Its assessment included changes in training size, errors in reported water-quality quantities, and physical violation diagnostics. This establishes a direct component-level activated sludge comparison and provides the foundation for the physical-enforcement and interpretation questions developed later in this chapter.

Table 2.1 summarizes this progression from monitoring selected indicators to approximating complete process responses. The distinction is not that one task is inherently

better than another. Rather, each prediction target establishes a different kind of evidence. The table therefore identifies what additional information is needed when the objective shifts from estimating a reported quantity to evaluating a complete activated sludge component state.

Table 2.1: Treatment prediction studies and the scope of their evidence.

Study and task	Contribution established	Additional question for this dissertation
Ching et al. (2022) Oxygen-demand soft sensing	Estimation across two plants with support for missing sensor observations.	Can a predictor estimate the complete effluent state from influent and proposed operating inputs?
Y. Wang et al. (2025) Effluent TN prediction	Comparison of six learned models with a mechanistic model and inspection of influential inputs.	Are the individual nitrogen and other component concentrations accurate and physically consistent?
P. Wang et al. (2024) Influent component estimation	Neural estimation of five COD-related quantities used in process modeling.	How accurately is the resulting effluent component state predicted after treatment?
Fang et al. (2011) Simulation-based operating search	Mechanistic simulation combined with a support vector surrogate and optimization.	Can connected component predictions be constrained and checked at the selected controls?
Selerio (2026) Complete reactor prediction	Structured prediction of 20 components with physical checks and training-size assessment.	How do common projection and plant connections affect prediction and operating assessment?

2.2.3 Model comparison, data availability, and extrapolation

The wastewater applications reviewed above use statistical learners with different assumptions about how nonlinear input–output relationships should be represented. Random forests, developed by Breiman (2001), average predictions from randomized decision trees. Chen and Guestrin (2016) developed XGBoost, while Ke et al. (2017) introduced the light gradient boosting machine (LightGBM). Smola and Schölkopf (2004) described support vector regression, which combines regularized fitting with kernel-based similarity. These methodological studies provide the statistical basis for several learners now used in wastewater treatment. Their differences justify comparing multiple model families rather than assuming that one fitting structure is best for every treatment response.

Inspectable linear and nonlinear models provide another part of that comparison. Wold et al. (2001) described partial least squares regression, which represents related variation in inputs and outputs through latent components. These latent statistical

components are weighted combinations of variables and should not be confused with physical wastewater constituents. Woo et al. (2009) applied a kernel extension of partial least squares to estimate COD, TN, and cyanide concentrations in industrial cokes wastewater treatment. The nonlinear extension outperformed the assessed linear and other nonlinear partial least squares formulations in that application. This supports the value of nonlinear structure for the studied response while leaving ordinary partial least squares as a distinct baseline for other comparison tasks.

Broader tabular-learning research reinforces the value of comparing model families. Shwartz-Ziv and Armon (2022) examined tabular machine learning, while Borisov et al. (2024) surveyed neural approaches for tabular data. Activated sludge surrogate inputs have a similar tabular form, with each case containing operating settings and influent concentrations. However, these methodological comparisons cannot determine the preferred learner for a specific treatment problem. That ranking must come from a common treatment target, a common data design, and a common evaluation procedure. The general literature therefore guides comparator selection, while the activated sludge benchmark supplies the process-specific evidence.

The amount of available training data adds another dimension to the comparison. Viering and Loog (2023) reviewed learning curves and showed that their shapes depend on the learner, the dataset, and the prediction task. A learning curve relates model performance to the number of training observations and therefore provides evidence about data efficiency. Selerio (2026) applied this type of assessment to activated sludge reactor surrogates. The result characterizes performance within the stated input range, but a broader comparison still needs to determine how complete-state error and physical adjustment change together when the number of simulated training cases becomes small.

Such comparisons also depend on strict separation between model development and assessment information. Kaufman et al. (2012) defined information leakage and described how information unavailable at the intended prediction time can inadvertently enter model construction. In wastewater applications, this distinction is particularly important because downstream measurements may be available for monitoring but not for predicting a proposed operating condition before the treatment response has occurred. Scaling, preprocessing, and model tuning must likewise exclude observations reserved for final assessment. These requirements determine what an error score means; they do not imply that downstream measurements are inappropriate in studies whose stated purpose is monitoring.

The remaining comparative question therefore combines three issues: learner choice, training support, and operating domain. Existing indicator and component studies demonstrate that useful prediction is possible within their assessed conditions. What is still needed here is a common complete-state comparison in which several learners use the same inputs, outputs, data partitions, and error definitions. That assessment must

also distinguish interpolation within the sampled domain from prediction beyond it. Only then can component error and physical validity be evaluated together as influent and operating conditions change.

2.3 Methods for imposing physical constraints on predictions

2.3.1 Physical guidance during fitting

Machine learning models can reproduce observed responses while violating the equations or bounds that define a physically acceptable process state. A major branch of physics-informed learning addresses this problem by incorporating governing relations directly into model fitting. Karniadakis et al. (2021) reviewed approaches that combine physical knowledge with machine learning. One influential example is the residual-based formulation of Raissi et al. (2019), in which discrepancies in governing differential equations are added to the training objective. Equation disagreement is therefore penalized alongside prediction error, allowing process knowledge to guide learning even when response observations are limited.

Wastewater applications provide direct evidence for this idea. Li et al. (2024) examined physics-informed neural models for water-treatment unit dynamics, including an activated sludge reactor. They compared physically informed models with conventional learned models under limited data and changing loading conditions. The physically informed models showed stronger predictive capability and generalization in the assessed systems. Their results demonstrate that governing equations can provide useful learning information when data alone are insufficient.

The distinction between physical guidance and physical enforcement is important, however. A finite residual penalty balances equation agreement against the other terms in the fitting objective. It can reduce violations without requiring every new prediction to satisfy the relevant equality exactly. The contribution of Li et al. (2024) therefore concerns improved learning through physical information. If the engineering requirement is that every returned component state satisfy mass conservation and concentration bounds, an additional output construction, projection, or numerical check is still required. Physical guidance during fitting and exact sample-specific enforcement address complementary questions.

2.3.2 Conservation through model construction

A second approach is to build conservation directly into the way model outputs are constructed. Beucler et al. (2021) introduced neural architectures that encode analytic constraints and compared them with constraints imposed through a loss function. In

their climate-convection application, the architectural formulation enforced the selected conservation laws to machine precision without degrading overall predictive performance. Their results show that, when the output representation is designed appropriately, known equalities can be satisfied by construction rather than encouraged through a penalty.

Related work has used stoichiometric structure to couple predicted species. Sturm and Wexler (2020) developed a conservation framework in which learned transfers are converted into species changes through balanced process relationships. Sturm and Wexler (2022) later placed the stoichiometric mapping in a fixed neural layer for a photochemical surrogate. This construction avoided the need to reconstruct all reaction transfers as separate learning targets while retaining atom conservation in the predicted species changes. The common idea is that conservation is carried by the mapping between process changes and the final species response.

That idea transfers naturally to activated sludge modeling because stoichiometry also defines allowable combinations of component changes. However, conservation alone does not guarantee a physically acceptable concentration state. A balanced change can still consume more of a component than is available and thereby produce a negative concentration. The distinction follows from the difference between constraining a reaction-induced change and constraining the final state. The physical claim must therefore be matched to the quantity that the model construction actually restricts.

Harder et al. (2022) illustrated this broader requirement in an aerosol microphysics surrogate by examining both physical penalties and constrained output completion. Their assessment treated mass conservation and positive mass as separate but simultaneous concerns. The lesson for activated sludge prediction is direct: the complete returned state must be assessed for both balance satisfaction and concentration bounds after every stage of prediction has been applied.

2.3.3 Projection after statistical prediction

Projection provides another route to physical enforcement. Instead of changing the fitted learner, a prediction is first produced and then adjusted to satisfy specified constraints. Sturm and Silva (2025) developed a species-weighted least-squares correction for atom conservation in an atmospheric chemistry surrogate. Their weights accounted for the magnitude and uncertainty of individual species changes. In the reported assessment, an unweighted correction could damage prediction of a small radical, whereas weighting retained its accuracy and slightly improved overall agreement. The enforced conservation relation was unchanged; what changed was the way the required adjustment was distributed among species.

The same issue arises in activated sludge prediction because component concentrations can differ by several orders of magnitude. A correction that is numerically small relative to an abundant substrate may be large relative to a residual nutrient concentration. The results of Sturm and Silva (2025) therefore motivate assessing

projection size and component error together rather than treating physical correction as neutral with respect to prediction quality. Their method addresses an equality constraint, however, so an additional construction is needed when non-negativity must also be guaranteed.

Kircher and Votsmeier (2025) addressed this joint requirement through a positivity-preserving projection followed by linear interpolation backtracking. They evaluated the method on atmospheric chemistry, heterogeneous catalysis, and synthetic reaction networks. The corrected predictions satisfied atom balances and remained positive in the tested systems without reducing overall prediction accuracy. This establishes a general procedure for imposing both conservation and positivity across different underlying surrogates. The remaining question for activated sludge analysis is how such a procedure behaves with wastewater component scales, reactor boundaries, limited training data, and inputs that move beyond the sampled domain.

A direct activated sludge example is provided by Selerio (2026). The structured regression in that study used physical penalties during fitting, then checked each raw prediction and projected it onto the mass conservation equalities. If non-negativity still failed, a linear program—an optimization problem with a linear objective and linear constraints—was used to complete the correction. The published assessment reported no final mass conservation or non-negativity violations. Because the raw regression itself did not satisfy both requirements, the final enforcement procedure was the stage responsible for the physical guarantee.

That result also illustrates why physical validity and predictive accuracy must be evaluated separately. The structured surrogate was not the most accurate method under every reported score, even though its final states satisfied the declared constraints. Physical enforcement therefore establishes that the returned state belongs to the specified feasible set; it does not establish that the state is the closest possible approximation to the mechanistic reference. The next comparative question is consequently a paired one: when the same raw predictions are held fixed, how does projection change error, adjustment size, and computational cost across several activated sludge learners?

Figure 2.1 places the reviewed methods along the prediction pipeline. Fitting penalties guide learning toward equation agreement, constrained output constructions impose selected relations through the model architecture, and final-state projection corrects the prediction after it has been produced. These approaches can all introduce physical information, but they make different guarantees because they act at different stages.

2.3.4 Constraint scope and recent extensions

Recent work has expanded constrained learning beyond deterministic point predictions. Hansen et al. (2024) used integral conservation relations to adjust probabilistic predictions while accounting for both predicted means and uncertainty. Their framework

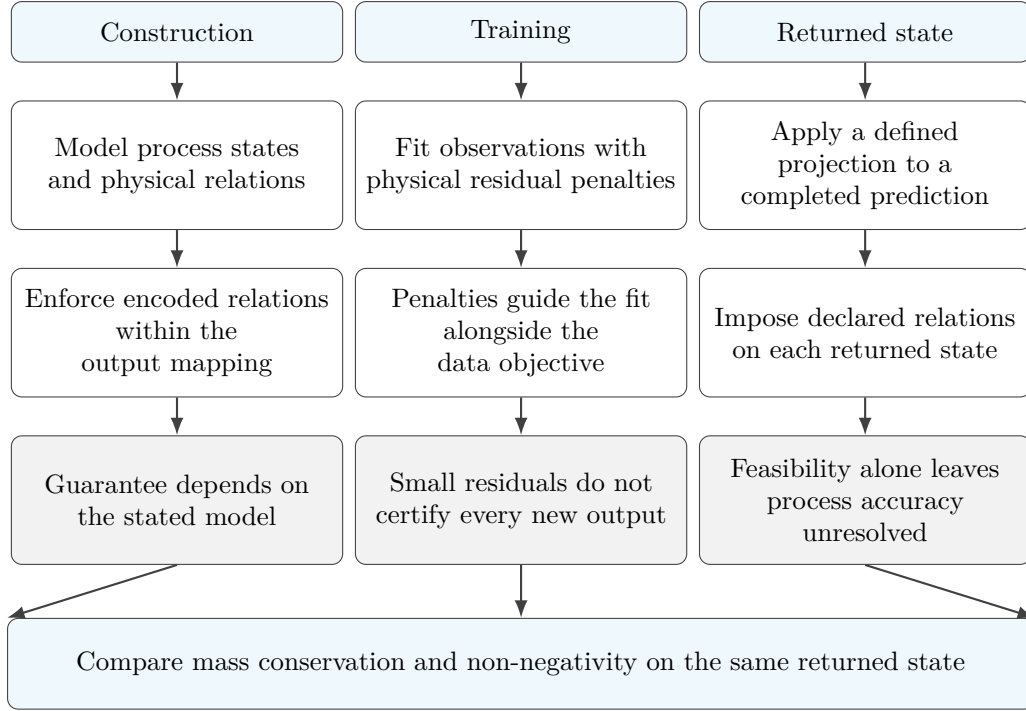


Figure 2.1: Stages at which the reviewed methods introduce physical information. Fitting penalties guide equation agreement. Output construction and projection impose the relations encoded in their formulations. Simultaneous mass conservation and non-negativity require an assessment of the final component state.

enforced conservation exactly when the imposed balance was assigned zero uncertainty and was evaluated on a family of transport equations. Utkarsh et al. (2025) trained a probabilistic predictor together with a differentiable projection and evaluated the approach on differential-equation and time-series problems. Because the projection is differentiable, its effect can participate directly in gradient-based learning. These studies extend physical enforcement from correcting one predicted state to representing uncertainty within the constrained prediction process.

Other work has examined the expressive capacity of hard-constrained models. Min et al. (2024) developed neural output layers for input-dependent linear and convex constraints and established approximation results under the stated mathematical conditions. A convex feasible set contains the straight line joining any two feasible states, which permits useful mathematical guarantees for constrained outputs. Valente et al. (2025) examined projection onto physical manifolds, or sets of outputs satisfying selected physical laws, using spring-mass and reactive-plasma examples. Their work extends correction beyond simple linear relations to nonlinear physical structures.

These contributions broaden the available tools for physically constrained machine learning, but they do not remove the need to define the wastewater-specific constraint set carefully. The literature already provides methods for enforcing conservation and, in some cases, conservation together with positivity. The gap addressed in this dissertation

is therefore not the absence of physical enforcement methods. It is the need to determine how common enforcement affects accuracy, adjustment size, and computational cost for a complete activated sludge component response and then to extend the material-accounting requirements from a single reactor to a connected plant. The imposed constraints remain intentionally narrower than the complete biological and settling equations.

2.4 Interpretable modeling of treatment responses

2.4.1 Inspection of inputs and fitted relationships

Interpretability concerns what can be learned about a model's response after it has been fitted. Rudin (2019) argued for using directly inspectable models in consequential applications when suitable interpretable alternatives are available. In wastewater treatment, this becomes relevant when an engineer needs to understand how influent composition, aeration, recycle, or another operating variable contributes to a predicted treatment outcome. The argument supports examining the model that informs the decision. It does not imply that every visible statistical coefficient has a direct biological interpretation.

Several treatment studies use post-fitting inspection to identify influential variables. Y. Wang et al. (2025) combined effluent TN prediction with feature-attribution analysis, while Ekinici et al. (2023) used feature selection and attribution in sludge prediction. These approaches help identify which observed variables contribute to a particular statistical output and can guide process investigation or reduce unnecessary inputs. Their explanations, however, apply to the fitted TN or sludge model. They do not describe how every component in a mechanistic activated sludge state responds.

Explicit algebraic models provide a complementary form of interpretability. Nazlabadi et al. (2021) reviewed response-surface methods in biological wastewater treatment, where polynomial terms describe operating effects and interactions over a fitted domain. Brunton et al. (2016) developed sparse equation discovery from candidate functions for nonlinear dynamical systems. The latter provides a methodological precedent for inspecting named terms, while response-surface applications provide a direct wastewater-treatment context. Together, they show why visible mathematical structure can be useful when the objective is to understand how inputs contribute to a predicted response.

For activated sludge analysis, the distinction among terms matters. An aeration term represents a fitted dependence on aeration, an influent substrate term represents a fitted dependence on loading, and their product allows the aeration response to change with the substrate condition. Such terms can make the statistical response easier to examine within the fitted domain. They do not, by themselves, establish causal

biological parameters. Demonstrating a microbial mechanism would require a different identification problem and different evidence, particularly when the model inputs are correlated.

2.4.2 Structured regression and the final projected state

Selerio (2026) addressed the need for an inspectable complete-component activated sludge surrogate by separating operating effects, influent effects, curvature, interactions, and learned dependence among effluent components. The explicit second-order terms allow the fitted input relationships to be examined in physical coordinates. Component coupling then contributes to the complete raw state, and a fixed composition map converts that state into reported water-quality quantities. The result is a treatment surrogate whose statistical structure can be traced from inputs to component predictions and then to aggregate treatment indicators.

The same study also shows why interpretability cannot stop at the fitted coefficients when the deployed prediction includes physical correction. Its displayed coefficient relationships describe the raw regression before final constraint enforcement. Mass conservation projection and the subsequent linear-program stage can change the concentrations ultimately returned to the engineer. If a concentration bound becomes active, the local sensitivity of the final prediction can also differ from the slope implied by the unconstrained equation. The study therefore distinguishes the displayed raw coefficient structure from derivatives of the final projected response.

This distinction becomes especially important when component coupling is present. A visible coefficient can affect one intermediate quantity, which then influences several components through the coupling relation before projection imposes the physical constraints. Different fitted coefficient combinations can also produce similar predictions. Selerio (2026) therefore identifies coefficient stability across resampled training data as an additional diagnostic rather than assuming that one fitted coefficient set uniquely represents the treatment process.

The broader literature on differentiable hybrid modeling supports this complete-response perspective. Shen et al. (2023) reviewed approaches that connect learned relationships with process equations and showed, through geoscience examples, how combined models can retain physical knowledge while remaining open to interpretation. They also identify verification of the physical significance of outputs as an ongoing need. For the present dissertation, this supports following the prediction through all of its stages rather than interpreting only the first fitted equation.

The remaining interpretive question is therefore an extension of an established contribution. The raw structured regression already exposes its terms. What remains is to connect those terms to component coupling, reporting, and the constraints that determine the final effluent response, and to examine how stable the inferred relationships remain as training cases change. This preserves the distinction between statistical

association and biological mechanism while making the complete deployed calculation available for engineering interpretation.

2.5 Surrogate modeling for operating decisions

2.5.1 Mechanistic and learned routes to operating optimization

Optimization of activated sludge operation predates the recent use of machine learning surrogates. Mechanistic studies established how treatment constraints, influent conditions, and operating degrees of freedom can be brought into one decision problem. de Araujo et al. (2013) optimized a steady-state process model and used sensitivity analysis to identify variables for economic control. Their analysis linked active effluent constraints and remaining operating degrees of freedom to the resulting control structure. Nazif et al. (2023) calibrated a treatment model to a plant in Tehran, grouped influents according to quantitative and qualitative characteristics, and optimized operating settings for each group. These studies provide examples of direct process-based optimization in which operating decisions are evaluated through the mechanistic treatment model itself.

Learned surrogates provide an alternative when repeated mechanistic calculations are expensive. Fang et al. (2011) trained a support vector surrogate on mechanistic simulations and then used the learned model in an optimization of a nutrient-removal system. Kusiak and Wei (2013) fitted neural models for airflow, effluent carbonaceous biochemical oxygen demand, and TSS from industrial data and used a multiobjective particle-swarm search to examine dissolved-oxygen settings. These studies demonstrate that a learned treatment response can be used directly in the search for operating conditions. The engineering meaning of the result, however, depends on the particular outputs that the surrogate predicts and on how those outputs represent the objectives and constraints of the decision.

Dynamic control studies provide related evidence at a different time scale. Bernardelli et al. (2020) developed a neuro-fuzzy predictive controller for aeration and field-tested it at a large municipal plant, reporting improved effluent quality and energy savings in that application. Croll et al. (2023) reviewed reinforcement learning for wastewater control and discussed the methods and challenges involved in changing operation over time. These contributions show that learned models can participate directly in treatment control. The present dissertation addresses a different question: fixed-input steady-state operating assessment. Dynamic control results therefore provide important context without replacing the need for a separate evaluation of the steady-state decision problem.

The operating literature establishes that both mechanistic and learned routes can be used to select treatment settings. The unresolved question for a new surrogate is whether its selected settings remain favorable when both routes are judged on the same

engineering basis. Such a comparison requires common treatment objectives, operating bounds, and reference calculations. It also requires separating the one-time cost of developing a surrogate from the computational cost of repeatedly using it during an operating search.

2.5.2 Integrated resource recovery and connected responses

Whole-system wastewater studies broaden the decision beyond the performance of a single biological reactor. Padrón-Páez et al. (2020) examined sustainable treatment design through multiple objectives, while Aboagye et al. (2021) combined wastewater-network design, optimization, and sustainability assessment. Their work shows how treatment quality and resource priorities can be considered together when several process choices are available. This provides the broader systems context for a connected plant objective in which treatment gains must be considered alongside the resources required to obtain them.

Durkin et al. (2024) extended this idea through surrogate-based process synthesis for resource recovery from brewery wastewater. Their framework combined high-fidelity simulations, regression surrogates, and classification models for process feasibility. The application considered biogas recovery, recoverable nutrient quality, aeration demand, and uncertainty in wastewater composition. It therefore demonstrates that learned approximations can support integrated decisions involving several resource pathways while accounting explicitly for whether a simulated configuration is feasible.

The decision problem in this dissertation is related but different. Durkin et al. (2024) considered a network-design problem with alternative process configurations. The present connected operating study retains a specified activated sludge topology and predicts the mixer, biological reactor stages, clarifier outlets, and stored solids jointly. Its additional requirement is that those predicted quantities remain mutually consistent at the fixed physical connections of the plant. A classifier indicating simulator feasibility and an explicitly imposed component balance therefore address different aspects of model reliability.

The single-reactor work of Selerio (2026) supplies a precedent for physical enforcement at one biological unit, while the benchmark of Alex et al. (2008) provides a mechanistic precedent for comparing connected plant responses and controls. Bringing these lines of research together leads to a specific remaining question: can a statistical plant response preserve the declared unit connections and solids inventories while still supporting useful operating decisions? Answering that question requires evidence beyond single-reactor feasibility and beyond the accuracy of one aggregate operating objective.

2.5.3 Approximation error at selected operating conditions

A surrogate can have good average predictive accuracy and still be inaccurate where the optimizer ultimately chooses to operate. Surrogate-optimization research addresses this distinction directly. Bhosekar and Ierapetritou (2018) reviewed surrogate modeling for feasibility analysis and optimization, while McBride and Sundmacher (2019) reviewed surrogate construction from experiments and simulations for process-system decisions. Both place prediction within a larger decision workflow: the surrogate must approximate the quantities and the operating region that matter to the optimization, not merely reproduce an average response over the original training design.

Alexandrov et al. (1998) developed a trust-region framework that manages approximation models during optimization by using reference information to assess local model quality and control the region in which the approximation is trusted. The contribution is methodological, but its implication for wastewater treatment is clear. A model used to select aeration, recycle, or wasting settings should be checked near the controls that the optimizer actually selects rather than judged only by an average error calculated elsewhere in the operating domain.

The distinction between predictive quality and optimization quality was examined explicitly by Pedrozo et al. (2025). They compared five surrogate approaches for a carbon-dioxide-stream pooling problem and assessed predictive accuracy, computational effort, and optimization behavior separately. Their comparison of one-shot optimization with a trust-region filter strategy showed that models with favorable fitting performance could still differ in convergence and reliability during search. Although the application concerns carbon capture rather than wastewater treatment, the methodological lesson is directly relevant: a good surrogate fit and a good engineering decision are separate outcomes.

Physical feasibility does not eliminate this distinction. Mass conservation and non-negativity remove certain impossible component states, but many different physically admissible states can still have different nitrogen distributions, biomass concentrations, and treatment performance. An optimizer can therefore select a prediction that satisfies the declared physical constraints yet remains inaccurate relative to the mechanistic reference. Physical-enforcement studies establish how to exclude impossible states; approximation-management studies establish why the response used by the optimizer still needs to be checked.

Together, these findings motivate independent mechanistic evaluation of the aeration, recycle, wasting, and other controls selected by the surrogate. The remaining decision gap concerns the complete body of evidence for a connected activated sludge surrogate. Its unit balances and stored solids must be mutually consistent. Its selected controls must satisfy the original treatment requirements when those controls are returned to the reference model. Its objective must be evaluated on the same mechanistic basis as the direct comparator. Search time and verification effort must also be reported on

a clearly defined basis. Only then can computational benefit and decision quality be assessed without assuming that physical constraint satisfaction establishes either one.

2.6 Synthesis of established contributions and remaining gaps

2.6.1 What the reviewed studies have addressed

The literature reviewed in this chapter shows a clear progression from process representation to engineering decision support. Mechanistic activated sludge models define components, biological reactions, separation, recirculation, and solids inventories. Calibration studies connect those model quantities to observations from real treatment systems. Data-driven soft sensors and indicator models improve estimation of selected treatment responses. Simulation-based surrogates extend learned prediction to repeated process calculations and optimization. Physics-informed and constrained-learning methods introduce governing relations into fitting or impose them on model outputs. Interpretable regression and surrogate-optimization research then address how fitted responses can be examined and how they should be evaluated when used to make operating decisions.

Several requirements relevant to this dissertation have therefore already been addressed in important forms. Kircher and Votsmeier (2025) established joint atom balance and positivity for specified reactive-chemistry surrogates. Selerio (2026) established final mass conservation and non-negativity for a structured activated sludge reactor surrogate and showed why interpretation must distinguish the raw and projected responses. Durkin et al. (2024) demonstrated surrogate-supported optimization for integrated wastewater resource recovery. These studies are not gaps that the dissertation claims to fill. They are the starting points from which its more specific questions arise.

The remaining gaps appear when these established contributions are brought into one complete-component activated sludge assessment. A comparison of learners must use the same component target. Physical enforcement must be evaluated not only for feasibility but also for the errors and adjustments it introduces. Interpretation must follow the prediction through coupling and projection rather than stopping at the visible regression terms. Plant-level decisions must preserve unit connections and then be checked on the original mechanistic basis. Table 2.2 summarizes this relationship between the established literature and the specific questions retained for the dissertation.

2.6.2 The four linked questions of the dissertation

The first gap concerns comparative prediction of the complete reactor response. Existing indicator studies show that machine learning can predict useful treatment

Table 2.2: Established studies and the remaining questions addressed by the dissertation.

Research need	Studies addressing the need	Remaining question
Complete treatment prediction	Y. Wang et al. (2025) assesses TN prediction. P. Wang et al. (2024) estimates influent components. Selerio (2026) predicts complete reactor components.	How do several learners compare on one complete-state target under training scarcity and extrapolation?
Physical validity of predictions	Beucler et al. (2021) and Sturm and Wexler (2022) impose conservation. Kircher and Votsmeier (2025) and Selerio (2026) address conservation and concentration bounds together.	How does common enforcement affect component accuracy, adjustment size, and cost across learners?
Interpretation of treatment responses	Y. Wang et al. (2025) inspects input contributions. Selerio (2026) exposes regression terms and distinguishes raw from final interpretation.	How do coupling, reporting, and active constraints affect sensitivities and stability of the explanation?
Connected operating decisions	Fang et al. (2011) and Durkin et al. (2024) use surrogates in treatment decisions. Alex et al. (2008) defines a connected plant benchmark.	Can joint component enforcement support settings that remain credible under independent plant evaluation?

quantities, while the structured-regression study of Selerio (2026) extends prediction to all modeled reactor components. What remains is a common comparison across learning families in which the target, partitions, and assessment scales are held fixed. That comparison must also show how performance changes as the number of training cases decreases and as influent or operating conditions move beyond the sampled domain. The purpose is not to identify one universally superior learner, but to determine how learner choice depends on the available data and the intended input range.

The second gap concerns what physical enforcement changes after a raw prediction has already been produced. Joint mass conservation and concentration bounds have been established by existing methods. The remaining question is how a common enforcement procedure affects the same raw predictions across several activated sludge learners. Component error, physical residuals, negative concentrations, projection displacement, and computational cost describe different consequences of that correction. Evaluating them together makes it possible to determine the value of enforcement without assuming that physical compliance must also improve predictive accuracy.

The third gap concerns interpretation of the complete returned treatment state. Structured regression already provides visible operating, loading, curvature, interaction, and coupling terms. However, the response ultimately used in engineering analysis can differ from the raw equation because coupling, reporting, and projection occur afterward. The remaining question is therefore how sensitivities pass through those stages, particularly when concentration bounds become active, and whether the resulting fitted relationships remain stable when the training cases change. This extends interpretation from the visible regression to the complete prediction actually returned to the engineer.

The fourth gap concerns connected prediction and operating assessment. Mechanistic plant models establish the shared streams, clarification behavior, recycle structure, and stored solids that connect treatment units. Surrogate studies establish that learned models can support operating and resource-recovery decisions. What remains here is to combine those requirements in one connected component response, impose the declared plant relations jointly, and then independently calculate treatment outcomes at the controls selected by the surrogate. Comparison with direct mechanistic optimization places both routes on the same treatment and resource basis and separates faster search from better decision quality.

Figure 2.2 shows why these four questions cannot be treated independently. The component concentrations being predicted are also the quantities adjusted by physical enforcement and examined during interpretation. Those same concentrations determine treatment indicators and enter the operating objective and constraints. A change introduced by projection can therefore affect component error, local sensitivity, and the settings selected during optimization. The integrated assessment must follow this shared treatment response through each stage rather than allowing evidence from one stage to stand in for another.

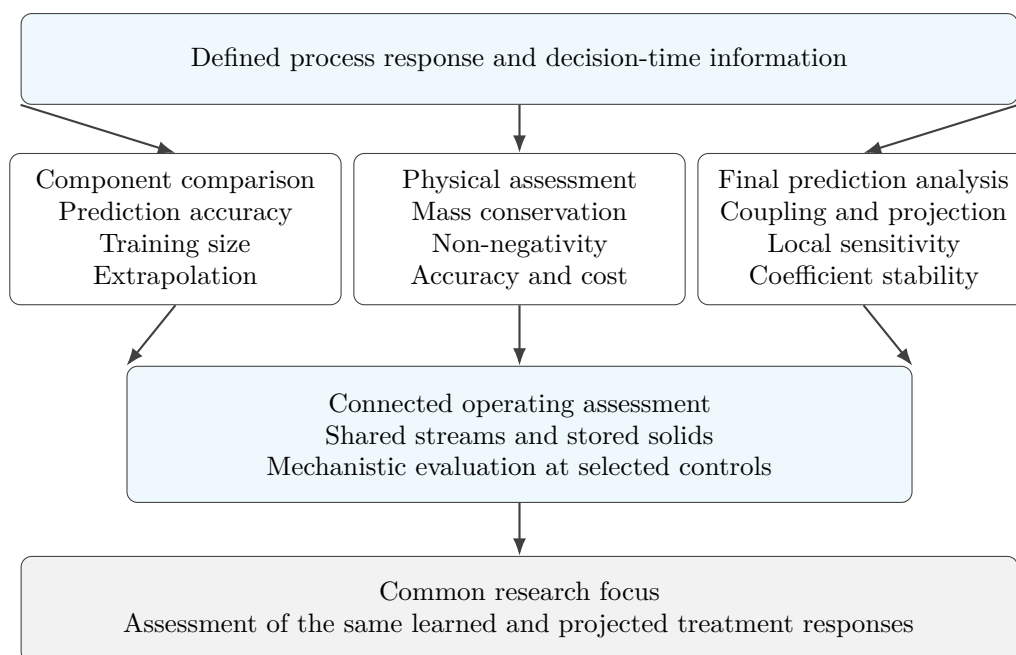


Figure 2.2: Links among the remaining comparative, physical, interpretive, and operating questions. Each question concerns the same defined activated sludge response. Established methods address parts of the problem, while their combined use requires common component and decision assessment.

The literature therefore supports a progression from a defined reactor response to a physically checked and independently evaluated plant decision. Chapter 3 develops the common component and mechanistic basis for that progression. Chapter 4 compares reactor predictors and examines the effect of physical enforcement on the same raw responses. Chapter 5 develops the structured regression and follows interpretation through component coupling and projection. Chapter 6 extends the framework to a connected activated sludge plant and independently evaluates the controls selected through surrogate-assisted and direct mechanistic optimization. Chapters 4–6 each present the rationale, theory, methodology, results, and discussion for their corresponding assessment, while Chapter 7 brings their findings together and identifies directions for future research.

Chapter 3

Mechanistic Foundations of Physically Admissible Prediction

3.1 The component state as the object of prediction

An activated sludge reactor does not simply remove material; it changes its form and distribution. Organic substrate can become biomass, ammonium can become nitrite and nitrate, and soluble phosphate can enter biological storage or a precipitate. Describing these transformations therefore requires more than a small set of aggregate effluent concentrations. The ASM framework addresses this need by separating wastewater material into components and linking their changes through biological and chemical processes (Henze et al., 2000). In this dissertation, that component representation also defines the physical meaning of a surrogate prediction: the surrogate estimates a complete component state from the influent state and operating conditions.

The mechanistic reference is Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). It combines the phosphorus-removal framework of Henze et al. (2000) with the nitrogen-transformation formulation used by Guerrero et al. (2011). The model contains 20 components and 28 processes, with ten dissolved and ten particulate components. Stoichiometry specifies the relative amounts of material consumed and produced by each process, while kinetics specifies how rapidly those processes proceed as functions of the reactor state and fixed parameters. The surrogate approximates the resulting steady-state concentrations. Projection then imposes selected physical accounting constraints on those predictions.

This chapter develops the mechanistic basis for that procedure. It first defines how the component state is represented and manipulated mathematically, then connects the components to the biological transformations of the activated sludge model. It subsequently derives the reactor balances, the mass conservation invariants used for physical enforcement, the geometry of physically admissible states, and the reporting map from component concentrations to familiar water quality quantities. These ideas

provide the common foundation for the surrogate models and optimization studies developed in Chapters 4–6.

3.1.1 Reading component vectors and matrix operations

The component formulation uses vectors and matrices to record many concentration and balance equations at once. Their meaning remains scalar, however: each coordinate still represents one physical quantity with a specified unit. For example, a concentration of 5 g N m^{-3} is numerically equal to 5 mg N L^{-1} because both the mass and volume units change by a factor of one thousand. Oxygen, COD, nitrogen, phosphorus, solids, and alkalinity nevertheless use different reporting bases. A quantity expressed as g COD m^{-3} , for instance, represents oxygen-demand equivalents rather than grams of one named organic molecule. These distinctions determine which coefficients are needed before quantities can be combined.

A vector collects related scalar values in a fixed order. A three-component column vector and its transpose can be written as

$$c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \quad c^\top = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}.$$

The superscript $\top$ denotes transpose. It changes a column into a row, or interchanges the rows and columns of a matrix, without changing the underlying component values. The notation $c \in \mathbb{R}^3$ states that the vector contains three real-valued coordinates. The notation $c \geq 0$ states that every coordinate is non-negative: $c_1 \geq 0$, $c_2 \geq 0$, and $c_3 \geq 0$. It does not merely require their sum to be non-negative.

A strictly positive coordinate satisfies $c_j > 0$, whereas non-negativity permits $c_j = 0$. This distinction becomes important when a logarithm, reciprocal, or other operation requires a strictly positive argument even though a physical concentration may legitimately be zero.

Subscripts identify particular coordinates or observations. The symbol c_j denotes component j . When several observations are involved, c_{ij} denotes component j in observation i . The labels *in* and *out* identify the two sides of the reactor boundary. Thus $c_{out,ij}$ is the effluent concentration of component j in observation i . Parenthetical superscripts used to identify a surrogate or prediction type are labels rather than powers.

Matrices collect coefficients in rows and columns. A matrix with two rows and three columns has dimension 2×3 . Matrix multiplication is defined when the number of columns of the first factor equals the number of rows of the second. Each output coordinate is obtained by multiplying corresponding entries from a matrix row and a

vector column and then summing those products. For hypothetical numerical values,

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 1(3) + 2(4) + 0(5) \\ 0(3) + 1(4) + 1(5) \end{bmatrix} = \begin{bmatrix} 11 \\ 9 \end{bmatrix}.$$

The result has two coordinates because the matrix has two rows. In a physical calculation, the coefficients also carry the conversions needed to express the result on the intended output basis. The numerical example illustrates the operation itself without assigning the coefficients to an activated sludge process.

A row-column product is therefore a weighted sum. For a coefficient column a , the scalar product $a^\top c$ means $a_1c_1 + a_2c_2 + a_3c_3$. The summation $\sum_{j=1}^3 a_jc_j$ expresses the same operation by explicitly identifying the indexed terms. An identity matrix, denoted I , leaves a vector unchanged, while a zero matrix contains only zero entries. These simple operations allow many component equations to be written compactly without losing the scalar accounting that gives them physical meaning.

Vector magnitude is described by a norm. The Euclidean norm is the square root of the sum of squared coordinates. A hypothetical vector $(3, -4)^\top$ therefore has Euclidean norm $\sqrt{3^2 + (-4)^2} = 5$. The sum of absolute coordinates gives its absolute-sum norm, $|3| + |-4| = 7$. Squaring a norm and squaring individual coordinates are consequently different operations and must be applied in the stated order. When vector coordinates use different physical bases, the numerical value of a norm is meaningful only under the adopted unit convention or after an explicit standardization.

The same column-vector convention is used for the 20-component activated sludge state. Its fixed ordering is

$$c = (S_O, S_F, S_A, S_{NH_4}, S_{NO_2}, S_{NO_3}, S_{N_2}, S_{PO_4}, S_I, S_{ALK}, X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH})^\top. \quad (3.1)$$

Each coordinate is represented numerically in its native component unit. The vector therefore combines several constituent bases. An organic component expressed as g COD m⁻³ is not on the same material basis as ammonium expressed as g N m⁻³. Any physical operator applied to the state must account for these differences through its coefficients.

The symbols in Table 3.1 are state variables rather than additional process abbreviations. For example, X_{AOB} is the concentration coordinate assigned to ammonia-oxidizing biomass. The numerical constituent contents and solids conversion factors belong to the fixed reporting map. They are not learned from the surrogate data, and their interpretation follows the component accounting of activated sludge modeling (Henze et al., 2000).

Retaining the complete state preserves distinctions that aggregate water quality quantities cannot recover. Ammonium, nitrite, and nitrate all contribute to reported

Table 3.1: Component meanings, native units, and fixed coefficients for the four reported water quality quantities. Each coefficient converts the native component basis to the corresponding reported basis.

Component	Meaning	Native unit	COD	TN	TP	TSS
S_O	Dissolved oxygen	$\text{g O}_2 \text{ m}^{-3}$	0	0	0	0
S_F	Fermentable substrate	g COD m^{-3}	1	0	0	0
S_A	Acetate and fermentation products	g COD m^{-3}	1	0	0	0
S_{NH4}	Ammonium nitrogen	g N m^{-3}	0	1	0	0
S_{NO2}	Nitrite nitrogen	g N m^{-3}	0	1	0	0
S_{NO3}	Nitrate nitrogen	g N m^{-3}	0	1	0	0
S_{N2}	Dissolved dinitrogen	g N m^{-3}	0	0	0	0
S_{PO4}	Soluble phosphate	g P m^{-3}	0	0	1	0
S_I	Soluble inert organic matter	g COD m^{-3}	1	0.01	0	0
S_{ALK}	Alkalinity	mol m^{-3}	0	0	0	0
X_I	Particulate inert organic matter	g COD m^{-3}	1	0.02	0.01	0.75
X_S	Slowly biodegradable substrate	g COD m^{-3}	1	0	0	0.75
X_H	Heterotrophic biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{PAO}	Phosphate-accumulating biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{PP}	Stored polyphosphate	g P m^{-3}	0	0	1	3.23
X_{PHA}	Stored polyhydroxyalkanoates	g COD m^{-3}	1	0	0	0.60
X_{AOB}	Ammonia-oxidizing biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{NOB}	Nitrite-oxidizing biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{MeP}	Precipitated metal phosphate	g TSS m^{-3}	0	0	1/4.87	1
X_{MeOH}	Metal hydroxide	g TSS m^{-3}	0	0	0	1

nitrogen, yet their transformations involve different microbial populations and oxygen requirements. Dissolved dinitrogen remains part of the modeled nitrogen inventory even though it has zero weight in reported TN. Likewise, soluble phosphate, stored polyphosphate, and metal phosphate play different process roles even when they all contribute to TP. These distinctions become important when a predicted state is used not only to report treatment performance, but also to interpret why an operating condition produces that performance.

3.2 Biological meaning of the component transformations

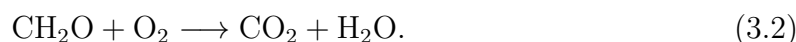
The component vector becomes useful only when its coordinates are connected to the transformations they represent. Activated sludge treatment redistributes carbon, nitrogen, phosphorus, oxygen demand, biomass, and solids through coupled biological and chemical pathways. The following discussion does not replace the full ASM2d-TSN stoichiometric formulation. Instead, it uses simplified reactions to show the material accounting that underlies the component model and the physical constraints later imposed on surrogate predictions.

3.2.1 Electron donors, electron acceptors, and biomass

Microorganisms use part of a substrate to obtain energy and another part to form cellular material. In an oxidation-reduction reaction, an electron donor supplies electrons and an electron acceptor receives them. Under aerobic conditions, organic material can serve as the donor and oxygen as the acceptor. Under anoxic conditions, some organisms can instead use nitrate or nitrite. Identifying the donor and acceptor therefore helps explain why the same organic substrate can support different treatment pathways (Narayanan & Narayan, 2019; Preena et al., 2021).

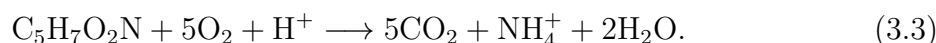
The carbon source and the energy source describe different properties of microbial growth. Heterotrophic biomass uses organic carbon to form cells. Chemolithoautotrophic nitrifying biomass obtains energy from inorganic nitrogen oxidation while using inorganic carbon for cell synthesis. Nitrification is therefore not simply another form of organic-substrate removal. The separate biomass coordinates in the adopted model preserve this distinction (Guerrero et al., 2011; Henze et al., 2000). Complete ammonia oxidation can also occur within a single microbial organism (van Kessel et al., 2015), but that biological observation does not replace the separate ammonium- and nitrite-oxidation processes used in the present model.

A simple organic unit, CH_2O , can illustrate oxidation without assigning one chemical formula to all wastewater organic matter. Its complete aerobic oxidation is



Both sides contain one carbon atom, two hydrogen atoms, and three oxygen atoms. Using approximate molar masses of 30 g mol^{-1} for the organic unit and 32 g mol^{-1} for oxygen gives an oxygen requirement of $32/30 = 1.067 \text{ g oxygen per g}$ of this idealized material. Thus, a hypothetical 30 g quantity would require 32 g oxygen for complete oxidation. The purpose of the calculation is to explain the oxygen-demand basis of organic components, not to prescribe the oxygen demand of an arbitrary wastewater mixture.

Cell synthesis changes this accounting because part of the available carbon and nitrogen becomes biomass rather than being completely oxidized. The empirical formula $\text{C}_5\text{H}_7\text{O}_2\text{N}$ provides a simplified representation of biomass composition, as discussed in the assimilation study of Hoover and Porges (1952). It represents an average composition rather than one molecular species. An idealized complete oxidation of this biomass with nitrogen released as ammonium can be written as

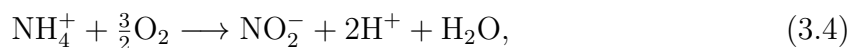


The carbon, hydrogen, oxygen, and nitrogen counts are 5, 8, 12, and 1 on both sides, and the net charge is +1 on both sides. Five moles of oxygen correspond to 160 g , while the empirical biomass unit has an approximate mass of 113 g . The resulting ratio is about $1.416 \text{ g oxygen per g}$ of this idealized biomass. This distinction is useful because

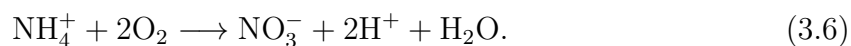
dry biomass mass and biomass COD are not interchangeable quantities. The adopted model nevertheless uses its specified yields, constituent contents, and lysis processes rather than replacing its stoichiometric entries with these teaching reactions.

3.2.2 Nitrogen transformations and their oxygen requirements

Nitrogen transformations provide a particularly clear example of why a component-resolved state is useful. Separating nitrification into two steps makes both the oxygen requirement and the nitrite intermediate explicit. Ignoring cell synthesis for this teaching calculation, ammonium oxidation and nitrite oxidation can be written as

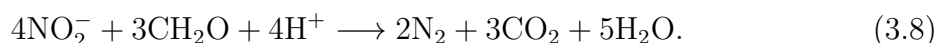
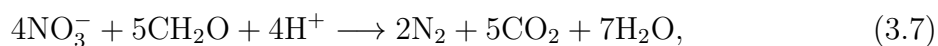


The first reaction preserves one nitrogen atom, four hydrogen atoms, and three oxygen atoms. Its right-hand charge is $-1 + 2 = +1$, equal to the left-hand charge. The second reaction preserves one nitrogen atom, three oxygen atoms, and a net charge of -1 . Adding the two reactions and cancelling the nitrite intermediate gives the overall reaction



One mole of nitrogen has an approximate mass of 14 g. The first step therefore requires $1.5(32)/14 = 3.43$ g oxygen per g nitrogen, while the second requires $0.5(32)/14 = 1.14$ g oxygen per g nitrogen. Together, the idealized overall requirement is 4.57 g oxygen per g nitrogen. A hypothetical conversion of 10 g ammonium nitrogen to nitrate would therefore require about 45.7 g oxygen under these assumptions. Actual process demand also depends on biomass synthesis and the adopted yields. The hydrogen ions produced in the first step further show why nitrification draws on the buffering capacity represented by alkalinity (Guerrero et al., 2011; Henze et al., 2000).

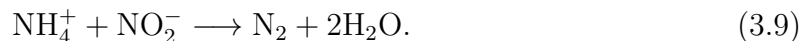
Denitrification follows a different pathway. Instead of oxygen, heterotrophic organisms can use oxidized nitrogen as an electron acceptor and convert it to dinitrogen. Using CH_2O as an illustrative electron donor and omitting cell synthesis, two balanced net reactions are



For the nitrate reaction, both sides contain four nitrogen atoms, five carbon atoms, fourteen hydrogen atoms, and seventeen oxygen atoms. The left-hand charge is $-4 + 4 = 0$, matching the uncharged products. The nitrite reaction can be checked in the same manner. Their illustrative donor requirements are $5/4$ and $3/4$ moles of organic unit per mole of nitrogen, respectively. The comparison shows why a pathway through nitrite can require less organic donor than a pathway through nitrate under the same

simplified assumptions. It does not determine the reaction rate, which also depends on carbon supply, donor identity, biomass, and process conditions (Lu et al., 2014).

Anaerobic ammonium oxidation provides another nitrogen pathway. An idealized net reaction that omits biomass synthesis and minor products is



The equation preserves two nitrogen atoms, four hydrogen atoms, two oxygen atoms, and zero net charge. In this simplified balance, ammonium and nitrite are coupled without an organic electron donor. The cultivation work of Strous et al. (1998) and the pathway studies of Dietl et al. (2015) and Maalcke et al. (2016) explain why this process differs biologically from ordinary heterotrophic denitrification. Its application also requires its own organisms and operating conditions (Vandekerckhove, 2018; Winkler & Straka, 2019; Xie, 2021). It is included here only to clarify the boundary of the nitrogen discussion; it is not an additional process in the adopted 20-component, 28-process reactor model.

3.2.3 From elemental accounting to the adopted state basis

The preceding teaching reactions can be checked directly by counting atoms and charge because each displayed species has a chemical formula. The activated sludge state is different. Its coordinates combine oxygen-demand equivalents, nitrogen mass, phosphorus mass, solids mass, and alkalinity. A biomass coordinate expressed in g COD m^{-3} therefore cannot simply be added to an ammonium coordinate expressed in g N m^{-3} to obtain a meaningful material total. The biomass contribution must first be converted to the nitrogen basis through its nitrogen-content coefficient.

This distinction determines what “mass conservation” means in the present work. Takács and Vanrolleghem (2006) distinguish balances on lumped wastewater components from complete elemental accounting. A full elemental balance may require water, carbon dioxide, gas exchange, and other species that lie outside a particular activated sludge state. The conservation conditions used in this dissertation are therefore derived from the adopted stoichiometric matrix and the declared flow and transfer boundary. They certify the quantities represented by that model. They do not imply that every unrepresented chemical species or external flux has been measured or balanced. The following sections derive exactly which combinations of the component state can be imposed without introducing a physical requirement unsupported by the model.

3.3 Process transformations and their rates

The mechanistic model combines many transformations of the component state. Hydrolysis moves slowly biodegradable material toward forms that microorganisms

can use more readily. Heterotrophic metabolism links substrate consumption with biomass growth and electron-acceptor use. Oxygen serves as the electron acceptor under aerobic conditions, whereas nitrate or nitrite can support the corresponding modeled transformations under anoxic conditions. Fermentation produces acetate from suitable organic material. Biomass lysis redistributes material from active populations to substrate and inert pools. Biological phosphorus removal introduces storage, growth, phosphate uptake, and release, while chemical precipitation and redissolution redistribute phosphorus between soluble and particulate forms (Guerrero et al., 2011; Henze et al., 2000).

Two-step nitrification represents ammonium oxidation to nitrite separately from nitrite oxidation to nitrate, and therefore retains separate ammonia-oxidizing and nitrite-oxidizing biomass. A one-step description may reproduce the overall conversion from ammonium to nitrate while hiding the nitrite intermediate. By retaining that intermediate, the state can distinguish a low-ammonium effluent that still contains nitrite from one in which nitrite has been oxidized further. This distinction belongs to the adopted reaction model; it does not imply that every nitrogen pathway occurring in a real treatment plant is represented. The broader diversity of microbial nitrogen-removal pathways reviewed by Duan et al. (2022) provides that context.

The stoichiometric matrix, also called the Petersen matrix, records how each modeled process consumes and produces the state components. A small hypothetical network illustrates the logic before the full matrix is introduced. Let three material pools be denoted by A , B , and C , and suppose that one unit of B contains twice the tracked inventory of one unit of A or C . Consider



Figure 3.1 connects these transformations to their component changes. The coefficients below each arrow describe how the three pools change, while the inventory weights show why both processes preserve the same underlying material quantity.

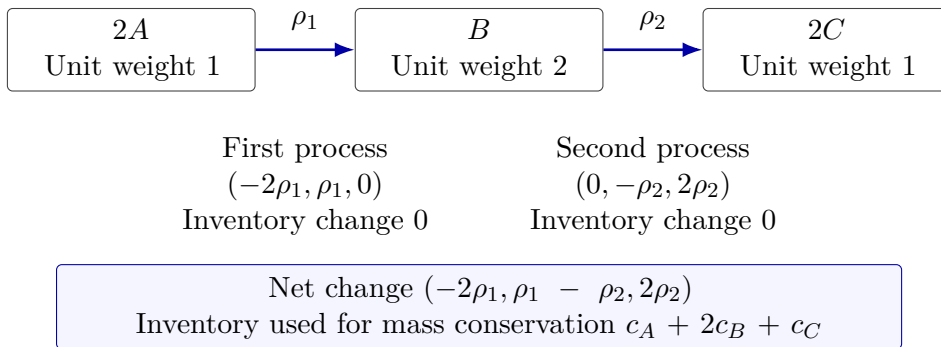


Figure 3.1: Hypothetical reaction accounting for three pools with inventory weights 1, 2, and 1. Each process redistributes the tracked inventory without creating or removing it.

Two units of A contain the same tracked inventory as one unit of B , and one unit of B contains the same inventory as two units of C . Pool B therefore acts as an intermediate: the first process produces it and the second consumes it. Its net concentration change depends on the difference between the two process rates even though each process individually preserves the weighted inventory.

If the process rates are ρ_1 and ρ_2 , the scalar reaction contributions are

$$\begin{aligned}\dot{c}_A \Big|_{\text{reaction}} &= -2\rho_1, \\ \dot{c}_B \Big|_{\text{reaction}} &= \rho_1 - \rho_2, \\ \dot{c}_C \Big|_{\text{reaction}} &= 2\rho_2.\end{aligned}$$

The dot denotes a time derivative. These expressions describe reaction contributions only; hydraulic transport is not yet included. The same equations can be collected into a process-by-component matrix and a rate column,

$$\nu_{\text{toy}} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \end{bmatrix}, \quad \rho_{\text{toy}} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}.$$

Transposing the stoichiometric matrix places the coefficients for each component in one row. Multiplication then reconstructs all three scalar rates at once:

$$\nu_{\text{toy}}^\top \rho_{\text{toy}} = \begin{bmatrix} -2 & 0 \\ 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} = \begin{bmatrix} -2\rho_1 \\ \rho_1 - \rho_2 \\ 2\rho_2 \end{bmatrix}.$$

For hypothetical rates of 3 and 1 in compatible amount-per-time units, the resulting contributions are -6 , 2 , and 2 . Pool B increases because its production exceeds its consumption. The positive changes in B and C do not mean that material has been created; they are accompanied by a corresponding loss from A .

The full activated sludge model uses the same convention. Its matrix orientation is fixed as

$$\nu \in \mathbb{R}^{P \times F}, \quad P = 28, \quad F = 20. \quad (3.10)$$

Row p represents process p , and column j represents component j . A negative entry ν_{pj} denotes consumption, a positive entry denotes production, and a zero entry means that the process has no direct stoichiometric contribution to that component. The coefficients also include the yields and constituent conversions required by the native component units.

For component j , the individual process contributions are $\nu_{1j}\rho_1$, $\nu_{2j}\rho_2$, and so on through $\nu_{28j}\rho_{28}$. Their sum gives the net reaction rate of that component. Writing the rate vector as $\rho(c, u)$ emphasizes that these rates depend on the current component state and operating inputs. For $\rho(c, u) \in \mathbb{R}^{28}$,

$$r_{\text{reaction}}(c, u) = \nu^\top \rho(c, u), \quad r_{\text{reaction}, j}(c, u) = \sum_{p=1}^{28} \nu_{pj} \rho_p(c, u). \quad (3.11)$$

This formulation separates two different roles. Stoichiometry specifies the proportions in which components are consumed and produced; kinetics specifies how quickly those transformations proceed at a particular state. Increasing a biomass concentration can therefore change a process rate without changing the corresponding stoichiometric row. Likewise, changing aeration can alter oxygen limitation and thus process kinetics without changing the material accounting of the reaction itself.

A simple saturation factor makes this distinction concrete. If a limiting dissolved substrate has concentration S and saturation constant $K_S > 0$, then

$$\frac{S}{K_S + S} \quad (3.12)$$

equals one half when $S = K_S$, approaches zero as S approaches zero, and approaches one as S becomes large. For illustration, if both S and K_S equal 4 g COD m^{-3} , the factor is $4/(4 + 4) = 0.5$. The concentration unit cancels because the numerator and denominator use the same constituent basis.

A representative growth expression combines a maximum specific growth rate, biomass abundance, and multiple limitation factors:

$$\rho_{\text{growth}} = \mu X \frac{S}{K_S + S} \frac{S_O}{K_O + S_O}. \quad (3.13)$$

Here μ is a maximum specific growth rate, X is the associated biomass coordinate, and K_O is an oxygen saturation constant. With hypothetical values $\mu = 2 \text{ d}^{-1}$, $X = 10 \text{ g biomass COD m}^{-3}$, a substrate factor of 0.5, and an oxygen factor of 0.25, the growth contribution is $2(10)(0.5)(0.25) = 2.5 \text{ g biomass COD m}^{-3} \text{ d}^{-1}$. The dimensionless limitation factors reduce the unrestricted contribution $\mu X = 20$ to 2.5.

Equation (3.13) is illustrative rather than a replacement for the complete 28-process formulation. The actual model contains process-specific yields, inhibition terms, and storage dependencies from Henze et al. (2000) and Guerrero et al. (2011). Its purpose here is to show why nonlinear interactions arise naturally. A hypothetical substrate concentration of $K_S/9$ gives a substrate factor of 0.1, whereas $9K_S$ gives a factor of 0.9. The substrate concentration changes by a factor of 81, but the saturation factor changes only by a factor of nine. Several such limitation terms and competing populations can act simultaneously. For this reason, a linear relationship between influent loading and effluent state is generally insufficient, and a surrogate may need to represent interactions without treating them as new stoichiometric laws.

3.4 The steady-state reactor boundary

The stoichiometric and kinetic relations describe transformation within the reactor. A reactor model also requires a boundary that accounts for material entering, leaving, and crossing that boundary through external transfer. The standalone prediction

problem considers a completely mixed reactor at steady state. Perfect mixing makes the reactor concentration equal to the effluent concentration. The boundary includes hydraulic inflow and outflow, the 28 modeled reaction processes, and dissolved-oxygen transfer. It excludes a clarifier, solids return, and other external additions. These exclusions define the specific balance enforced in the reactor study; they do not imply that such processes are absent from wastewater treatment plants.

Let c_{in} and c_{out} denote the influent and effluent component vectors, and let $D > 0$ denote the hydraulic dilution rate. For component j , hydraulic inflow contributes $Dc_{in,j}$ and hydraulic outflow contributes $-Dc_{out,j}$. Reaction contributes the sum of the 28 stoichiometric-rate products. Oxygen transfer contributes only to the dissolved-oxygen coordinate. The scalar steady-state balance is therefore

$$0 = Dc_{in,j} - Dc_{out,j} + \sum_{p=1}^{28} \nu_{pj} \rho_p + \omega(e_{S_O})_j.$$

Because S_O is the first component in Equation (3.1), the selector e_{S_O} has a first coordinate equal to one and nineteen remaining coordinates equal to zero. The oxygen balance consequently contains the direct transfer term $+\omega$, while, for example, the ammonium balance does not. Written separately, the dissolved-oxygen equation is

$$0 = D(S_{O,in} - S_{O,out}) + \sum_{p=1}^{28} \nu_{p,S_O} \rho_p + \omega.$$

Every term in this equation is expressed as a numerical rate on the oxygen basis. The native basis changes with the component row. Collecting all twenty scalar balances gives

$$0 = D(c_{in} - c_{out}) + \nu^\top \rho(c_{out}, u) + \omega e_{S_O}. \quad (3.14)$$

Equivalently, moving the hydraulic difference to the opposite side gives

$$D(c_{out} - c_{in}) = \nu^\top \rho(c_{out}, u) + \omega e_{S_O}. \quad (3.15)$$

For an individual component, $D(c_{out,j} - c_{in,j}) = \sum_p \nu_{pj} \rho_p + \omega(e_{S_O})_j$. The hydraulic term therefore has the same native component-unit-per-day basis as the corresponding reaction and transfer terms. A consistent numerical unit convention is required when the complete balance is treated as a vector equation.

The two operating inputs are HRT and the dimensionless aeration coordinate u_{air} . HRT specifies how long the influent flow would require to replace one reactor volume. A 12-hour HRT corresponds to two reactor-volume replacements per day because $24/12 = 2$, so the associated dilution rate has units of inverse time. Oxygen transfer uses a different inverse-time coefficient, which becomes a concentration-per-time rate when multiplied by the oxygen concentration deficit. The adopted parameterization is

$$D = \frac{24 \text{ h d}^{-1}}{\text{HRT}}, \quad (3.16)$$

$$k_{La} = (2 + 18u_{air}) \text{ d}^{-1}, \quad (3.17)$$

$$\omega = k_{La}(S_{O,sat} - S_{O,out}), \quad S_{O,sat} = 8.5 \text{ g O}_2 \text{ m}^{-3}. \quad (3.18)$$

The dilution rate places hydraulic replacement on the same time basis as the biological process rates. The oxygen-transfer term depends on both the selected aeration setting and the difference between saturation and the actual reactor oxygen concentration.

For a hypothetical HRT of 12 h, Equation (3.16) gives $D = 2 \text{ d}^{-1}$. A component difference of 5 g m^{-3} then corresponds to a net transformation or transfer of $10 \text{ g m}^{-3} \text{ d}^{-1}$ in that coordinate. For a hypothetical aeration setting of 1, the transfer coefficient is 20 d^{-1} . If dissolved oxygen is $2 \text{ g O}_2 \text{ m}^{-3}$, the oxygen-transfer rate is $130 \text{ g O}_2 \text{ m}^{-3} \text{ d}^{-1}$. These calculations illustrate the accounting implied by the equations; they are not simulated treatment outcomes.

Steady state means that the modeled inventories have zero time derivative under the stated equations and boundary. It does not establish that the steady state is unique, dynamically stable, or representative of a field-calibrated treatment plant. Sensitivity of operating choices to the process model therefore remains relevant even when the numerical balance closes (de Araujo et al., 2013). A small residual can certify agreement with the specified equations at a returned state. It cannot certify that those equations completely describe a real wastewater system.

3.5 From stoichiometry to mass conservation invariants

The next step is to identify combinations of the component state that the modeled reactions cannot change. These combinations provide the mass conservation relations used later for physically constrained prediction. An invariant should not be interpreted as a concentration that remains unchanged throughout the plant. Ammonium, for example, can decrease while nitrate increases. What remains unchanged is an appropriate weighted combination of the included nitrogen pools, provided that the corresponding boundary flows and external sources are accounted for.

A stoichiometric invariant is therefore a linear combination of component concentrations whose reaction contribution is always zero. The hypothetical three-pool network provides the simplest example. Adding A , B , and C without weights would incorrectly treat one unit of B as if it contained the same inventory as one unit of A . The correct tracked inventory is

$$I_{\text{toy}} = c_A + 2c_B + c_C.$$

Substituting the scalar reaction rates gives

$$\frac{dI_{\text{toy}}}{dt} = (-2\rho_1) + 2(\rho_1 - \rho_2) + (2\rho_2) = 0.$$

The ρ_1 and ρ_2 terms cancel independently. The equality therefore holds for every combination of process rates. For the hypothetical component change $(-6, 2, 2)^\top$, the weighted change is likewise $-6 + 2(2) + 2 = 0$.

The invariant weights can also be derived rather than guessed. Let $a = (a_A, a_B, a_C)^\top$ be an unknown coefficient column. Requiring each process to make zero weighted contribution gives

$$\begin{aligned} -2a_A + a_B &= 0, \\ -a_B + 2a_C &= 0. \end{aligned}$$

The first equation gives $a_B = 2a_A$, and the second gives $a_C = a_A$. Choosing $a_A = 1$ yields $(1, 2, 1)^\top$. Choosing $a_A = 3$ yields $(3, 6, 3)^\top$, which represents the same conservation law multiplied by three. These columns lie in the null space of ν_{toy} : they are component-space directions that the stoichiometric matrix sends to zero.

The full activated sludge model follows the same principle. If $a \in \mathbb{R}^{20}$ satisfies $\nu a = 0$, then

$$a^\top \nu^\top \rho = (\nu a)^\top \rho = 0 \quad (3.19)$$

for every possible process-rate vector. Importantly, this identity does not depend on estimating the rates correctly or on choosing a particular operating condition. It follows from the stoichiometric accounting itself. Mass conservation-aware surrogate methods exploit this separation by preserving balances without reproducing all original kinetic calculations during prediction (Hansen et al., 2024; Sturm & Wexler, 2020, 2022).

The number of independent conservation relations depends on the number of independent reaction directions. In the toy system, neither stoichiometric row is a multiple of the other. If $s(-2, 1, 0) + t(0, -1, 2) = (0, 0, 0)$, the first coordinate requires $s = 0$ and the third requires $t = 0$. The two rows are therefore independent, so the matrix has rank two. With three component coordinates and two independent reaction directions, one independent invariant remains. This is the rank–nullity relation,

$$\text{number of component coordinates} = \text{rank} + \text{null-space dimension}.$$

The free coefficient in the toy derivation is the arbitrary scale assigned to a_A . A basis is a set of independent vectors from which every vector in the space can be formed by weighted addition. Because the toy invariant space is one-dimensional, one basis column is sufficient.

For the full model, singular-value decomposition provides a numerically stable way to identify these directions. It factors a matrix into orthonormal directions and non-negative singular values. Orthonormal columns have unit Euclidean norm and zero scalar product with one another. A zero singular value marks a direction that the matrix sends to zero. For the toy matrix,

$$\nu_{\text{toy}} \nu_{\text{toy}}^\top = \begin{bmatrix} 5 & -1 \\ -1 & 5 \end{bmatrix}.$$

This square matrix has eigenvalues 4 and 6, giving two nonzero singular values, 2 and $\sqrt{6}$. The remaining component-space direction is the invariant direction. An eigenvalue describes the factor by which a square matrix stretches a corresponding direction, while

singular values provide the analogous non-negative stretching magnitudes for matrices that may be rectangular.

For the full matrix, U has dimension 28×28 , Σ has dimension 28×20 , and V has dimension 20×20 . The rectangular matrix Σ contains the singular values on its diagonal and zeros elsewhere. Their product reconstructs the original 28×20 stoichiometric matrix:

$$\nu = U\Sigma V^\top. \quad (3.20)$$

Finite arithmetic requires a numerical definition of rank. A mathematically zero singular value may appear as a very small nonzero number after floating-point calculations. Rank is therefore determined using

$$\tau_{\text{rank}} = \max(P, F)\epsilon_{\text{mach}}\|\nu\|_2. \quad (3.21)$$

Here ϵ_{mach} is machine precision and $\|\nu\|_2$ is the largest singular value. The largest singular value establishes the scale of the matrix, while the dimension factor accounts for the size of the calculation. Under this criterion, the adopted stoichiometric model has rank 15. With 20 component coordinates, rank-nullity therefore gives five independent invariants.

The toy invariant $(1, 2, 1)^\top$ illustrates how the basis is normalized. Its Euclidean length is $\sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$, so its unit-length form is

$$V_{0,\text{toy}} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Its scalar product with each stoichiometric row is zero: $(-2 + 2 + 0)/\sqrt{6} = 0$ and $(0 - 2 + 2)/\sqrt{6} = 0$. For the full model, five orthonormal right singular vectors perform the same role. Let $V_0 \in \mathbb{R}^{20 \times 5}$ contain the right singular vectors spanning $\text{null}(\nu)$, and define

$$A = V_0^\top \in \mathbb{R}^{5 \times 20}. \quad (3.22)$$

The operator then satisfies

$$A\nu^\top = 0, \quad AA^\top = I_5. \quad (3.23)$$

The first identity states that every invariant row has zero contribution from every reaction direction. The second states that the invariant rows have unit length and are mutually orthogonal. These rows need not correspond one-to-one with named elemental inventories. Their purpose is to provide a fixed, numerically scaled basis for the same conservation space.

The matrix orientation is important. Because the processes are rows of ν , the component combinations that define conservation lie in the right null space of ν . By contrast, the right null space of $\nu^\top$ lies in process coordinates and describes dependencies

among process contributions. These are different objects. Checking matrix dimensions prevents the two interpretations from being confused.

Reaction conservation alone, however, is not enough to establish conservation across an open reactor boundary. The inflow, outflow, and external transfer terms must also be considered. For invariant row h , multiply the component balance by A_{hj} and sum over the twenty components:

$$\begin{aligned} D \sum_{j=1}^{20} A_{hj}(c_{out,j} - c_{in,j}) &= \sum_{p=1}^{28} \rho_p \sum_{j=1}^{20} A_{hj} \nu_{pj} + \omega \sum_{j=1}^{20} A_{hj}(e_{SO})_j \\ &= 0 + \omega A_{h1}. \end{aligned}$$

The reaction contribution vanishes because of the invariant construction. The oxygen-transfer selector leaves only the oxygen-column coefficient. Premultiplying Equation (3.15) by A gives the same result in vector form:

$$DA(c_{out} - c_{in}) = \omega Ae_{SO}. \quad (3.24)$$

Reaction cancellation and external-forcing compatibility are therefore separate requirements.

The distinction can again be seen in the toy system. Adding one unit directly to pool A corresponds to a forcing column $(1, 0, 0)^\top$. Its invariant product is $1/\sqrt{6}$, so the forcing changes the tracked inventory. A forcing column $(-2, 1, 0)^\top$, in contrast, has invariant product $(-2 + 2)/\sqrt{6} = 0$ and lies within the reaction-change space. This abstract example illustrates why an external forcing must be checked against the conservation space rather than assumed to preserve it.

The range of a matrix is the set of vectors that it can produce by multiplication with all possible coordinate columns. Any forcing in $\text{range}(\nu^\top)$ can be written as a combination of reaction-change directions and is therefore annihilated by every invariant row. For the adopted stoichiometric matrix, the dissolved-oxygen selector satisfies this condition. There exists a process-coordinate vector $\lambda_O \in \mathbb{R}^{28}$ such that

$$e_{SO} = \nu^\top \lambda_O, \quad Ae_{SO} = A\nu^\top \lambda_O = 0. \quad (3.25)$$

The coordinates in λ_O establish a subspace relation only. They do not mean that aeration is itself a biological process or that the entries of λ_O are non-negative process rates.

Because $Ae_{SO} = 0$, Equation (3.24) reduces to $DA(c_{out} - c_{in}) = 0$. Since $D > 0$,

$$Ac_{out} = Ac_{in}. \quad (3.26)$$

This equality follows jointly from the stoichiometric model, the direction of the oxygen forcing, and the stated reactor boundary. It should not be transferred unchanged to a system with an unaccounted phosphorus dose, sludge withdrawal, or another external source that changes the conserved inventories.

More generally, let g denote an additional component source expressed on a concentration-per-time basis. Its contribution to invariant row h is $\sum_j A_{hj}g_j$. Dividing by the positive dilution rate converts that contribution to a concentration-equivalent invariant change. When aeration remains compatible,

$$A(c_{out} - c_{in}) = D^{-1}Ag. \quad (3.27)$$

A phosphate addition with a nonzero phosphorus-inventory contribution would therefore change the right-hand side. Imposing the original influent equality in such a system would incorrectly remove a real external contribution. This example illustrates a general principle used throughout the dissertation: a physical constraint must be derived from the actual model boundary, not selected merely because it produces a convenient statistical result.

All numerical calculations use the full-precision operator. Its construction is checked through $A\nu^\top$, $AA^\top - I_5$, and Ae_{S_O} . Rounded coefficients displayed for interpretation cannot replace the original operator in computation. Any nonsingular row transformation T produces $A' = TA$ and defines the same equality set, but an arbitrary transformation changes the numerical scale of residuals. The orthonormal construction therefore fixes both the invariant space and the scaling used for mass conservation diagnostics.

3.6 Reaction progress and the feasible state geometry

The invariant relation describes what cannot change. A complementary view describes the directions along which the component state can change while preserving those invariants. This reaction-progress view is useful because it connects the mechanistic stoichiometric model to the geometry of the feasible states later used for projection.

Consider again the hypothetical three-pool system. Suppose the first process advances by one unit and the second by one half-unit on a compatible concentration-equivalent basis. The accumulated component changes are

$$\begin{aligned} \Delta c_A &= -2(1) + 0(0.5) = -2, \\ \Delta c_B &= 1(1) - 1(0.5) = 0.5, \\ \Delta c_C &= 0(1) + 2(0.5) = 1. \end{aligned}$$

Their weighted sum is $-2 + 2(0.5) + 1 = 0$. Starting from the hypothetical state $(8, 1, 1)^\top$ gives $(6, 1.5, 2)^\top$, and both states have weighted inventory 11. The accumulated progress is conceptually different from an instantaneous process rate: multiplying a rate by an appropriate time scale produces a concentration-equivalent state change.

For the steady reactor, that natural time scale is D^{-1} . Using $\omega e_{s_O} = \nu^\top (\omega \lambda_O)$ allows reaction and compatible oxygen transfer to be represented in the same component-change space. Rearranging the steady-state balance gives

$$c_{out} - c_{in} = \nu^\top r, \quad r = D^{-1}(\rho + \omega \lambda_O). \quad (3.28)$$

The vector r provides one set of process-space coordinates for the total component change. It should not be interpreted as a unique reconstruction of the biological reaction rates. The matrix $\nu^\top$ has rank 15 but 28 columns, so multiple process-coordinate vectors can produce the same component change. Moreover, r contains the algebraic representation of oxygen transfer. Its entries are therefore not independently measured reaction extents or microbial activities.

A mass-conserving component change also need not be represented using the original process coordinates. In the toy system, every admissible change v satisfies $v_A + 2v_B + v_C = 0$. Two independent unit-length directions satisfying this condition are

$$z_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad z_2 = \frac{1}{\sqrt{30}} \begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix}.$$

Their invariant products are zero, their mutual scalar product is zero, and both have unit length. Thus $Z_{\text{toy}} = [z_1 \ z_2]$ is an orthonormal basis of the two-dimensional change space. Any toy-state change that preserves the weighted inventory can be written as $v = z_1 q_1 + z_2 q_2$.

The change basis and the invariant basis answer complementary questions. The invariant rows identify combinations that cannot change. The columns of Z identify directions along which the state can move without changing those combinations. In the full model, five independent invariant rows leave fifteen independent change directions. Let $Z \in \mathbb{R}^{20 \times 15}$ contain orthonormal columns spanning $\text{null}(A)$. The corresponding subspace relation is

$$\text{range}(\nu^\top) = \text{null}(A), \quad AZ = 0, \quad Z^\top Z = I_{15}. \quad (3.29)$$

Every component change satisfying the conservation equalities can therefore be written uniquely as Zq for some $q \in \mathbb{R}^{15}$ in the selected orthonormal basis. These coordinates describe admissible material redistribution. They do not identify a unique sequence of the 28 kinetic processes responsible for that redistribution.

The same relation defines the feasible component states used in prediction. For sample i , the invariant target is $b_i = Ac_{in,i}$. A physically admissible effluent must satisfy the five invariant equalities and the twenty non-negativity bounds:

$$\mathcal{F}_i = \{c \in \mathbb{R}^{20} \mid Ac = b_i, \ c \geq 0\}. \quad (3.30)$$

The equality $Ac = b_i$ defines a 15-dimensional affine space before the concentration bounds are applied. An affine space is a linear change space translated by a fixed

reference state; the influent provides one such reference here. Intersecting that affine space with the non-negative orthant retains only states whose component concentrations are all non-negative.

The resulting set has several useful properties. It is closed because the equalities and inequalities include their boundary points. It is convex because any weighted average of two feasible states, using weights between zero and one, preserves both the invariant target and component non-negativity. It is also nonempty whenever the influent is non-negative, because the influent itself satisfies $Ac_{in,i} = b_i$ and provides a feasible anchor.

The toy example makes this geometry visible. For the hypothetical influent $(8, 1, 1)^\top$, the conservation condition is $c_A + 2c_B + c_C = 11$, together with $c_A, c_B, c_C \geq 0$. If only one pool is nonzero, the three boundary states are $(11, 0, 0)^\top$, $(0, 5.5, 0)^\top$, and $(0, 0, 11)^\top$. The feasible region is the triangle connecting these states within the inventory plane. The state $(6, 1.5, 2)^\top$ lies inside this triangle. The state $(12, -0.5, 0)^\top$ lies in the same inventory plane because $12 - 1 = 11$, but it is not physically admissible because one coordinate is negative. Mass conservation selects the plane; non-negativity selects the admissible portion of that plane.

Figure 3.2 shows the same region using c_A and c_B as plotting axes. The conservation equation supplies the third coordinate through $c_C = 11 - c_A - 2c_B$, so every plotted point still represents a complete three-component state.

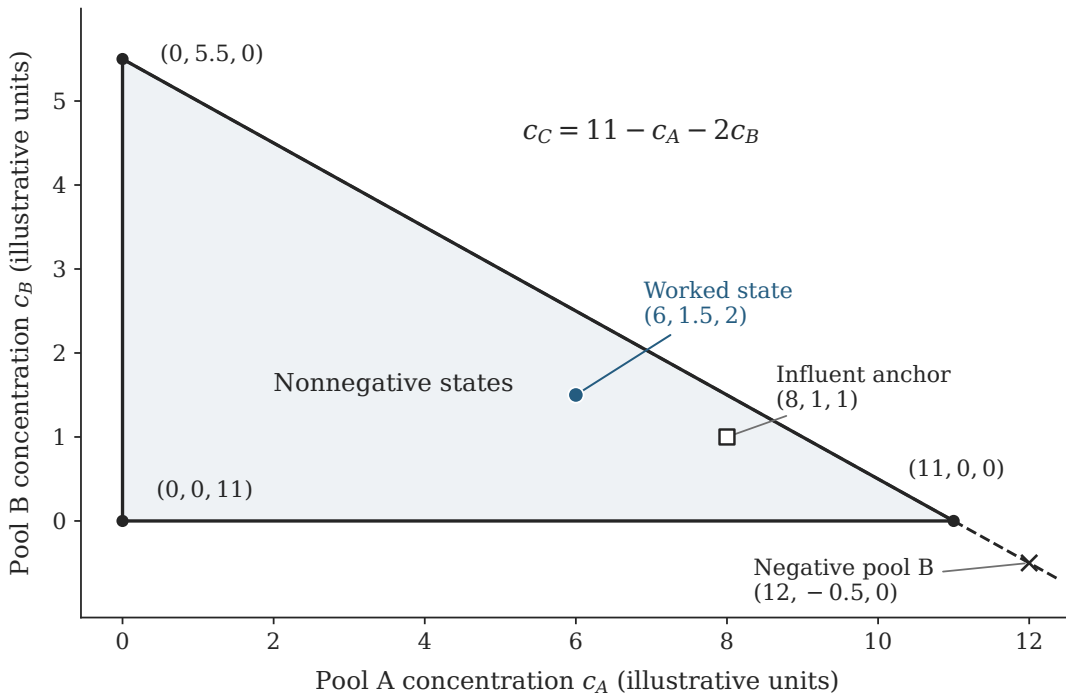


Figure 3.2: Hypothetical non-negative states for the weighted inventory $c_A + 2c_B + c_C = 11$. The shaded triangle includes its boundary. Full state labels retain the third component even though only two coordinates are plotted.

The sloping edge corresponds to $c_C = 0$, while the two coordinate edges correspond to $c_A = 0$ or $c_B = 0$. All three components are positive in the interior, allowing the same conserved inventory to be distributed among them in many different proportions. The cross-marked point preserves the weighted inventory but has a negative B concentration. The figure therefore shows that conservation and concentration bounds define different requirements.

A two-component example makes the same point even more directly. Suppose conservation fixes $c_1 + c_2 = 10$. The equality alone permits states such as $(-1, 11)$, $(4, 6)$, and $(12, -2)$. Non-negativity reduces this infinite line to the segment between $(0, 10)$ and $(10, 0)$. Simply clipping $(-1, 11)$ to $(0, 11)$ removes the negative concentration but changes the conserved sum from 10 to 11. Enforcing the equality and enforcing non-negativity are therefore not interchangeable operations; simultaneous enforcement requires their intersection.

The geometry also clarifies the limitation of a physically admissible prediction. Many non-negative component states can share the same five invariant values. A state with an incorrect distribution between ammonium and nitrate may still satisfy all conservation equalities. The same can occur for an implausible biomass concentration. The feasible set therefore establishes the declared material accounting and sign restrictions, but it does not establish that the complete 28-process kinetics balance hydraulics and aeration at that state. This distinction is central to the constraint methods of Kircher and Votsmeier (2025) and to the broader physical-consistency perspective of Valente et al. (2025).

3.7 Reported water quality quantities and hidden component errors

Engineering decisions are often communicated through aggregate quantities such as COD, TN, TP, and TSS rather than through the complete 20-component state. These quantities are important summaries, but they are calculated from the component state after prediction. The distinction matters because aggregation can hide component-level errors and because a reporting total is not necessarily a conserved inventory.

Using the coefficients in Table 3.1, the four reported quantities are

$$\begin{aligned}
 \text{COD} &= S_F + S_A + S_I + X_I + X_S + X_H + X_{PAO} \\
 &\quad + X_{PHA} + X_{AOB} + X_{NOB}, \\
 \text{TN} &= S_{NH_4} + S_{NO_2} + S_{NO_3} + 0.01S_I + 0.02X_I \\
 &\quad + 0.07(X_H + X_{PAO} + X_{AOB} + X_{NOB}), \\
 \text{TP} &= S_{PO_4} + 0.01X_I + 0.02(X_H + X_{PAO} + X_{AOB} + X_{NOB}) \\
 &\quad + X_{PP} + X_{MeP}/4.87, \\
 \text{TSS} &= 0.75(X_I + X_S) + 0.90(X_H + X_{PAO} + X_{AOB} + X_{NOB}) \\
 &\quad + 3.23X_{PP} + 0.60X_{PHA} + X_{MeP} + X_{MeOH}.
 \end{aligned}$$

Each coefficient converts a component from its native basis to the output basis of the corresponding reported quantity. For example, the factor 0.07 has units of g nitrogen per g biomass COD. Multiplying it by X_H therefore gives a nitrogen concentration. Likewise, the factor 3.23 converts stored phosphorus to its suspended-solids contribution. Every term within one reported quantity is consequently expressed on a common basis before being summed.

A sparse hypothetical state containing only $X_H = 100$ g COD m^{-3} and $X_{PP} = 2$ g P m^{-3} gives

$$\begin{aligned}
 \text{COD} &= 100, & \text{TN} &= 0.07(100) = 7, \\
 \text{TP} &= 0.02(100) + 2 = 4, & \text{TSS} &= 0.90(100) + 3.23(2) = 96.46.
 \end{aligned}$$

Each result has its own constituent unit. The example illustrates how reporting coefficients combine different state components; it is not intended to represent a reactor steady state.

The same four sums can be written compactly using a fixed reporting matrix:

$$y = I_{comp}C, \quad y = (\text{COD}, \text{TN}, \text{TP}, \text{TSS})^\top, \quad I_{comp} \in \mathbb{R}^{4 \times 20}. \quad (3.31)$$

The first row contains the COD coefficients, the second the TN coefficients, and so forth. The rows convert the component state to g COD m^{-3} , g N m^{-3} , g P m^{-3} , and g TSS m^{-3} . Because all reporting coefficients are non-negative, a non-negative component state necessarily gives non-negative reported quantities.

A reported total should not, however, be confused with a conserved inventory. Dissolved dinitrogen has zero weight in the adopted TN reporting map. A hypothetical transfer of 3 g N m^{-3} from nitrate to dissolved dinitrogen therefore reduces reported TN by 3 g N m^{-3} even though the complete modeled nitrogen inventory can remain unchanged. A phosphorus reporting row may coincide more closely with a modeled phosphorus inventory, but that relationship must still be checked from the coefficients. The word “total” alone does not establish conservation.

The reporting coefficients also make the constituent conversions transparent. Under the adopted map, a hypothetical $X_H = 100 \text{ g COD m}^{-3}$ contributes 100 g COD m^{-3} to COD, 7 g N m^{-3} to TN, 2 g P m^{-3} to TP, and 90 g TSS m^{-3} to TSS. A hypothetical $X_{PP} = 2 \text{ g P m}^{-3}$ contributes 2 g P m^{-3} to TP and $6.46 \text{ g TSS m}^{-3}$ to TSS. These values arise from fixed reporting assumptions rather than from the surrogate model or from an observed treatment outcome.

The main consequence of aggregation is that different component errors can cancel. Equal and opposite errors in ammonium and nitrate disappear in reported TN because both have unit nitrogen weights. A negative ammonium prediction can even be obscured by a sufficiently large nitrate prediction. Likewise, biomass and substrate errors can compensate in COD while remaining visible in TSS because their solids conversion factors differ. Accurate COD, TN, TP, and TSS values therefore do not establish either an accurate or a physically admissible 20-component state.

The reporting map and the invariant operator consequently answer different questions. The reporting map summarizes treatment performance in familiar engineering quantities. The invariant operator tests whether the component state satisfies the declared material accounting. Neither one reconstructs the detailed component distribution lost through aggregation. Component prediction, composite reporting, mass conservation, and non-negativity must therefore remain separate objects of assessment.

3.8 Mixing and spatial transport as model boundaries

The standalone reactor model assumes one completely mixed concentration for each component. This removes spatial coordinates from the reactor state and allows the treatment response to be represented by one component vector. The assumption is a modeling boundary rather than a statement that every biological treatment unit is spatially uniform. Biofilms, trickling filters, and incompletely mixed aerated vessels can exhibit concentration gradients and transport limitations (Sadino-Riquelme et al., 2023; Wik, 2003; T. C. Zhang & Bishop, 1994). Recognizing these effects helps define what the mixed-reactor surrogate does and does not represent.

A small spatial control volume illustrates the additional accounting required when concentrations vary in space. For one component, let V denote the fixed control-volume size and $\bar{c}$ its average concentration. Let $F_{\text{adv,in}}$ and $F_{\text{adv,out}}$ denote advective mass flow rates, and let $F_{\text{diff,in}}$ and $F_{\text{diff,out}}$ denote diffusive mass flow rates. The finite-volume balance is

$$V \frac{d\bar{c}}{dt} = F_{\text{adv,in}} - F_{\text{adv,out}} + F_{\text{diff,in}} - F_{\text{diff,out}} + Vr. \quad (3.32)$$

The quantity r is the net reaction production rate per unit volume and becomes negative when the component is consumed. Every term on the right-hand side has units of

component mass per time.

For a hypothetical volume of 2 m^3 , advective inflow and outflow of 5 and 3 g d^{-1} , diffusive inflow and outflow of 2 and 0 g d^{-1} , and total reaction consumption of 3 g d^{-1} , the net accumulation is $5 - 3 + 2 - 0 - 3 = 1 \text{ g d}^{-1}$. Dividing by the control-volume size gives $d\bar{c}/dt = 0.5 \text{ g m}^{-3} \text{ d}^{-1}$. The example shows why a local concentration change cannot be inferred from the reaction term alone: transport can be equally important.

In a one-dimensional domain with coordinate z , the advective flux is vc and the diffusive flux is $-D_z \partial c / \partial z$, where v is velocity and D_z is a dispersion or diffusion coefficient. Dividing the balance of a thin slice by its volume and taking the limit as its thickness approaches zero gives

$$\frac{\partial c_j}{\partial t} = -\frac{\partial(vc_j)}{\partial z} + \frac{\partial}{\partial z} \left(D_z \frac{\partial c_j}{\partial z} \right) + \sum_{p=1}^{n_p} \nu_{pj} \rho_p + s_j. \quad (3.33)$$

The first two terms describe spatial transport, the summation retains the same component-wise reaction accounting used in the mixed reactor, and s_j represents a specified external source per unit volume. Boundary conditions then determine how material enters and leaves the spatial domain. Applying an invariant weighted sum can cancel compatible reaction terms, but transport and external-source contributions remain in the overall balance.

This formulation shows how the same theoretical ideas can extend beyond a completely mixed reactor without implying that a spatial reactor is added to the present study. Chapter 6 instead uses a sequence of mixed biological reactors and a layered clarifier. The reactor stages represent distinct treatment conditions, while the clarifier layers represent solids transport and storage. Their boundaries are different, and those differences must be respected when the connected plant constraints are derived.

3.9 A common basis for assessing surrogate responses

The preceding sections establish a common component representation for the rest of the dissertation. The mechanistic model supplies the reference treatment state. A statistical surrogate approximates that state from the influent and operating controls. Projection adjusts the prediction so that it lies within the specified physical set. Interpretation examines how the prediction is produced, and optimization uses the returned response to select operating conditions. The same component state therefore connects prediction, physical enforcement, interpretation, and decision assessment.

This connection is useful because these questions require different evidence. Prediction error measures agreement with the mechanistic reference. Physical admissibility concerns the imposed equalities, bounds, and inventories. Interpretability concerns the structure of the fitted response and how that structure changes through subsequent

transformations. Operating quality concerns the treatment outcome obtained at the selected controls. One type of evidence cannot substitute for another.

The same distinction applies when raw and projected states are compared. A paired raw–projected assessment keeps the fitted learner fixed and isolates the effect of physical enforcement. Comparing separately trained predictors also includes differences in model structure, hyperparameter selection, and fitting. The structured regression developed later introduces additional intermediate states, including coupled and affine predictions, so that the origin of a change can be traced through the complete prediction procedure.

At plant scale, the accounting extends across locations. The mixer, reactor stages, clarifier outlets, and stored solids each contribute part of the connected response. Location-specific errors show where joint projection changes the predicted state, while plant-level summaries describe the overall approximation. The quantities used in the intended operating decision then determine which parts of that response require the closest scrutiny (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019).

Table 3.2 summarizes this assessment logic. Input transformations, learned scales, regularization, and hyperparameter selection use development information. Reserved responses supply the final prediction assessment. This separation applies to the complete procedure, including both the fitted statistical model and the constrained calculation that follows it (Kaufman et al., 2012).

Table 3.2: Assessment questions and the evidence used to connect surrogate responses to engineering decisions.

Assessment question	Evidence	Engineering quantity
Predictive agreement	Held-out component and composite errors with defined scales	Substrate, nutrients, biomass, and solids concentrations
Physical admissibility	Equality residuals, non-negativity checks, and inventory bounds	Conserved inventories and feasible component states
Sample efficiency	Common learning curves and variation across partitions	Error at the available simulation budget
Inspectable structure	Named coefficient blocks, physical units, and complete-response sensitivities	Dependence on influent and operating inputs
Operating quality	Independent mechanistic evaluation at selected controls	Treatment outcomes, engineering requirements, and objective values
Computational effort	Development, search, prediction, projection, and verification times	Time required to deliver an evaluated decision

The numerical checks are reported using explicit tolerances and residual definitions.

The equalities define the mathematical constraints; the residuals quantify how closely the returned numerical state satisfies them. Reporting these quantities in their original units or under a clearly defined scaling makes the physical calculation open to inspection (Boyd & Vandenberghe, 2004; Buzzi-Ferraris & Manenti, 2013). Chapters 4–6 apply this common basis to the standalone reactor benchmark, the structured regression, and the connected-plant optimization study.

Chapter 4

Learning the Reactor Response and Enforcing Physical Admissibility

4.1 Rationale

Repeated activated sludge calculations are needed to compare influent conditions and operating settings. A surrogate can reduce the cost of these calculations by approximating responses learned from simulated cases. Its engineering value depends on what it predicts and how that prediction is checked. Organic substrate, nitrogen species, phosphorus, biomass, and solids all contribute to the treatment response and reactor material accounting (Henze et al., 2000). Predicting a few reported totals does not establish agreement with this complete state. Two effluents with the same TN can contain different amounts of ammonium and nitrate. A model that predicts the total correctly can therefore still misrepresent a treatment process of interest.

The first gap concerns comparative evidence for the complete activated sludge response. The soft sensor of Ching et al. (2022) estimates oxygen demand from available measurements, while Y. Wang et al. (2025) compare methods for effluent TN prediction. The structured reactor study of Selerio (2026) retains the complete effluent component state. These contributions support useful but different prediction tasks. Their findings do not by themselves establish which learning method best approximates the same complete state under a shared assessment. A meaningful comparison requires common influent and operating inputs, component targets, assessment cases, and error scales. Otherwise, differences in the prediction task can be mistaken for differences between learners.

Data availability is an important part of this gap. Generating training cases requires mechanistic reactor calculations, so the available simulation budget affects the practical choice of a surrogate. A flexible learner may perform well with many cases without being the best choice when only a small design is available. The learning-curve review of Viering and Loog (2023) explains why performance should be followed across data

budgets. The tabular comparisons of Borisov et al. (2024) and Shwartz-Ziv and Armon (2022) also support comparing several learning families rather than assuming that one family is consistently preferable. For activated sludge analysis, this comparison must retain the substrate, nutrient, biomass, and solids responses that the engineer intends to use.

Prediction beyond the sampled influent and operating ranges poses a separate question. A held-out case within those ranges tests a different use from an unfamiliar loading or operating condition. The study therefore examines learning-domain and exterior inputs separately. This distinction is important when a surrogate is used to explore alternatives rather than reproduce familiar cases. Neither a favorable learning-domain error nor a physically admissible exterior prediction establishes accurate treatment prediction beyond the sampled range (Kircher & Votsmeier, 2025; Selerio, 2026).

The second gap concerns simultaneous physical enforcement and its effect on the same reactor predictions. Ordinary regression approximates the statistical response without necessarily preserving the inventories or concentration bounds of a new influent. Physical guidance during fitting and explicit constraints on the returned state serve different purposes (Hansen et al., 2024; Kashinath et al., 2021). A balance check alone can accept a state with a negative component. Replacing that negative concentration with zero can then disturb the balance. Both requirements must therefore be checked on each returned state. The projection of Kircher and Votsmeier (2025) provides a method for enforcing mass conservation and non-negativity together. The unresolved application question is how this common adjustment behaves across activated sludge learners, component scales, and input conditions.

Physical compliance is only one part of that assessment. Projection changes the component vector and can redistribute error among its entries. Those changes also alter COD, TN, TP, and TSS, which are reported treatment quantities rather than interchangeable measures of reactor conservation. The species-weighted correction of Sturm and Silva (2025) shows why the scale assigned to an adjustment matters. An improvement in nutrient prediction can accompany a deterioration in another quantity. A single aggregate error or a count of feasible states cannot describe this tradeoff. Component errors, reported treatment errors, and physical checks are therefore needed together.

The adjustment also has a computational cost. A surrogate intended for repeated operating calculations must be assessed as the complete prediction procedure, including projection. Large changes from a raw state can reveal disagreement with the imposed physical relations, but small changes do not establish accurate reaction behavior. Paired raw and projected predictions allow error, physical compliance, displacement, and prediction cost to be examined without changing the fitted learner. This separation follows the need to assess approximation quality and computational usefulness together

in surrogate applications (Bhosekar & Ierapetritou, 2018).

The study addresses the first and second research questions in Section 1.3. It compares complete reactor predictions across learning methods, simulation budgets, and input-domain conditions. It then assesses the effect of a common projection on those same predictions. The contribution is an activated sludge benchmark that separates learner performance from physical enforcement and identifies their joint consequences for treatment analysis. It does not introduce a new projection method or presume that projection improves every error measure. Its practical purpose is to support a choice of learner and adjustment procedure for specified data availability, treatment quantities, and operating ranges.

4.2 Theory

4.2.1 Reactor inputs and raw predictions

For sample i , the two operating values are entered first. They are followed by the twenty influent concentrations in the established component order. Written as a list, the input begins with $(\text{HRT}_i, u_{\text{air},i}, S_{O,\text{in},i}, S_{F,\text{in},i})$ and ends with the remaining influent coordinates through $X_{\text{MeOH},\text{in},i}$. The component list and its units are fixed before any learner is fitted. Transposing the two input blocks into rows permits their horizontal concatenation. The final transpose restores the column convention. The complete input for sample i is

$$x_i = (u_i^\top, c_{\text{in},i}^\top)^\top \in \mathbb{R}^{22}, \quad u_i = (\text{HRT}_i, u_{\text{air},i})^\top. \quad (4.1)$$

A fitted surrogate is a function that takes these twenty-two numbers and returns twenty component values. The subscript m identifies which fitted method is used. For component j , its prediction can be written as $c_{\text{raw},ij}^{(m)} = f_{m,j}(x_{i1}, \dots, x_{i22})$. The superscript (m) identifies the model and is not an exponent. Collecting its twenty scalar output functions, a fitted surrogate f_m produces the raw component prediction

$$c_{\text{raw},i}^{(m)} = f_m(x_i) \in \mathbb{R}^{20}. \quad (4.2)$$

The raw prediction is assessed in its returned physical units. A post prediction projection then produces a second prediction for the same sample. Pairing the two states permits the effect of physical enforcement to be assessed separately from the choice of learner, data partition, and preprocessing (Kircher & Votsmeier, 2025; Sturm & Silva, 2025).

4.2.2 Statistical response models

Boosting and randomized tree ensembles

XGBoost builds an additive sequence of regression trees. A regression tree applies a sequence of input comparisons and assigns the value associated with the reached

terminal region. For one output, an additive tree model has the scalar form

$$f_j(x) = b_j + \beta_1 h_{1j}(x) + \beta_2 h_{2j}(x) + \cdots + \beta_T h_{Tj}(x).$$

Here h_{tj} is the response from tree t , β_t determines its contribution, and b_j is a baseline. The summation notation $b_j + \sum_{t=1}^T \beta_t h_{tj}(x)$ records the same additions. A hypothetical baseline of 10 and two tree contributions of -2 and 1 produce 9. Each new tree addresses the error left by the current ensemble while regularization discourages an unnecessarily complex fit. Its split structure represents changes in response across input regions, and repeated additions permit interactions and nonlinear response shapes. The computational design and regularized objective of Chen and Guestrin (2016) motivate its inclusion as a flexible tabular baseline.

LightGBM also uses gradient-boosted trees, but grows trees through leaf-wise expansion. This strategy can allocate more detail to regions where an additional split offers a larger loss reduction. The efficiency-oriented design of Ke et al. (2017) makes the method relevant when each candidate fit must be repeated over several validation partitions. Its learning rate, leaf count, and minimum leaf support jointly control how closely it follows the training response.

CatBoost uses symmetric boosted trees and a Bayesian bootstrap in the specified regression setting. The work of Prokhorenkova et al. (2018) addresses bias associated with categorical-feature handling and boosting. The reactor benchmark evaluates this learner on continuous inputs. Each level of a symmetric tree uses the same splitting condition across nodes at that level. This structural restriction gives a different approximation class from the leaf-wise trees of LightGBM.

Adaptive boosting (AdaBoost) regression combines shallow trees through adaptive emphasis on observations that are difficult to predict. The base trees have depth three. The estimator count, learning rate, and loss choice control how strongly the ensemble responds to poorly fitted examples. Drucker (1997) develops the regression extension of boosting that supports this comparator. These four boosting methods fit one regressor for each effluent component. All 20 regressors within a method use one shared selected hyperparameter vector.

Random Forest averages randomized regression trees. For component j , the average of T tree responses is $f_j(x) = [h_{1j}(x) + \cdots + h_{Tj}(x)]/T$. Hypothetical tree values of 2, 3, and 4 give a mean of 3. Variation in sampled observations and candidate features reduces dependence among trees, so the average is less sensitive to a particular tree fit (Breiman, 2001). Extra Trees adds randomization of split thresholds (Geurts et al., 2006). Both methods fit the 20 outputs jointly in native multiresponse ensembles. Joint output handling lets the same tree partition support several component predictions. The projection subsequently enforces their stoichiometric relationships.

Tree averaging can preserve non-negativity when all leaf responses are non-negative. Mass conservation also requires the balance target of the new influent. For a hypothetical

two-component system, two training responses with totals of 10 and 20 have an equally weighted average with total 15. If a new influent requires total 12, the average is positive but inconsistent with mass conservation for that sample. This distinction is caused by the sample-specific target b_i , rather than by whether the predictor averages positive observations.

Kernel, neighborhood, and latent-variable regression

Support vector regression (SVR) fits a function with an error-insensitive region around each training response. For scalar error e and positive insensitive width ϵ , its error contribution is zero when $|e| \leq \epsilon$ and $|e| - \epsilon$ otherwise. This can be recorded as $\max(0, |e| - \epsilon)$. With a hypothetical width of 0.2, an error of 0.1 contributes zero and an error of -0.5 contributes 0.3. Predictions within that region incur no corresponding error penalty. A linear kernel gives a globally linear input mapping. A radial kernel allows a nonlinear mapping through similarity to training inputs. Its similarity has the form

$$K(z, z') = \exp \left[-\gamma \sum_{\ell=1}^{22} (z_{\ell} - z'_{\ell})^2 \right].$$

Here z and z' contain the standardized input coordinates used by this learner. The positive coefficient γ controls how quickly similarity decreases with squared distance. Identical inputs give similarity one. Greater distance gives a smaller value. The penalty parameter balances fit and regularity, while the insensitive width sets the tolerated response discrepancy. Smola and Schölkopf (2004) explain these mechanisms and the importance of their parameterization. The benchmark fits one regressor per component with one shared setting vector.

The k -nearest-neighbor (k -NN) predictor finds training observations close to a new input and combines their 20-component responses. For one component and a selected neighborhood $\mathcal{N}(x)$, weighted averaging means

$$f_j(x) = \frac{\sum_{i \in \mathcal{N}(x)} w_i c_{out,ij}}{\sum_{i \in \mathcal{N}(x)} w_i}.$$

The neighborhood size controls the degree of smoothing. Uniform weights give every selected neighbor equal influence, whereas distance weights give closer observations greater influence. Hypothetical neighbor values 2 and 6 give 4 under equal weights. Weights 3 and 1 instead give $[3(2) + 1(6)]/(3 + 1) = 3$. The local behavior analyzed by Mack (1981) explains why input scaling and distance choice matter. An unscaled alkalinity coordinate spanning hundreds of numerical units could dominate a much smaller aeration coordinate even though both influence the reactor. Standardization avoids using native numerical range as an implicit measure of scientific importance.

Partial least squares (PLS) regression represents the inputs and joint response through latent directions chosen to capture their covariance. One latent score is a weighted input sum $t_h = w_{1h}z_1 + \dots + w_{22h}z_{22}$, where z contains standardized inputs.

An output is then a baseline plus a weighted sum of the retained scores. Substituting each score's input sum shows why ordinary partial least squares remains a linear input mapping. It can summarize strongly related input variables with a relatively small number of latent components. These weighted statistical combinations differ from the physical effluent components predicted by the model. The chemometric formulation described by Wold et al. (2001) provides an inspectable linear baseline. Its wastewater use is illustrated by Woo et al. (2009), who employs a kernel extension for process-variable estimation, and by C. Yang et al. (2021), who combines partial least squares with another predictor for adaptive effluent-quality prediction. The benchmark retains ordinary partial least squares with one to six latent components while predicting all 20 physical effluent components.

Shared sparse coefficient models

Multi-task Lasso fits one coefficient matrix for all component outputs. A predicted output is a baseline plus a sum of input-coefficient products. For two hypothetical inputs and two outputs, the coefficient rows can be $(3, 4)$ and $(0, 0)$. The first input contributes to both outputs and the second contributes to neither. The Euclidean length of the first row is $\sqrt{3^2 + 4^2} = 5$, and the length of the second row is zero. Adding row lengths gives 5. This operation penalizes an input's complete coefficient group rather than adding the absolute value of every coefficient separately. If row $B_{j,\cdot}$ contains the coefficients of input j across those outputs, the row-group penalty is

$$\|B\|_{2,1} = \sum_j \|B_{j,\cdot}\|_2. \quad (4.3)$$

This penalty can set an entire input row to zero. It therefore promotes a shared selection of inputs rather than 20 unrelated selections. The feature-sharing principle of Argryriou et al. (2008), the multi-task Lasso algorithm of Liu et al. (2009), and the grouped-variable framework of Yuan and Lin (2006) provide the relevant foundations.

Multi-task Elastic Net combines the row-group penalty with a squared Frobenius penalty. The squared Frobenius norm is the sum of the squares of all matrix entries. For the hypothetical coefficient rows just given, it is $3^2 + 4^2 + 0^2 + 0^2 = 25$. With penalty strength $\lambda > 0$ and mixing ratio $r \in (0, 1)$, the combined penalty has the form

$$\lambda r \sum_j \sqrt{\sum_{f=1}^{20} B_{jf}^2} + \frac{\lambda(1-r)}{2} \sum_j \sum_{f=1}^{20} B_{jf}^2.$$

For the two-output illustration only, taking $\lambda = 0.2$ and $r = 0.5$ gives $0.2(0.5)(5) + 0.2(0.5)(25)/2 = 1.75$. The row-group part encourages selection and the squared part shrinks the retained coefficients. This extends the elastic-net principle of Zou and Hastie (2005) to a joint response structure, with multiresponse computation described by Simon et al. (2013). Both methods use standardized inputs and targets. Their

global coefficient matrices can be inspected to identify fitted associations. A nonzero row identifies an input that contributes to the fitted prediction.

These two predictors and partial least squares are grouped as structurally inspectable surrogates. The grouping identifies their globally available coefficient structures (Rudin, 2019). Physical projection enforces the influent-specific balances and non-negativity of their returned predictions.

Neural predictors

The multilayer perceptron (MLP) uses three fully connected hidden layers with 128, 64, and 32 units and one joint 20-output layer. One unit first forms a weighted sum, such as $b + w_1 z_1 + w_2 z_2$, and then applies an activation. A rectified linear activation returns the larger of zero and that sum. For hypothetical values $b = 0.2$, $w_1 = 0.5$, $w_2 = -1$, $z_1 = 2$, and $z_2 = 0.3$, the sum is $0.2 + 0.5(2) - 0.3 = 0.9$ and the activated value is 0.9. If the sum were -0.9 , the activated value would be zero. Each hidden layer applies such operations to the values supplied by the preceding layer. Backpropagation updates the weights according to the prediction loss (Rumelhart et al., 1986). The approximation result of Hornik et al. (1989) motivates the expressive capacity of feedforward networks. The topology considerations discussed by Curteanu and Cartwright (2011) motivate the stated architecture and training controls.

TabNet applies sequential attentive feature selection. At a decision step, a feature mask multiplies each input coordinate by its selected attention weight. If two hypothetical standardized inputs are $(2, 1)$ and their mask weights are $(0.8, 0.2)$, the masked inputs are $(1.6, 0.2)$. A different step can use different weights and therefore examine a different input emphasis. The selected features then contribute to a joint output layer. Arik and Pfister (2021) develop this approach for tabular learning. The attention masks show how the model weights its inputs during prediction. TabNet remains a neural comparator even though its internal masks can be inspected.

An epoch is one pass through the training data during neural fitting. Patience is the allowed number of consecutive epochs without the specified improvement before stopping. These fitting quantities describe the use of simulated reactor cases.

Both neural predictors start from random initialization and use only the simulated training partition available to that fit. The multilayer perceptron uses all available training rows and stops on training-loss convergence or at 250 epochs. Validation-based early stopping is disabled. Its training-loss patience is 15 epochs. TabNet's epoch count is selected in the inner search and ranges from 20 to 50. Its patience is zero, so a refit runs the selected epoch count without reserving an additional response-based stopping partition.

4.2.3 Simultaneous physical requirements

Now let the first hypothetical concentration be 11 and the second be -1 . Their sum is $11 + (-1) = 10$, so the inventory equality is satisfied. The second component fails non-negativity. Replacing it by zero changes the sum to $11 + 0 = 11$. To return to the non-negative part of the line defined by mass conservation, both coordinates can be changed. The state with first concentration 10 and second concentration zero has sum ten and passes both checks.

The individual adjustments make the distinction explicit,

$$\begin{aligned}\text{raw state} &= (11, -1)^\top, \\ \text{clipped state} &= (11 + 0, -1 + 1)^\top = (11, 0)^\top, \\ \text{jointly feasible state} &= (11 - 1, -1 + 1)^\top = (10, 0)^\top.\end{aligned}$$

Clipping changes only the second coordinate. The joint projection also compensates in the first coordinate. An equality projection alone can retain a negative component, while clipping alone can violate an equality. The final assessment therefore checks both properties on the same returned state (Haddad et al., 2010; Kircher & Votsmeier, 2025).

Figure 4.1 places the three states on the line defined by mass conservation and its non-negative segment. Their positions show which requirement each state satisfies.

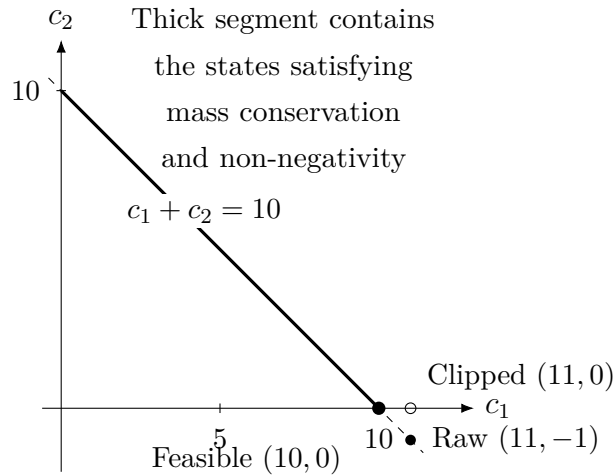


Figure 4.1: Hypothetical two-component geometry showing why clipping a negative concentration can violate a sum fixed by mass conservation.

The raw point is on the extended line defined by mass conservation but below the concentration boundary. The clipped point reaches a non-negative second coordinate while leaving that line. The jointly projected point lies at the end of the thick feasible segment. The geometry shows the two properties checked on each returned state.

Separate marginal summaries can also be misleading. If one prediction satisfies only mass conservation and another satisfies only non-negativity, neither is jointly feasible. For a hypothetical collection of two such states, each marginal compliance frequency

is one half, while the joint compliance frequency is zero. The correct denominator is the number of states checked, and both conditions are tested within each state before counting joint compliance.

4.2.4 The Kircher–Votsmeier projection for mass conservation and non-negativity

A positive provisional target and relative-error geometry

Kircher and Votsmeier (2025) develop a projection that combines mass conservation with non-negativity for mechanistic-kinetics surrogates. Its transfer to the reactor prediction problem uses the invariant operator A , the basis Z , and the feasible influent anchor. The projection depends on the completed component prediction rather than on predictor internals. It can therefore be applied to every comparator without changing its training objective or architecture. Its inputs are the full component vector, the corresponding non-negative influent, and the physical operators.

The first step compares each raw component with its positive floor. If a raw component is above the floor, its provisional value is the raw value. Otherwise, its provisional value is the floor. For component j , the operation is $c_{+,ij}^{(m)} = \max(c_{raw,ij}^{(m)}, \varepsilon_{c,j})$. A hypothetical raw pair $(-0.2, 3)$ becomes $(10^{-8}, 3)$ when both native-unit floors have value 10^{-8} . The corresponding inverse values are 10^8 and $1/3$. The floor gives a positive denominator for every provisional concentration. The subsequent projection enforces mass conservation.

A diagonal weighting matrix places each inverse concentration on its diagonal and zero elsewhere. For the hypothetical positive pair $(1, 2)^\top$, its action on a discrepancy column h is

$$W = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}, \quad Wh = \begin{bmatrix} h_1 \\ h_2/2 \end{bmatrix}.$$

No component is mixed with another in this operation. Each component discrepancy is divided by its own provisional concentration, so both weighted coordinates are relative discrepancies. The first step forms a strictly positive provisional target,

$$c_{+,i}^{(m)} = \max(c_{raw,i}^{(m)}, \varepsilon_c), \quad W_i^{(m)} = \text{Diag}(\mathbf{1} \oslash c_{+,i}^{(m)}). \quad (4.4)$$

The maximum and division are componentwise. Each entry of ε_c is 10^{-8} in its component's native unit. The provisional target supplies both a reference state and finite positive weights. The subsequent mass conservation and non-negativity stages determine the final physical prediction.

For a component change h_j , the weighted discrepancy is $h_j/c_{+,ij}^{(m)}$. The geometry therefore penalizes relative movement. A hypothetical movement of 1 g m^{-3} corresponds to a relative discrepancy of 0.01 at a component value of 100 g m^{-3} and 10 at a component value of 0.1 g m^{-3} . Small coordinates receive strong protection. Large coordinates

can absorb larger absolute changes at the same relative cost. Relative-error projection and variance-standardized accuracy therefore assess redistribution using different scales (Kircher & Votsmeier, 2025; Sturm & Silva, 2025).

Selecting a candidate that satisfies mass conservation

Changes consistent with mass conservation can first be written one component at a time. If Z has fifteen change columns, coordinate j of a change is $Z_{j1}q_1 + \cdots + Z_{j,15}q_{15}$. Adding that change to $c_{in,ij}$ gives the candidate component. Since each column of Z has zero invariant product, every such combination preserves the influent target.

A complete two-component hypothetical calculation shows how these coordinates are chosen. Both components use the same abstract concentration basis. Let the required sum be four and the feasible anchor be $(2, 2)^\top$. The raw pair is $(1, 2)^\top$, which sums to three. The provisional pair is unchanged because both values exceed the floor. A normalized invariant row and a normalized change column are

$$A_{\text{ex}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad Z_{\text{ex}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Their product is $(1-1)/2 = 0$. A scalar change amount t gives the candidate $(2+t, 2-t)^\top$. In the normalized basis its coordinate is $q = \sqrt{2}t$, because $Z_{\text{ex}}q = (t, -t)^\top$. The sum remains four for every t . All states consistent with mass conservation for sample i have the form

$$c = c_{in,i} + Zq, \quad q \in \mathbb{R}^{15}. \quad (4.5)$$

For the hypothetical pair, the provisional target change from the anchor is $(-1, 0)^\top$. The difference between an allowed change and this target change is $(t+1, -t)^\top$. Applying the inverse provisional concentrations gives $(t+1, -t/2)^\top$. Its squared Euclidean norm is

$$E(t) = (t+1)^2 + \frac{t^2}{4}.$$

At $t = 0$, this value is one. Choosing a negative t moves the first component toward its raw value while requiring an increase in the second to preserve the sum. A derivative describes the change in the objective for a small change in t . The derivative of $(t+1)^2$ is $2(t+1)$ and that of $t^2/4$ is $t/2$. Adding them gives the rate of change of the complete objective. The best compromise can be found by differentiating,

$$\frac{dE}{dt} = 2(t+1) + \frac{t}{2} = \frac{5}{2}t + 2.$$

Setting the derivative to zero gives $t = -4/5$. The second derivative is $5/2 > 0$, so this point is the unique minimum. The resulting candidate is $(6/5, 14/5)^\top = (1.2, 2.8)^\top$. Its relative discrepancies from the provisional target are 0.2 and 0.4, giving squared objective $0.2^2 + 0.4^2 = 0.20$. A candidate $(1.5, 2.5)^\top$ instead gives $0.5^2 + 0.25^2 = 0.3125$.

The weighting favors the first candidate because the smaller provisional component receives stronger protection against absolute movement.

Figure 4.2 shows this calculation as a geometric comparison. Curves around the raw pair connect states with the same projection objective. The line defined by mass conservation restricts the candidate to states whose two components sum to four.

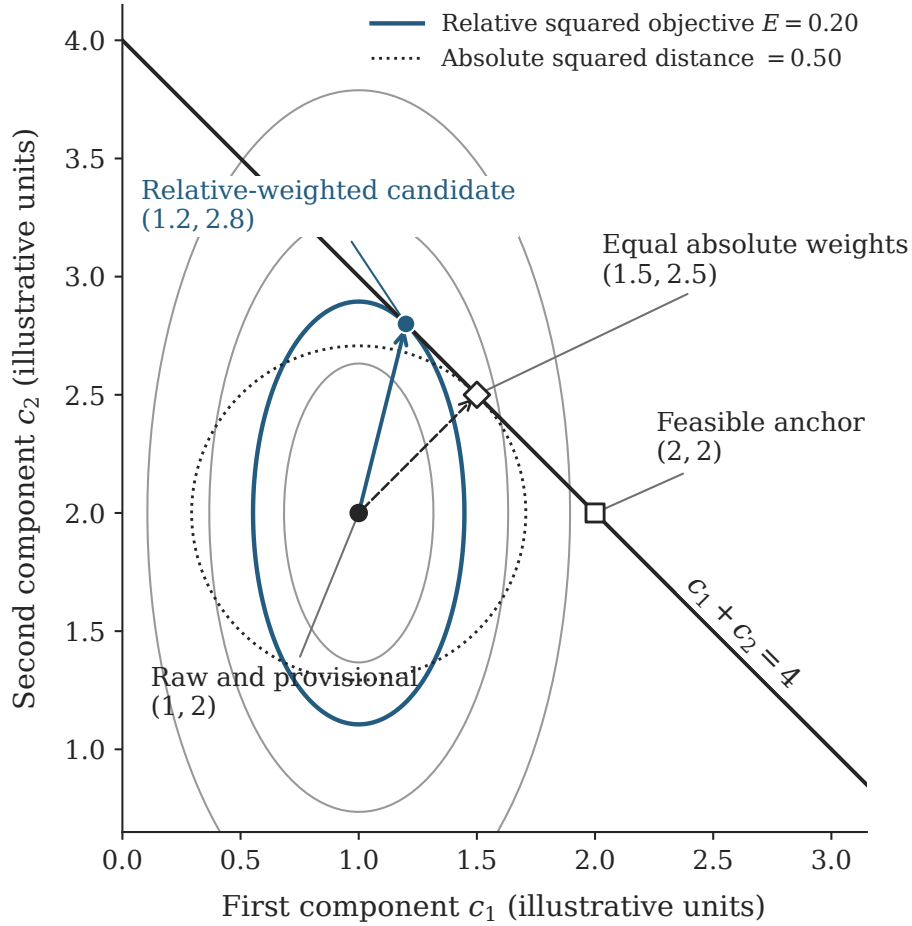


Figure 4.2: Hypothetical projection for mass conservation with relative and equal absolute weights. The highlighted ellipse first touches the line defined by mass conservation at the relative-weighted candidate. The dotted circle touches it at the equal-absolute-weight candidate.

The relative-weighted ellipse is taller than it is wide because movement in the second component is divided by two. A larger absolute change in that component can therefore have the same weighted cost as a smaller change in the first. The circular geometry assigns equal cost to equal absolute changes. Both chosen points satisfy the sum fixed by mass conservation, but the distance definition determines where each geometry touches that line. The feasible anchor locates the reference from which changes satisfying mass conservation are constructed and need not be the nearest candidate.

For the full problem, let $d_j = c_{+,ij}^{(m)} - c_{in,ij}$ denote the provisional change and let

$w_j = 1/c_{+,ij}^{(m)}$. The scalar objective adds all twenty squared relative discrepancies,

$$E(q) = \sum_{j=1}^{20} w_j^2 \left(\sum_{k=1}^{15} Z_{jk} q_k - d_j \right)^2.$$

The inner sum builds one component of the change satisfying mass conservation. Subtracting d_j gives its discrepancy from the target change. Multiplication by w_j^2 scales its square. The outer sum combines these component contributions. The notation argmin identifies the coordinate vector at which this objective is smallest, rather than the value of the objective itself. The projection selects the change satisfying mass conservation closest to the provisional target change in the relative-error geometry,

$$q_i^{*,(m)} = \operatorname{argmin}_{q \in \mathbb{R}^{15}} \left\| W_i^{(m)} [Zq - (c_{+,i}^{(m)} - c_{in,i})] \right\|_2^2. \quad (4.6)$$

The squared objective is a quadratic function of q . To minimize it, differentiate with respect to each coordinate q_ℓ . A partial derivative, written with ∂ , varies this one coordinate while holding the others fixed. Differentiating the square and then its inner sum gives

$$\frac{\partial E}{\partial q_\ell} = 2 \sum_{j=1}^{20} w_j^2 Z_{j\ell} \left(\sum_{k=1}^{15} Z_{jk} q_k - d_j \right).$$

Set this derivative to zero, divide by two, and move the target-change contribution to the right-hand side. For every ℓ , the resulting equation is

$$\sum_{k=1}^{15} \left(\sum_{j=1}^{20} w_j^2 Z_{j\ell} Z_{jk} \right) q_k = \sum_{j=1}^{20} w_j^2 Z_{j\ell} d_j.$$

The coefficient of q_k is a weighted dot product of columns ℓ and k of Z . Because $W_i^{(m)}$ is diagonal, its square places w_j^2 on the diagonal. Collecting these fifteen equations gives a 15×15 coefficient matrix and a fifteen-coordinate right-hand side. Differentiation gives

$$Z^\top (W_i^{(m)})^2 Z q_i^{*,(m)} = Z^\top (W_i^{(m)})^2 (c_{+,i}^{(m)} - c_{in,i}). \quad (4.7)$$

For the hypothetical pair, the compact coefficient calculation gives $Z_{\text{ex}}^\top W^2 Z_{\text{ex}} = (1 + 1/4)/2 = 5/8$. Its right-hand side is $-1/\sqrt{2}$. Dividing gives $q^* = -8/(5\sqrt{2}) = -4\sqrt{2}/5$, which equals $\sqrt{2}t$ for the scalar minimum already obtained. The scalar and matrix calculations therefore return the same candidate. For any nonzero vector v , the full problem's associated quadratic form equals $\|W_i^{(m)} Z v\|_2^2 > 0$. If Zv were zero, multiplying by $Z^\top$ would give $v = Z^\top Z v = 0$, contradicting the choice of v . Positive diagonal weights cannot turn a nonzero column into zero. The squared norm is consequently strictly positive. A positive-definite matrix has a strictly positive such product in every nonzero direction. The matrix is therefore positive definite and the minimizer is unique.

The selected coordinates build the component change through the fifteen sums $\sum_k Z_{jk} q_k^*$. Adding each sum to its matching influent coordinate constructs the state.

In the hypothetical pair, this operation is $(2, 2)^\top + (-0.8, 0.8)^\top = (1.2, 2.8)^\top$. Both entries are non-negative and their invariant value is $(1.2 + 2.8)/\sqrt{2} = 4/\sqrt{2}$, equal to the anchor value. This example needs no projection interpolation for non-negativity. The candidate satisfying mass conservation is

$$c_{cand,i}^{(m)} = c_{in,i} + Zq_i^{*,(m)}. \quad (4.8)$$

Since $AZ = 0$, it satisfies $Ac_{cand,i}^{(m)} = Ac_{in,i} = b_i$ in exact arithmetic. If the provisional target is already consistent with mass conservation, its change lies in $\text{null}(A)$. The least-squares residual can then be zero and the candidate reproduces that target to numerical tolerance.

Equation (4.7) provides the derivation rather than the preferred numerical solution. The weighted design $W_i^{(m)}Z$ is solved through an orthogonal–triangular factorization. Singular-value decomposition with relative cutoff 10^{-12} is the fallback. No explicit inverse of the normal-equation matrix is formed. Near-zero provisional targets can produce very unequal weights, so solving the original weighted system avoids the extra conditioning penalty associated with forming a normal-equation inverse (Buzzi-Ferraris & Manenti, 2013).

The largest feasible interpolation toward the candidate

The candidate is also checked for non-negativity. A different hypothetical three-component example isolates the boundary problem. Its quantity used for mass conservation is the unweighted sum, with feasible influent $(8, 1, 1)^\top$ and candidate satisfying mass conservation $(4, -0.5, 6.5)^\top$. Both sums are ten. Subtracting the influent from the candidate gives the direction $(-4, -1.5, 5.5)^\top$. A fraction α of that direction gives individual concentrations

$$c_1(\alpha) = 8 - 4\alpha, \quad c_2(\alpha) = 1 - 1.5\alpha, \quad c_3(\alpha) = 1 + 5.5\alpha.$$

The first component requires $8 - 4\alpha \geq 0$, so $\alpha \leq 2$. The second requires $1 - 1.5\alpha \geq 0$, so $\alpha \leq 2/3$. The third increases and gives no upper bound for $\alpha \geq 0$. Restricting α to $[0, 1]$ makes $2/3$ the first boundary point. Define

$$\delta_i^{(m)} = c_{cand,i}^{(m)} - c_{in,i} = Zq_i^{*,(m)}. \quad (4.9)$$

The segment $c_{in,i} + \alpha\delta_i^{(m)}$, with $0 \leq \alpha \leq 1$, satisfies mass conservation because $A\delta_i^{(m)} = 0$. For a general decreasing coordinate, start with $c_{in,ij} + \alpha\delta_{ij}^{(m)} \geq 0$. Moving the negative change to the other side gives $c_{in,ij} \geq \alpha(-\delta_{ij}^{(m)})$. The factor $-\delta_{ij}^{(m)}$ is positive, so dividing by it preserves the inequality direction. For every decreasing coordinate, non-negativity requires

$$\alpha \leq \frac{c_{in,ij}}{-\delta_{ij}^{(m)}} \quad \text{when } \delta_{ij}^{(m)} < 0. \quad (4.10)$$

The minimum of these bounds is the first point at which the segment reaches the non-negative-orthant boundary.

If every candidate coordinate is non-negative, the entire direction is allowed and $\alpha = 1$. If at least one candidate coordinate is negative, the smallest decreasing-coordinate ratio supplies a bound below one. Taking that bound reaches the concentration boundary. Multiplication by $1 - \eta$ retreats slightly from the boundary to limit a sign change caused by rounding. With numerical safety margin $\eta = 10^{-12}$, the accepted interpolation factor and final state are

$$\alpha_i^{(m)} = \begin{cases} 1, & c_{cand,i}^{(m)} \geq 0, \\ (1 - \eta) \min_{j | \delta_{ij}^{(m)} < 0} \frac{c_{in,ij}}{-\delta_{ij}^{(m)}}, & \text{otherwise,} \end{cases} \quad (4.11)$$

$$\tilde{c}_i^{(m)} = c_{in,i} + \alpha_i^{(m)} \delta_i^{(m)}.$$

If the candidate is negative somewhere, at least one bound lies below one. Non-negativity of the influent makes every bound non-negative. The factor is restricted to $[0, 1]$ against roundoff. Decreasing coordinates satisfy $\tilde{c}_{ij}^{(m)} \geq \eta c_{in,ij} \geq 0$, and increasing coordinates remain non-negative. For invariant row h , the final weighted component sum is $\sum_j A_{hj} c_{in,ij} + \alpha \sum_j A_{hj} \delta_{ij}^{(m)}$. The second sum is zero and the first is b_{ih} . Mass conservation follows from

$$A \tilde{c}_i^{(m)} = A c_{in,i} + \alpha_i^{(m)} A Z q_i^{*,(m)} = b_i. \quad (4.12)$$

The unrestricted boundary factor is the largest feasible interpolation factor. The safety margin makes the returned factor slightly smaller when projection is necessary.

A hypothetical three-component system with sum fixed by mass conservation 10 illustrates this safeguard. Let the feasible influent be $(8, 1, 1)^\top$ and a candidate satisfying mass conservation be $(4, -0.5, 6.5)^\top$. The direction is $(-4, -1.5, 5.5)^\top$. Its decreasing-coordinate bounds are 2 and $2/3$, so the largest feasible factor is $2/3$. Ignoring the numerical safety margin only for this illustration gives $(16/3, 0, 14/3)^\top$. The sum remains 10 and every component is non-negative. The first coordinate changes even though the second coordinate determines the boundary. One scalar shortens the complete direction satisfying mass conservation.

If an influent coordinate is zero and the candidate direction decreases that coordinate, its bound is zero. The final state then returns to the influent anchor. This returns a feasible state. Likewise, a component that is absent from the invariant rows can still move during interpolation because the complete direction is shortened. Recording the factor and activation frequency is therefore necessary for interpreting the projection.

Numerical acceptance and returned predictions

Numerical acceptance tests the returned state rather than assuming that the exact algebra eliminates rounding error. For row h , form the residual $r_h = \sum_j A_{hj} \tilde{c}_{ij}^{(m)} - b_{ih}$.

Square its five values, add them, and take the square root to obtain the mass conservation norm. Separately compare each of the twenty concentrations with the negative of its own tolerance. A hypothetical concentration of -5×10^{-11} passes a 10^{-10} sign tolerance, whereas -2×10^{-10} fails it. These examples illustrate the numerical sign check. A returned projection is accepted only when

$$\|A\tilde{c}_i^{(m)} - b_i\|_2 \leq \tau_{mc}, \quad \tilde{c}_{ij}^{(m)} \geq -\tau_{nn,j} \quad \text{for every } j, \quad (4.13)$$

with $\tau_{mc} = 10^{-10}$ in the fixed invariant-coordinate convention and $\tau_{nn,j} = 10^{-10}$ in the native unit of component j . These checks distinguish the exact mathematical construction from its finite-precision realization. A finite raw vector is required.

The final state satisfies the declared invariants and sign restrictions. The least-squares stage minimizes weighted distance over the affine space, and the interpolation stage selects a feasible point on the influent-to-candidate line segment. The prediction error uses training standard deviations, while the projection weights use the positive provisional concentrations. These definitions give the two calculations distinct objectives.

Figure 4.3 brings the projection steps and their checks into one prediction procedure. The non-negativity decision concerns the candidate satisfying mass conservation. The final numerical audit applies to the state returned by either branch.

The provisional positive target supplies a reference and weights for the mass conservation solve. A non-negative candidate can retain the full change satisfying mass conservation, while a candidate crossing a concentration boundary requires shortening toward the feasible influent. Both branches reach the same final checks. This arrangement applies the same numerical acceptance criteria to each returned state.

Each raw or projected composite is obtained by multiplying the relevant component values by the coefficients already defined in Table 3.1 and adding them. The coefficients do not change between the two prediction types. A paired comparison therefore changes only the component state supplied to the four scalar sums. Raw and projected composites are calculated with the same map,

$$y_{raw,i}^{(m)} = I_{comp}c_{raw,i}^{(m)}, \quad \tilde{y}_i^{(m)} = I_{comp}\tilde{c}_i^{(m)}. \quad (4.14)$$

Feasibility is checked on the complete component state before this aggregation.

4.3 Methodology

4.3.1 Constructing a mechanistic learning domain

The single-reactor dataset is designed to contain 10,000 accepted steady states. Each candidate varies all 22 input coordinates. A randomized, scrambled, strength-one Latin hypercube places one sample in each stratum of every marginal input distribution. A stratum is one interval in a partition of an input range. A marginal distribution

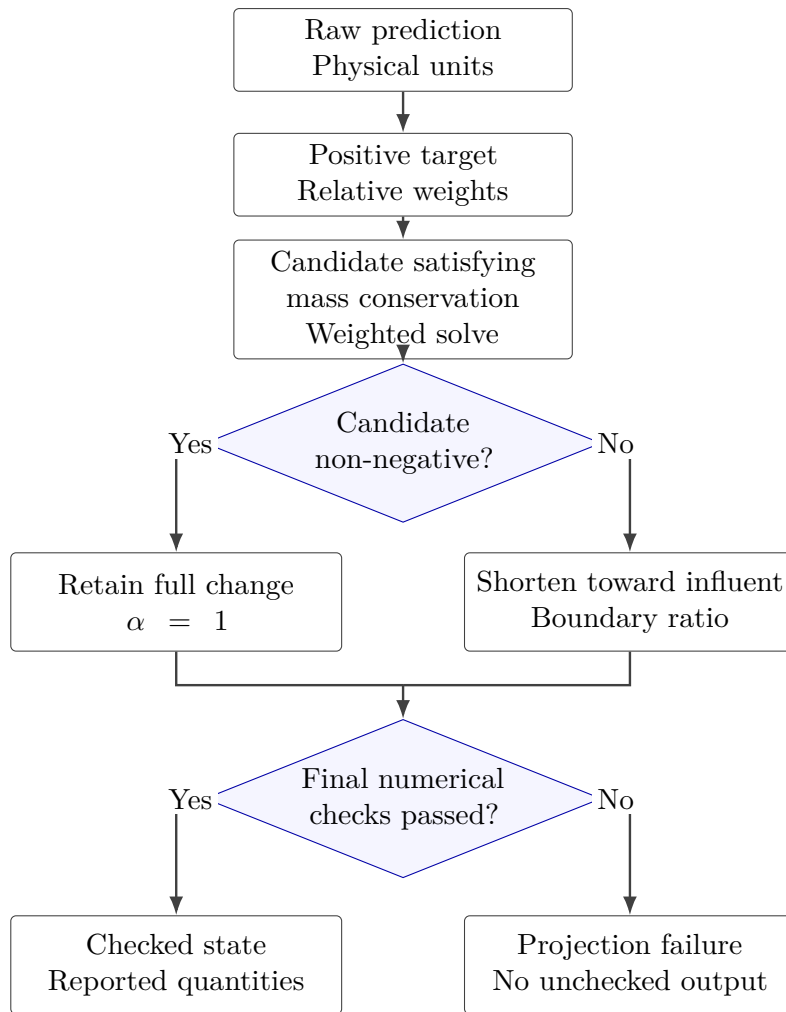


Figure 4.3: Post prediction mass conservation and non-negativity procedure. Positive-target construction, projection for mass conservation, and final-state acceptance have separate roles. The diagram shows the sequence used for each prediction.

describes one coordinate, such as influent ammonium, considered separately from the other inputs. The construction uses seed 42 and no subsequent design optimization. Linear scaling maps the unit-hypercube samples to the bounds in Table 4.1. Latin hypercube sampling distributes coverage across marginal ranges without requiring a full Cartesian grid (McKay et al., 1979).

Table 4.1: Design bounds for the standalone reactor learning domain. The intervals define the sampled input ranges.

Input	Meaning	Interval	Unit
HRT	Hydraulic operating input	[6, 36]	h
u_{air}	Dimensionless aeration input	[0.5, 2.5]	1
S_O	Influent dissolved oxygen	[0, 0.5]	g O ₂ m ⁻³
S_F	Influent fermentable substrate	[20, 180]	g COD m ⁻³
S_A	Influent acetate	[5, 80]	g COD m ⁻³
S_{NH_4}	Influent ammonium nitrogen	[12, 55]	g N m ⁻³
S_{NO_2}	Influent nitrite nitrogen	[0, 3]	g N m ⁻³
S_{NO_3}	Influent nitrate nitrogen	[0, 8]	g N m ⁻³
S_{N_2}	Influent dissolved dinitrogen	[0, 2]	g N m ⁻³
S_{PO_4}	Influent soluble phosphate	[2, 18]	g P m ⁻³
S_I	Influent soluble inert organic matter	[10, 90]	g COD m ⁻³
S_{ALK}	Influent alkalinity	[80, 260]	mol m ⁻³
X_I	Influent particulate inert matter	[20, 120]	g COD m ⁻³
X_S	Influent slowly biodegradable substrate	[60, 280]	g COD m ⁻³
X_H	Influent heterotrophic biomass	[15, 100]	g COD m ⁻³
X_{PAO}	Influent phosphate-accumulating biomass	[5, 60]	g COD m ⁻³
X_{PP}	Influent stored polyphosphate	[2, 20]	g P m ⁻³
X_{PHA}	Influent stored polyhydroxyalkanoates	[1, 30]	g COD m ⁻³
X_{AOB}	Influent ammonia-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{NOB}	Influent nitrite-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{MeP}	Influent precipitated metal phosphate	[0, 12]	g TSS m ⁻³
X_{MeOH}	Influent metal hydroxide	[0, 12]	g TSS m ⁻³

The standalone domain varies the operating and influent inputs within the declared simulation bounds. Its alkalinity interval uses the stated native coordinate. The coupled plant uses its own operating and influent bounds.

For each candidate input, nonlinear least squares solves Equation (3.14) with numerical state bounds $[10^{-8}, 1500]$ in each component's native unit. The cost-change, step-size, and gradient convergence tolerances are 10^{-9} . Each solve allows at most 1200 function evaluations. The generation procedure has an overall attempt cap of 25 times the target retained count. A state is retained only if the solver converges and the maximum absolute component-balance residual is no greater than 10^{-9} on the fixed native component-unit-per-day coordinates. The residual criterion uses the native component-unit-per-day coordinates.

Two physically informed starting states are tried before dynamic relaxation. The first blends the sampled influent with a preceding accepted effluent when one is available. It lowers readily consumed substrate coordinates, retains biomass pools, and estimates dissolved oxygen from dilution and transfer. The second applies stronger substrate depletion and biomass enrichment. Starting vectors are restricted to $[10^{-6}, 1500]$. If both steady solves fail, a stiff backward differentiation formula integrates the dynamic balance for 20 days and supplies a new initial state. The state bounds and starting-state restrictions apply to concentrations. The 28 kinetic process rates are evaluated without an artificial ceiling or clipping.

Solver acceptance, invariant residual, and minimum concentration are recorded separately. A converged bounded least-squares calculation can terminate without meeting the required balance residual. A positive state can also fail mass conservation. Retaining these separate diagnostics prevents one successful numerical check from standing in for all others. The surrogate inputs contain the sampled operating and influent coordinates. Initialization choices and solver history are not predictive features.

4.3.2 Preprocessing and validation without response leakage

A fold is one reserved group of cases in cross-validation. Successive fits change which group is withheld. Nested cross-validation separates an inner comparison used to select hyperparameters from an outer comparison used to assess the selected model. Here each case contains influent and operating inputs paired with the mechanistic effluent state.

The evaluation uses nested subsets at

$$N \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (4.15)$$

A seed-42 permutation determines the nested row order and persistent master fold identifiers before subsets are formed. Every smaller subset is contained in each larger subset. A row retains its outer-fold role at every scale. Five outer folds train on $0.8N$ observations and score the remaining $0.2N$. Thus a design size of 500 gives 400 training rows and 100 scored rows per outer iteration. A design size of 10,000 gives 8000 training rows and 2000 scored rows. The five iterations change which fold is withheld, so each

row supplies a scored prediction from a fit that excludes it. The same rows and fold identifiers are used by every surrogate and both prediction types.

Hyperparameters are selected using four inner folds within the largest outer-training partition. The scored outer fold is absent from the search. Within an 8000-row outer-training pool, an inner iteration fits on 6000 rows and scores 2000 different rows from that pool. Those inner-validation rows are not the 2000 rows reserved for outer assessment. The inner score selects the setting, and the outer score assesses a model fitted with that selected setting. For each outer fold, its selected setting is retained over the smaller nested subsets. Learning curves therefore examine the response to sample size under a fixed fold-specific setting. The bias–variance discussion of Belkin et al. (2019) cautions against assuming that error must vary monotonically with model capacity or available data. The learning-curve review of Viering and Loog (2023) supports treating the complete curve as an object of assessment.

Every transformation is fitted within the applicable training partition. During inner selection, that partition is the inner-training split. During outer scoring, it is the outer-training subset at the current size. During extrapolation assessment, it is the full in-domain training pool. A training coordinate has one mean and one standard deviation for that fit. If its M training values are $v_1, \dots, v_M$, these quantities are calculated in three steps,

$$\mu = \frac{v_1 + \dots + v_M}{M}, \quad \sigma^2 = \frac{(v_1 - \mu)^2 + \dots + (v_M - \mu)^2}{M}, \quad \sigma = \sqrt{\sigma^2}.$$

The second quantity is the population variance. The population convention divides by the training count M . A z-score subtracts the mean and divides by the standard deviation, giving $z = (v - \mu)/\sigma$. Subtracting the mean centers the coordinate and division puts its numerical variation on a unit-standard-deviation scale. The transformation requires $\sigma > 0$.

For hypothetical training values 2, 2, 6, and 6, the mean is 4. Their centered values are $-2, -2, 2$, and 2 , so the variance is $(4 + 4 + 4 + 4)/4 = 4$ and the standard deviation is 2. A validation value of 6 therefore becomes $(6 - 4)/2 = 1$. The validation value is not added to the training values before calculating the mean or variance. If this coordinate is an output, a predicted z-score of 0.5 returns to physical units as $4 + 2(0.5) = 5$.

The mean and population standard deviation of a training coordinate define its z-score transformation. Validation inputs are transformed with those training quantities, and target-scaled predictions are restored to physical units before projection or physical diagnostics. Every target has its own inverse transformation $\hat{c}_j = \mu_j + \sigma_j \hat{z}_j$. A single common inverse scale would be incorrect for components with different units or variability.

Tree partitions depend on coordinate ordering, so a positive affine input rescaling preserves candidate threshold partitions. Distance-based, covariance-based, and regularized models require more deliberate scaling because relative numerical magnitudes affect

Table 4.2: Output handling and preprocessing for the statistical comparators. Scaling uses only the training partition applicable to each fit.

Surrogates	Output handling	Scaling
XGBoost, LightGBM, Cat-Boost	Separate component regressors	None
AdaBoost	Separate component regressors	Targets only
Random Forest	Joint tree response	None
Extra Trees	Joint tree response	Targets only
Support vector regression	Separate component regressors	Inputs and targets
k -nearest neighbors	Joint neighborhood response	Inputs and targets
Partial least squares, Multi-task Lasso, Multi-task Elastic Net	Joint coefficient structure	Inputs and targets
Multilayer perceptron, TabNet	Joint neural output	Inputs and targets

their objectives. Separate-output tree regressors do not combine different component units within one joint fitting loss. The target-only scaling of Extra Trees and AdaBoost is retained as a specified benchmark choice. Independent of those training choices, every scored component error is standardized with the designated training reference. Training target scaling and evaluation normalization are distinct operations.

No missing-value imputation, response-based outlier removal, or target filtering is applied. No validation or extrapolation response is used to estimate a scaling parameter. A hypothetical decision to standardize a validation target with the standard deviation of its own fold would change the scoring reference using data outside training. Retaining a training reference instead asks how large an error is relative to the variability the learner was allowed to observe.

4.3.3 Hyperparameter selection

Each surrogate uses a seeded tree-structured Parzen search with 100 trials per study and no pruning or timeout. One trial proposes a complete parameter setting, fits it within the prescribed inner-training partitions, and calculates its validation objective. The objective is an error value to be minimized. It differs from a parameter value such as tree depth or penalty strength. The search mechanism is supported by the optimization framework of Akiba et al. (2019). The objective is the mean squared standardized component error over the inner validation folds. Five studies select the settings for the five largest-scale outer-training partitions. A sixth, independent four-fold study

selects the setting for a final full-domain fit. The design specifies $6(100) = 600$ trials per surrogate and $13(600) = 7800$ across all 13 surrogates.

Table 4.3: Search domains and fixed controls for tree ensemble surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
XGBoost	80–320 trees. Learning rate 0.01–0.2 (log). Depth 3–8. Minimum child weight 0.5–8 (log). Row and feature fractions 0.6–1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-3} –10 (log). Histogram or approximate trees. Squared-error objective.
LightGBM	80–320 trees. Learning rate 0.01–0.2 (log). 15–63 leaves. Depth unbounded or in $\{4, 6, 8, 10\}$. Minimum child rows 5–40. Row and feature fractions 0.6–1 with subsampling frequency 1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-6} –10 (log). Regression objective.
CatBoost	80–320 iterations. Learning rate 0.01–0.2 (log). Depth 4–8. Squared penalty 0.01–20 (log). Bagging temperature 0–5. Random strength 10^{-3} –5 (log). Root mean squared error loss and Bayesian bootstrap.
AdaBoost	50–250 depth-three trees. Learning rate 0.01–1 (log). Linear, squared, or exponential regression loss.
Random Forest	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.
Extra Trees	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.

The search ranges, fixed controls, and 100-trial allocation define a common fitting protocol for the thirteen predictors.

4.3.4 Evaluation of error, feasibility, and projection size

Component errors on a training-defined scale

Let $\mathcal{T}$ denote the applicable training reference and let $\sigma_j^{(\mathcal{T})}$ be the population standard deviation of target component j in that reference. For one assessed concentration, subtract the reference effluent value from the predicted value to obtain its signed physical error. Divide this error by the training standard deviation of the same component. For example, a hypothetical prediction of 3 g N m^{-3} against a reference of

Table 4.4: Search domains and fixed controls for kernel, neighbor, multiresponse, and neural surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
Support vector regression	Penalty 0.1–10 (log). Error-insensitive width 0.01–0.5 (log). Linear or radial kernel. Radial coefficient set from standardized input variance or inverse input dimension. Tolerance 10^{-3} – 10^{-2} (log). Shrinking enabled. At most 20,000 iterations. Kernel cache of 256 megabytes per output regressor.
k -nearest neighbors	1–15 neighbors. Uniform or distance weights. Automatic, ball-tree, k -dimensional tree, or brute-force search. Leaf size 15–60. Minkowski, Euclidean, or Manhattan distance. Minkowski order in $\{1, 2, 3\}$.
Partial least squares	1–6 latent components. 200–1000 maximum iterations. Tolerance 10^{-8} – 10^{-4} (log). Internal scaling disabled because external training-partition scaling is applied.
Multi-task Elastic Net	Penalty strength 10^{-6} –10 (log). Row-group mixing ratio 0.05–0.95. Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multi-task Lasso	Penalty strength 10^{-6} –10 (log). Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multilayer perceptron	Hidden widths (128, 64, 32). Rectified linear, hyperbolic tangent, or logistic activation. Adaptive moment optimization. Squared-weight penalty 10^{-6} – 10^{-2} (log). Batch size in $\{64, 128, 256\}$. Learning rate 5×10^{-4} – 10^{-2} (log). Training-loss tolerance 10^{-4} – 10^{-3} (log). Shuffling enabled. Training-loss patience 15 epochs. At most 250 epochs.
TabNet	Equal decision and attention widths in $\{8, 16, 24\}$. Three or four attention steps. Relaxation coefficient 1–2. Sparsity coefficient 10^{-6} – 10^{-2} (log). Adaptive moment learning rate 0.003–0.03 (log). Sparsemax or entmax-1.5 masks. 20–50 selected epochs. Batch size in $\{512, 1024\}$. Virtual batch size 128. Patience zero.

2 g N m^{-3} has signed error 1 g N m^{-3} . A training standard deviation of 2 g N m^{-3} gives standardized error 0.5. The constituent units cancel in the division. For prediction type t , which is either raw or projected, define

$$\xi_{ij}^{(m,t)} = \frac{c_{ij}^{(m,t)} - c_{out,ij}}{\sigma_j^{(\mathcal{T})}}. \quad (4.16)$$

The normalized mean squared error (nMSE), normalized root mean squared error (nRMSE), and normalized mean absolute error (nMAE) combine these standardized errors. A two-observation, two-component hypothetical example shows their order of calculation. Suppose the training standard deviations are 2 and 4. Let the reference and predicted values be

Observation	Reference		Prediction	
	Component 1	Component 2	Component 1	Component 2
1	2	4	3	2
2	4	8	2	12

Values within each component use that component's own physical basis. The four physical errors are 1, -2 , -2 , and 4. Dividing each by its corresponding training standard deviation gives

$$\xi = \begin{bmatrix} 1/2 & -2/4 \\ -2/2 & 4/4 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -1 & 1 \end{bmatrix}.$$

There are four scored component-observation pairs. Their squared errors sum to $0.25 + 0.25 + 1 + 1 = 2.5$. Their mean squared standardized error is $2.5/4 = 0.625$, and its square root is approximately 0.7906. Their absolute errors sum to $0.5 + 0.5 + 1 + 1 = 3$, so the mean absolute standardized error is $3/4 = 0.75$. The square is taken before averaging for nMSE. The square root is taken after that average for nRMSE. Absolute values are taken before averaging for nMAE.

For the twenty-component assessment, the first summation accumulates errors across observations and the second accumulates them across components. Division by $20n$ gives every scored pair equal weight after standardization. The compact definitions are

$$\text{nMSE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} (\xi_{ij}^{(m,t)})^2, \quad (4.17)$$

$$\text{nRMSE}_{m,t} = \sqrt{\text{nMSE}_{m,t}}, \quad (4.18)$$

$$\text{nMAE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} |\xi_{ij}^{(m,t)}|. \quad (4.19)$$

The component state is the primary scoring space because it is the prediction target and the space in which constraints are imposed. The root mean squared score emphasizes larger standardized errors. The absolute score gives a complementary average deviation.

The normalized measures require a positive training standard deviation for every component.

A hypothetical error of 1 g m^{-3} has standardized magnitude 0.1 for a component with training standard deviation 10 g m^{-3} . The same physical error has magnitude 10 for a component with training standard deviation 0.1 g m^{-3} . The score gives equal numerical weight to standardized component variation.

The coefficient of determination compares squared prediction error with the error of predicting one constant value for every observation. That constant is the true mean within the scored partition. For component 1 of the hypothetical example, the true mean is $(2 + 4)/2 = 3$. Predicting 3 for both observations would give squared errors $(2 - 3)^2 + (4 - 3)^2 = 2$. The actual predictions give squared errors $(3 - 2)^2 + (2 - 4)^2 = 5$. Dividing 5 by 2 and subtracting from one gives $R^2 = 1 - 5/2 = -1.5$. For component 2, the true mean is 6. The corresponding baseline square sum is $(4 - 6)^2 + (8 - 6)^2 = 8$, and the prediction square sum is $(2 - 4)^2 + (12 - 8)^2 = 20$. Its score is also $1 - 20/8 = -1.5$. The negative scores mean that the predictions are worse than their scored-partition constant baselines.

The mean used in this comparison is $\bar{c}_{out,j} = (c_{out,1j} + \dots + c_{out,nj})/n$. It belongs to scoring and is not used to fit the model. The denominator of R^2 therefore differs from the training-defined scale of nMSE. For a scored partition with true mean $\bar{c}_{out,j}$, the component coefficient of determination and its equal-weight average are

$$R_{j,m,t}^2 = 1 - \frac{\sum_i (c_{ij}^{(m,t)} - c_{out,ij})^2}{\sum_i (c_{out,ij} - \bar{c}_{out,j})^2}, \quad \bar{R}_{m,t}^2 = \frac{1}{20} \sum_{j=1}^{20} R_{j,m,t}^2. \quad (4.20)$$

The macro-average requires a positive scored-partition variance for each component. The coefficient can be negative when the prediction is worse than the scored-partition mean. It complements the standardized error but uses a different response-variability reference.

The macro-average adds the twenty component R^2 values and divides by twenty. Likewise, pooled nRMSE is the square root of the mean of component squared standardized errors. It is generally different from the arithmetic mean of component root errors. Hypothetical component mean squared standardized errors of 1 and 9 give pooled root error $\sqrt{(1 + 9)/2} = \sqrt{5}$, whereas the mean of their separate roots is $(1 + 3)/2 = 2$. This distinction prevents differently constructed aggregate scores from being treated as interchangeable.

Physical-unit errors are retained for individual components. Secondary composite error is calculated separately for COD, TN, TP, and TSS. For a composite, calculate one signed error per observation, square the errors, average them over observations, and take the square root. Dividing that physical-unit root error by the training standard deviation of the composite gives its normalized root error. A hypothetical root error of 5 g COD m^{-3} divided by a training standard deviation of 10 g COD m^{-3} gives 0.5. The normalized root mean squared error of composite q is its physical-unit root

mean squared error divided by $\sigma_q^{(\mathcal{T})}$, the training standard deviation of that composite. This composite score uses the variability of one reported quantity. Composite standard deviations are calculated after applying the common reporting map to the training targets.

Violation frequencies and their magnitudes

The mass conservation residual begins with five scalar errors. For invariant row h , its error is $r_h = \sum_j A_{hj}c_j - b_{ih}$. Its Euclidean magnitude is $\sqrt{r_1^2 + \dots + r_5^2}$. A hypothetical two-row residual of $(3, 4)^\top$ has magnitude five in its fixed invariant-coordinate convention. This example explains the norm.

The negative magnitude uses a different operation. First divide each concentration by its training standard deviation. Then retain a negative coordinate if one is present and replace every non-negative coordinate by zero. Finally add the absolute values of the retained negatives. Hypothetical standardized values $(-2, 1.5, -2)$ become $(-2, 0, -2)$ and have absolute-sum magnitude four. A positive component cannot cancel a negative component in this diagnostic. The mass conservation residual and standardized total negative magnitude of a state c are

$$v_{mc,i}(c) = \|Ac - b_i\|_2, \quad v_{nn,i}^{\text{std}}(c) = \|\min(c \oslash \boldsymbol{\sigma}^{(\mathcal{T})}, 0)\|_1. \quad (4.21)$$

The minimum is componentwise. The mass conservation residual uses the same full-precision orthonormal basis for every sample. Its numerical value uses the adopted invariant-coordinate convention. The negative magnitude uses training standard deviations to compare different component bases without summing incompatible physical units.

A frequency counts observations satisfying a stated failure condition and divides that count by the assessed observation count. The indicator $\mathbb{K}[\cdot]$ returns one when its bracketed condition is true and zero otherwise. For a hypothetical four-observation assessment, suppose the mass conservation failure indicators are $(1, 0, 1, 0)$ and the non-negativity failure indicators are $(0, 1, 1, 0)$. The two frequencies are $(1+0+1+0)/4 = 0.5$ and $(0+1+1+0)/4 = 0.5$. An observation with several negative components is still one failing observation in this row-level count. The two row-level violation frequencies are

$$\pi_{mc} = \frac{1}{n} \sum_i \mathbb{K}[v_{mc,i}(c_i) > \tau_{mc}], \quad \pi_{nn} = \frac{1}{n} \sum_i \mathbb{K}[\min_j (c_{ij} + \tau_{nn,j}) < 0]. \quad (4.22)$$

A state is counted as non-negative only when every component passes its tolerance. Component-specific violation frequencies identify which coordinates cross the boundary. In the four-observation example, only the fourth observation passes both conditions, so its joint feasible fraction is one quarter. The third fails both conditions and appears in both marginal failure counts. Adding the two failure frequencies would therefore count

it twice. A joint indicator applies both pass conditions to the same observation before averaging. The joint feasible fraction is

$$\pi_{\text{feasible}} = \frac{1}{n} \sum_i \mathbb{I}[v_{mc,i}(c_i) \leq \tau_{mc} \text{ and } \min_j (c_{ij} + \tau_{nn,j}) \geq 0]. \quad (4.23)$$

Mass conservation and non-negativity frequencies are not added to calculate this fraction because the same state can violate both requirements.

Mean, median, and maximum violation magnitudes are retained along with the frequencies. A hypothetical set of many concentrations just below zero can have a high violation frequency but small total negative magnitude. Another set with one strongly negative state can have low frequency and a large maximum magnitude. The two summaries answer different questions. Physical-unit negative magnitudes are also retained separately for each component.

Paired accuracy effects and displacement

The paired accuracy effect subtracts the raw score from the projected score on the same assessment observations. If hypothetical raw and projected nRMSE values are 0.40 and 0.35, the difference is $0.35 - 0.40 = -0.05$. If the projected value were 0.45, the difference would be $+0.05$. This sign convention expresses whether error decreases or increases. Component movement and mass conservation residuals are recorded separately. For each outer fold, the effect on the primary error score is

$$\Delta \text{nRMSE}_{m,N,g} = \text{nRMSE}_{m,\text{projected},N,g} - \text{nRMSE}_{m,\text{raw},N,g}. \quad (4.24)$$

A negative difference indicates lower error after projection, and a positive difference indicates higher error. Differences are calculated within paired folds before summary. Component and composite differences are retained because a single aggregate score can conceal opposite changes among outputs.

Displacement measures the difference between the projected prediction and its own raw prediction. Calculate one change per component, divide it by that component's training standard deviation, square it, sum the squares, and take the square root. For the hypothetical weighted-projection pair already calculated, the changes are $1.2 - 1 = 0.2$ and $2.8 - 2 = 0.8$. If its illustrative training standard deviations are 0.1 and 0.4, the standardized changes are 2 and 2. Their Euclidean displacement is $\sqrt{2^2 + 2^2} = 2\sqrt{2}$. The example deliberately uses a training scale different from the provisional concentration scale, as the two operations serve different purposes. The sample-level standardized displacement is

$$d_{c,i}^{\text{std},(m)} = \left\| (\tilde{c}_i^{(m)} - c_{\text{raw},i}^{(m)}) \oslash \boldsymbol{\sigma}^{(\mathcal{T})} \right\|_2. \quad (4.25)$$

This norm measures how far the projection moves the state relative to training variability. It measures movement rather than accuracy. A large displacement can reduce

violations of the specified physical constraints while moving away from the true kinetic response (Kircher & Votsmeier, 2025; Sturm & Silva, 2025). Mean, median, and maximum displacement, componentwise physical-unit movement, the fraction requiring backtracking, and the distribution of α are therefore assessed together.

Learning curves retain the full error trajectory over Equation (4.15). An area summary connects the errors at neighboring sample sizes by straight segments. The area under one segment is its width multiplied by the average of its two endpoint heights. For hypothetical error heights 0.8 and 0.6 at design sizes 500 and 1450, the segment area is $(1450 - 500)(0.8 + 0.6)/2 = 665$. If the next hypothetical height is 0.4 at size 2400, the next area is $(2400 - 1450)(0.6 + 0.4)/2 = 475$. The total area over those two segments is 1140. Dividing by the size interval $2400 - 500 = 1900$ gives an average height of 0.6.

The full design has ten adjacent intervals. Calculate the trapezoidal area of each interval, add those areas, and divide by $10000 - 500 = 9500$. Division by the total interval converts the area into an average error height. As a descriptive summary, the normalized area under an error curve is calculated by trapezoidal integration,

$$\mathcal{A}_{m,t} = \frac{1}{N_{11} - N_1} \sum_{\ell=1}^{10} \frac{E_{m,t}(N_\ell) + E_{m,t}(N_{\ell+1})}{2} (N_{\ell+1} - N_\ell), \quad (4.26)$$

where $E_{m,t}(N)$ is the mean held-out error at size N . This quantity has the same scale as the selected error measure. It summarizes behavior across the declared size interval alongside the individual size scores. Training and validation error at the largest size provide a separate descriptive gap. The complete learning curve shows how error responds to sample size (Viering & Loog, 2023).

Each in-domain metric is summarized as the arithmetic mean and sample standard deviation of five outer-fold values. If these values are $v_1, \dots, v_5$, first calculate $\bar{v} = (v_1 + \dots + v_5)/5$. Then calculate $s = \sqrt{[(v_1 - \bar{v})^2 + \dots + (v_5 - \bar{v})^2]/4}$. The denominator is four because this summary uses the sample standard deviation of five fold values. It differs from the population standard deviation used to standardize a training coordinate. The standard deviation describes partition-to-partition variation.

4.3.5 Testing inputs beyond the learning domain

Extrapolation assessment uses separately simulated inputs outside the training hyperrectangle. The final setting for each surrogate is selected with four folds on all 10,000 in-domain states. Its preprocessing and predictor are then refitted on that complete pool. No extrapolation response enters selection, fitting, scaling, or calibration. This separation addresses the domain-shift concern discussed in scientific machine learning by Karniadakis et al. (2021) and Shen et al. (2023).

Four inputs can exceed their upper learning bounds. Exceedance is measured relative to the width of the original input interval. For HRT, the width is $36 - 6 = 30$

h. A hypothetical input of 39 h exceeds the upper bound by 3 h, giving exceedance $3/30 = 0.10$. Dividing by the original width allows a comparable severity description for inputs with different units. For input ℓ with bounds $[L_\ell, U_\ell]$, define its normalized exceedance as

$$e_{i\ell} = \frac{x_{i\ell} - U_\ell}{U_\ell - L_\ell}. \quad (4.27)$$

The mild design uses $e_{i\ell} \in [0.10, 0.25]$ and the severe design uses $e_{i\ell} \in [0.35, 0.60]$. Reversing the exceedance calculation gives $x_{i\ell} = U_\ell + e_{i\ell}(U_\ell - L_\ell)$. For HRT, the mild endpoints are $36 + 0.10(30) = 39$ h and $36 + 0.25(30) = 43.5$ h. The severe endpoints are $36 + 0.35(30) = 46.5$ h and $36 + 0.60(30) = 54$ h. The generator targets a mixture of 70% one-input excursions and 30% two-input excursions. All remaining inputs remain within the original bounds. Every candidate therefore lies outside the training hyperrectangle and consequently outside the training-point convex hull. The hyperrectangle is the set satisfying every original coordinate interval. Every weighted average of training points remains inside those intervals, so a point violating even one interval is also outside their convex hull.

Table 4.5: Fixed exterior intervals for the two extrapolation designs. The intervals define the exterior input conditions.

Input	Learning interval	Mild interval	Severe interval	Unit
HRT	[6, 36]	[39, 43.5]	[46.5, 54]	h
u_{air}	[0.5, 2.5]	[2.7, 3.0]	[3.2, 3.7]	1
S_{NH_4}	[12, 55]	[59.3, 65.8]	[70.1, 80.8]	g N m ⁻³
X_S	[60, 280]	[302, 335]	[357, 412]	g COD m ⁻³

Each severity design targets 600 accepted steady states and retains the same solver, initialization sequence, and balance-residual criterion. Candidate, accepted, failed, valid-but-unselected, and retained counts are distinguished. The realized one-input and two-input proportions are also distinguished from their design targets. The exterior assessment measures prediction error across the four selected inputs.

Extrapolation errors are standardized with population component standard deviations from all 10,000 in-domain targets. Composite normalization uses the corresponding full-domain composite standard deviations. The nMSE, nRMSE, nMAE, mass conservation and non-negativity diagnostics, displacement, and backtracking diagnostics are calculated on identical raw and projected cases. The coefficient of determination is secondary because the deliberately shifted response distribution changes its denominator.

Each exterior metric summarizes 600 cases from one full-domain refit. Satisfaction of the projected constraints is interpreted separately from the predictive accuracy of extrapolation. A state can remain non-negative and consistent with mass conservation

while its estimated nitrogen distribution or biomass abundance is inaccurate (Henze et al., 2000; Selerio, 2026).

4.3.6 Computational cost assessment

Computation is partitioned into hyperparameter search, preprocessing and fitting, raw inference, projection alone, and complete projected inference. Search time is reported once for each distinct study because a largest-scale outer setting is reused at all eleven sizes. Summing that same search duration over the nested size rows would count the selection cost repeatedly. Fitting time excludes the search and reflects the preparation required for the specified training partition.

Inference uses a deterministic batch of 512 rows constructed by cycling through the relevant validation inputs. The batch size stays fixed even when fewer than 512 unique validation observations are available. Cycling reuses input rows to make a fixed-sized timing workload. Prediction errors retain their original assessment counts. Five warm-up calls precede 30 timed repetitions for each path. Sorting the thirty batch durations puts them in increasing order. Their median is the average of the fifteenth and sixteenth values. The median batch latency is divided by 512 and expressed as milliseconds per sample. For a hypothetical median batch duration of 102.4 ms, the reported per-sample quantity would be $102.4/512 = 0.2$ ms. For in-domain assessment, the five fold medians are summarized by their arithmetic mean and sample standard deviation. Exterior timing uses the same batching and repetitions for one final fit.

Projection timing includes positive-target construction, relative weighting, the weighted solve, mass conservation and non-negativity checks, and any projection interpolation for non-negativity. Complete projected inference includes both raw prediction and projection. All physical operators, returned component states, preprocessing, and metrics use double precision. TabNet’s internal tensors use single precision before its predictions return to the common physical-coordinate calculations. Pseudorandom generation and stochastic fitting use streams derived from seed 42. Deterministic coordinate-descent settings are retained for the multi-task models.

The timing measures describe the declared batch protocol. Complete prediction time includes the physical projection that accompanies each statistical prediction.

4.4 Results

4.4.1 Mechanistic reference-state checks

The learning domain contained 10,000 accepted steady states. Each exterior design contained 600 accepted states. Every attempted candidate passed the specified steady-state balance check, and every retained concentration was strictly positive. The realized

exterior designs contained 70% one-input excursions and 30% two-input excursions. Table 4.6 reports the invariant residual separately from solver acceptance.

Table 4.6: Accepted mechanistic reference states and their maximum invariant residuals. The residual uses the fixed invariant-coordinate convention. Solver acceptance uses the component-balance residual.

Assessment domain	Attempted	Accepted	Maximum $\ Ac_{out} - Ac_{in}\ _2$
Learning domain	10,000	10,000	4.90×10^{-11}
Mild exterior inputs	600	600	5.67×10^{-11}
Severe exterior inputs	600	600	2.91×10^{-10}

The largest invariant residual in the severe design was 2.91×10^{-10} , compared with the 10^{-10} tolerance imposed on projected predictions. The mechanistic acceptance rule used the maximum component-balance residual of 10^{-9} in the native component-unit-per-day coordinates. These two residuals have different definitions. The full-precision physical operators satisfied $\max_{h,p} |(A\nu^\top)_{hp}| = 4.25 \times 10^{-16}$. The small invariant residuals were consistent with numerical rather than structural departures from the specified balances.

4.4.2 Prediction within the learning domain

At $N = 10,000$, the multilayer perceptron had the lowest observed raw nRMSE of 0.150 ± 0.012 . CatBoost followed at 0.230 ± 0.006 , then LightGBM, XGBoost, and TabNet. Table 4.7 gives the paired results for all thirteen predictors. The comparisons use the same five outer folds.

Projection reduced the mean nRMSE for five predictors and increased it for eight. The largest reductions were approximately 0.020 for each multi-task linear model. Random Forest had the largest increase of 0.095. The multilayer perceptron retained the lowest projected error of 0.149 ± 0.012 . Its macro-average R^2 increased from 0.977 to 0.978. Random Forest’s macro-average R^2 decreased from 0.484 to 0.332. However, its nMAE decreased from 0.497 to 0.467. Its squared-error and absolute-error scores therefore changed in opposite directions.

Figure 4.4 shows the three component-level measures together. The relative positions of raw and projected bars differ across measures. This is most apparent for the randomized tree ensembles and partial least squares.

4.4.3 Response to the amount of training data

Every predictor had lower raw nRMSE at $N = 10,000$ than at $N = 500$. Figure 4.5 shows the complete trajectories. XGBoost and LightGBM had the lowest observed mean raw errors at $N = 500$, with values of 0.435 and 0.439. The multilayer perceptron

Table 4.7: Component prediction error at $N = 10,000$. Raw and projected nRMSE values are five-fold means with sample standard deviations. The paired difference is projected minus raw and is calculated before rounding.

Surrogate	Raw nRMSE	Projected nRMSE	Difference
XGBoost	0.254 ± 0.006	0.255 ± 0.006	+0.0006
LightGBM	0.249 ± 0.008	0.250 ± 0.007	+0.0007
CatBoost	0.230 ± 0.006	0.229 ± 0.006	-0.0013
AdaBoost	0.525 ± 0.004	0.581 ± 0.028	+0.0561
Random Forest	0.718 ± 0.005	0.813 ± 0.020	+0.0947
Extra Trees	0.563 ± 0.005	0.608 ± 0.025	+0.0445
Support vector regression	0.364 ± 0.011	0.356 ± 0.011	-0.0082
k -nearest neighbors	0.607 ± 0.007	0.621 ± 0.021	+0.0142
Partial least squares	0.721 ± 0.006	0.745 ± 0.031	+0.0239
Multi-task Elastic Net	0.536 ± 0.008	0.516 ± 0.008	-0.0200
Multi-task Lasso	0.536 ± 0.008	0.516 ± 0.008	-0.0200
Multilayer perceptron	0.150 ± 0.012	0.149 ± 0.012	-0.0016
TabNet	0.272 ± 0.027	0.275 ± 0.024	+0.0032

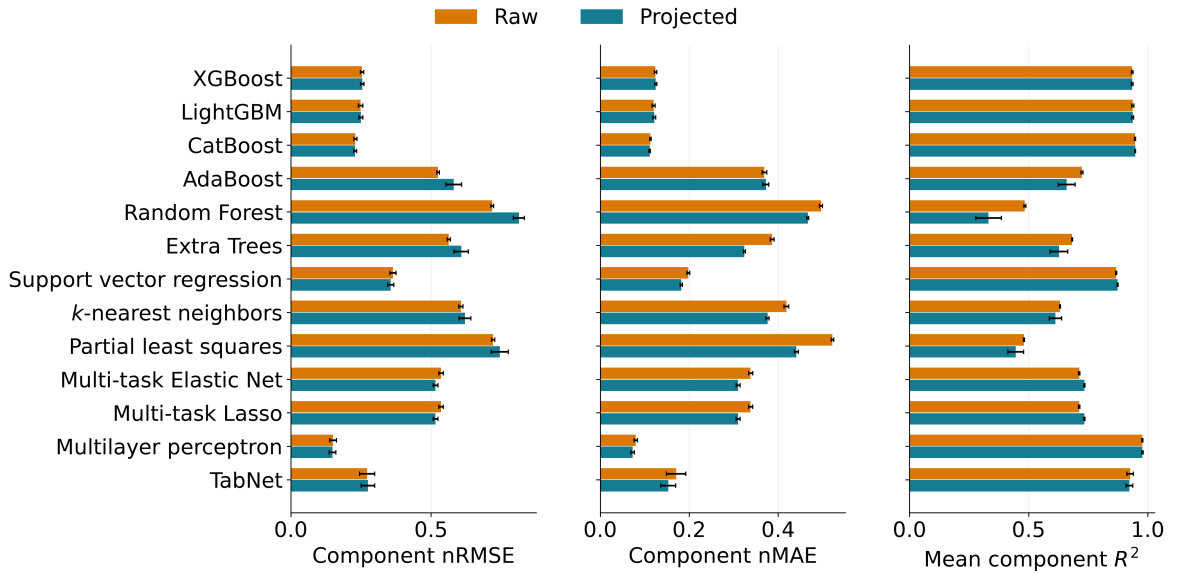


Figure 4.4: Raw and projected component accuracy at $N = 10,000$. Panels show nRMSE, nMAE, and the mean component R^2 . Error bars show one sample standard deviation across five outer folds.

improved from 0.524 to 0.150. TabNet improved from 0.938 to 0.272 and showed the largest endpoint reduction among the predictors.

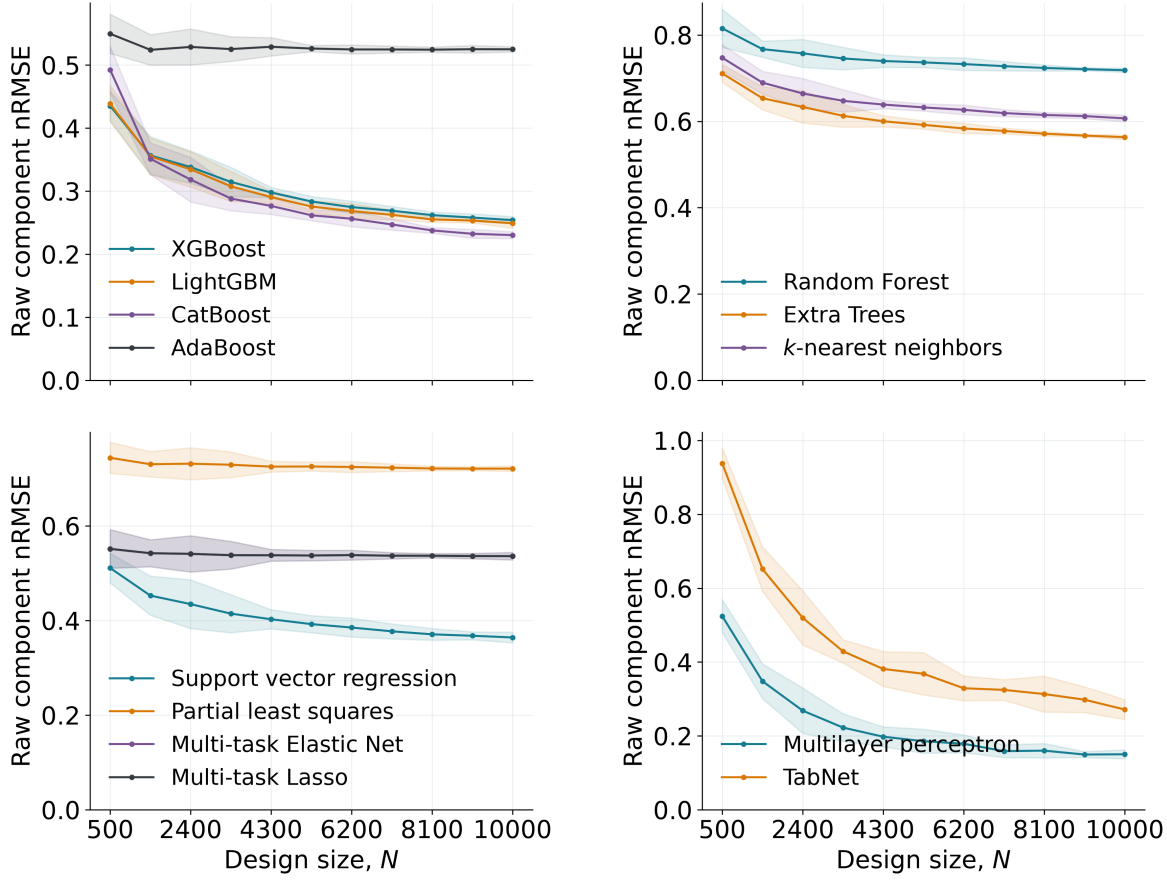


Figure 4.5: Raw component prediction error across the eleven nested design sizes. Separate panels group the predictors by their response representation. Lines show the five-fold mean nRMSE. Shaded regions show one sample standard deviation across folds.

The lowest-error predictor changed with the data budget. The boosted trees performed well at the smallest design size. The multilayer perceptron attained the lowest error at the largest size. TabNet still improved from 0.298 at $N = 9050$ to 0.272 at $N = 10,000$. Its final reduction was approximately 0.0266. Its error continued to decrease at the largest evaluated sizes.

The effect of projection also varied with design size. Both multi-task linear models benefited at all eleven sizes. Support vector regression benefited at every size from $N = 5250$ onward. The projection effect for the multilayer perceptron and TabNet changed sign across the size sequence. These trajectories were obtained with the prescribed fold-specific settings selected at the largest size.

Figure 4.6 separates this adjustment effect from the raw learning curves. A negative cell indicates lower error after projection. The persistent reductions for the multi-task models contrast with the mixed effects for the more flexible predictors.

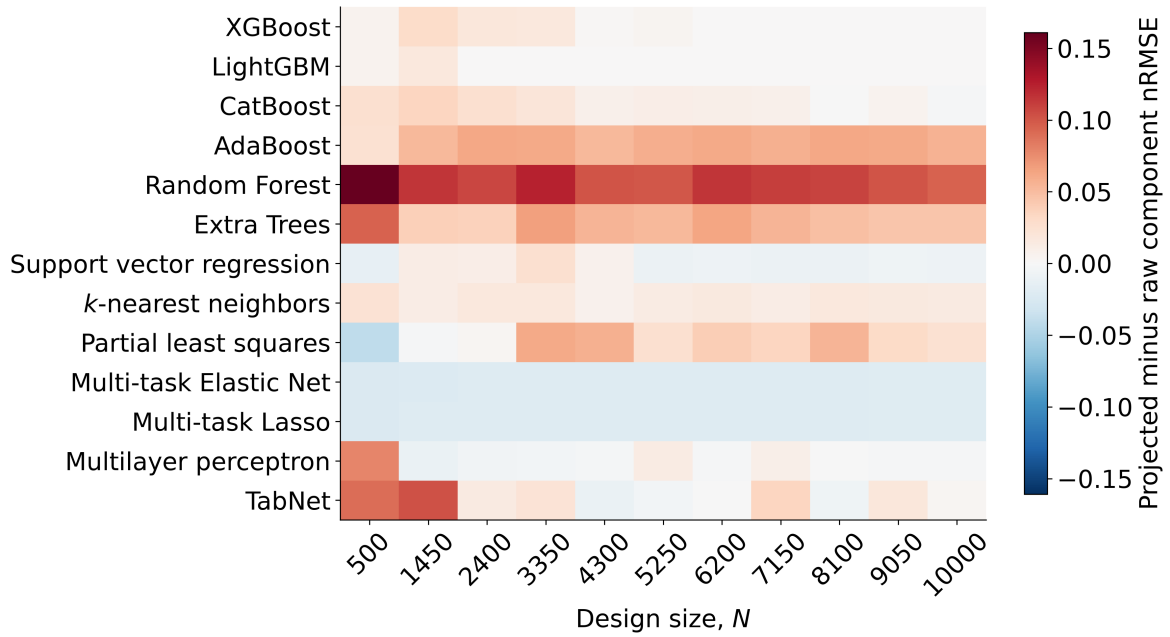


Figure 4.6: Projected minus raw mean component nRMSE for every predictor and design size. Negative values indicate improvement. Both prediction types use the same assessment folds and training-defined component scales.

4.4.4 Physical enforcement and component redistribution

Every raw assessment state failed the declared mass conservation tolerance at every design size. At $N = 10,000$, AdaBoost, Random Forest, Extra Trees, and the nearest-neighbor predictor nevertheless returned only non-negative raw concentrations. The other predictors had row-level non-negativity violation rates from 39.0% to 76.8%. Table 4.8 reports their rates and the size of the subsequent adjustment.

Projection produced no mass conservation or non-negativity failures in the 750,750 prediction instances evaluated across all thirteen predictors and eleven design sizes. These prediction instances include repeated assessment across nested sizes. The maximum projected invariant residual was 2.06×10^{-13} . Interpolation for non-negativity was activated in 2489 instances, or 0.332%. Random Forest, partial least squares, and Extra Trees accounted for 904, 574, and 566 activations, respectively. Together they accounted for 82.1% of the activations.

Figure 4.7 compares negative raw states with adjustment size and interpolation use. These are separate properties. Random Forest had no negative raw states but required the largest mean displacement and the most frequent interpolation at the largest design size.

At the largest design size, projection reduced component nRMSE in 188 of the 260 predictor–component comparisons. Both multi-task linear models improved eighteen of twenty components. The multilayer perceptron improved seventeen. CatBoost, partial least squares, and support vector regression each improved sixteen. In contrast, the

Table 4.8: Physical diagnostics at $N = 10,000$. Entries are five-fold means. Every raw mass conservation violation rate was 100%. Every projected mass conservation and non-negativity violation rate was zero.

Surrogate	Raw negative states (%)	Interpolation activated (%)	Mean d_c^{std}
XGBoost	39.72	0.00	0.300
LightGBM	38.95	0.00	0.301
CatBoost	59.06	0.00	0.242
AdaBoost	0.00	0.30	0.901
Random Forest	0.00	1.42	2.002
Extra Trees	0.00	0.88	1.528
Support vector regression	76.78	0.01	0.363
k -nearest neighbors	0.00	0.25	1.192
Partial least squares	54.31	0.93	1.808
Multi-task Elastic Net	71.06	0.00	0.482
Multi-task Lasso	71.05	0.00	0.482
Multilayer perceptron	58.58	0.00	0.189
TabNet	60.48	0.04	0.545

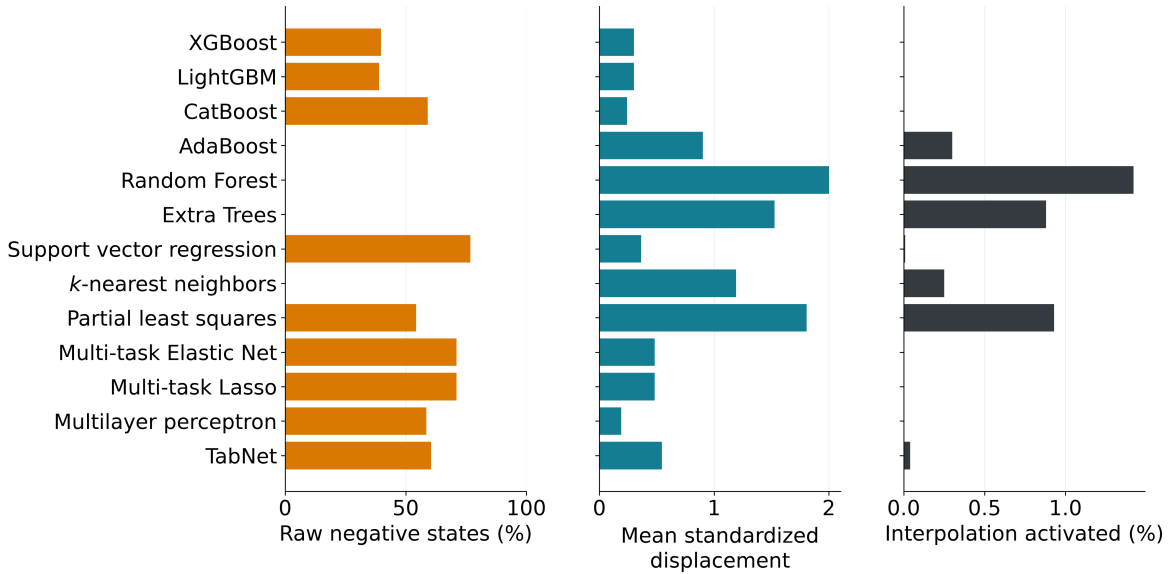


Figure 4.7: Physical diagnostics at $N = 10,000$. Panels show the fraction of raw states with negative components, mean standardized displacement, and interpolation activation. Values are five-fold means. All raw states failed the mass conservation tolerance, while every projected state passed both imposed physical checks.

heterotrophic biomass component X_H had higher error for every predictor. Random Forest’s fermentable-substrate component S_F increased from nRMSE 0.887 to 1.993. Its 142 interpolated predictions accounted for that component’s error increase, while the other 9858 predictions retained essentially the same squared error.

Figure 4.8 locates the component-level changes. The larger positive changes in a few components explain why an aggregate squared-error measure can increase even when most component scores improve.

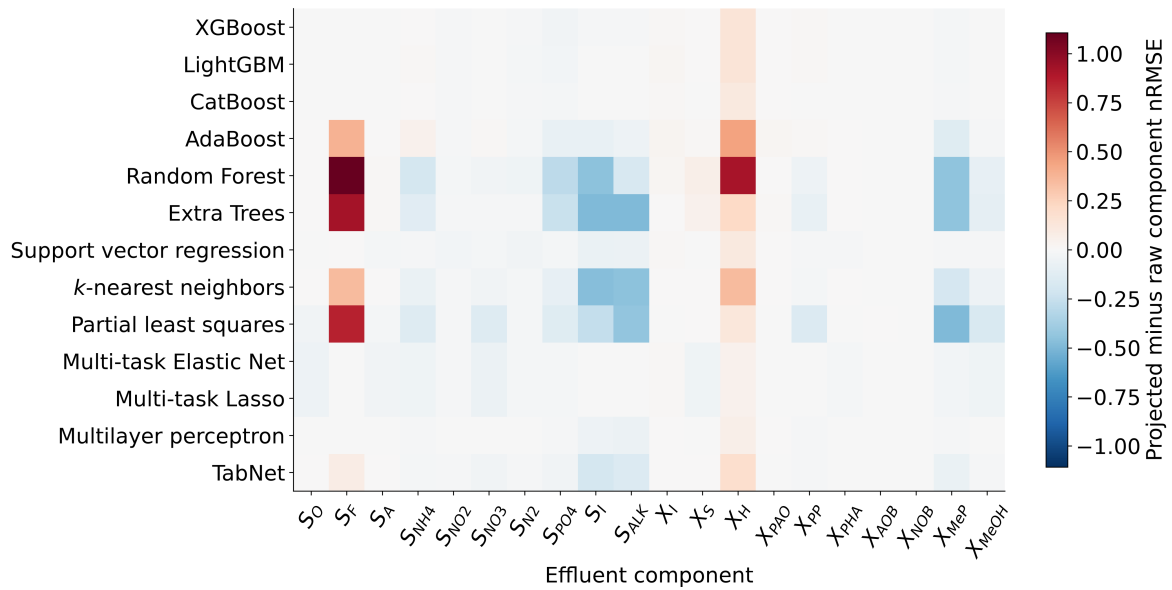


Figure 4.8: Projected minus raw component nRMSE at $N = 10,000$. Each cell is the difference between five-fold mean scores for one predictor and one component. Negative values indicate improvement. The same color limits apply to all cells.

The composite results also differed by quantity. Every predictor had lower TN and TP normalized root error after projection. Every predictor had higher COD and TSS normalized root error. The reported composites therefore responded differently to the same component adjustment.

Figure 4.9 presents the corresponding root errors in physical units. TN and TP improved while COD and TSS worsened for every predictor. The component projection thus changed the accuracy of different reporting quantities in different directions.

4.4.5 Prediction beyond the learning domain

Both raw and projected error increased from mild to severe exterior inputs for every predictor. The multilayer perceptron had the lowest error in both designs. Its projected nRMSE was 0.251 for mild inputs and 0.521 for severe inputs. TabNet followed at 0.333 and 0.547. Table 4.9 reports the full comparison under the common full-domain normalization.

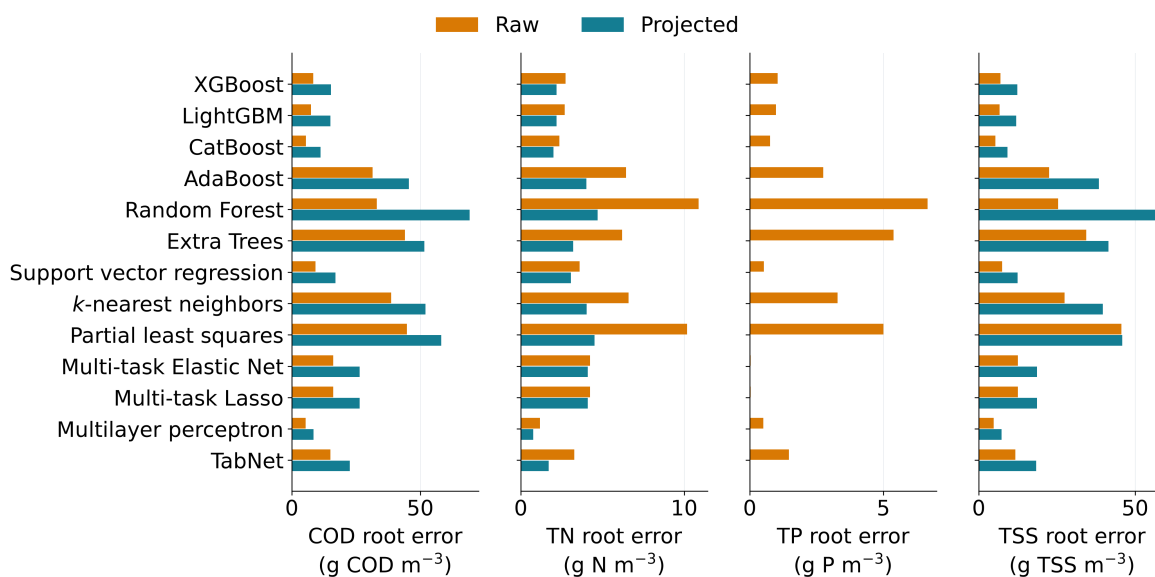


Figure 4.9: Raw and projected root mean squared errors for COD, TN, TP, and TSS at $N = 10,000$. Bars show means of five fold-specific root errors. Each panel uses the physical unit of its own reported quantity. These errors are distinct from the standardized component nRMSE.

Table 4.9: Component nRMSE beyond the learning domain. Each entry summarizes 600 accepted mechanistic states from one full-domain refit.

Surrogate	Mild exterior inputs		Severe exterior inputs	
	Raw	Projected	Raw	Projected
XGBoost	0.424	0.420	0.861	0.853
LightGBM	0.419	0.413	0.848	0.842
CatBoost	0.393	0.387	0.836	0.828
AdaBoost	0.692	0.711	1.036	1.139
Random Forest	0.908	0.996	1.242	1.489
Extra Trees	0.717	0.679	1.084	1.222
Support vector regression	0.555	0.538	0.989	0.948
k -nearest neighbors	0.760	0.732	1.113	1.147
Partial least squares	0.908	0.841	1.267	1.190
Multi-task Elastic Net	0.750	0.688	1.146	1.044
Multi-task Lasso	0.751	0.688	1.146	1.044
Multilayer perceptron	0.255	0.251	0.531	0.521
TabNet	0.336	0.333	0.563	0.547

Projection reduced aggregate nRMSE in eleven of thirteen mild comparisons and nine of thirteen severe comparisons. AdaBoost and Random Forest worsened in both designs. Extra Trees and the nearest-neighbor predictor also worsened in the severe design. Component error decreased in 192 of 260 mild comparisons and 193 of 260 severe comparisons. Every predictor improved a majority of its component scores in each exterior design.

Figure 4.10 uses a common horizontal scale for the two severity levels. The larger severe-domain errors remain visible even for the most accurate predictors. Both prediction types had larger errors under severe exterior inputs.

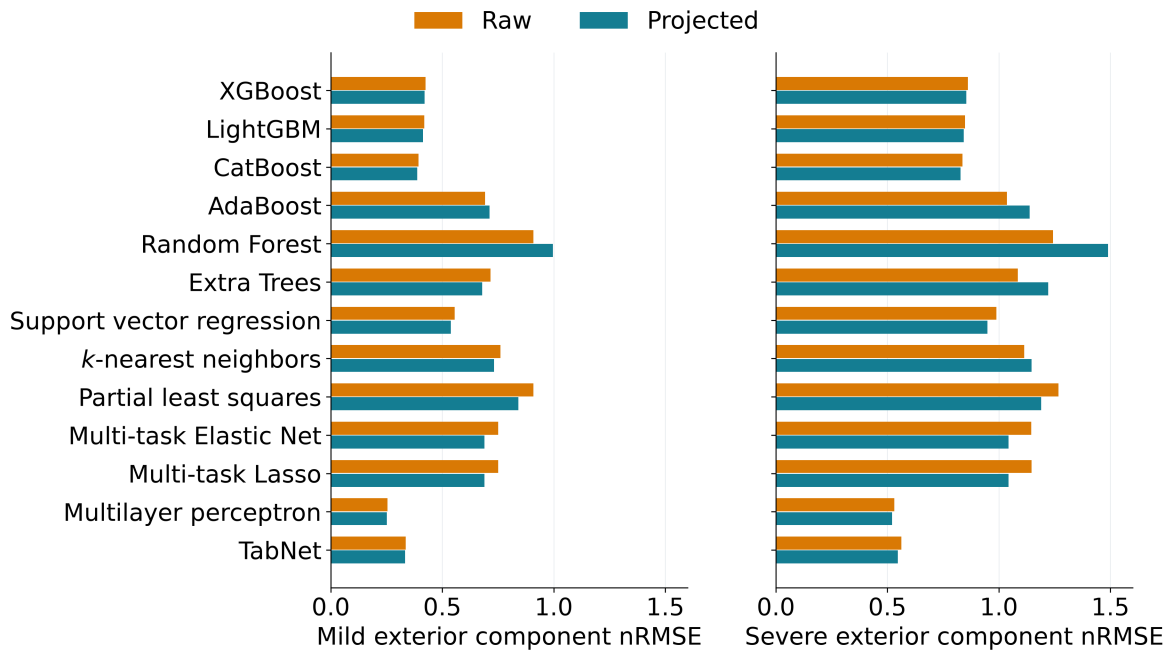


Figure 4.10: Raw and projected component nRMSE for mild and severe exterior inputs. Each severity includes 600 independently simulated reference states. Both panels use full-domain training scales and a common error-axis range.

All 15,600 exterior projected prediction instances passed the imposed physical checks. Their maximum invariant residual was 1.72×10^{-13} . Interpolation was activated in 68 instances, or 0.436%. Combining the interior and exterior assessments gave 766,350 projected prediction instances with no failures of the declared mass conservation and non-negativity checks. The increase in severe-domain error occurred despite this complete constraint compliance.

4.4.6 Fitting and prediction costs

Table 4.10 separates model preparation from repeated prediction. At the largest design size, mean fitting time ranged from approximately 0.024 s for Multi-task Elastic Net to 56.02 s for support vector regression. Mean search time ranged from 14.21 s for

Multi-task Lasso to 11,082.77 s for support vector regression. These preparation costs were incurred before repeated use of the fitted predictor.

Figure 4.11 shows these differences on logarithmic axes. Projection dominated the inference cost of the fastest linear predictors. The two panels report preparation time and repeated prediction time separately.

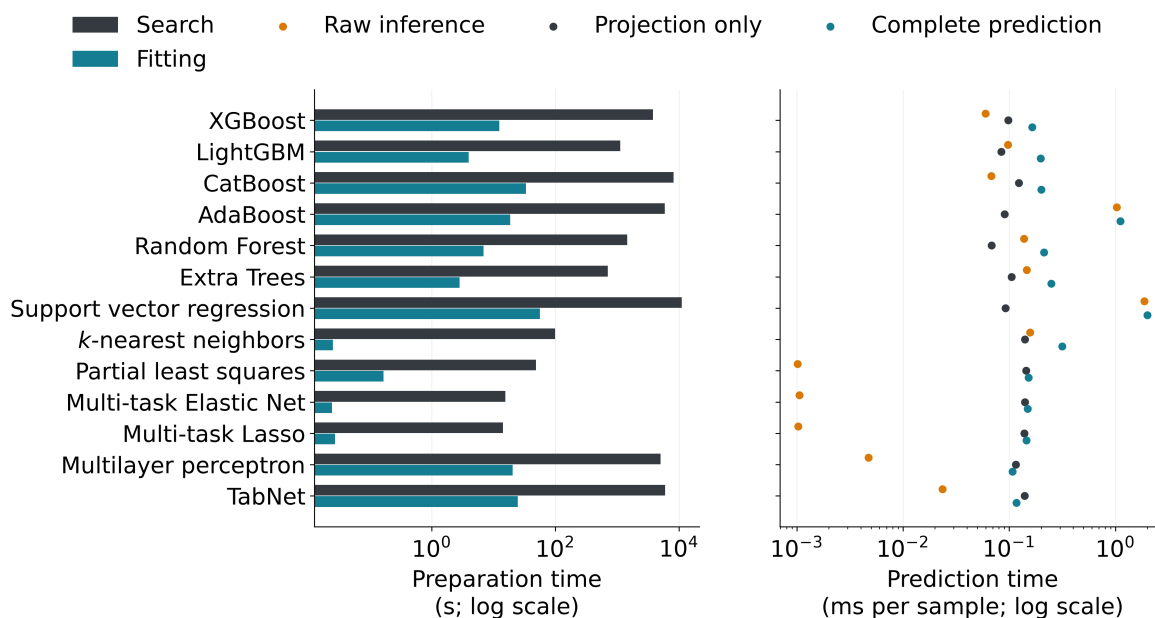


Figure 4.11: Model preparation and repeated prediction costs at $N = 10,000$. The left panel separates search and fitting in seconds. The right panel separates raw inference, projection alone, and complete prediction in milliseconds per sample under the fixed batch protocol. Both panels show five-fold means on logarithmic axes. Complete prediction was timed separately.

The multilayer perceptron had the lowest observed complete batch-normalized prediction cost of 0.1071 ms per sample. Support vector regression had the largest at 2.0008 ms. Projection alone ranged from 0.0681 to 0.1442 ms per sample. It dominated the cost of the fastest raw linear and neural predictions. The complete multilayer perceptron time of 0.1071 ms was smaller than the separately measured projection-only time of 0.1151 ms. Each path has its own repeated batch measurements, and the complete prediction cost uses the directly measured complete path.

4.5 Discussion

4.5.1 Mass conservation and prediction error

Consider a hypothetical two-component prediction problem in which both components use one material basis. The first reference concentration is 6 and the second is 4 in arbitrary concentration units. Their sum is $6 + 4 = 10$. The raw prediction gives 8

Table 4.10: Five-fold mean computational costs at $N = 10,000$. Search and fitting are in seconds. Prediction costs are in milliseconds per sample under the fixed 512-row batch protocol. Complete prediction includes raw inference and projection.

Surrogate	Search	Fitting	Raw	Projection	Complete
XGBoost	3810.95	12.46	0.0597	0.0976	0.1649
LightGBM	1126.04	3.96	0.0971	0.0840	0.1974
CatBoost	8165.42	33.30	0.0677	0.1235	0.1997
AdaBoost	5894.14	18.55	1.0310	0.0901	1.1101
Random Forest	1452.33	6.89	0.1376	0.0681	0.2117
Extra Trees	705.21	2.83	0.1461	0.1053	0.2484
Support vector regres- sion	11082.77	56.02	1.8736	0.0922	2.0008
k -nearest neighbors	99.34	0.03	0.1574	0.1401	0.3148
Partial least squares	48.75	0.17	0.0010	0.1442	0.1515
Multi-task Elastic Net	15.59	0.02	0.0011	0.1406	0.1493
Multi-task Lasso	14.21	0.03	0.0010	0.1387	0.1447
Multilayer perceptron	5016.24	20.46	0.0047	0.1151	0.1071
TabNet	5971.04	24.73	0.0235	0.1398	0.1170

for the first component and 3 for the second. Its sum is $8 + 3 = 11$, so its inventory residual is $11 - 10 = 1$. Both coordinates are non-negative, but the state is inconsistent with mass conservation under the specified equality.

Suppose a hypothetical projection for mass conservation lowers the first prediction by one unit and leaves the second unchanged. The projected sum is $7 + 3 = 10$. The component errors can be calculated directly,

$$\begin{aligned}
 \text{raw errors} &= (8 - 6, 3 - 4) = (2, -1), \\
 \text{projected errors} &= (7 - 6, 3 - 4) = (1, -1), \\
 \text{raw squared distance} &= 2^2 + (-1)^2 = 5, \\
 \text{projected squared distance} &= 1^2 + (-1)^2 = 2.
 \end{aligned}$$

The corresponding Euclidean distances are $\sqrt{5} \approx 2.236$ and $\sqrt{2} \approx 1.414$. Their values follow from comparison with the reference state. The mass conservation test instead uses the predicted sums. In compact component notation, the raw state is $(8, 3)^\top$, the reference state is $(6, 4)^\top$, and the chosen projected state is $(7, 3)^\top$. This calculation assesses feasibility and error on the same hypothetical prediction.

4.5.2 Statistical accuracy and physical admissibility

The paired assessment gave a direct counterexample to using low error as a physical acceptance rule. The multilayer perceptron had the lowest raw component error at the largest size, yet every raw state failed the mass conservation tolerance and 58.58%

contained a negative component beyond tolerance. The positive outputs of the averaging-based predictors also failed mass conservation. Their responses combined cases with different influent-specific balance targets. Positivity of the contributing states did not preserve the target of a new case.

This result supports the distinction between learning a response and enforcing a constraint discussed by Kashinath et al. (2021) and Hansen et al. (2024). The projection of Kircher and Votsmeier (2025) addressed the second task across all thirteen predictor types. Its observed constraint compliance did not depend on the predictor having the lowest statistical error. A final physical acceptance check is therefore justified when a subsequent calculation requires the full non-negative component state and the adopted invariant balances.

Violation frequency and magnitude also require separate interpretation. A state was counted as failing when any one of twenty components crossed the tolerance. For CatBoost, raw non-negativity incidence increased from 12.8% at $N = 500$ to 59.1% at $N = 10,000$, even as aggregate error decreased. Its overall fit improved over the same interval. The negative predictions were concentrated in small substrate and nitrogen pools. A more accurate aggregate response can still contain small departures at a zero boundary.

4.5.3 Why enforcement can increase prediction error

The projection minimized relative movement from a positive provisional target. The accuracy score measured error relative to training variability. These were different objectives. The method also shortened one complete direction when a candidate crossed a concentration boundary. Neither operation was designed to minimize the assessed nRMSE. The species-weighted conservation correction of Sturm and Silva (2025) provides a related example of how the adjustment metric affects component accuracy.

The S_F result illustrates the effect of shared interpolation. Its column in the invariant operator was numerically zero. The mass conservation solve consequently left it essentially unchanged before interpolation. When another component required a shorter direction, S_F was also moved toward its much larger influent value. A safeguard needed by one component could therefore worsen an otherwise acceptable prediction of another. Random Forest’s 142 affected full-size states provide a concrete example of this mechanism.

The X_H losses had a different explanation. Its relatively large concentrations made an absolute adjustment less costly under inverse-concentration weighting. Such an adjustment could satisfy the invariants while moving away from the mechanistic biomass target. The opposing changes in TN, TP, COD, and TSS further showed that mass conservation did not determine every reported composite. Near-alignment of the TP reporting vector with the phosphorus invariant explained its very small projected error.

COD and TSS remained sensitive to the redistribution of organic and particulate components (Henze et al., 2000).

Both multi-task linear models benefited consistently. The largest aggregate losses occurred in predictors with larger displacements and more interpolation activations. Paired error, displacement, and activation summaries identified this trade-off more clearly than a feasible fraction alone.

4.5.4 Data support, extrapolation, and practical use

The strong small-data performance of boosted trees is consistent with the tabular comparisons discussed by Borisov et al. (2024) and Shwartz-Ziv and Armon (2022). The multilayer perceptron had the lowest error at the largest data budget. TabNet also continued to improve as more mechanistic states became available. Learning-curve interpretation requires the full trajectory and its validation design (Viering & Loog, 2023).

Exterior inputs increased error even when every projected state remained feasible. The changed error under exterior inputs is consistent with the role of input-domain coverage discussed by Shen et al. (2023) and Valente et al. (2025). The projection maintained the imposed component balances and non-negativity throughout both exterior designs.

The timing results supported rapid repeated evaluation after preparation. They also showed that a very fast raw predictor could spend most of its complete prediction time on physical enforcement. A deployment choice must therefore compare complete prediction cost rather than raw inference alone (Bhosekar & Ierapetritou, 2018; Blik et al., 2023).

The evidence establishes reliable enforcement of the specified mass conservation and non-negativity conditions for the evaluated simulated steady states. It also establishes that the statistical cost of enforcement depends on the predictor, component, and assessment domain.

4.5.5 Data Availability

The simulation data and numerical results supporting this study are publicly available at <https://github.com/egbertoselerio1997/surrogate-projection>.

Chapter 5

Interpretable Regression with Checked Physical Predictions

5.1 Rationale

An activated sludge surrogate used for process analysis should allow an engineer to examine how operating settings and influent composition contribute to its prediction. An error score alone cannot explain why a predicted nutrient concentration changes when aeration or loading changes. The response must also retain the substrate, nutrient, biomass, and solids components needed for treatment reporting and material accounting (Henze et al., 2000). The research problem is therefore to obtain an inspectable complete-state approximation whose returned concentrations satisfy the adopted physical requirements.

Directly inspectable models provide a basis for scrutinizing predictions used in consequential decisions (Rudin, 2019). Polynomial response-surface methods offer a treatment-specific precedent for representing operating effects and interactions (Nazlabadi et al., 2021). The nonlinear libraries of Brunton et al. (2016) and the combined process and learning models reviewed by Shen et al. (2023) provide related ways to connect an explicit mathematical structure with process knowledge. These ideas do not by themselves establish an interpretable prediction of the full activated sludge component state after physical enforcement. The first gap concerns how operating and loading terms can remain identifiable within that complete prediction calculation.

The distinction between operation and loading is important for nutrient removal. Carbon availability influences denitrification and phosphorus storage, while oxygen availability influences carbon oxidation and nitrification. Wentzel et al. (1992) describe these connected pathways, and Guerrero et al. (2011) examine competition for carbon in nitrogen and phosphorus removal. For example, the predicted effect of increasing aeration may depend on incoming ammonium and organic substrate. Separate operating, loading, curvature, and interaction terms allow this dependence to be examined. Their

interpretation remains a statement about the fitted response rather than proof of a unique biological mechanism.

Relationships among effluent components introduce another level of interpretation. A term associated with influent substrate can influence several predicted concentrations through learned component coupling. Its coefficient in the regression driver is therefore not necessarily the final substrate or biomass sensitivity. Coefficient units and the position of a term in the calculation both matter. The structured treatment model of Selerio (2026) supplies the basis for separating the explicit driver from the coupled raw effluent. Examining both stages makes the fitted associations visible without treating the coupling coefficients as kinetic parameters.

The second gap concerns interpretation after physical enforcement. Projection adjusts the raw component state before COD, TN, TP, and TSS are calculated. A concentration bound can become active at one operating point and inactive at another. The resulting local response can differ from the slope suggested by the raw regression. Interpretation must therefore follow the driver, component coupling, and projection to the returned effluent (Selerio, 2026). A visible polynomial is not sufficient if the explanation stops before the operation that changes the prediction used by the engineer.

Physical requirements also need to be distinguished from fitting preferences. Penalties can discourage balance violations or negative concentrations during fitting, but they do not establish that every new prediction satisfies both conditions (Hansen et al., 2024). An explicit polynomial can return negative components even when all training states are non-negative. Mass conservation and non-negativity therefore require a defined enforcement procedure and checks on its final output. The joint projection of Kircher and Votsmeier (2025) demonstrates the importance of treating these requirements together. For an inspectable activated sludge regression, the enforcement stage must be both physically checked and included in the explanation of the response.

The practical value of this structure also depends on the available training data. A compact description of operating effects may be useful when mechanistic calculations limit the number of training cases. It need not outperform a more flexible learner when a larger design is available. Learning curves make this distinction measurable (Viering & Loog, 2023). Held-out errors and fitting and prediction effort are therefore assessed alongside coefficient structure and physical checks. This comparison tests whether inspectability is accompanied by useful predictive performance rather than assuming that a transparent equation is automatically the best approximation.

The study presents invariant-constrained second-order regression (ICSOR), which combines an explicit polynomial driver, learned component coupling, and checked prediction (Selerio, 2026). Deployment here means evaluating a fitted model at a new influent and operating setting and returning its physically checked component state. The study addresses the third research question in Section 1.3 by examining the complete response, its fitted terms, and its sensitivities before and after projection. Its

contribution is an inspectable component-state formulation with a traceable physical enforcement procedure. The assessment identifies the data conditions and interpretation limits under which this structure can support treatment analysis. It does not presume causal coefficient meanings or superior accuracy at every training size.

5.2 Theory

The prediction target is the effluent component vector $c_{out} \in \mathbb{R}^F$. The adopted process basis has $F = 20$ components and $P = 28$ reactions. The operating vector $u \in \mathbb{R}^{M_{op}}$ has $M_{op} = 2$ entries for HRT and aeration level. The full influent vector is $c_{in} \in \mathbb{R}^F$. Component concentrations retain their physical units. The four reported composites are obtained from a fixed map $I_{comp} \in \mathbb{R}^{4 \times F}$. That map is applied after component prediction.

5.2.1 Mass conservation in the adopted state space

The link between a component balance and a regression constraint can first be seen in a hypothetical conversion of one component into another. Suppose that one unit of component 1 is converted into one unit of component 2. The two components are expressed in a common inventory unit for this illustration. Their changes satisfy

$$c_{out,1} - c_{in,1} = -\xi, \quad c_{out,2} - c_{in,2} = \xi. \quad (5.1)$$

Adding these equations eliminates the unknown conversion amount. The relation available to the regression is $c_{out,1} + c_{out,2} = c_{in,1} + c_{in,2}$. With an illustrative influent $(6, 4)^\top$ and conversion amount 2, the effluent is $(4, 6)^\top$. Both totals are 10. Many other non-negative pairs also have that total. The balance defines a set of regression outputs with the same total inventory.

The same calculation can be written using the small matrices

$$\nu = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad A\nu^\top = 1(-1) + 1(1) = 0. \quad (5.2)$$

These hypothetical matrices use reduced dimensions to make each operation visible. The adopted reactor uses the stated 20-component basis, reaction matrix, and operating domain.

The stoichiometric matrix $\nu \in \mathbb{R}^{P \times F}$ records reactions as rows and components as columns. Let $V_0 \in \mathbb{R}^{F \times K}$ span its right null space, where $K = 5$. The invariant operator is

$$A = V_0^\top \in \mathbb{R}^{K \times F}, \quad \nu V_0 = 0, \quad A\nu^\top = 0. \quad (5.3)$$

The columns of V_0 are independent, so A has full row rank. With an orthonormal null-space basis, $AA^\top = I_K$. A different full-rank basis describes the same mass

conservation conditions, although its row scaling changes the numerical size of a soft invariant penalty.

For the control volume and boundary accounting represented by the adopted process model, a reaction-induced change has the form

$$c_{out} - c_{in} = \nu^\top \xi, \quad \xi \in \mathbb{R}^P. \quad (5.4)$$

The reaction-progress vector ξ expresses the net contribution of the reactions in concentration-equivalent units. Multiplication by A removes that unknown contribution and gives

$$A(c_{out} - c_{in}) = 0. \quad (5.5)$$

This construction uses the mass conservation logic of the photochemical surrogate framework of Sturm and Wexler (2020). Sturm and Wexler (2022) further show how a stoichiometric transformation can connect learned reaction information to state changes satisfying mass conservation. The same linear algebra applies here, but the component combinations used for mass conservation are those of the activated sludge reaction network.

Equation (5.5) uses the modeled boundary in Equation (5.4). External gas transfer, chemical dosing, solids withdrawal, and other boundary fluxes enter the balance where they occur. With an explicit boundary contribution s , the relation is $A(c_{out} - c_{in} - s) = 0$.

For a new influent state, the component set used by checked deployment is

$$\mathcal{F}(c_{in}) = \{c \in \mathbb{R}^F \mid Ac = Ac_{in}, c \geq \mathbf{0}\}. \quad (5.6)$$

When $c_{in} \geq \mathbf{0}$, this set contains c_{in} and is therefore nonempty. Its members satisfy the modeled linear invariants and component non-negativity within the declared boundary.

The reported composites provide another reason to work in component space. Distinct component states can have the same COD, TN, TP, and TSS values. A negative component can also be hidden by positive contributions to an aggregate. When I_{comp} is entrywise non-negative, a non-negative component state produces non-negative composites. The reverse implication does not generally hold. Mass conservation and non-negativity are therefore tested before the component state is collapsed into the four reported quantities (Henze et al., 2000; Selerio, 2026).

5.2.2 A partitioned second-order driver

Separating operation from loading

Operating conditions and influent composition play different engineering roles. HRT and aeration define aspects of the reactor environment. Influent components define the material presented to that environment. This distinction is consistent with the operating-sensitivity analysis of de Araujo et al. (2013) and the separation of operational and influent uncertainty examined by Machado et al. (2014).

Start with two hypothetical operating entries u_1, u_2 and two influent entries c_1, c_2 . A second-order model uses the individual entries, their squares, and their pairwise products. The operating products are $u_1^2, u_1u_2, u_2u_1, u_2^2$. The feature vector keeps both orders as an accounting convention, with $u_1u_2 = u_2u_1$.

For two column vectors, the Kronecker operation multiplies the entire second vector by the first entry of the first vector, then by its second entry. Thus

$$\begin{aligned} u \otimes u &= \begin{bmatrix} u_1u_1 \\ u_1u_2 \\ u_2u_1 \\ u_2u_2 \end{bmatrix}, & c \otimes c &= \begin{bmatrix} c_1c_1 \\ c_1c_2 \\ c_2c_1 \\ c_2c_2 \end{bmatrix}, \\ u \otimes c &= \begin{bmatrix} u_1c_1 \\ u_1c_2 \\ u_2c_1 \\ u_2c_2 \end{bmatrix}. \end{aligned} \tag{5.7}$$

For the illustrative values $u = (2, 3)^\top$ and $c = (4, 5)^\top$, these three blocks are respectively $(4, 6, 6, 9)^\top$, $(16, 20, 20, 25)^\top$, and $(8, 10, 12, 15)^\top$. Including the baseline 1 and the four first-order entries gives 17 ordered features in this reduced example. The same construction applies to the larger adopted input vector.

The feature vector retains both roles through the ordered expansion

$$\phi(u, c_{in}) = \begin{bmatrix} 1 \\ u \\ c_{in} \\ u \otimes u \\ c_{in} \otimes c_{in} \\ u \otimes c_{in} \end{bmatrix} \in \mathbb{R}^D, \quad D = 1 + M_{op} + F + M_{op}^2 + F^2 + M_{op}F. \tag{5.8}$$

The symbol $\otimes$ denotes the Kronecker product. The adopted dimensions give $D = 467$. The ordered expansion includes both u_iu_j and u_ju_i , and both $c_{in,i}c_{in,j}$ and $c_{in,j}c_{in,i}$. These pairs are equal as numerical features. Their interpretation requires the symmetry convention developed below.

Multiplication by regression coefficients can also be read one entry at a time. For an illustrative first driver, take the baseline coefficient as 1, operating coefficients as $(2, -1)$, and influent coefficients as $(0.5, 0)$. Let its operating quadratic row be $(0.1, 0.1, 0.1, 0)$, influent quadratic row be $(0, 0, 0, -0.05)$, and operating-load row be

$(0, 0, 0.25, 0)$. Applying these rows to the numerical feature blocks gives

$$\begin{aligned}
 r_1 &= 1 + \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \\ 6 \\ 9 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 0 & 0 & 0 & -0.05 \end{bmatrix} \begin{bmatrix} 16 \\ 20 \\ 20 \\ 25 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 0 & 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 8 \\ 10 \\ 12 \\ 15 \end{bmatrix} \\
 &= 1 + 1 + 2 + 1.6 - 1.25 + 3 = 7.35.
 \end{aligned} \tag{5.9}$$

Each row-vector multiplication is a sum of coefficient times feature. It is the same operation as an ordinary linear regression, but the supplied features include nonlinear products. A second hypothetical driver $r_2 = -u_1 + u_2 + 0.2c_2$ would equal 2 at the same inputs. The coupled relation converts the two driver values into predicted concentrations.

Before grouping the coefficients into matrices, the first driver in this illustration can be written as

$$r_1 = 1 + 2u_1 - u_2 + 0.5c_1 + 0.1u_1^2 + 0.2u_1u_2 - 0.05c_2^2 + 0.25u_2c_1. \tag{5.10}$$

The coefficient 0.2 of u_1u_2 is the sum of the two ordered coefficients 0.1. In a general row, the scalar calculation is $r_f = \sum_{\ell=1}^D B_{f\ell}\phi_\ell$. Carrying out this sum for all components gives the driver $r \in \mathbb{R}^F$ through

$$r(u, c_{in}) = B\phi(u, c_{in}), \quad B \in \mathbb{R}^{F \times D}. \tag{5.11}$$

The driver is an intermediate regression quantity supplied to the coupled prediction relation. Its coefficients have no sign restriction, so the regression can represent both increasing and decreasing effects. Non-negativity is imposed on component states where it has a physical meaning. The driver block form is

$$\begin{aligned}
 r(u, c_{in}) &= b + W_u u + W_{in} c_{in} + \Theta_{uu}(u \otimes u) \\
 &\quad + \Theta_{cc}(c_{in} \otimes c_{in}) + \Theta_{uc}(u \otimes c_{in}).
 \end{aligned} \tag{5.12}$$

Here $b \in \mathbb{R}^F$ is a baseline term. The matrices $W_u \in \mathbb{R}^{F \times M_{op}}$ and $W_{in} \in \mathbb{R}^{F \times F}$ contain first-order operating and influent terms. The matrices $\Theta_{uu} \in \mathbb{R}^{F \times M_{op}^2}$ and $\Theta_{cc} \in \mathbb{R}^{F \times F^2}$

contain curvature within each input group. The matrix $\Theta_{uc} \in \mathbb{R}^{F \times M_{op} F}$ contains operating-load interactions.

Illustrative operating and loading effects

Consider an illustrative driver for one component with aeration level a and influent ammonium concentration n . A term of the form

$$r_f = \beta_0 + \beta_a a + \beta_n n + \beta_{aa} a^2 + \beta_{an} an \quad (5.13)$$

has an aeration derivative $\partial r_f / \partial a = \beta_a + 2\beta_{aa} a + \beta_{an} n$. The effect described by this derivative changes with both aeration level and ammonium loading. The derivative of this hypothetical polynomial combines the linear aeration term, curvature, and the aeration-ammonium interaction.

The same reasoning applies to simultaneous influent loads. A carbon-ammonium interaction allows the modeled association with carbon availability to depend on nitrogen loading. A phosphorus-storage interaction allows the modeled association with dissolved phosphate to depend on a particulate storage pool. These terms provide a transparent approximation to a nonlinear response involving the microbial pathways described by Henze et al. (1999).

The second-order library represents smooth local curvature and pairwise input interactions. Explicit nonlinear libraries make candidate response structures available for inspection, as illustrated by Brunton et al. (2016). The response-surface applications reviewed by Nazlabadi et al. (2021) provide a wastewater treatment precedent for using polynomial operating terms and interactions.

Symmetry and the meaning of cross terms

For a two-entry vector, direct multiplication gives

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = q_{11}x_1^2 + (q_{12} + q_{21})x_1x_2 + q_{22}x_2^2. \quad (5.14)$$

Only the sum of the two off-diagonal coefficients enters this polynomial. Replacing them by their average leaves that sum unchanged. For example,

$$Q = \begin{bmatrix} 0.1 & 0.3 \\ -0.1 & 0 \end{bmatrix}, \quad \frac{Q + Q^\top}{2} = \begin{bmatrix} 0.1 & 0.1 \\ 0.1 & 0 \end{bmatrix}. \quad (5.15)$$

At $x = (2, 3)^\top$, the first matrix gives $0.4 + 1.8 - 0.6 = 1.6$. The symmetric matrix gives $0.4 + 0.6 + 0.6 = 1.6$. The asymmetric coefficients cannot be distinguished from these predictions. Their squared sums are different. The original matrix has squared sum $0.01 + 0.09 + 0.01 = 0.11$, while its symmetric part has squared sum $0.01 + 0.01 + 0.01 = 0.03$. The removed part has squared sum 0.08 and contributes no

prediction. This is why a positive coefficient penalty favors the symmetric coefficient matrix.

For component f , reshape the corresponding rows of Θ_{uu} and Θ_{cc} into square matrices $Q_{uu}^{(f)}$ and $Q_{cc}^{(f)}$. A quadratic term can then be written as $x^\top Qx$. For any real square matrix,

$$x^\top Qx = x^\top \text{sym}(Q)x, \quad \text{sym}(Q) = \frac{Q + Q^\top}{2}. \quad (5.16)$$

The skew-symmetric part contributes nothing to the prediction. It can nevertheless increase the coefficient norm because

$$\|Q\|_F^2 = \|\text{sym}(Q)\|_F^2 + \|\text{skew}(Q)\|_F^2. \quad (5.17)$$

Replacing each square slice by its symmetric part preserves every fitted value and cannot increase a positive ridge penalty. The projection is applied after each driver update. With a strictly positive driver ridge weight, the exact minimum-norm solution has no skew-symmetric part.

Symmetry also determines how a coefficient display is read. If $q_{ij} = q_{ji}$, the two displayed off-diagonal cells together contribute $2q_{ij}x_ix_j$. A displayed cell is therefore half the coefficient of the corresponding unordered cross term. For an illustrative matrix with $q_{12} = q_{21} = 0.3$, the polynomial contribution is $0.6x_1x_2$. Reading either cell as the entire interaction coefficient would halve its contribution.

Each component has $M_{op}(M_{op} + 1)/2 = 3$ distinct operating quadratic terms and $F(F + 1)/2 = 210$ distinct influent quadratic terms. Together with the baseline, linear terms, and operating-load interactions, the library has 276 distinct polynomial terms per component. These counts define the structural dimensions of the polynomial library.

5.2.3 Learned coupling among effluent components

Consider two hypothetical component equations in which each channel has a coupling coefficient 0.2 to the other channel. With driver values 4 and 2, their equations are

$$c_1 = 4 + 0.2c_2, \quad c_2 = 2 + 0.2c_1. \quad (5.18)$$

Substituting the second equation into the first gives $c_1 = 4 + 0.2(2 + 0.2c_1)$. Moving the remaining c_1 term to the left gives $0.96c_1 = 4.4$. Therefore $c_1 = 55/12$ and $c_2 = 35/12$. The solution is approximately $(4.5833, 2.9167)^\top$. These hypothetical values demonstrate the arithmetic of a simultaneous regression relation.

Moving the coupled terms to the left makes the same calculation a matrix solve,

$$\Gamma = \begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & -0.2 \\ -0.2 & 1 \end{bmatrix}, \quad R \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}. \quad (5.19)$$

The determinant of R is $1 - 0.2^2 = 0.96$. Its inverse is

$$R^{-1} = \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}. \quad (5.20)$$

Multiplying this inverse by the driver gives the same two scalar solutions. The inverse also explains why a change in one driver can change both predicted components.

The regression includes a matrix $\Gamma \in \mathbb{R}^{F \times F}$ that describes fitted dependence among the effluent components. Define

$$R = I_F - \Gamma, \quad Rc = r(u, c_{in}). \quad (5.21)$$

Equivalently, the equation for component f is $c_f = r_f + \sum_{j \neq f} \Gamma_{fj} c_j$. This form separates the input-dependent driver from dependence among the predicted component channels. The diagonal of Γ is fixed to zero to avoid a redundant self-coupling coefficient.

The sign of Γ_{fj} describes reinforcement or compensation within the fitted simultaneous relation. Activated sludge reactions involve several components at once, and the stoichiometric matrix specifies those transformations (Henze et al., 2000). The coupling matrix is learned from state patterns. Its entries describe fitted dependence associated with correlated transformations, loading patterns, and the allocation of effects between the driver and coupling blocks (Selerio, 2026).

When R is nonsingular, the raw component prediction is $R^{-1}r$. Multiplication by R^{-1} redistributes or amplifies the response represented by the driver. For interpretation, the driver, coupling matrix, and complete raw mapping are examined together.

The adopted coupling constraints require

$$\Gamma_{ff} = 0, \quad |\Gamma_{fj}| \leq \gamma_{abs} \quad (f \neq j), \quad \kappa_2(I_F - \Gamma) \leq \kappa_{max}. \quad (5.22)$$

The condition number is the ratio of the largest to smallest singular value of the coupled-system matrix. A nearly singular matrix can amplify small driver perturbations and numerical errors (Buzzi-Ferraris & Manenti, 2013). The condition check controls the complete matrix separately from the bounds on individual entries.

The small symmetric matrix in Equation (5.19) illustrates this measure without a large singular-value calculation. In the direction $(1, 1)^\top$, multiplication by R scales the vector by 0.8. In the direction $(1, -1)^\top$, it scales the vector by 1.2. These orthogonal directions give singular values 0.8 and 1.2, so the condition number is $1.2/0.8 = 1.5$.

If the two illustrative couplings were 0.99, the same directions would have scales 0.01 and 1.99. The condition number would be 199. A driver error of magnitude ϵ in each component along the first direction would produce a state error of 100ϵ in each component. This hypothetical choice illustrates amplification at a coupling magnitude of 0.99, above the adopted bound.

Even a small entry bound can permit singularity in a larger matrix. In a hypothetical four-component matrix with every off-diagonal entry equal to $1/3$, every row sum of Γ is one. Hence $\Gamma \mathbf{1} = \mathbf{1}$ and $(I - \Gamma)\mathbf{1} = 0$. Each entry is smaller than the adopted bound 0.368, but the coupled-system matrix is singular. This example explains why entry bounds and the condition-number test perform separate tasks.

Shrinking a candidate coupling matrix toward zero can provide a safe candidate because $I_F - 0 = I_F$. A zero coupling matrix gives a condition number of one. A

prescribed $\kappa_{max} > 1$ therefore permits this fallback. The nonlinear condition-number requirement defines a generally nonconvex admissible coupling set. It is evaluated after the convex box-constrained estimation problem proposes a coupling update.

5.3 Methodology

5.3.1 Fitting in prediction space

The coupled training objective

The fitting problem adjusts coefficients to approximate the mechanistic effluent concentrations. It also uses auxiliary fitted states, which are concentration variables optimized for the training cases. These variables help express fitting constraints and penalties. They differ from the component predictions calculated for a new influent after fitting.

The squared error of one prediction is the square of its difference from the reference. For several components and observations, fitting adds these squared differences. A compact matrix norm represents this sum. For any matrix M with entries M_{ij} ,

$$\|M\|_F^2 = \sum_i \sum_j M_{ij}^2. \quad (5.23)$$

For example, a residual matrix $\begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ has squared norm $1^2 + (-2)^2 + 3^2 + 0^2 = 14$.

The norm is $\sqrt{14}$, while a least-squares objective uses 14. Squaring prevents positive and negative errors from canceling when they are added.

Let i identify an observation, f a component, k an invariant, and ℓ a feature. The entry $\hat{c}_{if}$ is the fitted concentration, while $\phi_{i\ell}$ is the corresponding feature entry. Before grouping the five training contributions into norms, their scalar form is

$$\begin{aligned} J = & \sum_{i=1}^N \sum_{f=1}^F (c_{out,if} - \hat{c}_{if})^2 \\ & + \lambda_{inv} \sum_{i=1}^N \sum_{k=1}^K \left[\sum_{f=1}^F A_{kf} (\hat{c}_{if} - c_{in,if}) \right]^2 \\ & + \lambda_{sys} \sum_{i=1}^N \sum_{f=1}^F \left[\hat{c}_{if} - \sum_{j=1}^F \Gamma_{fj} \hat{c}_{ij} - \sum_{\ell=1}^D B_{f\ell} \phi_{i\ell} \right]^2 \\ & + \lambda_B \sum_{f=1}^F \sum_{\ell=1}^D B_{f\ell}^2 + \lambda_\Gamma \sum_{f=1}^F \sum_{j=1}^F \Gamma_{fj}^2. \end{aligned} \quad (5.24)$$

The second contribution first computes an inventory residual by adding component changes with their invariant weights. It then squares that residual. The third contribution compares the fitted component state, after coupling, with the driver generated by

the inputs. The last two sums penalize coefficient magnitude rather than prediction residuals.

The dimensions of the matrices show how these sums are assembled. The entry in row i , column k of $(\hat{C} - C_{in})A^\top$ is the inventory residual in the second sum. The entry in row i , column f of $\hat{C}R^\top$ is $\hat{c}_{if} - \sum_j \Gamma_{fj}\hat{c}_{ij}$. The transpose appears because observations are stored as rows, while the single-observation equation $Rc = r$ uses column vectors.

For N training observations, $\Phi \in \mathbb{R}^{N \times D}$ contains feature vectors as rows. The matrices C_{in} and C_{out} contain influent and reference effluent states in the same row convention. An auxiliary fitted-state matrix $\hat{C} \in \mathbb{R}_+^{N \times F}$ separates the predicted training states from the regression coefficients. The training criterion is

$$\begin{aligned} J(B, \Gamma, \hat{C}) = & \|C_{out} - \hat{C}\|_F^2 + \lambda_{inv} \|(\hat{C} - C_{in})A^\top\|_F^2 \\ & + \lambda_{sys} \|\hat{C}(I_F - \Gamma)^\top - \Phi B^\top\|_F^2 \\ & + \lambda_B \|B\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2. \end{aligned} \quad (5.25)$$

The estimation problem minimizes this criterion subject to non-negative fitted training states, zero coupling diagonal, coupling box bounds, and the conditioning safeguard. All penalty weights are non-negative. The adopted ridge weights and coupled-system weight are strictly positive.

The first term measures component prediction error. The second term discourages disagreement with the modeled invariants. The third term encourages agreement between the fitted states and the coupled regression equation. The final two terms limit coefficient magnitude. The non-negative fitted training state balances prediction error, invariant mismatch, and coupled-system mismatch through these weighted terms.

An illustrative one-observation calculation makes the five contributions concrete. Take $c_{out} = (3, 1)^\top$, $c_{in} = (2, 2)^\top$, $\hat{c} = (2.5, 1.2)^\top$, and $A = [1 \ 1]$. Use the hypothetical coupling matrix in Equation (5.19). For a single baseline feature, take $B = (2, 1)^\top$, so the driver is $(2, 1)^\top$. The five unweighted contributions are

$$\begin{aligned} \text{prediction error} &= (3 - 2.5)^2 + (1 - 1.2)^2 = 0.29, \\ \text{inventory error} &= (2.5 + 1.2 - 2 - 2)^2 = 0.09, \\ \text{system error} &= (2.5 - 0.2(1.2) - 2)^2 \\ &\quad + (1.2 - 0.2(2.5) - 1)^2 = 0.1576, \\ \text{driver squared sum} &= 2^2 + 1^2 = 5, \\ \text{coupling squared sum} &= 0.2^2 + 0.2^2 = 0.08. \end{aligned} \quad (5.26)$$

With illustrative weights $\lambda_{inv} = 2$, $\lambda_{sys} = 3$, $\lambda_B = 0.1$, and $\lambda_\Gamma = 0.5$, the objective is $0.29 + 2(0.09) + 3(0.1576) + 0.1(5) + 0.5(0.08) = 1.4828$. These hypothetical weights illustrate the relative cost assigned to each discrepancy. Table 5.1 gives the adopted numerical fitting weights.

The auxiliary state is a variable in the fitting problem. The raw prediction for a new input is obtained by solving the learned coupled relation. Residual penalties guide this fit using the general penalty principle of physics-informed training developed by Raissi et al. (2019). Checked deployment then enforces mass conservation and non-negativity on the returned state.

The objective is expressed in physical component coordinates. Components have different units and numerical scales, so the magnitudes of their residuals and coefficients reflect those conventions. The four penalty weights also depend on the fixed feature and invariant scaling. Consistent rescaling transforms the features, coefficients, invariant operator, reporting map, and penalty weights together. Pinheiro et al. (2025) discuss the influence of feature scaling on learning. For physical interpretation, unit consistency is as necessary as numerical conditioning.

Deterministic block updates

The term $\hat{C}\Gamma^\top$ makes the joint objective nonconvex. Holding two variable blocks fixed gives a simpler problem in the remaining block. The fitting procedure updates the driver, coupling matrix, and fitted states in that order within each cycle.

This distinction can be checked with the reduced residual $(1 - \gamma z)^2$, where γ is a coupling and z a fitted component. If z is fixed, this is a quadratic function of γ . If γ is fixed, it is a quadratic function of z . It is not jointly convex. At $(\gamma, z) = (0.2, 5)$ and $(0.1, 10)$, the residual is zero. At the midpoint $(0.15, 7.5)$, it is $(1 - 1.125)^2 = 1/64$. A convex function could not exceed the average of the two zero endpoint values at their midpoint. The product between a fitted state and a coupling coefficient accounts for the nonconvex joint objective.

For fixed Γ and $\hat{C}$, the driver update is a ridge-regularized linear least-squares problem. Define $T = \hat{C}R^\top$. For one output row, write its coefficient column as β and its target column from T as t . The part of the objective that depends on this row is

$$\lambda_{sys} \sum_{i=1}^N \left(t_i - \sum_{\ell=1}^D \phi_{i\ell} \beta_\ell \right)^2 + \lambda_B \sum_{\ell=1}^D \beta_\ell^2. \quad (5.27)$$

Differentiate with respect to one coefficient β_h . The derivative of the squared residual contributes its residual times the feature ϕ_{ih} , while the ridge term contributes $2\lambda_B\beta_h$. Setting the derivative to zero gives

$$\lambda_{sys} \sum_{i=1}^N \phi_{ih} \left(\sum_{\ell=1}^D \phi_{i\ell} \beta_\ell - t_i \right) + \lambda_B \beta_h = 0. \quad (5.28)$$

Writing this equation for every coefficient gives

$$\left(\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D \right) \beta = \lambda_{sys} \Phi^\top t. \quad (5.29)$$

The matrix $\Phi^\top \Phi$ contains sums of products of feature columns. The vector $\Phi^\top t$ contains sums of feature times target. Adding the positive ridge weight to the diagonal prevents a zero-curvature direction when feature columns are duplicated.

For a hypothetical calculation with one coefficient and no intercept, let $\Phi = (1, 2)^\top$, $t = (2, 3)^\top$, and both weights equal one. The objective becomes

$$(2 - \beta)^2 + (3 - 2\beta)^2 + \beta^2 = 13 - 16\beta + 6\beta^2. \quad (5.30)$$

Its derivative is $-16 + 12\beta$, so the minimizer is $\beta = 4/3$. The matrix calculation gives $\Phi^\top \Phi = 1^2 + 2^2 = 5$, $\Phi^\top t = 1(2) + 2(3) = 8$, and $(5 + 1)\beta = 8$. Both calculations agree. Without the penalty, the coefficient would be $8/5$. The reduction illustrates ridge shrinkage for one coefficient. The adopted driver still includes its baseline and all stated features.

Assembling the normal equations for all output rows gives the exact solution

$$B^\top = \left(\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D \right)^{-1} \lambda_{sys} \Phi^\top \hat{C} R^\top. \quad (5.31)$$

The positive driver ridge weight makes the matrix in this expression positive definite, even when ordered quadratic columns are duplicated. In computation, the corresponding linear system is solved. An explicit matrix inverse is unnecessary. The square quadratic slices are then symmetrized without changing their predictions.

For fixed B and $\hat{C}$, define $E = \hat{C} - \Phi B^\top$.

The coupling calculation can be read one output row at a time. For row f , the residual at observation i is $E_{if} - \sum_{j \neq f} \hat{c}_{ij} \Gamma_{fj}$. Its minimization uses the other fitted component columns as predictors of E_{if} , with a ridge penalty and box bounds. In a hypothetical row with one permitted coupling γ , two predictor values 1 and 2, and residual targets 0.4 and 0.8, equal unit weights give

$$(0.4 - \gamma)^2 + (0.8 - 2\gamma)^2 + \gamma^2 = 0.8 - 4\gamma + 6\gamma^2. \quad (5.32)$$

The derivative $-4 + 12\gamma$ vanishes at $\gamma = 1/3$. This value lies below the adopted absolute entry bound 0.368. If the illustrative box bound were instead 0.25, the convex scalar problem would attain its minimum at 0.25. The conditioning test is applied to the complete coupling matrix after these row-wise calculations.

Collecting these scalar residuals across all output rows gives the coupling proposal

$$\min_{\Gamma} \quad \lambda_{sys} \|E - \hat{C} \Gamma^\top\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2 \quad (5.33)$$

subject to its zero diagonal and box bounds. This is a convex quadratic program. Its candidate is subsequently screened using Equation (5.22). A candidate that fails the condition check is shrunk toward zero or replaced by an admissible fallback. The screened proposal is evaluated against the previous admissible coupling matrix using the complete objective.

For fixed B and Γ , the fitted-state update separates over the training rows. For sample i , write $\hat{c}_i$ as a column vector.

Expand the three residual sums for one observation before collecting their coefficients. The data term contributes $\hat{c}^\top \hat{c} - 2c_{out}^\top \hat{c}$. The invariant term contributes $\lambda_{inv} \hat{c}^\top A^\top A \hat{c} -$

$2\lambda_{inv}c_{in}^\top A^\top A\hat{c}$. The system term contributes $\lambda_{sys}\hat{c}^\top R^\top R\hat{c} - 2\lambda_{sys}(B\phi)^\top R\hat{c}$. Terms independent of $\hat{c}$ do not affect its minimizer. Collecting its quadratic terms gives H_C , and collecting its linear terms gives h_i .

Write the expanded quadratic as $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$. The entry $(H_C)_{fg}$ multiplies the ordered product $\hat{c}_{if}\hat{c}_{ig}$, and $(h_i)_f$ supplies its linear term. For distinct components, the two mirrored entries contribute together, following the quadratic symmetry convention. Let δ_{fg} equal one when $f = g$ and zero otherwise. Reading the coefficients entry by entry gives

$$\begin{aligned} (H_C)_{fg} &= \delta_{fg} + \lambda_{inv} \sum_{k=1}^K A_{kf} A_{kg} + \lambda_{sys} \sum_{j=1}^F R_{jf} R_{jg}, \\ (h_i)_f &= c_{out,if} + \lambda_{inv} \sum_{k=1}^K A_{kf} \sum_{g=1}^F A_{kg} c_{in,ig} \\ &\quad + \lambda_{sys} \sum_{j=1}^F R_{jf} \sum_{\ell=1}^D B_{j\ell} \phi_{i\ell}. \end{aligned} \tag{5.34}$$

For example, the invariant contribution to $(H_C)_{12}$ adds the product of the first and second component weights over the invariant rows. The system contribution adds the product of columns 1 and 2 of R . These products explain why correcting one fitted component can change the preferred value of another.

Completing the square explains what the non-negative solve is adjusting. With $c_0 = H_C^{-1}h_i$,

$$\hat{c}^\top H_C \hat{c} - 2h_i^\top \hat{c} = (\hat{c} - c_0)^\top H_C (\hat{c} - c_0) - h_i^\top H_C^{-1} h_i. \tag{5.35}$$

The unrestricted minimizer is c_0 . When it contains a negative entry, the constrained problem seeks the closest non-negative state in the quadratic geometry defined by H_C . Off-diagonal entries tilt that geometry, so ordinary coordinate clipping need not give the closest state.

For example, take hypothetical $H_C = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and $h_i = (-1, 2)^\top$. These coefficients can arise from unit data, invariant, and system weights with $A = \begin{bmatrix} 1 & 1 \end{bmatrix}$ and $R = I$. A signed driver can make the first entry of h_i negative. The unrestricted solution is $c_0 = (-5/8, 7/8)^\top$. Clipping gives $(0, 7/8)^\top$. Instead, set the constrained first component to zero and minimize the remaining expression $3c_2^2 - 4c_2$. Its derivative $6c_2 - 4$ gives $c_2 = 2/3$.

At $(0, 2/3)^\top$, the derivative in the first coordinate is $2(c_2 + 1) = 10/3$, which is positive. Increasing that zero component would increase the objective, so the boundary value is appropriate. The objective is $-4/3$ at this state and $-77/64$ at the clipped state. The constrained solution has the smaller objective. The negative objective values are possible because the constants removed during expansion are absent. The complete residual-based training criterion remains non-negative.

The general coefficient arrays derived above are assembled in the compact form

$$\begin{aligned} H_C &= I_F + \lambda_{inv} A^\top A + \lambda_{sys} R^\top R, \\ h_i &= c_{out,i} + \lambda_{inv} A^\top A c_{in,i} + \lambda_{sys} R^\top B \phi_i. \end{aligned} \quad (5.36)$$

Each row minimizes $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$ over $\hat{c}_i \geq \mathbf{0}$. The identity term makes H_C positive definite, so the row minimizer is unique. Clipping the unconstrained solution componentwise generally does not solve this problem because the off-diagonal terms in H_C connect component adjustments.

The quadratic-program method of Stellato et al. (2018) provides an algorithmic foundation for the constrained coupling and fitted-state blocks. These blocks use their quadratic structure directly. A generic stochastic first-order method could also optimize a related criterion, but it would require separate mechanisms to control non-negativity, coupling bounds, and conditioning. The block formulation keeps each requirement attached to an explicit subproblem.

Descent, stopping, and numerical safeguards

Exact minimization over a block cannot increase the objective when its admissible set contains the previous block value. Since $J \geq 0$, an objective sequence that is monotone non-increasing converges in value. The convergence analysis of Razaviyayn et al. (2013) relates blockwise minimization to stationary solutions under its stated conditions.

The conditioning screen requires particular care. Shrinking a coupling proposal can increase the training criterion relative to the previous admissible coupling matrix. The procedure therefore evaluates the objective after screening and retains the previous admissible matrix when a screened candidate cannot provide an acceptable non-increasing update. Solver tolerances also require an explicit allowance for numerical error. Monotone descent is assessed for the accepted objective-decreasing updates after screening.

Each restart begins with an admissible coupling matrix and non-negative fitted states. A ridge solve supplies the initial driver. Full cycles continue until an iteration limit is reached or the slope of the trailing objective history is sufficiently small. A running minimum records the best admissible state reached. When several restarts are used, the admissible state with the smallest recorded objective is retained. A nearly flat objective triggers the prescribed stopping rule.

Figure 5.1 separates the three fitting updates from the checks that make an update acceptable. The scalar calculations in Equations (5.30) and (5.32) illustrate the first and third computational boxes in the left column. The non-negative-state example explains the return path in the right column.

The cycle returns to the driver because each block changes the conditions faced by the others. The conditioning screen acts on the coupling proposal before the fitted-state

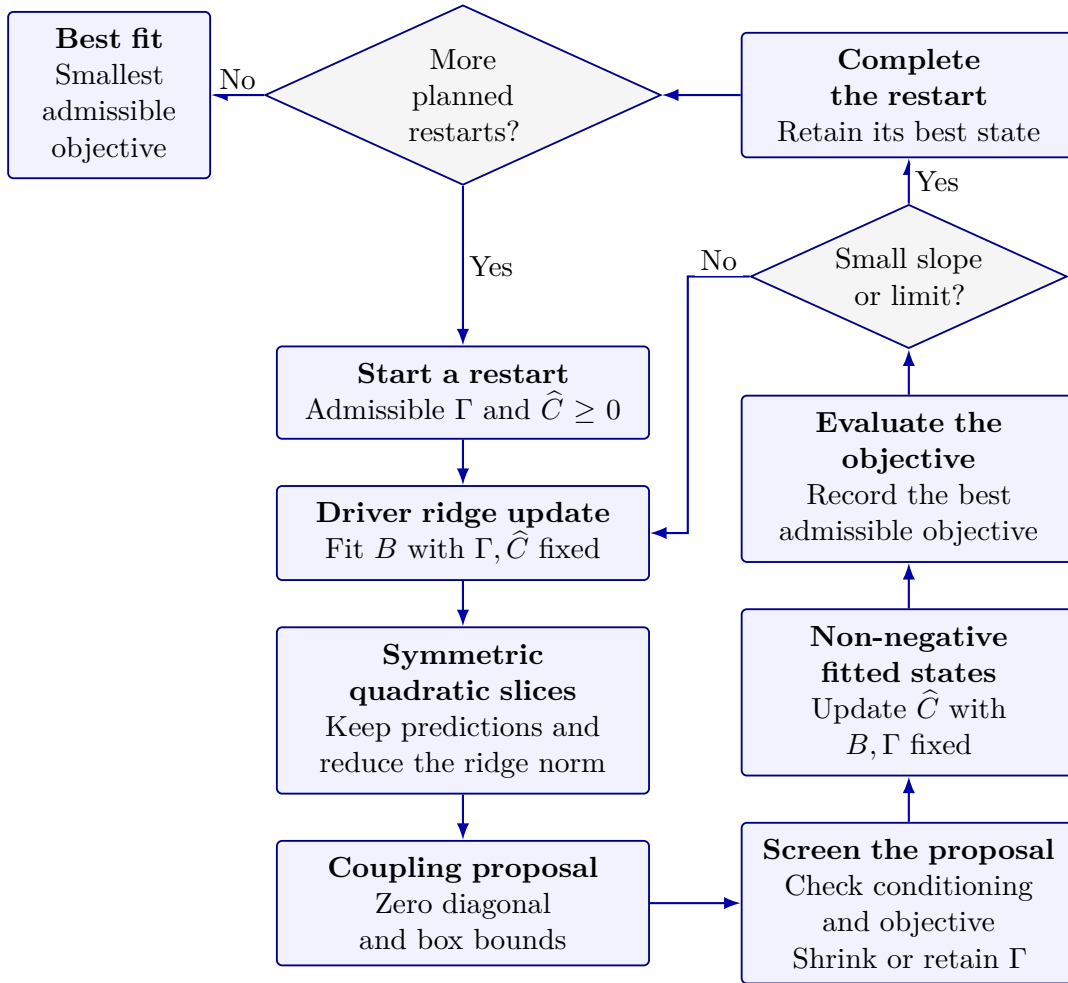


Figure 5.1: Conceptual fitting cycle with explicit symmetry, conditioning, descent, and non-negativity checks. The prescribed restart count can be one.

update. The full objective is then recorded after the cycle. The restart decision concerns the prescribed search procedure, while the final selection compares admissible objective values. These steps define the accepted sequence of updates for the nonconvex joint fit.

Table 5.1: Specified fitting settings for the structured regression.

Setting	Value
Baseline feature	Included
Invariant penalty λ_{inv}	8.68×10^{-3}
Coupled-system penalty λ_{sys}	1.27×10^{-3}
Driver ridge penalty λ_B	6.09×10^{-5}
Coupling ridge penalty λ_Γ	6.55×10^{-4}
Off-diagonal bound γ_{abs}	0.368
Maximum full fitting cycles	60
Restarts	1
Trailing objective window	12 cycles
Objective slope tolerance	2.85×10^{-7}

Table 5.1 specifies numerical fitting choices. The condition threshold κ_{max} remains an additional prescribed numerical safeguard and is checked independently of the coefficient bound. Hyperparameter assessment uses validation observations within the training pool. Held-out test observations do not determine penalty weights or stopping choices (Kaufman et al., 2012). Fitting settings remain fixed within a sample-size comparison so that changes in the learning curve reflect changing sample availability under a common model specification.

5.3.2 Checked deployment

The raw coupled state

For a new pair $(u_*, c_{in,*})$, the fitted parameters give

$$\hat{R} = I_F - \hat{\Gamma}, \quad d_* = \hat{B}\phi(u_*, c_{in,*}), \quad \hat{R}c_{raw} = d_*. \quad (5.37)$$

The notation $c_{raw} = \hat{R}^{-1}d_*$ describes the mapping mathematically. A linear-system solve evaluates it numerically. The accepted conditioning bound limits the sensitivity of that solve. The raw state is returned only when it satisfies both the invariant and non-negativity checks.

For a vector, the Euclidean norm is the square root of the sum of squared entries. Thus an illustrative invariant-residual vector $(3, 4)^\top$ has norm $\sqrt{3^2 + 4^2} = 5$. The absolute norm adds the absolute entries instead. For an illustrative concentration vector $(-0.3, 0.1, -0.4)^\top$, retaining only negative entries gives $(-0.3, 0, -0.4)^\top$, whose absolute

norm is 0.7. The first calculation measures the size of a mass conservation discrepancy. The second measures total negative concentration in the stated numerical coordinates.

For any candidate c , define the diagnostic residuals

$$v_{mc}(c) = \|A(c - c_{in,*})\|_2, \quad v_{nn}(c) = \|\min(c, \mathbf{0})\|_1. \quad (5.38)$$

The minimum is taken componentwise. The adopted diagnostic tolerance is 10^{-10} . Algebraic equality and non-negativity define the mathematical requirement. Residuals below the numerical tolerance define its computational assessment. Invariant residual magnitudes use the stated scaling of A .

Affine projection for mass conservation

The nearest state satisfying mass conservation can first be derived for the hypothetical two-component inventory. Write the raw prediction as $(p_1, p_2)^\top$ and the required total as T . The two unknown projected concentrations minimize $(c_1 - p_1)^2 + (c_2 - p_2)^2$ while satisfying $c_1 + c_2 = T$. Introduce a multiplier η for that equality and write

$$\mathcal{L} = \frac{1}{2}(c_1 - p_1)^2 + \frac{1}{2}(c_2 - p_2)^2 + \eta(c_1 + c_2 - T). \quad (5.39)$$

The factor one-half simplifies differentiation without changing the minimizer. Differentiating with respect to the two concentrations and the multiplier gives

$$c_1 - p_1 + \eta = 0, \quad c_2 - p_2 + \eta = 0, \quad c_1 + c_2 = T. \quad (5.40)$$

The first two equations give $c_1 = p_1 - \eta$ and $c_2 = p_2 - \eta$. Substituting them into the inventory equality gives $p_1 + p_2 - 2\eta = T$. Thus

$$\begin{aligned} \eta &= \frac{p_1 + p_2 - T}{2}, \\ c_1 &= p_1 - \frac{p_1 + p_2 - T}{2}, \\ c_2 &= p_2 - \frac{p_1 + p_2 - T}{2}. \end{aligned} \quad (5.41)$$

The unweighted distance shares the inventory discrepancy equally between these two coordinates. With $p_1 = 12$, $p_2 = 1$, and $T = 10$, the discrepancy is 3 and the multiplier is 1.5. The projected state is $(10.5, -0.5)^\top$. The equality is satisfied even though the second component is negative. This example identifies the inequality that is absent from an equality-only projection.

When the raw state fails its checks, the first projection solves

$$c_{aff} = \arg \min_{c \in \mathbb{R}^F} \|c - c_{raw}\|_2^2 \quad \text{subject to} \quad Ac = Ac_{in,*}. \quad (5.42)$$

Introduce a Lagrange multiplier $\eta \in \mathbb{R}^K$. Stationarity gives $c - c_{raw} + A^\top \eta = 0$. Substitution into the equality gives $AA^\top \eta = A(c_{raw} - c_{in,*})$. The full row rank of A therefore gives the unique solution

$$c_{aff} = c_{raw} - A^\top (AA^\top)^{-1} A(c_{raw} - c_{in,*}). \quad (5.43)$$

The fixed matrix $A^\top(AA^\top)^{-1}A$ can be prepared once. The affine projection is then a matrix-vector operation for each new prediction.

Each matrix step follows the same scalar derivation. The multiplier column contains one multiplier per independent inventory. The vector $A^\top\eta$ converts these inventory projections into component adjustments. Solving $AA^\top\eta = A(c_{raw} - c_{in,*})$ determines the adjustments needed to remove the inventory residual. For the two-component illustration,

$$AA^\top = 2, \quad A^\top(AA^\top)^{-1}A = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}. \quad (5.44)$$

Taking the illustrative influent as $(6, 4)^\top$, the raw-minus-influent difference is $(6, -3)^\top$. Multiplying it by this projection matrix gives $(1.5, 1.5)^\top$, which is subtracted from the raw state. This recovers the scalar calculation without guessing how much either component should change.

This projection is an orthogonal projection onto the affine set defined by mass conservation in the chosen numerical component coordinates. It removes the part of the prediction error normal to that set and retains the part tangent to it. The projection distance uses the stated component unit convention. Sturm and Silva (2025) develop related projections for mass conservation and explain why species-dependent weights matter when concentration scales differ.

The affine state is checked for both mass conservation residuals and non-negativity before deployment. The equality follows analytically from Equation (5.43), but checking its residual catches numerical or operator inconsistency. An affine state that satisfies mass conservation can still contain a negative concentration. That possibility requires a projection with inequality constraints.

Joint mass conservation and non-negativity

The absolute-distance projection can also be written before introducing compact vector inequalities. For the state that satisfies mass conservation but violates non-negativity $(10.5, -0.5)^\top$, require $c_1 + c_2 = 10$ and $c_1, c_2 \geq 0$. Introduce non-negative adjustment bounds ρ_1, ρ_2 . The constraints

$$\begin{aligned} \rho_1 &\geq c_1 - 10.5, & \rho_1 &\geq 10.5 - c_1, \\ \rho_2 &\geq c_2 + 0.5, & \rho_2 &\geq -0.5 - c_2 \end{aligned} \quad (5.45)$$

make ρ_1 at least the absolute first adjustment and ρ_2 at least the absolute second adjustment. Minimizing $w_1\rho_1 + w_2\rho_2$ with positive weights makes each bound equal to the corresponding absolute adjustment. No absolute-value function remains in the optimization constraints.

Set $c_2 = t$ to impose the inventory equality, so $c_1 = 10 - t$ and $0 \leq t \leq 10$. Both absolute adjustments then equal $t + 0.5$. The objective is $(w_1 + w_2)(t + 0.5)$, which increases with t . Its minimum is at $t = 0$, giving $(10, 0)^\top$. For equal unit weights, its

cost is 1. At $t = 1$, the state would be $(9, 1)^\top$, with cost 3. This direct comparison explains the optimal projection in the reduced problem.

When the affine state fails the non-negativity check, the final projection solves

$$\begin{aligned} \hat{c} = \arg \min_{c, \rho \in \mathbb{R}^F} \quad & w_c^\top \rho \\ \text{subject to} \quad & Ac = Ac_{in,*}, \\ & c \geq \mathbf{0}, \\ & -\rho \leq c - c_{aff} \leq \rho, \\ & \rho \geq \mathbf{0}. \end{aligned} \tag{5.46}$$

For strictly positive fixed weights w_c , minimization makes $\rho_f = |c_f - c_{aff,f}|$ at an optimum. The objective is therefore the weighted absolute adjustment $\sum_f w_{c,f} |c_f - c_{aff,f}|$. The program minimizes adjustment from the affine state while retaining the fitted regression coefficients.

The program is linear because the absolute values are represented by auxiliary variables and linear inequalities. The revised-simplex method described by Huangfu and Hall (2018) provides an algorithmic foundation for solving this type of projection. The mathematical feasible set is nonempty whenever the supplied influent state is non-negative under the adopted boundary relation. The positive weights make the adjustment objective bounded below and discourage unconstrained movement of any component. A minimizer exists, although several minimizing states can exist when the absolute-distance objective has a flat face.

Large weights make changes to a component more costly. Small weights permit larger changes when those changes help satisfy the specified constraints. Weights are fixed before deployment using the intended component scaling and scientific priorities. A sensitivity analysis of the weights can examine how much the returned state depends on that modeling choice. Equal numerical weights also embody a choice because component units and typical magnitudes differ.

Three hypothetical components show how weights can change an admissible allocation. Let the inventory row be $[1 \ 1 \ 1]$ and the affine reference satisfying mass conservation be $(-1, 4, 7)^\top$, whose total is 10. Restoring its first component to zero requires removal of one unit from another component. Two feasible candidates are $(0, 3, 7)^\top$ and $(0, 4, 6)^\top$. For weights $(1, 10, 1)^\top$, their costs are respectively $1(1) + 10(1) = 11$ and $1(1) + 1(1) = 2$. The second component is more costly to change, so removing material from the third is preferable. With weights $(1, 1, 10)^\top$, the two costs become 2 and 11. The preferred allocation reverses. Both allocations preserve the same inventory, while the weights determine the preferred adjustment.

Kircher and Votsmeier (2025) show why a projection for mass conservation needs simultaneous non-negativity enforcement in reaction-kinetics surrogates. Their scaled-space treatment protects low-concentration species during projection. Equation (5.46)

addresses the same joint requirement through a weighted absolute-distance problem for activated sludge components. It imposes the adopted linear invariants and non-negativity on the returned state.

Illustrative projection and the failure of clipping

Consider an illustrative two-component system with $A = [1 \ 1]$, a total influent inventory of 10, and a raw prediction $c_{raw} = (12, 1)^\top$. The raw total is 13. The affine formula subtracts 1.5 from each component and gives $c_{aff} = (10.5, -0.5)^\top$. The total is now 10, but the second component is negative.

Clipping that negative value to zero produces $(10.5, 0)^\top$, which violates the total inventory. With equal projection weights, Equation (5.46) instead returns $(10, 0)^\top$. Both constraints hold. These hypothetical values illustrate the projection geometry.

The deployment order uses this geometry directly. A feasible raw state needs no adjustment. An infeasible raw state is first projected onto the mass conservation set. A feasible affine state can then be returned. The linear program is required only when the affine state still fails non-negativity. Every returned candidate is checked under the declared numerical tolerances. An unsuccessful numerical solve is recorded as a prediction failure.

Figure 5.2 makes the branch conditions explicit. Both physical checks appear at the raw and affine stages. The last projection branch also requires successful optimization and an independent admissibility check.

The hypothetical state $(12, 1)^\top$ follows the two No branches because its total is incorrect and its affine projection contains a negative component. The weighted projection supplies $(10, 0)^\top$, which reaches the return branch only after the stated checks pass. A state that satisfies both conditions at an earlier branch reaches the same reporting step. The different paths therefore change the projection effort while retaining the same acceptance requirements.

The component state returned by this rule is the deployed output. Its reported quantities are

$$\hat{y}_{\text{comp}} = I_{\text{comp}} \hat{c}. \quad (5.47)$$

The deployment constraints enforce admissibility, and the stage-wise assessment measures how often each projection is needed.

5.3.3 Interpreting the complete prediction map

Driver coefficients and back projection

The transformation from driver coefficients to output coefficients can be computed with small matrices. Use the hypothetical coupling matrix in Equation (5.19), a reduced

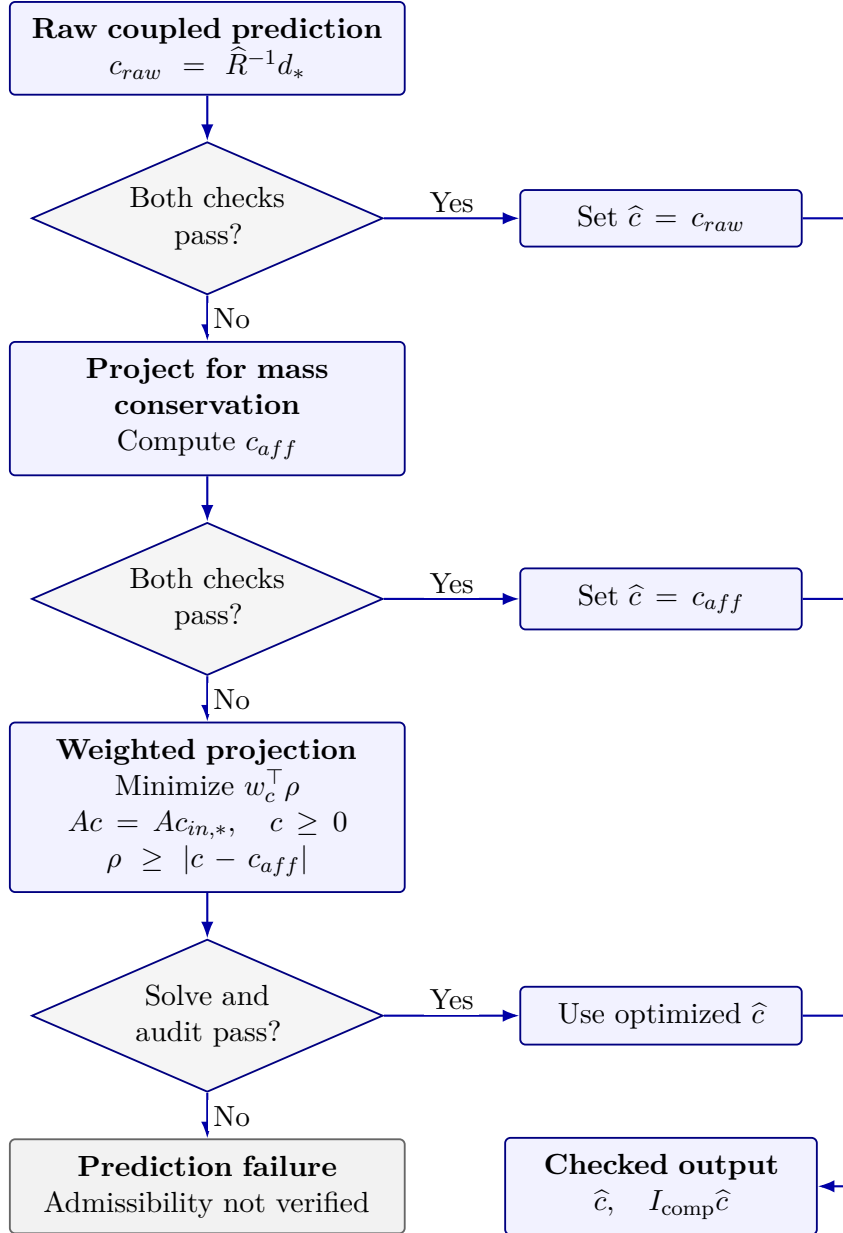


Figure 5.2: Checked deployment of one predicted component state. Accepted branches merge before the common reporting map is applied. A failed final audit produces a reported prediction failure.

feature vector $(1, a)^\top$, and driver coefficients

$$B = \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix}, \quad r = \begin{bmatrix} 4 + a \\ 2 \end{bmatrix}. \quad (5.48)$$

The first coefficient column describes the baseline, and the second describes dependence on the hypothetical scalar input a . Multiplication through the coupling inverse gives

$$\begin{aligned} R^{-1}B &= \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 2 & 0 \end{bmatrix} \\ &= \frac{1}{0.96} \begin{bmatrix} 4.4 & 1 \\ 2.8 & 0.2 \end{bmatrix} = \begin{bmatrix} 55/12 & 25/24 \\ 35/12 & 5/24 \end{bmatrix}. \end{aligned} \quad (5.49)$$

The second component now depends on a , even though its driver coefficient for a was zero. This dependence is introduced by the simultaneous coupling solve.

For a hypothetical reported quantity $y = c_1 + 2c_2$, its reporting row is $[1 \ 2]$. Its transformed coefficient row is

$$[1 \ 2] R^{-1}B = [125/12 \ 35/24]. \quad (5.50)$$

At $a = 2$, the driver is $(6, 2)^\top$, the raw component state is $(20/3, 10/3)^\top$, and the reported quantity is $40/3$. The transformed row gives the same value through $125/12 + (35/24)2 = 40/3$. The hypothetical reporting row provides a reduced arithmetic example.

The rows of $\hat{B}$ refer to component drivers. Coupling and the reporting map convert these rows into COD, TN, TP, and TSS coefficients. The raw component coefficient matrix is

$$B_{\text{raw}} = \hat{R}^{-1}\hat{B}, \quad c_{\text{raw}} = B_{\text{raw}}\phi. \quad (5.51)$$

The corresponding map in reported-composite space is

$$B_{\text{raw,comp}} = I_{\text{comp}}\hat{R}^{-1}\hat{B}, \quad y_{\text{raw,comp}} = B_{\text{raw,comp}}\phi. \quad (5.52)$$

This transformation carries the driver coefficients through the learned coupling and then through the reporting operator. Inspection of a COD quadratic block uses its row in $B_{\text{raw,comp}}$, rather than an arbitrary row of $\hat{B}$.

Figure 5.3 shows the complete raw prediction chain using the named feature groups. It keeps the component driver, simultaneous coupling solve, and reporting operation visible as separate mathematical steps.

In the two-feature example, the second driver has no direct coefficient for the scalar input, but the coupling path gives its raw component a nonzero input coefficient. The reporting path combines both component coefficients. Reading the paths in this order explains the back projection in Equations (5.49) and (5.50). Checked deployment acts on the raw component state before the final reported quantities are calculated.

The named blocks can be displayed as coefficient maps. Operating terms are grouped separately from influent terms. Symmetric quadratic maps display both ordered cells

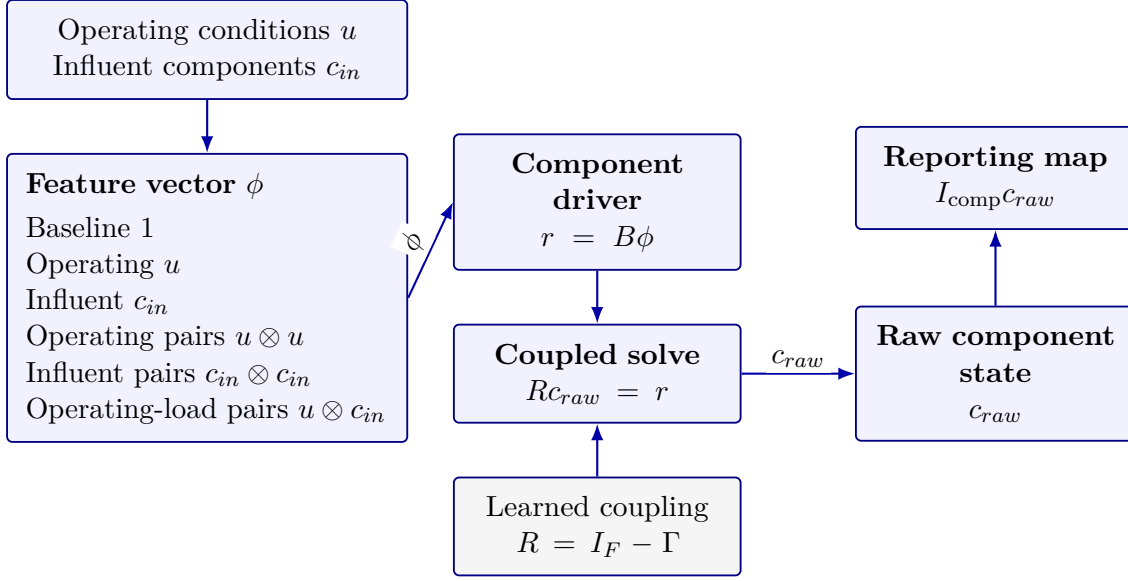


Figure 5.3: Named feature blocks and the transformation from driver to raw component and reported-composite predictions. Composing the three operations gives the raw reported coefficient map $I_{\text{comp}}R^{-1}B$.

while reporting unique cross terms consistently. Operating-load maps show which operating variables interact with which influent components. The coupling matrix is displayed separately because it has a different interpretation from the input-driver terms.

A coefficient-display threshold is used only to simplify reporting after fitting. For each reported-composite raw-regression map, the threshold is 10^{-4} times its largest absolute coefficient. For the coupling matrix, the threshold is 10^{-4} times its largest absolute off-diagonal entry. The fitted mapping is retained in full, and the threshold determines which entries appear in the coefficient display. Symmetric quadratic summaries count unique upper-triangular terms. A small coefficient can still have a large contribution when multiplied by a large feature. Interpretation therefore examines coefficient magnitude, input scale, and the contribution of the corresponding term over the declared domain (Nazlabadi et al., 2021; Selerio, 2026).

Analytical jacobians of the raw regression

Differentiation of a scalar polynomial shows why a coefficient and a marginal response differ. For the illustrative driver in Equation (5.10),

$$\begin{aligned}
 \frac{\partial r_1}{\partial u_1} &= 2 + 0.2u_1 + 0.2u_2, \\
 \frac{\partial r_1}{\partial u_2} &= -1 + 0.2u_1 + 0.25c_1, \\
 \frac{\partial r_1}{\partial c_1} &= 0.5 + 0.25u_2, & \frac{\partial r_1}{\partial c_2} &= -0.1c_2.
 \end{aligned} \tag{5.53}$$

At $u = (2, 3)^\top$ and $c = (4, 5)^\top$, these derivatives are 3, 0.4, 1.25, and -0.5 . In particular, the first derivative is 3 although the linear coefficient of u_1 is 2. The operating quadratic term and interaction with u_2 supply the remaining contribution.

The derivative of the second illustrative driver $r_2 = -u_1 + u_2 + 0.2c_2$ with respect to u_1 is -1 . Differentiate the coupled equations with the coupling held fixed. Their derivative equations are $\partial c_1/\partial u_1 - 0.2\partial c_2/\partial u_1 = 3$ and $-0.2\partial c_1/\partial u_1 + \partial c_2/\partial u_1 = -1$. Thus the same inverse used for prediction gives

$$\begin{bmatrix} \partial c_1/\partial u_1 \\ \partial c_2/\partial u_1 \end{bmatrix} = \frac{1}{0.96} \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 35/12 \\ -5/12 \end{bmatrix}. \quad (5.54)$$

For the hypothetical reporting row $[1 \ 2]$, the reported derivative is $35/12 + 2(-5/12) = 25/12$. Differentiating the driver, solving through the coupling, and applying the reporting weights are three separate operations.

Putting several derivatives in one matrix gives a Jacobian. Its rows identify the output being differentiated, and its columns identify the input being varied. At the illustrative inputs, the operating-driver Jacobian and its coupled transformation are

$$J_{r,u} = \begin{bmatrix} 3 & 0.4 \\ -1 & 1 \end{bmatrix}, \quad R^{-1}J_{r,u} = \begin{bmatrix} 35/12 & 5/8 \\ -5/12 & 9/8 \end{bmatrix}. \quad (5.55)$$

The first column is the derivative calculation above. The second follows by solving the same coupled system for the derivative column $(0.4, 1)^\top$. A Jacobian therefore collects several scalar sensitivity calculations rather than introducing a different kind of response.

A coefficient describes a term in the polynomial. A Jacobian describes the change in the whole mapping at an operating point. For component f , let $H_{uc}^{(f)} \in \mathbb{R}^{M_{op} \times F}$ reshape its interaction coefficients so that the cross term is $u^\top H_{uc}^{(f)} c_{in}$. The driver can then be expressed as

$$\begin{aligned} r_f = & b_f + (W_u)_{f,:} u + (W_{in})_{f,:} c_{in} + u^\top Q_{uu}^{(f)} u \\ & + c_{in}^\top Q_{cc}^{(f)} c_{in} + u^\top H_{uc}^{(f)} c_{in}. \end{aligned} \quad (5.56)$$

With symmetric square slices, its gradients are

$$\begin{aligned} \nabla_u r_f &= (W_u)_{f,:}^\top + 2Q_{uu}^{(f)} u + H_{uc}^{(f)} c_{in}, \\ \nabla_{c_{in}} r_f &= (W_{in})_{f,:}^\top + 2Q_{cc}^{(f)} c_{in} + (H_{uc}^{(f)})^\top u. \end{aligned} \quad (5.57)$$

The quadratic gradients in Equation (5.57) follow by differentiating each scalar product. For $x^\top Q x = \sum_i \sum_j q_{ij} x_i x_j$, differentiation with respect to x_h collects the terms with $i = h$ and those with $j = h$. Their sum is $\sum_j (q_{hj} + q_{jh}) x_j$. It becomes the h th entry of $2Qx$ when Q is symmetric. For $u^\top H c = \sum_i \sum_j h_{ij} u_i c_j$, differentiation with respect to u_i gives $\sum_j h_{ij} c_j$, while differentiation with respect to c_j gives $\sum_i h_{ij} u_i$. These are the entries of Hc and $H^\top u$.

Stacking the transposed gradients gives $J_{r,u} \in \mathbb{R}^{F \times M_{op}}$ and $J_{r,c} \in \mathbb{R}^{F \times F}$. The raw-state and raw-composite Jacobians follow through the same back projection as the coefficients,

$$\begin{aligned} J_{raw,u} &= \hat{R}^{-1} J_{r,u}, & J_{raw,c} &= \hat{R}^{-1} J_{r,c}, \\ J_{raw,comp,u} &= I_{comp} J_{raw,u}, & J_{raw,comp,c} &= I_{comp} J_{raw,c}. \end{aligned} \quad (5.58)$$

These derivatives describe the complete fitted raw regression at the specified inputs, including curvature and interactions. The projection-stage derivatives are calculated below.

An illustrative aeration sensitivity can be inspected at low and high ammonium loading while holding the remaining inputs fixed. A difference in the two derivatives identifies a loading-dependent association within the fitted model. Oxygen-transfer limitations and heterotrophic competition provide process context for assessing such a pattern, as discussed by Stenstrom and Song (1991). Comparison with that process context provides a plausibility check on the fitted response pattern.

Sensitivity after the affine projection

Return to the two-feature example in Equation (5.49). Its raw derivative with respect to a is $(25/24, 5/24)^\top$, whose total is $5/4$. This raw response changes the total component inventory. At fixed influent inventory, an affine projection must remove that total change. For the two-component invariant,

$$\begin{aligned} P_A &= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \\ \frac{\partial c_{aff}}{\partial a} &= P_A \begin{bmatrix} 25/24 \\ 5/24 \end{bmatrix} = \begin{bmatrix} 5/12 \\ -5/12 \end{bmatrix}. \end{aligned} \quad (5.59)$$

The two entries now sum to zero. For the reporting row $[1 \ 2]$, the derivative is $5/12 + 2(-5/12) = -5/12$. The corresponding raw derivative was $35/24$. The projection changes both its magnitude and sign because it enforces a different material-accounting condition on the complete returned state.

The influent reference also changes when an influent component is perturbed. In this reduced example, the driver contains no influent term. Increasing either influent component by δ increases the required total by δ . The unweighted affine projection then adds $\delta/2$ to each predicted component. Its influent derivative is the matrix $\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, even though the raw influent derivative is zero. This contribution comes from the right-hand side of the mass conservation equality.

Define the complementary projection matrices

$$Q_A = A^\top (A A^\top)^{-1} A, \quad P_A = I_F - Q_A. \quad (5.60)$$

Then $c_{aff} = P_A c_{raw} + Q_A c_{in}$. With the fitted coefficients and invariant operator held fixed, its Jacobians are

$$J_{aff,u} = P_A J_{raw,u}, \quad J_{aff,c} = P_A J_{raw,c} + Q_A. \quad (5.61)$$

The second expression includes the change in the mass conservation reference when the influent changes. The term Q_A accounts for the changing influent reference.

Multiplication by A gives $AJ_{aff,u} = 0$ and $AJ_{aff,c} = A$. An operating perturbation changes the affine prediction only along directions that preserve its fixed influent inventories. An influent perturbation changes those inventories consistently with the influent reference. These relations provide analytical checks on the sensitivity calculation.

When the affine branch is used over a neighborhood of inputs, its coefficients and Jacobians describe that branch. The complete deployment rule includes transitions between raw, affine, and linear-program branches, each with its corresponding sensitivity (Selerio, 2026).

Active constraints in the final projection

An explicit boundary in the same hypothetical model shows the change in sensitivity caused by non-negativity. Keep the total influent inventory equal to 10. The affine components obtained from Equation (5.49) are

$$c_{aff,1} = \frac{35}{6} + \frac{5a}{12}, \quad c_{aff,2} = \frac{25}{6} - \frac{5a}{12}. \quad (5.62)$$

Their sum is 10. Near $a = 10$, the first component is positive, while the second reaches zero at $a = 10$. Immediately below this boundary the affine state is feasible and its derivative is $(5/12, -5/12)^\top$. Above the boundary the affine second component is negative. The positive-weight absolute projection returns $(10, 0)^\top$, so its derivative with respect to a is $(0, 0)^\top$. The final projected state therefore has a different derivative on each side of the boundary. The boundary is characterized by its two one-sided derivatives.

In the projected branch, the binding equalities are $c_1 + c_2 = 10$ and $c_2 = 0$. Differentiating them gives $\partial c_1 / \partial a + \partial c_2 / \partial a = 0$ and $\partial c_2 / \partial a = 0$. These two equations directly give the zero derivative. The general active-set calculation uses the same idea with all binding inventory and adjustment constraints.

The linear program can make a component concentration equal to zero or bind one of the absolute-adjustment inequalities. These binding constraints define an active set. If the optimal solution is unique and its nondegenerate optimal basis remains unchanged over a local neighborhood, the solution varies affinely with the relevant right-hand sides in that neighborhood. A branch-specific sensitivity can then be derived from that fixed basis.

At an active-set transition, the derivative can change abruptly. When several optimal projected states exist, the returned solution can also depend on the rule used to select

among them. Local deployment sensitivity is assessed by differentiating a verified fixed basis where appropriate or by perturbing admissible inputs and resolving the entire deployment problem. The perturbation size and any branch changes are recorded as part of that sensitivity assessment.

This distinction prevents three different quantities from being conflated. A driver coefficient belongs to $\hat{B}$. A raw-state sensitivity passes through $\hat{R}^{-1}$. A deployed-state sensitivity also depends on the mass conservation reference and active projection constraints. Each quantity answers a different question about the model.

Following sensitivity through projection

Interpretation follows the complete prediction map. Coupling transforms the driver into component predictions. The reporting map combines components, and active output constraints change their local derivatives. Physical-unit interpretation identifies a driver coefficient, a raw prediction derivative, or a derivative of the final projected response (Brunton et al., 2016; Shen et al., 2023).

A second hypothetical calculation isolates the effect of projection using different raw equations. Let a dimensionless operating input be a , and let two hypothetical raw concentration equations be

$$c_{raw,1}(a) = 4 + a, \quad c_{raw,2}(a) = 3 + 2a.$$

The raw first-component slope is one and the second-component slope is two. At $a = 2$, the concentrations are 6 and 7, with a total of 13. Suppose the required total is ten and the projection minimizes the sum of squared adjustments with equal weights. If the first adjustment is d_1 and the second is d_2 , mass conservation requires $d_1 + d_2 = -3$. Substitute $d_2 = -3 - d_1$ into the squared-adjustment objective and differentiate,

$$\begin{aligned} d_1^2 + d_2^2 &= d_1^2 + (-3 - d_1)^2, \\ \frac{d}{dd_1} [d_1^2 + (-3 - d_1)^2] &= 4d_1 + 6 = 0. \end{aligned}$$

Both adjustments are -1.5 , giving a state satisfying mass conservation with concentrations 4.5 and 5.5. For a general a , the excess total is $(4 + a) + (3 + 2a) - 10 = 3a - 3$. Equal-weight projection subtracts one half of that excess from each coordinate,

$$\begin{aligned} c_{aff,1}(a) &= 4 + a - \frac{3a - 3}{2} = 5.5 - 0.5a, \\ c_{aff,2}(a) &= 3 + 2a - \frac{3a - 3}{2} = 4.5 + 0.5a. \end{aligned}$$

The projected slopes are -0.5 and 0.5 . The first changes sign even though its raw slope is positive. The sum of the projected slopes is zero, consistent with a total that does not depend on a .

Non-negativity introduces another stage. The affine first component reaches zero at $a = 11$. For $a > 11$, the affine state lies outside the segment satisfying mass conservation

and non-negativity. In this equal-weight two-component example, the nearest jointly feasible state is then $(0, 10)^\top$. Both local output slopes are zero while that boundary remains active. The slope changes at the transition point. This calculation illustrates why a raw coefficient, an affine sensitivity, and a final constrained sensitivity have distinct meanings.

Figure 5.4 plots the hypothetical first-component equations and the slopes calculated from them. The left panel shows the concentration returned at each stage. The right panel shows the corresponding local rate of change with respect to the operating input.

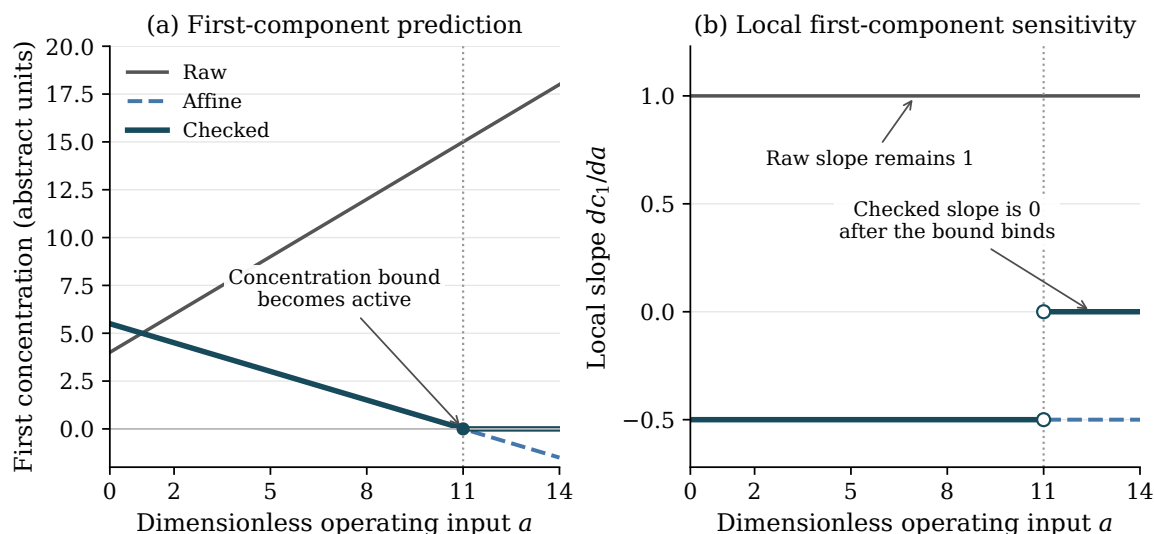


Figure 5.4: Hypothetical first-component response and sensitivity before and after mass conservation and non-negativity enforcement. Open circles mark the two one-sided checked slopes at the active-boundary transition.

Before the boundary becomes active, the affine and checked curves coincide. Their negative first-component slope differs from the positive raw slope because mass conservation redistributes the operating response between the two components. After the boundary becomes active, the checked concentration stays at zero and its local slope is zero. The transition is characterized by the two one-sided checked slopes.

5.3.4 Assessment design for the structured regression

Common inputs and comparator models

The structured-regression assessment uses a design collection of 10,000 accepted steady-state reactor cases. Every method receives the same 22 physical inputs and predicts the same 20 effluent components. Its reported composites are obtained from the common composition matrix. The input domain, reaction model, simulation bounds, and acceptance criterion remain those of the declared reactor benchmark. The comparison therefore changes the predictor while holding the modeled process and prediction task

fixed.

The nine unconstrained comparators are XGBoost, LightGBM, CatBoost, AdaBoost, random forest, support vector regression, k -nearest neighbors, partial least squares, and a fully connected multilayer perceptron. These methods represent boosted and randomized tree ensembles, kernel regression, local neighbor averaging, latent linear regression, and nonlinear neural regression. Their foundations include the ensemble construction of Breiman (2001), the boosting formulations of Drucker (1997), Chen and Guestrin (2016), Ke et al. (2017), and Prokhorenkova et al. (2018), and the latent regression of Wold et al. (2001). The comparators have no invariant penalty, auxiliary non-negative fitted-state matrix, or learned component-coupling equation during fitting. The latent components used by partial least squares are statistical input combinations. The shared 20-component prediction target remains the physical effluent state for every comparator.

The neural comparator has hidden layers with 128, 64, and 32 units and logistic activation. Dense connections map the 22 inputs through these layers to 20 component outputs. Its capacity is evaluated under the same samples and reported-composite map as the other predictors. A transparent polynomial and a neural network can therefore differ in their internal structure without changing the supervised target.

Training partitions and hyperparameter selection

The primary assessment uses a row-wise 80% training and 20% held-out split with random seed 42. The specified collection gives a training pool of 8000 rows and a held-out set of 2000 rows. Hyperparameter selection uses only the training pool. Half of that pool forms a 4000-row tuning subset. A further 80% and 20% split gives 3200 tuning-training rows and 800 tuning-validation rows. The tuning design uses one internal validation split.

Each predictor receives 100 hyperparameter trials under a seeded tree-structured Parzen estimator with seed 42. No trial pruning or wall-clock timeout is imposed. Bergstra et al. (2011) develop this search approach, while Akiba et al. (2019) provide the associated hyperparameter optimization framework. The selection criterion is mean squared error in the shared physical component space on the internal validation set. Selected settings are held fixed for the primary assessment and the repeated sample-size design. The final primary fit uses the full 8000-row training pool.

For the unconstrained comparators, input and output standardization parameters are estimated from the relevant training partition. Predictions are transformed back to physical component coordinates before any invariant check, reporting map, or error assessment. ICSOR remains in physical coordinates so that its driver, invariant operator, and deployed constraints use the same component convention. Internal-validation transformations are estimated from tuning-training rows only. The training partitions determine the means, standard deviations, and hyperparameters. Projection

weights are fixed before held-out assessment.

Table 5.2 gives the specified numerical configurations. These values define fitted model settings rather than measured predictive performance. The internal search compares candidate configurations according to the stated validation objective.

Table 5.2: Specified configurations of the nine unconstrained comparators.

Comparator	Configuration
XGBoost	309 trees, learning rate 0.125, maximum depth 3, minimum child weight 6.84, row fraction 0.915, feature fraction 0.828, absolute penalty 1.18×10^{-3} , and squared penalty 0.382.
LightGBM	302 trees, learning rate 0.0809, 16 leaves, maximum depth 8, minimum leaf count 14, row fraction 0.845, feature fraction 0.869, absolute penalty 3.17×10^{-3} , and squared penalty 3.36×10^{-4} .
CatBoost	236 boosting iterations, learning rate 0.172, depth 4, leaf squared penalty 0.0266, bagging temperature 0.0277, and random strength 0.0657.
AdaBoost	236 estimators, learning rate 0.999, and squared regression loss.
Random forest	390 trees, maximum depth 14, minimum split count 3, minimum leaf count 1, feature fraction 0.500, and no bootstrap resampling.
Support vector regression	Polynomial kernel of degree 4, regularization parameter 0.118, insensitive-error width 0.0458, constant offset 0.889, and solver tolerance 8.96×10^{-4} . The kernel scale is the reciprocal of the product of input dimension and training input variance.
k -nearest neighbors	15 neighbors, inverse-distance weighting, Manhattan distance, and a k -dimensional search tree with leaf size 40.
Partial least squares	6 latent components, maximum 887 fitting iterations, and tolerance 9.91×10^{-5} .
Multilayer perceptron	Hidden widths (128, 64, 32), logistic activation, adaptive moment estimation, squared penalty 6.33×10^{-4} , batch size 64, initial learning rate 1.08×10^{-3} , maximum 500 iterations, and tolerance 1.63×10^{-5} . Training rows are not shuffled, and early stopping is disabled.

Scoring states and physical diagnostics

The scored ICSOR output is the final checked component state. For comparator accuracy assessment, the raw component prediction receives the same affine projection for mass conservation in Equation (5.43). The composition map is then applied to the projected state. This comparison describes the accuracy of a checked structured prediction and comparator predictions after affine projection. Comparator accuracy scoring uses the affine state, while ICSOR accuracy scoring uses the final state from checked deployment.

The numerical assessment used an operator A_{num} obtained by rounding the null-space basis, setting negligible coefficients to zero, and dividing each row by its first nonzero coefficient. The fitting, affine projection, and constrained deployment calculations used this operator in place of the ideal operator A . The reported violation rates use these adopted numerical balances and the stated operator scaling.

Physical diagnostics use the direct raw states of the comparators. For ICSOR, diagnostics distinguish the raw coupled state, affine state, and final deployed state. This stage separation identifies the component state assessed by each diagnostic. The empirical balance diagnostic is $v_{mc}^{num}(c) = \|A_{num}(c - c_{in})\|_2$. The non-negativity diagnostic remains v_{nn} in Equation (5.38). At each stage, the violation frequency for diagnostic v is

$$f_v = \frac{100}{n} \sum_{i=1}^n \mathbf{1}\{v(c_i) > 10^{-10}\}, \quad (5.63)$$

where n is the number of assessed observations and v is either v_{mc}^{num} or v_{nn} . Frequencies, residual magnitudes, adjustment distances, and deployment-branch use describe different aspects of the same prediction procedure. Predictive error is reported alongside these physical diagnostics.

Composite error measures

The differences among the five measures can first be computed for two hypothetical observations of one reported quantity. Take reference values (2, 4) and predictions (3, 2). Their residuals are (1, -2). Squaring them gives (1, 4), while taking absolute values gives (1, 2). The mean squared error is $(1 + 4)/2 = 2.5$, its square root is approximately 1.5811, and the mean absolute error is $(1 + 2)/2 = 1.5$. These measures describe different summaries of the same two errors.

The absolute relative errors are $1/2$ and $2/4$, so their mean is 0.5. The reference mean is 3. Its total squared variation is $(2 - 3)^2 + (4 - 3)^2 = 2$. The prediction residual squared sum is 5. Therefore the coefficient of determination is $1 - 5/2 = -1.5$. Its negative value says that the hypothetical predictions have greater squared error than assigning the reference mean to both observations. These values illustrate the definitions on a hypothetical pair of observations.

For reported quantity $q \in \{1, 2, 3, 4\}$, define the prediction residual $e_{iq} = \hat{y}_{iq} - y_{iq}$. The five measures are mean squared error, root mean squared error, mean absolute error, mean absolute relative error, and the coefficient of determination. They are computed per reported quantity and summarized across the four quantities. The squared and absolute measures are

$$E_{2,q} = \frac{1}{n} \sum_i e_{iq}^2, \quad E_{root,q} = \sqrt{E_{2,q}}, \quad E_{1,q} = \frac{1}{n} \sum_i |e_{iq}|. \quad (5.64)$$

For strictly nonzero reference values, the relative measure is

$$E_{rel,q} = \frac{1}{n} \sum_i \frac{|e_{iq}|}{|y_{iq}|}. \quad (5.65)$$

It is reported as a ratio. Multiplication by 100 converts it to a percentage. A zero reference value makes this expression undefined, and a very small value can dominate it. Any zero-reference convention is declared explicitly before comparing predictors. Targets with undefined relative error are identified under that convention.

The per-target coefficient of determination is

$$R_q^2 = 1 - \frac{\sum_i e_{iq}^2}{\sum_i (y_{iq} - \bar{y}_q)^2}. \quad (5.66)$$

A target with zero reference variance makes this expression undefined. A negative value is possible when the predictor has a greater squared error than the reference-mean predictor. This coefficient measures prediction error relative to the target variance.

The aggregate squared error pools the four composite residuals. Its root is

$$E_{root} = \left(\frac{1}{4n} \sum_{q=1}^4 \sum_{i=1}^n e_{iq}^2 \right)^{1/2}. \quad (5.67)$$

Aggregate absolute and relative errors use the corresponding mean over observations and composites. Aggregate R^2 uses the mean of the four target-specific coefficients. The composite coordinates retain their own physical units. Their pooled error is a numerical benchmark summary under that coordinate convention. Per-target scores remain necessary to show which reported quantities contribute to an aggregate comparison.

Pooling before taking the square root also matters. If four hypothetical target mean squared errors are 1, 9, 1, and 9, the pooled mean squared error is 5 and its root is $\sqrt{5}$, approximately 2.2361. The arithmetic mean of the four target root errors is instead $(1 + 3 + 1 + 3)/4 = 2$. These are different summaries. Equation (5.67) specifies the first calculation.

The primary ordering measure is aggregate root mean squared error. The other measures provide complementary assessments. Squared error emphasizes larger residuals, absolute error gives average deviation, relative error emphasizes deviations in relation to the reference magnitude, and R^2 compares the squared residual with reference variation. Predictor orderings are compared separately under each measure.

Repeated sample-size assessment

The sample-size design uses

$$n \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (5.68)$$

At each size, ten subsamples are drawn without replacement from the design collection. Deterministic seeds are derived from seed 42. Each run uses a row-wise 80% training and 20% held-out split. All ten predictors receive the same sampled rows and partitions. The design specifies 110 fits per predictor across the size schedule. The assessment uses repeated random splitting.

The fixed hyperparameters isolate sensitivity to available observations under one model specification. The mean and spread of each held-out score are summarized across the ten runs. Learning curves describe predictive performance across the entire sample-size schedule. Geman et al. (1992) explain the relation between flexibility and the bias-variance problem, and Viering and Loog (2023) review the varied shapes of learning curves.

The learning-curve area can first be calculated over a reduced hypothetical schedule. Suppose an error measure has values 6, 4, and 3 at sample sizes 100, 200, and 400. Between the first two sizes, a trapezoid has width 100 and mean height $(6 + 4)/2 = 5$, giving area 500. Between the next two, its width is 200 and mean height $(4 + 3)/2 = 3.5$, giving area 700. The total area is 1200. Dividing by the total width $400 - 100 = 300$ gives a normalized area of 4. A larger interval contributes more to this average than a smaller interval, even when each has one pair of endpoints. This hypothetical schedule illustrates the area calculation for unequal sample-size intervals.

For successive retained sizes $n_1, \dots, n_S$, write the mean error of predictor m under measure j as $\overline{M}_{m,j}(n_s)$. The general numerical calculation is

$$\frac{1}{n_{\max} - n_{\min}} \sum_{s=1}^{S-1} (n_{s+1} - n_s) \frac{\overline{M}_{m,j}(n_s) + \overline{M}_{m,j}(n_{s+1})}{2}. \quad (5.69)$$

Let $\overline{M}_{m,j}(n)$ denote the mean held-out measure j for predictor m at sample size n . The normalized area under its learning curve is

$$L_{m,j} = \frac{1}{n_{\max} - n_{\min}} \int_{n_{\min}}^{n_{\max}} \overline{M}_{m,j}(n) \, dn. \quad (5.70)$$

The trapezoidal rule evaluates this integral over the stated sample sizes. Normalization by the sample-size interval gives an average height of the learning curve and preserves the units of measure j . The normalization denominator is the sample-size interval. Cawley (2011) and Guyon et al. (2011) provide related trajectory summaries in active learning. The present design varies the number of randomly sampled observations.

A rank discards the numerical distance between scores and keeps their order. For four hypothetical predictor errors 3.0, 3.0, 4.5, and 6.0, the dense ranks are 1, 1, 2, and 3. The tied first two predictors receive the same rank. There is no skipped rank after the tie. The difference between the third and fourth errors does not influence the difference between their ranks. If one predictor receives ranks 1, 2, and 2 in three illustrative comparisons, its mean rank is $(1 + 2 + 2)/3 = 5/3$. The actual assessment averages over the stated combinations of target, measure, and size.

For each measure j , reported quantity q , and retained size n_s , predictors receive a dense rank $r_{m,j,q}(n_s)$ from their mean held-out score. Lower squared, root, absolute, and relative errors receive better ranks. Larger R^2 values receive better ranks. Equal scores receive equal ranks, and the next distinct score receives the next consecutive

rank. With $S = 11$ sizes and five measures, the summaries are

$$\bar{r}_{m,j} = \frac{1}{4S} \sum_{q=1}^4 \sum_{s=1}^S r_{m,j,q}(n_s), \quad \bar{r}_m = \frac{1}{5} \sum_{j=1}^5 \bar{r}_{m,j}. \quad (5.71)$$

These summaries describe relative ordering across the stated comparisons. Demšar (2006) develops broader rank-based model comparisons, while Shevchenko et al. (2024) emphasize variability in benchmarking. Mean ranks and absolute errors are reported together to describe both ordering and error magnitude.

At the largest sample size, the generalization gap is

$$\Delta E_{root,m} = E_{root,m}^{held-out} - E_{root,m}^{training}. \quad (5.72)$$

A positive gap indicates greater held-out error. A small gap can accompany either accurate prediction or high-error underfitting, so it is interpreted with the corresponding absolute scores (Hastie et al., 2009). Training time is recorded for every repeated fit and summarized as a distribution across runs. Deployment effort is assessed separately through prediction time, projection-branch use, and constrained-solve effort. This separation makes the cost of learning distinct from the cost of returning a checked state.

For a hypothetical predictor with training error 0.5 and held-out error 0.8, the gap is 0.3. Another with errors 5.0 and 5.1 has a gap of only 0.1. The second has the smaller gap but much larger absolute errors. This arithmetic shows how absolute error and the training-to-held-out gap describe different aspects of prediction.

5.3.5 Interpretability assessment

The interpretability assessment examines the six driver blocks, the coupling matrix, and the corresponding component and composite raw-regression maps. It compares fitted response patterns with the declared process domain. The reported coefficient summaries describe the response structure of one fitted model. Brunton et al. (2016) connect interpretable nonlinear terms to the problem of recovering structure from data, while Machado et al. (2014) show how operational and influent uncertainty complicate parameter identification in activated sludge models.

The driver and coupling parameters can trade off in the relation $Rc = B\phi$. Different joint stationary solutions can predict similar component states with different coefficients (Selerio, 2026). Symmetry removes the redundancy of ordered self-quadratic pairs. Coefficient signs and magnitudes are interpreted together with the complete fitted response.

Prediction error and physical diagnostics are assessed separately for the raw regression, the affine state, and the final deployed state. The comparison distinguishes what the regression learns from what the projection enforces (Karniadakis et al., 2021; Kircher & Votsmeier, 2025). The magnitude of the projection is also examined because a feasible deployed state can lie far from its raw prediction. Computational assessment

records fitting effort and deployment effort separately. These measures describe the cost of fitting the structured regression and enforcing its physical conditions.

The assessment evaluates steady-state component prediction in the stated input domain and control volume. The raw regression supplies an inspectable approximation, and checked deployment enforces the adopted numerical invariants and non-negativity on the returned state.

5.4 Results

5.4.1 Accuracy on the fixed held-out partition

Table 5.3 reports the four-composite scores on the 2000 held-out cases. ICSOR used its final checked state. The other predictors used their affine-aligned states for accuracy scoring. The multilayer perceptron had the lowest pooled root mean squared error of 4.38. LightGBM followed at 5.30. Support vector regression and ICSOR were closely spaced at 5.97 and 5.98. These scores pool the native composite coordinates under Equation (5.67). The pooled scores retain the native composite-coordinate convention.

Table 5.3: Fixed-split held-out composite performance. ICSOR is scored after checked deployment. Comparators are scored after affine alignment without a non-negativity constraint. Relative absolute error is a ratio rather than a percentage.

Predictor	E_{root}	E_1	Mean R^2	E_{rel}
Multilayer perceptron	4.38	2.55	0.9915	0.0123
LightGBM	5.30	3.21	0.9806	0.0203
Support vector regression	5.97	3.49	0.9713	0.0244
ICSOR	5.98	3.70	0.9700	0.0256
XGBoost	6.12	3.81	0.9733	0.0244
CatBoost	6.45	4.07	0.9727	0.0250
AdaBoost	21.27	13.15	0.8643	0.0653
k -nearest neighbors	23.61	13.87	0.8535	0.0557
Partial least squares	28.24	17.05	0.7808	0.0681
Random Forest	29.00	16.93	0.7934	0.0632

ICSOR's pooled root error was 36.5% higher than the multilayer perceptron's and 12.8% higher than LightGBM's. It remained below XGBoost and CatBoost on this measure. The mean absolute relative error of 0.0256 corresponded to 2.56%. The fixed-split comparison quantified the accuracy trade-off associated with the structured prediction procedure.

Figure 5.5 compares root error, absolute error, and mean composite R^2 . ICSOR ranked fourth by pooled root error on this partition.

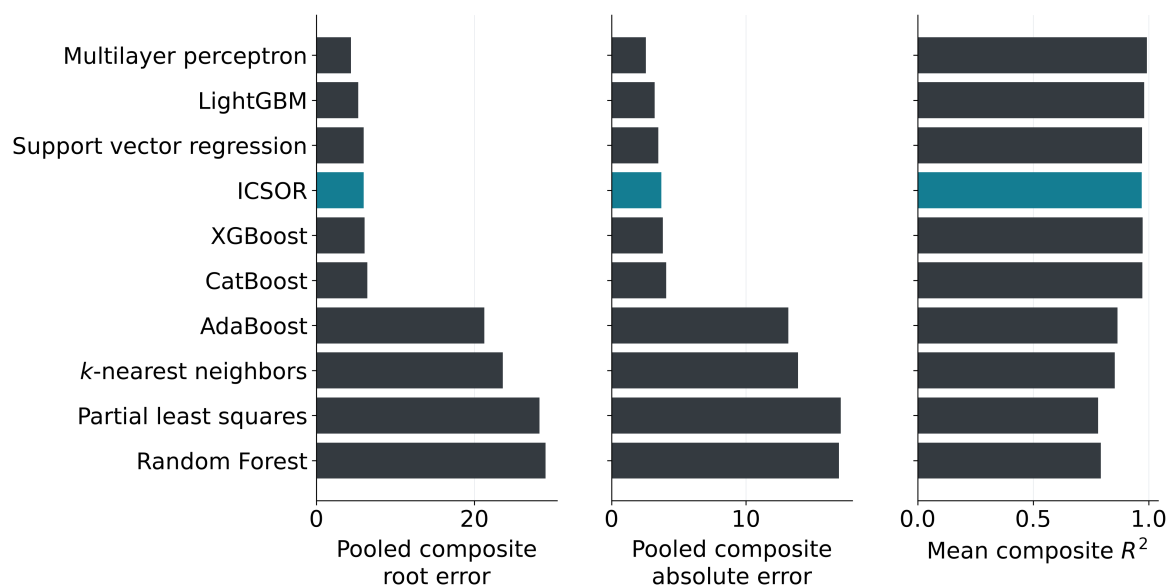


Figure 5.5: Fixed-split composite accuracy on 2000 held-out cases. Teal bars identify ICSOR after checked deployment. Gray bars identify comparators after affine alignment. Pooled root and absolute errors retain the native composite-coordinate convention. The right panel averages four composite-specific R^2 values.

Table 5.4 separates the composite contributions. ICSOR’s COD root error of 7.77 was 0.18 above LightGBM’s and below that of the other non-neural predictors. Its TN and TSS errors were 3.55 and 8.36. The multilayer perceptron had the lowest COD, TN, and TSS errors. Every method had a TP error of 0.0361 after the specified alignment and reporting operations. TP therefore did not distinguish the predictors in this assessment.

Figure 5.6 shows why the aggregate comparison needs separate target scores. COD and TSS contributed much larger numerical errors than TP. The separate target panels retain the constituent units used for each reported quantity.

5.4.2 Sample efficiency and generalization

ICSOR had the lowest mean held-out root error at the smallest design size of 500 cases. Its value was 10.79, compared with 11.26 for XGBoost and 11.45 for LightGBM. The multilayer perceptron ranked fifth at 13.26. At the largest design size, the repeated-split means were 4.62 for the multilayer perceptron, 5.27 for LightGBM, 5.84 for support vector regression, and 5.94 for ICSOR. The ordering therefore changed as more training cases became available.

ICSOR had the lowest reported normalized root-error curve area of 6.33. LightGBM was close at 6.35 and had the best mean dense rank of 2.2. The multilayer perceptron had the second-best mean rank of 2.5. ICSOR and XGBoost each had a reported mean rank of 4.3. ICSOR led the curve-area comparison, while LightGBM led the mean-rank

Table 5.4: Fixed-split root mean squared error for each composite. COD, TN, TP, and TSS use g COD m^{-3} , g N m^{-3} , g P m^{-3} , and g TSS m^{-3} , respectively.

Predictor	COD	TN	TP	TSS
Multilayer perceptron	6.48	1.58	0.0361	5.66
LightGBM	7.59	2.77	0.0361	6.85
Support vector regression	8.87	3.50	0.0361	7.20
ICSOR	7.77	3.55	0.0361	8.36
XGBoost	8.78	3.27	0.0361	7.88
CatBoost	9.13	3.23	0.0361	8.53
AdaBoost	38.30	5.58	0.0361	17.68
k -nearest neighbors	38.24	4.23	0.0361	27.37
Partial least squares	43.22	5.09	0.0361	36.00
Random Forest	45.87	4.00	0.0361	35.26

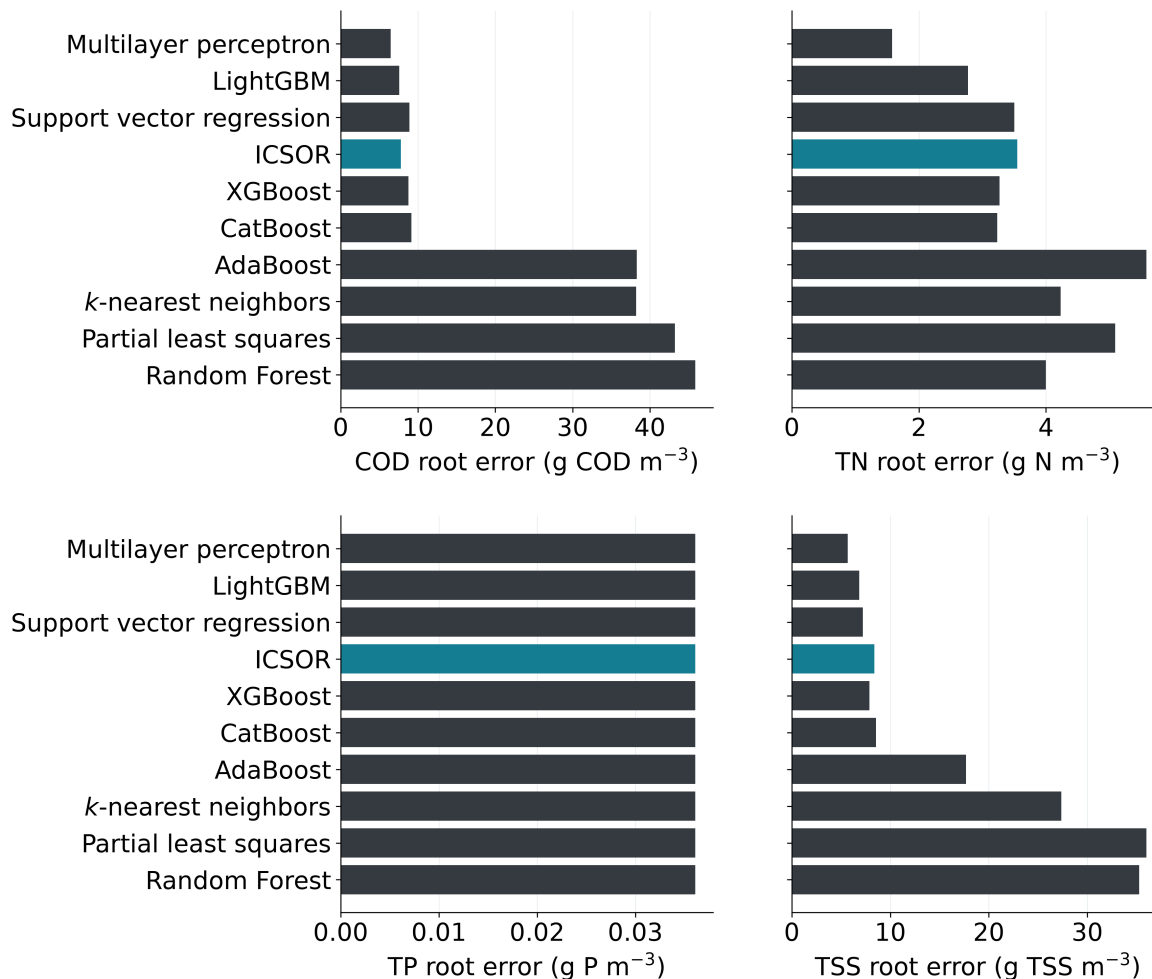


Figure 5.6: Fixed-split root mean squared error for COD, TN, TP, and TSS. Teal identifies ICSOR and gray identifies the comparators. Each panel has its own constituent unit and horizontal scale. All methods have the same reported TP error under the specified alignment and reporting procedure.

Table 5.5: Repeated-split composite assessment over eleven design sizes and ten runs per size. Final error and the held-out minus training gap refer to $n = 10,000$. The learning-curve area is normalized by the design-size interval. Lower error and curve-area values indicate better performance. The gap measures the separation between training and held-out error.

Predictor	Final E_{root}	Normalized curve area	Mean rank	ΔE_{root}
Multilayer perceptron	4.62	6.75	2.5	0.65
LightGBM	5.27	6.35	2.2	1.99
Support vector regression	5.84	6.90	4.0	1.19
ICSOR	5.94	6.33	4.3	0.22
XGBoost	6.10	7.01	4.3	1.44
CatBoost	6.26	7.60	5.0	0.69
AdaBoost	21.42	20.29	8.2	0.23
k -nearest neighbors	23.76	25.77	7.0	23.66
Partial least squares	27.96	30.17	9.4	0.38
Random Forest	28.78	30.31	8.2	18.71

comparison.

ICSOR's held-out minus training root-error gap was 0.22. This was smaller than the gaps for the multilayer perceptron, LightGBM, XGBoost, and support vector regression. Partial least squares also had a relatively small gap of 0.38, but its held-out root error was 27.96. ICSOR combined a small gap with substantially lower absolute error than partial least squares.

Figure 5.7 presents the three trajectory summaries separately. ICSOR had the lowest root-error curve area, whereas LightGBM had the best mean rank. The two summaries answer different comparative questions.

Figure 5.8 compares the smallest and largest design sizes for four leading predictors. ICSOR led this group at 500 cases, while the multilayer perceptron led at 10,000 cases. The two endpoints show the change in the leading predictor between the smallest and largest designs.

5.4.3 Physical checks and deployment branches

Table 5.6 distinguishes the final ICSOR output from the raw outputs of the comparators. The final structured predictions had no violations of the adopted numerical balances or non-negativity tolerance. Every comparator raw state violated the adopted balance tolerance. AdaBoost, Random Forest, and the nearest-neighbor model nevertheless had no negative raw states. The other raw non-negativity violation rates ranged from 46.00% to 70.10%.

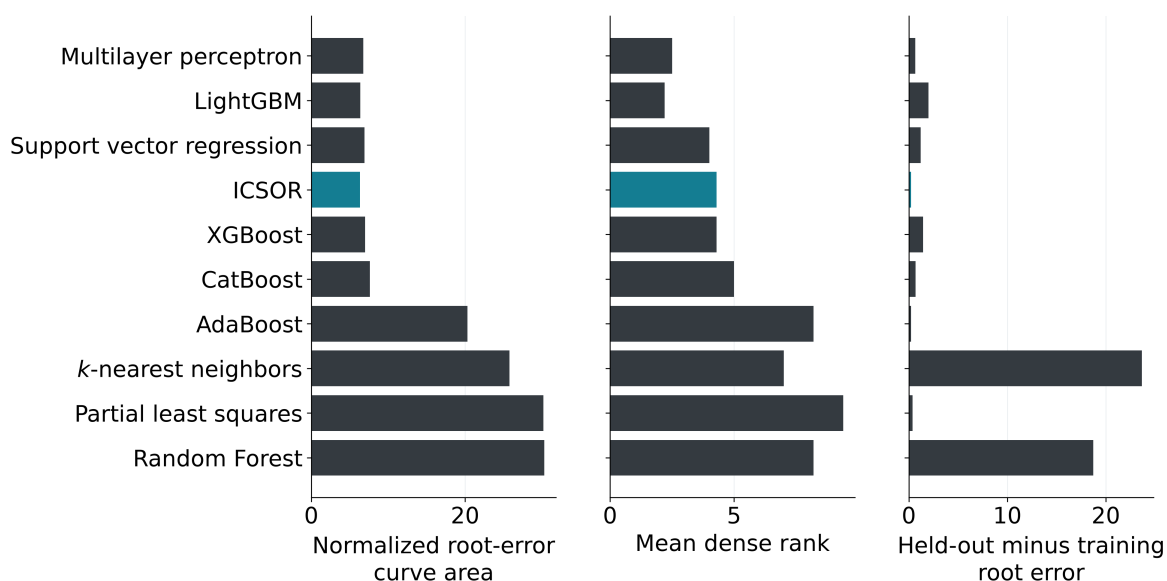


Figure 5.7: Normalized root-error curve area, mean dense rank, and held-out minus training root-error gap. Curve areas and ranks summarize eleven design sizes with ten repeated splits per size. The gap refers to the largest design size. Teal identifies ICSOR. The gap is reported alongside the held-out error.

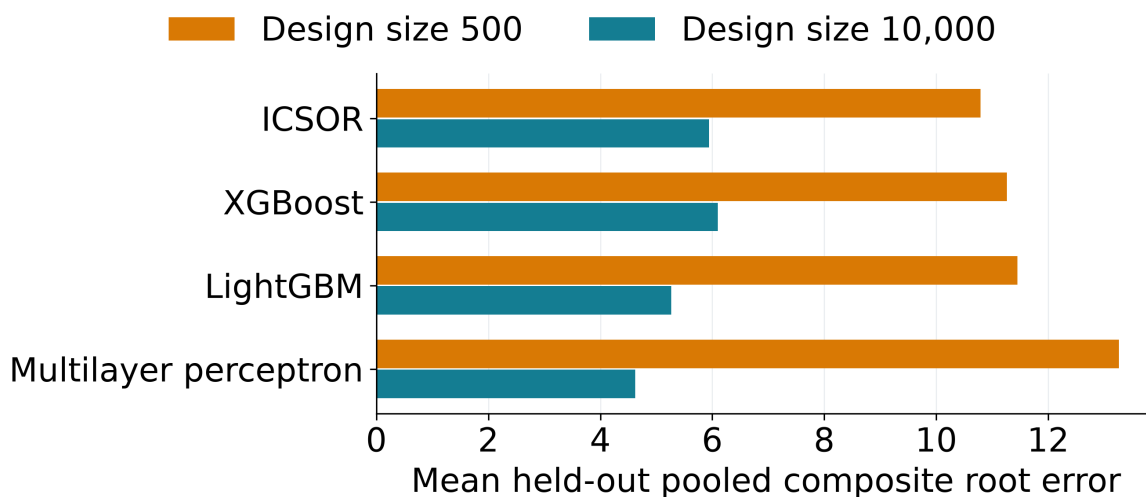


Figure 5.8: Mean held-out pooled composite root error at design sizes 500 and 10,000 for ICSOR, XGBoost, LightGBM, and the multilayer perceptron. Values summarize ten repeated splits. The panels show repeated-split endpoint means at the two specified design sizes.

Table 5.6: Component-state violation frequencies on the 2000 held-out cases. ICSOR uses its final checked state. Comparators use their raw state, not the affine-aligned state used for accuracy scoring. Balance violations refer to A_{num} and the declared 10^{-10} tolerance.

Predictor	Balance violations (%)	Negative states (%)
Multilayer perceptron	100.0	58.20
LightGBM	100.0	46.00
Support vector regression	100.0	68.65
ICSOR	0.0	0.00
XGBoost	100.0	64.95
CatBoost	100.0	70.10
AdaBoost	100.0	0.00
k -nearest neighbors	100.0	0.00
Partial least squares	100.0	48.70
Random Forest	100.0	0.00

No raw coupled ICSOR prediction passed both physical checks. All failed the balance tolerance, although 21.7% were non-negative. Affine projection restored the adopted balances for every case and also returned a non-negative state for 436 cases, or 21.8%. Those cases were returned after the affine stage. The remaining 1564 cases, or 78.2%, required the weighted linear projection. The solve used a mean of 23.5 iterations and a maximum of 40. Every final state passed both checks.

Figure 5.9 keeps the comparator diagnostics and ICSOR’s deployment stages separate. The rise from zero raw balance compliance to complete final compliance occurred through projection. The branch counts show that the constrained solve was used for most deployed predictions.

These stage counts identify the source of the final compliance. The affine stage restored the adopted balances, and the constrained projection supplied non-negativity for the remaining cases. The complete deployment procedure returned zero final violation rates.

5.4.4 Computational effort

At the smallest design size, mean ICSOR fitting time was 21.5 s. LightGBM, XGBoost, and the multilayer perceptron required 4.72, 3.16, and 4.99 s, respectively. At the largest size, ICSOR required 94.4 s. The faster tree ensembles required 4.13–8.13 s, while partial least squares and the nearest-neighbor model required 0.228 and 2.70 s. ICSOR was nevertheless faster to fit than AdaBoost at 117.7 s, support vector regression at 136.8 s, and the multilayer perceptron at 161.6 s.

The recursive constrained fitting procedure was more costly than the fastest com-

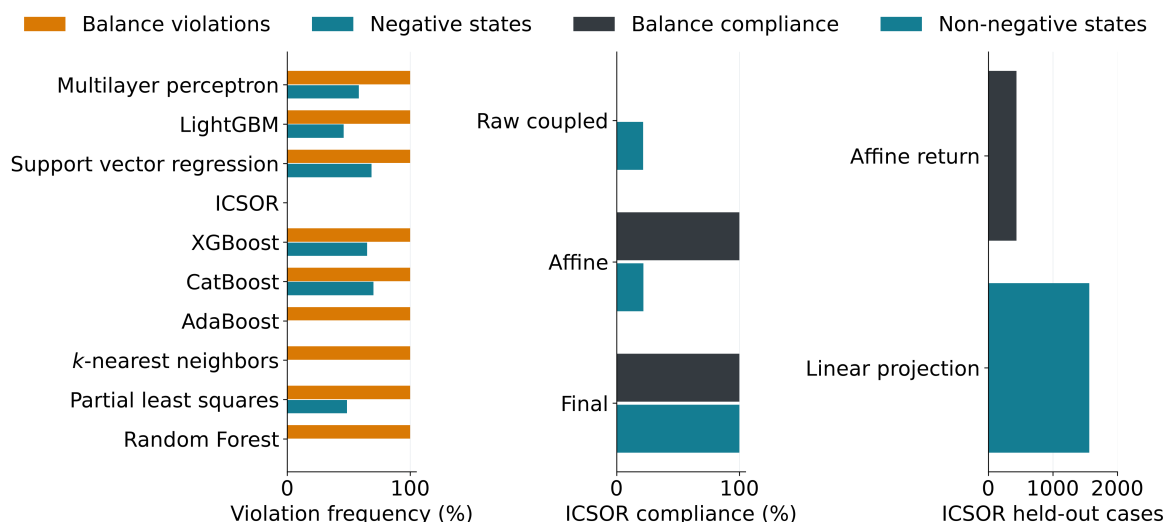


Figure 5.9: Physical diagnostics and ICSOR deployment on 2000 held-out cases. The left panel compares final ICSOR states with raw comparator states. The middle panel separates raw coupled, affine, and final ICSOR compliance. The right panel gives the numbers returned after affine projection and after constrained linear projection. Balance checks use the adopted numerical operator A_{num} .

parators and faster than three comparators at the largest design size. Fitting time was assessed separately from deployment effort. The 78.2% linear-projection activation rate showed that most fixed-split predictions required an optimization step after regression evaluation. The reported iteration counts described this deployment workload.

Figure 5.10 separates the two design sizes on logarithmic time axes. The smaller-size panel compares four predictors, and the larger-size panel compares ICSOR with five alternatives. The comparison reports the time required for the repeated model fits.

5.4.5 Inspectable coefficient structure

Table 5.7 reports the COD raw-composite coefficient counts and the shared coupling entries. The polynomial summary counted 276 unique candidates. Adding the 380 off-diagonal coupling candidates gave 656 entries in this display. The display combined the COD polynomial row with the shared coupling entries. The display retained 469 entries, or 71.5%, under the specified post-fitting thresholds.

The influent curvature block retained 44 of 210 unique terms. In contrast, 371 of 380 coupling entries exceeded their reporting threshold. The fitted structure therefore combined selective influent curvature with dense component coupling. Selectivity was concentrated in the influent-curvature block, while the coupling matrix was dense.

The largest absolute COD influent-curvature entry was the dissolved-oxygen self term of -5.8097 . Each mirrored dissolved-oxygen and dinitrogen cell had value -1.7411 . Their combined unordered interaction coefficient was -3.4822 . Other prominent terms

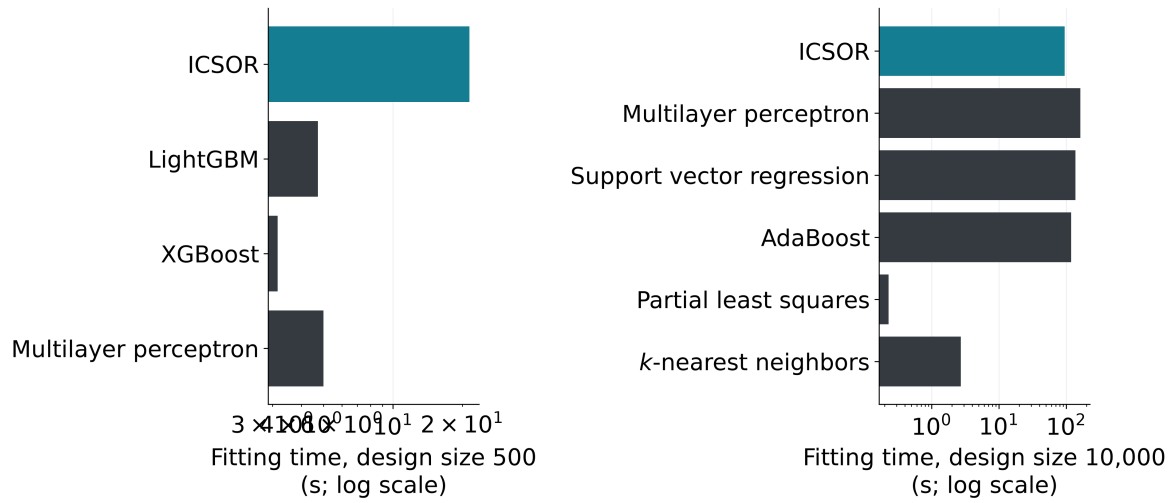


Figure 5.10: Mean fitting times for the specified predictors at design sizes 500 and 10,000. Teal identifies ICSOR. Each panel uses a logarithmic seconds axis. The values concern repeated model fits and exclude deployment calculations.

Table 5.7: Coefficients exceeding the reporting thresholds in one fitted model. Polynomial blocks describe the COD row of $B_{\text{raw,comp}}$. The coupling matrix is shared across component outputs and uses its own threshold. Symmetric quadratic blocks count unique upper-triangular terms.

Coefficient block	Candidates	Retained	Retained (%)
Baseline b	1	1	100.0
Operating terms W_u	2	2	100.0
Influent terms W_{in}	20	19	95.0
Operating curvature Θ_{uu}	3	3	100.0
Operating-load interactions Θ_{uc}	40	29	72.5
Influent curvature Θ_{cc}	210	44	21.0
Shared off-diagonal coupling Γ	380	371	97.6
Combined display	656	469	71.5

linked dissolved oxygen with metal phosphate, ammonia-oxidizing biomass, soluble phosphate, nitrite-oxidizing biomass, and nitrite. These values described the raw composite map after coupling and reporting.

Figure 5.11 displays the retained counts and selected oxygen-related curvature terms. Its interaction values combine the two mirrored cells. The comparison describes the fitted polynomial under its native input units.

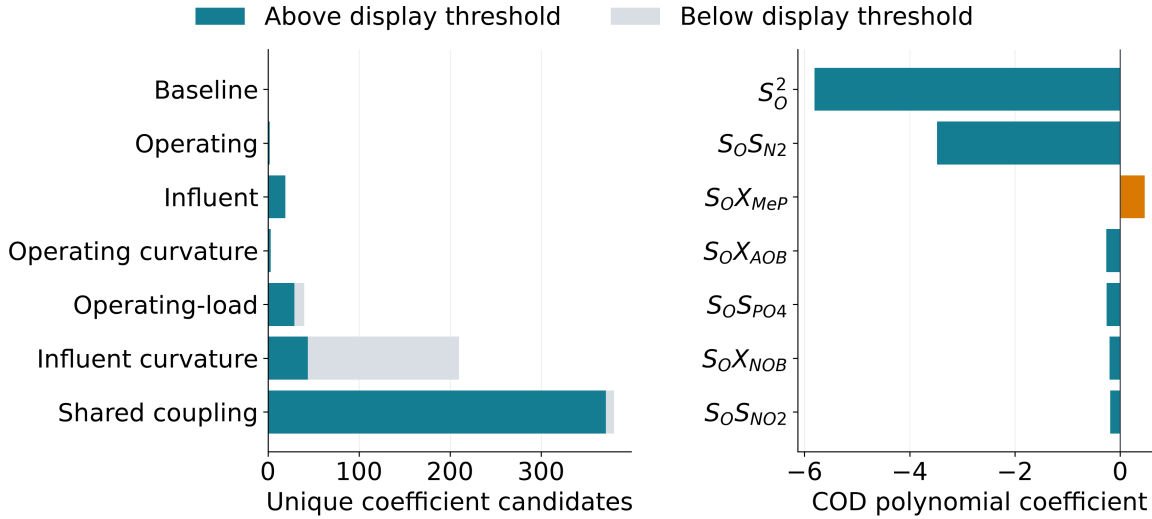


Figure 5.11: Coefficient structure of one fitted ICSOR model. The left panel separates entries above and below the display thresholds for the COD polynomial blocks and shared coupling. The right panel shows selected COD influent-curvature coefficients with mirrored interaction entries combined. Coefficient units depend on the corresponding input terms. The displayed values describe the raw composite mapping.

5.5 Discussion

5.5.1 A structured approximation under limited data

ICSOR’s lowest small-design error and lowest reported root-error curve area supported a data-efficiency benefit in the evaluated domain. Its explicit second-order terms defined a structured response approximation. The invariant-aware fitting objective and learned component coupling supplied additional structure. The combined design performed well when fewer simulated states were available. This interpretation is consistent with the role of model structure in hybrid learning discussed by Shen et al. (2023) and with the changing model orderings described by Viering and Loog (2023).

The leading predictor depended on the assessment measure. LightGBM had the best mean rank across targets, metrics, and sizes. The multilayer perceptron had the lowest full-size error. ICSOR’s curve area was 6.33, close to LightGBM’s 6.35. Selection based on full-size composite error favored the multilayer perceptron under this protocol.

An inspectable equation, data efficiency, and checked final states provided additional engineering criteria for selecting ICSOR (Bhosekar & Ierapetritou, 2018; Rudin, 2019).

The training-to-held-out gap and absolute error described complementary properties. Partial least squares combined a small gap with high prediction error. ICSOR combined a gap of 0.22 with substantially lower held-out error, while the leading flexible predictors achieved still lower absolute error. Reporting both measures clarified the relation between generalization and accuracy.

5.5.2 Fitting guidance and final-state enforcement

The physical assessment showed how the complete procedure supplied its returned state. The soft invariant penalty guided the fitted training states, and the learned coupled equation generated the raw prediction for each new input. Every fixed-split raw prediction required affine projection. The affine state passed both checks in 21.8% of cases, and the weighted linear projection handled the remaining 78.2%.

The complete deployment procedure returned zero final balance and non-negativity violations. Stage-wise diagnostics identified the affine and constrained projections as the source of this compliance. This separation between constrained fitting and output enforcement follows the approaches examined by Hansen et al. (2024) and Kircher and Votsmeier (2025).

The numerical assessment used the rounded and normalized operator A_{num} with a tolerance of 10^{-10} . Accuracy scoring used checked ICSOR states and affine-aligned comparator states. Physical comparator diagnostics used their raw states. These specified procedures produced the 36.5% root-error difference between ICSOR and the multilayer perceptron. The shared alignment and reporting convention also produced the common TP error of 0.0361.

5.5.3 What the coefficient structure explains

The named coefficient blocks exposed operating dependence, loading dependence, curvature, and component coupling for direct inspection. This form of interpretability follows the preference for transparent models discussed by Rudin (2019). Coupling and the reporting map translated the component-driver coefficients into composite response equations.

The retained counts showed where the fitted representation was selective and where it remained dense. Most unique influent-curvature terms were below the display threshold, while almost all coupling entries exceeded their separate threshold. The display retained 469 of 656 entries. The complete fitted mapping retained its coefficients, with reporting thresholds controlling the detail visible in the display.

The prominent oxygen-related curvature terms were consistent with oxygen-sensitive reaction pathways in the adopted process model. Stenstrom and Song (1991) provide

process context for oxygen dependence in nitrification, while Wentzel et al. (1992) describe interactions among nutrient-removal pathways. The dissolved-oxygen self coefficient and the oxygen-dinitrogen interaction therefore identified fitted response patterns that could be compared with the reaction model. Their contributions were interpreted in the native input units and through the complete coupled mapping.

The polynomial supplied analytical raw sensitivities, while the affine projection modified them to preserve the influent inventories. Equations (5.58) and (5.61) described these two stages. The frequent constrained branch made active-set sensitivity relevant to the final returned response. Figure 5.4 demonstrated this change through an explicit example in which the first-component slope changed from positive to negative and then became zero at an active non-negativity boundary.

5.5.4 Engineering implications

The structured model combined competitive composite prediction, favorable low-data behavior, and final states that passed the imposed numerical checks. Its second-order terms and coupling map made the fitted response available for inspection. Checked deployment then enforced the adopted material accounting and component non-negativity before reporting COD, TN, TP, and TSS.

These properties involved distinct computational costs. ICSOR required more fitting time than the fastest tree and linear comparators. At 10,000 cases, it fitted faster than AdaBoost, support vector regression, and the multilayer perceptron. The constrained deployment solve was used for most fixed-split predictions. Separate fitting and deployment measurements therefore described the effort required to construct the approximation and return a checked state for engineering analysis. Comparing preparation and repeated use follows the computational assessment principles discussed by Bhosekar and Ierapetritou (2018) and Blik et al. (2023).

5.5.5 Data Availability

The simulation data and numerical results supporting this study are publicly available at <https://github.com/egbertoselerio1997/icsor>.

Chapter 6

Connected plant surrogates and operating optimization

6.1 Rationale

Activated sludge process optimization concerns a connected response rather than an isolated effluent prediction. Biological reactors transform material, secondary clarification separates solids, and recycle returns material to earlier treatment locations. Mixed liquor contains wastewater and suspended biomass. Clarifier overflow carries treated water, while underflow supplies concentrated sludge for return and wasting. Their concentrations and flow rates determine the material discharged, circulated, and removed. Aeration, recycle, and wasting therefore affect both effluent quality and the solids retained within the plant (Härtel & Pöpel, 1992; Henze et al., 2008).

Mechanistic plant models provide a basis for these connections. The benchmark of Alex et al. (2008) links biological reactors, secondary clarification, internal recycle, return sludge, and wasting. The component-transport analysis of Jeppsson and Diehl (1996) explains why clarifier outlets must retain a consistent biological composition. The structured reactor prediction of Selerio (2026) supplies an inspectable approximation at a single treatment boundary. The first gap addressed here is how to extend component prediction and physical enforcement to shared streams and stored solids across a connected plant. Accurate or physically checked predictions at individual units do not establish consistency among those units.

The return-sludge connection makes this gap concrete. A predicted underflow concentration enters the mixer through the return flow. If the mixer prediction implies a different contribution, the two outputs describe different amounts of the same transported material. Each output can remain non-negative while their connection is inconsistent. Clarifier storage adds a further requirement because retained solids contribute to the plant inventory and solids retention time. A plant response must therefore retain internal concentrations, separator outlets, and stored solids rather than

predict final-effluent totals alone (Alex et al., 2008; Jeppsson & Diehl, 1996).

Joint projection can impose the declared relations on these outputs together. Its value needs to be tested at the treatment locations where the predictions will be used. An adjustment that reconciles recycle and separator balances can change both internal biomass and final nutrient concentrations. Location-specific errors, physical residuals, and computational effort are therefore needed to assess the connected prediction. Satisfaction of selected flow, inventory, and concentration constraints does not establish agreement with every nonlinear reaction or settling equation. This distinction limits what physical enforcement alone can demonstrate (Kircher & Votsmeier, 2025; Selerio, 2026).

The second gap concerns the operating decision obtained from the connected response. Surrogates can reduce repeated process evaluation effort (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019). The simulation-based search of Fang et al. (2011) and the resource-recovery framework of Durkin et al. (2024) illustrate their use in wastewater decisions. However, agreement over a holdout design does not establish agreement at the controls favored by an optimizer. The search can concentrate on a region where treatment is predicted too favorably. A low predicted nutrient concentration at that setting can then influence the choice of capacity, aeration, or recycle even when average prediction error is small.

Independent evaluation of selected controls is therefore a separate requirement. The surrogate comparisons of Pedrozo et al. (2025) distinguish predictive accuracy from optimization behavior. The approximation-management methods of Hameed et al. (2026) retain checks against the original model. These contributions motivate calculating the mechanistic plant response at the controls selected by each route. The relevant evidence includes the resulting effluent concentrations, the objective on a common reference basis, and satisfaction of the original engineering requirements. An available decision and a successfully certified numerical search are also different outcomes. Failed searches cannot be omitted when the practical usefulness of a route is assessed.

Operating comparisons also depend on the priorities built into the objective. de Araujo et al. (2013) examine constrained economic operation, while Padrón-Páez et al. (2020) and Aboagye et al. (2021) connect treatment choices with resource use. Two searches can prefer different settings because one accepts greater resource demand to improve treatment. A meaningful comparison therefore uses common priorities, control bounds, and mechanistic evaluation. It also examines the quality and resource contributions separately. A lower objective is a result under those declared preferences, not evidence that every effluent quantity or resource demand is lower.

The capacity interpretation is particularly important in this study. At fixed fresh-influent flow, varying HRT varies the biological reactor volume. The operating problem therefore combines capacity selection with aeration, recycle, and wasting choices. It is not solely an operating adjustment for an installed plant with fixed tanks. The

resource terms provide engineering proxies for comparing the declared choices rather than measured electricity bills or full facility costs. These boundaries define the practical meaning of the resulting decisions and prevent a numerical objective from being interpreted as a demonstrated field saving.

The study addresses the fourth research question in Section 1.3. The connected surrogate is termed Extended ICSOR. The direct route solves a smooth nonlinear programming (NLP) problem and is termed Smooth NLP. Their selected controls are compared through independent mechanistic evaluation and a common engineering objective. The contribution is a joint component-state formulation that connects physical enforcement to explicit decision verification. Its assessment distinguishes network consistency, prediction accuracy, selected treatment outcomes, search reliability, and computational effort. This supplies evidence about where rapid approximation can support plant analysis and where mechanistic evaluation remains necessary.

6.2 Theory

6.2.1 Hydraulic connections and reactor inventories

Flow through the treatment train

Let Q_0 denote the fresh-influent flow in $\text{m}^3 \text{d}^{-1}$ and let $c_{\text{in}} \in \mathbb{R}_+^F$ denote its component concentrations. The plant has n_r completely mixed reactors and an L -layer clarifier. The reactor concentrations are $c_1, \dots, c_{n_r}$. The mixer concentration is m . Clarifier overflow and underflow concentrations are c_E and c_U . The overflow is the final plant outlet, so $c_{\text{out}} = c_E$.

The dimensionless ratios r_I , r_R , and w describe mixed-liquor recycle, return activated sludge, and waste activated sludge flow relative to Q_0 . Mixed-liquor recycle leaves the final reactor before the clarifier. Return and waste sludge share the clarifier underflow concentration. Each ratio is an actual flow divided by Q_0 . Denote the train, clarifier-feed, underflow, and overflow flows by Q_P, Q_C, Q_U, Q_E . The mixer, reactor splitter, underflow splitter, and external water balance respectively give

$$Q_P = Q_0 + r_I Q_0 + r_R Q_0, \quad (6.1)$$

$$Q_C = Q_P - r_I Q_0, \quad (6.2)$$

$$Q_U = r_R Q_0 + w Q_0, \quad (6.3)$$

$$Q_E = Q_0 - w Q_0. \quad (6.4)$$

The second equation removes the mixed-liquor recycle before clarification. The third includes both underflow destinations. Dividing each hydraulic balance by Q_0 gives

$$q_P = 1 + r_I + r_R, \quad q_C = 1 + r_R, \quad q_U = r_R + w, \quad q_E = 1 - w. \quad (6.5)$$

Here q_P is the reactor-train flow ratio and q_C is the clarifier-feed flow ratio. The identity $q_C = q_E + q_U$ confirms the clarifier water balance. The adopted domain requires $r_I \geq 0$, $r_R \geq 0$, $0 \leq w < 1$, and $q_U > 0$. The outlet concentrations are then defined by positive outlet flows.

Figure 6.1 places these flow balances on the physical connections of the five-reactor plant. The two recycle routes return material to the mixer, while treated water and waste sludge cross the external boundary.

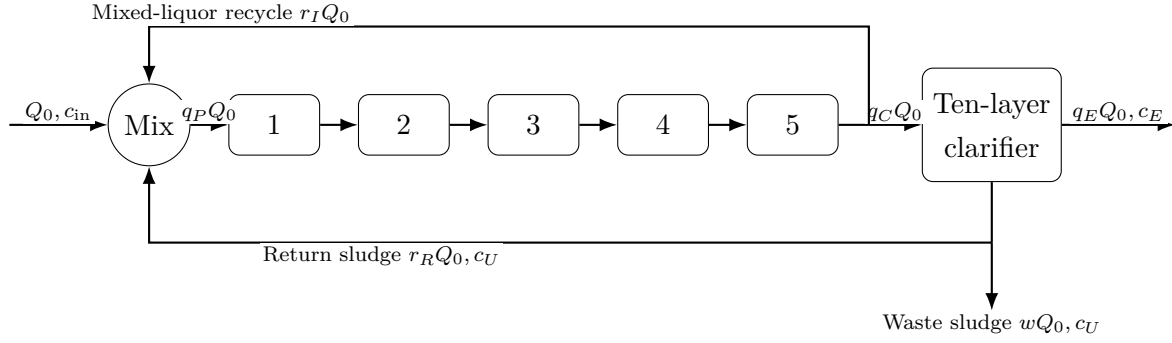


Figure 6.1: Connected plant configuration used for the operating study. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls.

The diagram shows why the reactor-train and clarifier-feed flows differ. Mixed-liquor recycle leaves before the clarifier, while return sludge leaves the clarifier underflow. Their concentrations consequently come from different locations. The mixer calculation uses each stream's own flow and concentration rather than assigning one shared recycle concentration.

The mixer balance is a component balance rather than a balance of a single water-quality total. For component f , its incoming and outgoing mass rates satisfy

$$Q_P m_f = Q_0 c_{in,f} + r_I Q_0 c_{n_r,f} + r_R Q_0 c_{U,f}. \quad (6.6)$$

Each product has units of g d^{-1} when the component basis is g m^{-3} . Divide by Q_0 and then by q_P to obtain

$$m_f = \frac{c_{in,f} + r_I c_{n_r,f} + r_R c_{U,f}}{1 + r_I + r_R}. \quad (6.7)$$

For two components, the same calculation is performed independently in each row,

$$\begin{bmatrix} q_P & 0 \\ 0 & q_P \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} = \begin{bmatrix} c_{in,1} + r_I c_{n_r,1} + r_R c_{U,1} \\ c_{in,2} + r_I c_{n_r,2} + r_R c_{U,2} \end{bmatrix}. \quad (6.8)$$

The complete concentration vector therefore satisfies

$$q_P m = c_{in} + r_I c_{n_r} + r_R c_U. \quad (6.9)$$

A high mixer concentration can therefore result from internal circulation. It does not imply that an external pollutant load has been created. The component-resolved

framework of X. Zhang et al. (2026) treats these recirculation and discharge streams explicitly.

A hypothetical mixing calculation makes the units and recycle effect explicit. Let $Q_0 = 10,000 \text{ m}^3 \text{ d}^{-1}$, $r_I = 2$, $r_R = 0.5$, and $w = 0.02$. The four physical flows are 35,000, 15,000, 5,200, and 9,800 $\text{m}^3 \text{ d}^{-1}$ in the order Q_P, Q_C, Q_U, Q_E . For one particulate component, specify fresh, final-reactor, and underflow concentrations of 200, 1,000, and 2,000 g m^{-3} . The mixer calculation is

$$q_P m_f = 200 + 2(1,000) + 0.5(2,000) = 3,200, \quad (6.10)$$

$$m_f = 3,200/3.5 = 914.29 \text{ g m}^{-3}. \quad (6.11)$$

The physical incoming mass rates are 2,000, 20,000, and 10,000 kg d^{-1} . Their sum is the 32,000 kg d^{-1} leaving the mixer. The mixer concentration exceeds the fresh concentration because the two return streams contain concentrated material. These hypothetical streams illustrate the flow-weighted mixer calculation.

The nominal stage HRT is measured on the fresh-flow basis. A volume divided by a flow gives time. Thus V_i/Q_0 has units of days, and multiplying by 24 converts it to hours. The actual flow through the reactor is $Q_P = Q_0 q_P$, so its actual residence time is V_i/Q_P (Alex et al., 2008; Henze et al., 2008). The inverse of that actual time is the dilution rate. With reactor volume V_i ,

$$h_i = \frac{24V_i}{Q_0}, \quad H = \sum_{i=1}^{n_r} h_i, \quad d_i = \frac{Q_0 q_P}{V_i} = \frac{24q_P}{h_i}. \quad (6.12)$$

The factors h_i and H are in hours. The dilution rate d_i is in d^{-1} . The actual residence time during one passage through stage i is h_i/q_P . For a hypothetical plant with $H = 24$ h, $r_I = 2$, and $r_R = 0.5$, the reactor-train flow ratio is 3.5. Its train residence time per passage is about 6.86 h. Equal tanks then have a residence time per passage of about 1.37 h. These illustrative numbers explain why nominal HRT and recycle flow must be interpreted together.

Fixed stage fractions can be represented by $h_i = f_i^V H$, where $f_i^V > 0$ and $\sum_i f_i^V = 1$. The case study uses equal fractions. Varying H at fixed Q_0 varies biological reactor volume. The resulting problem combines capacity selection and operating decisions. For an existing plant with installed tanks, the volumes would be fixed and H would be removed from the free-control vector.

Let ϑ collect only settings that vary freely. A constant setting remains a parameter. Treating it as a varying regression coordinate would create a zero-variance feature. A carbon or chemical addition may also enter the control vector. An addition represented only by a source term has negligible liquid carrier flow. An appreciable carrier flow requires a new hydraulic stream and revised downstream dilution rates.

Biological conversion and non-negativity

The reaction network uses the component and process model of Chapter 3. Its stoichiometric matrix is $\nu \in \mathbb{R}^{P \times F}$ and its process-rate vector in stage i is $\rho_i \in \mathbb{R}_+^P$. For component f , the dynamic component equation is

$$V_i \frac{dc_{i,f}}{dt} = Q_{Pc_{i-1,f}} - Q_{Pc_{i,f}} + V_i \sum_{p=1}^P \nu_{pf} \rho_{i,p} + V_i \omega_i(e_{SO})_f + V_i \xi_{i,f}. \quad (6.13)$$

The reaction sum combines the change caused by each process. A negative ν_{pf} consumes component f , while a positive entry produces it. The oxygen selector is one for the oxygen component and zero elsewhere. At steady state, the accumulation term is zero. Dividing the remaining equation by V_i gives

$$0 = d_i(c_{i-1,f} - c_{i,f}) + \sum_{p=1}^P \nu_{pf} \rho_{i,p} + \omega_i(e_{SO})_f + \xi_{i,f}. \quad (6.14)$$

Putting all F component equations into one vector gives the steady reactor equation, with $c_0 = m$,

$$0 = d_i(c_{i-1} - c_i) + \nu^\top \rho_i(c_i, \vartheta) + k_L a_i(a_i)(S_O^* - c_{i,SO})e_{SO} + \xi_i(\vartheta). \quad (6.15)$$

The vector e_{SO} selects dissolved oxygen. The parameter S_O^* is its saturation concentration. The vector ξ_i represents declared stage additions. Each term has units of component concentration per day. The treatment-model framework of Henze et al. (2000) separates hydraulic transport, stoichiometric conversion, and external sources in this way.

At a zero component concentration, the combined reaction and external-source terms must not direct that component below zero. With non-negative inflow and a locally Lipschitz balance function defined on the boundary, this condition preserves the non-negative state space of the dynamic model (Haddad et al., 2010). Numerical acceptance checks concentrations and reaction rates independently of the solver's stopping message.

The same distinction matters for a surrogate. A fitted quadratic equation can predict a negative concentration even when every training state is non-negative. Polynomial approximation imposes no sign restriction. The work of Kircher and Votsmeier (2025) addresses the simultaneous requirements of mass conservation and non-negativity in kinetic surrogates. Both requirements need explicit treatment.

Stage invariants and the external plant balance

Consider a hypothetical reaction that transfers material from component 1 to component 2 without changing their combined amount. Its stoichiometric and inventory rows are

$$\nu = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}. \quad (6.16)$$

The factor $1/\sqrt{2}$ gives the inventory row unit length. Matrix multiplication shows the cancellation directly,

$$A\nu^\top = \frac{1}{\sqrt{2}} [1(-1) + 1(1)] = 0. \quad (6.17)$$

At an illustrative dilution rate of 2 d^{-1} , take upstream concentrations $(12, 3)$ and downstream concentrations $(8, 7) \text{ g m}^{-3}$, with a reaction rate of $8 \text{ g m}^{-3} \text{ d}^{-1}$. The component equations are

$$0 = 2(12 - 8) - 8, \quad (6.18)$$

$$0 = 2(3 - 7) + 8. \quad (6.19)$$

Adding them removes the unknown reaction rate. Both upstream and downstream totals are 15 g m^{-3} . The normalized invariant is that total divided by $\sqrt{2}$. The full reactor model applies the same cancellation to more components and more processes.

Let $A \in \mathbb{R}^{K \times F}$ contain independent inventory rows such that

$$A\nu^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K. \quad (6.20)$$

The rows are normalized to unit Euclidean length. Unit row normalization does not make them orthogonal. Physically named inventories and an orthonormal basis can span the same subspace defined by mass conservation while using different coordinates. The stoichiometric fixed-layer formulation of Sturm and Wexler (2022) illustrates the value of separating quantities used for mass conservation from the rates that redistribute them.

For inventory row k , multiply component equation f by A_{kf} and sum over f . The reaction contribution becomes

$$\sum_{f=1}^F A_{kf} \sum_{p=1}^P \nu_{pf} \rho_{i,p} = \sum_{p=1}^P \left(\sum_{f=1}^F A_{kf} \nu_{pf} \right) \rho_{i,p} = 0. \quad (6.21)$$

The inner sum is entry (k, p) of $A\nu^\top$. Oxygen transfer also disappears because $Ae_{S_O} = 0$. The remaining scalar equation is

$$d_i \left(\sum_f A_{kf} c_{i-1,f} - \sum_f A_{kf} c_{i,f} \right) + \sum_f A_{kf} \xi_{i,f} = 0. \quad (6.22)$$

Moving the transport difference to the other side and dividing by the positive d_i gives the remaining stage relation,

$$A(c_i - c_{i-1}) = d_i^{-1} A\xi_i(\vartheta). \quad (6.23)$$

If there is no stage addition, $Ac_i = Ac_{i-1}$. For example, nitrification can transfer nitrogen from ammonium to oxidized forms while denitrification transfers it to dissolved nitrogen gas. These changes do not remove nitrogen from the model's complete nitrogen inventory. A reporting quantity that excludes dissolved nitrogen gas can decrease

while the inventory used for mass conservation remains unchanged. Nutrient-form interpretation in U.S. Environmental Protection Agency (2009) supports the need to inspect individual species as well as reported totals.

In the two-component example, adding component 1 at $2 \text{ g m}^{-3} \text{ d}^{-1}$ changes the downstream concentrations to (9, 7) when the same reaction rate is used. The two balances are $2(12 - 9) - 8 + 2 = 0$ and $2(3 - 7) + 8 = 0$. Their total changes by $2/2 = 1 \text{ g m}^{-3}$. Thus the invariant transition must include a dose whenever that dose carries the quantity used for mass conservation. A zero invariant difference would incorrectly remove the added material.

Internal recycle streams also cancel from an external invariant balance. First add the stage transitions. Every intermediate Ac_i appears once with a plus sign and once with a minus sign. This gives

$$A(c_{n_r} - m) = \sum_i \frac{V_i}{Q_0 q_P} A\xi_i, \quad (6.24)$$

$$q_P Ac_{n_r} - q_P Am = D_{\text{add}}, \quad D_{\text{add}} = \sum_i \frac{V_i}{Q_0} A\xi_i. \quad (6.25)$$

Next multiply the mixer balance by A and substitute it for $q_P Am$,

$$q_P Ac_{n_r} = Ac_{\text{in}} + r_I Ac_{n_r} + r_R Ac_U + D_{\text{add}}, \quad (6.26)$$

$$(q_P - r_I) Ac_{n_r} = Ac_{\text{in}} + r_R Ac_U + D_{\text{add}}. \quad (6.27)$$

Since $q_P - r_I = q_C$, the left-hand side is the clarifier-feed inventory flow. The clarifier balance gives a second expression for that same quantity,

$$q_C Ac_{n_r} = q_E Ac_E + (r_R + w) Ac_U. \quad (6.28)$$

Equating the expressions and subtracting $r_R Ac_U$ cancels the return-sludge flow. The mixed-liquor recycle already canceled in $q_P - r_I$. The remaining external relation is

$$Ac_{\text{in}} + \sum_{i=1}^{n_r} \frac{V_i}{Q_0} A\xi_i = q_E Ac_E + w Ac_U. \quad (6.29)$$

Only fresh feed, stage additions, effluent, and waste sludge remain. The recycle ratios still influence the state and treatment processes. Their cancellation means that they are internal transfers rather than external sources or sinks.

For ASM2d-TSN, $F = 20$, $P = 28$, and $K = 5$. The five inventories concern soluble inert material, total nitrogen including dissolved nitrogen gas, total phosphorus, an alkalinity-related combination, and a metal-related combination. The last two unnormalized combinations are

$$I_{\text{alk}} = S_{\text{ALK}} - \frac{S_{\text{NH4}}}{14} + \frac{S_{\text{NO2}}}{14} + \frac{S_{\text{NO3}}}{14} - \frac{S_{\text{PO4}}}{31}, \quad (6.30)$$

$$I_{\text{metal}} = 3.45 X_{\text{MeP}} + 4.87 X_{\text{MeOH}}. \quad (6.31)$$

The nitrogen and phosphorus rows use the constituent weights in Table 3.1. The nitrogen inventory also gives dissolved nitrogen gas unit weight. The stoichiometric rank is 15 under the stated parameterization. The inventory identities are audited rather than inferred from a reporting label such as TN.

6.2.2 Clarification and solids storage

A nonreactive separator with two outlet flows

The clarifier uses a nonreactive settling model. Nonreactive means that the adopted clarifier equations transport and separate components without biological or chemical conversion within the unit. Soluble components follow the liquid. Particulate components can settle and concentrate in the underflow. Takács et al. (1991) provide the settling and thickening formulation used for the aggregate solids profile. Jeppsson and Diehl (1996) explain why component propagation through a clarifier requires attention beyond a single aggregate solids concentration.

Selection means taking designated entries of a vector without changing them. For the illustrative order $(S_A, S_{NH_4}, X_H, X_I)$, the two selector matrices are

$$E_S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad E_X = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (6.32)$$

Thus $E_S c = (S_A, S_{NH_4})^\top$ and $E_X c = (X_H, X_I)^\top$. Transposing a selector puts selected entries back into their original positions and fills the other positions with zeros. In this example,

$$E_S^\top E_S = \text{diag}(1, 1, 0, 0), \quad E_X^\top E_X = \text{diag}(0, 0, 1, 1). \quad (6.33)$$

Their sum is the identity matrix because every component belongs to exactly one group. Their cross-product is zero because the groups do not overlap. For the complete ordered model, $E_S = [I_{10} \ 0]$ and $E_X = [0 \ I_{10}]$. In general, let $E_S \in \mathbb{R}^{F_S \times F}$ and $E_X \in \mathbb{R}^{F_X \times F}$ select soluble and particulate components. They satisfy

$$E_S^\top E_S + E_X^\top E_X = I_F, \quad E_S E_X^\top = 0. \quad (6.34)$$

The column t_X maps component concentrations to TSS. Its entries are non-negative and its soluble entries are zero.

For the four-component order above, $t_X = (0, 0, 0.90, 0.75)^\top$. If $X_H = 100$ and $X_I = 40$ g COD m⁻³, the TSS calculation is $0.90(100) + 0.75(40) = 120$ g TSS m⁻³. The two factors convert component COD bases to suspended-solids mass. Adding the native numerical concentrations without these conversion factors would give a different quantity.

For one component, mass conservation across the clarifier gives

$$Q_C c_{nr,f} = Q_{EC} c_{E,f} + Q_{UC} c_{U,f}. \quad (6.35)$$

The statistical response uses component flows divided by Q_0 ,

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (6.36)$$

Their units are the same as concentration because the flow ratios are dimensionless. The actual component discharge rates are $Q_0 g_E$ and $Q_0 g_U$. Writing balances in these coordinates avoids dividing by a small underflow ratio inside the equality system. The component balance and soluble pass-through are

$$g_E + g_U = q_C c_{n_r}, \quad E_S g_U = q_U E_S c_{n_r}. \quad (6.37)$$

The two relations imply $E_S g_E = q_E E_S c_{n_r}$. Settling therefore cannot remove a soluble nutrient directly in this model.

For a soluble component f , pass-through fixes $c_{U,f} = c_{n_r,f}$. Substitute $g_{U,f} = q_U c_{n_r,f}$ into the total balance to obtain

$$g_{E,f} = (q_C - q_U) c_{n_r,f} = q_E c_{n_r,f}, \quad (6.38)$$

$$c_{E,f} = g_{E,f} / q_E = c_{n_r,f}. \quad (6.39)$$

For a hypothetical feed concentration of 20 g m^{-3} and flow ratios $q_C = 1.5$, $q_U = 0.52$, and $q_E = 0.98$, the feed, underflow, and overflow component flows are 30, 10.4, and 19.6 g m^{-3} on the fresh-flow basis. Both outlet concentrations remain 20 g m^{-3} .

The particulate densification requirement is

$$E_X g_U \geq q_U E_X c_{n_r}. \quad (6.40)$$

After division by q_U , each underflow particulate concentration is at least its feed concentration. Non-negative overflow and the total component balance also prevent the underflow from recovering more than the incoming particulate flow.

For an illustrative particulate component, take $q_C = 1.5$, $q_U = 0.52$, and feed concentration 100 g m^{-3} . Densification requires $g_U \geq 52 \text{ g m}^{-3}$ on the fresh-flow basis. Mass conservation and non-negative overflow require $g_U \leq 150 \text{ g m}^{-3}$. Thus the underflow recovery fraction lies between $0.52/1.5$ and one. A soluble component with the same feed concentration instead has $g_U = 52$ and unchanged soluble outlet concentrations. These hypothetical examples describe the soluble and particulate constraints.

An admissible illustrative choice within that particulate interval is $g_U = 140$. Mass conservation then gives $g_E = 150 - 140 = 10$. Dividing by the respective flow ratios gives $c_U = 140/0.52 = 269.23$ and $c_E = 10/0.98 = 10.20 \text{ g m}^{-3}$. The underflow has recovered $140/150 = 93.33\%$ of the incoming component flow. The concentrations alone do not give that recovery fraction because the outlet water flows differ.

For any particulate component with positive feed concentration, the same reasoning gives

$$\frac{q_U}{q_C} \leq \frac{g_{U,f}}{q_C c_{n_r,f}} \leq 1. \quad (6.41)$$

Multiplication by the non-negative TSS weights preserves inequalities. Hence $X_U \geq X_N$. The solids balance $q_C X_N = q_E X_E + q_U X_U$ then gives

$$q_E X_E \leq (q_C - q_U) X_N = q_E X_N. \quad (6.42)$$

Division by $q_E > 0$ establishes the other endpoint ordering.

Applying t_X gives feed and outlet solids concentrations

$$X_N = t_X^\top c_{n_r}, \quad X_E = \frac{t_X^\top g_E}{q_E}, \quad X_U = \frac{t_X^\top g_U}{q_U}. \quad (6.43)$$

The declared constraints imply $X_E \leq X_N \leq X_U$. They bound separation but do not determine its magnitude. The nonlinear settling calculation and an empirical endpoint model supply additional information through different routes.

The layer-resolved mechanistic model

Clarifier layers are numbered from the surface to the hopper. Their TSS concentrations are $s_1, \dots, s_L$, with $s_1 = X_E$ and $s_L = X_U$. The exact settling velocity at concentration s is

$$v(s, X_N) = \max \left\{ 0, \min \left[v_{\max}, v_0 \left(e^{-r_h(s - f_{\text{ns}} X_N)} - e^{-r_p(s - f_{\text{ns}} X_N)} \right) \right] \right\}. \quad (6.44)$$

The gravitational solids flux is $sv(s, X_N)$. The nonsettleable fraction f_{ns} makes the settling offset depend on the feed concentration. At an interface, a receiver concentration above X_t activates a limitation by the smaller adjacent gravitational flux. Otherwise, the upper-layer gravitational flux passes to the receiver (Takács et al., 1991). Hydraulic transport uses the overflow velocity above the feed layer and the underflow velocity below it.

Each finite-volume layer equation is its incoming solids flux minus its outgoing solids flux, multiplied by $A_{\text{cl}}/V_{\text{cl},\ell}$. The feed layer also receives $Q_0 q_C X_N / V_{\text{cl},\ell_F}$. Internal interface fluxes cancel when these equations are summed. At steady state, the resulting total solids balance agrees with Equation (6.37) after application of t_X .

A three-layer illustration shows that cancellation without hiding it inside a summation. Place the feed in layer 2. Let B_{12} be the net solids rate from layer 2 toward layer 1 and B_{23} the net rate from layer 2 toward layer 3. These signed rates combine hydraulic and gravitational transport in g d^{-1} . The layer mass equations are

$$V_1 \dot{s}_1 = B_{12} - Q_E s_1, \quad (6.45)$$

$$V_2 \dot{s}_2 = Q_C X_N - B_{12} - B_{23}, \quad (6.46)$$

$$V_3 \dot{s}_3 = B_{23} - Q_U s_3. \quad (6.47)$$

Adding them removes both interface rates. At steady state, the remaining equation is $Q_C X_N = Q_E X_E + Q_U X_U$. This cancellation holds even though the interface rates

depend nonlinearly on the neighboring concentrations. Mass conservation of the whole clarifier therefore does not determine its internal fluxes or profile.

The mechanistic state retains one aggregate solids concentration per layer. At a steady state, particulate outlet components are reconstructed with the final-reactor particulate composition,

$$E_X c_E = \frac{s_1}{X_N} E_X c_{n_r}, \quad E_X c_U = \frac{s_L}{X_N} E_X c_{n_r}. \quad (6.48)$$

Soluble outlet components equal their feed concentrations. The steady component mapping is defined for $X_N > 0$ and uses the final-reactor particulate composition (Jeppsson & Diehl, 1996).

An exact interval for aggregate inventory under the endpoint envelope

Inventory is concentration multiplied by volume, summed over the vessel. For an illustrative four-layer clarifier, it is

$$M = V_1 X_E + V_2 s_2 + V_3 s_3 + V_4 X_U. \quad (6.49)$$

For example, four 100-m³ layers with concentrations (10, 200, 600, 1,000) g m⁻³ store 100(10 + 200 + 600 + 1,000) = 181,000 g, or 181 kg. The general clarifier's stored solids mass is

$$M_{\text{cl}} = \sum_{\ell=1}^L V_{\text{cl},\ell} s_{\ell}, \quad V_{\text{cl}} = \sum_{\ell=1}^L V_{\text{cl},\ell}. \quad (6.50)$$

An outlet concentration cannot determine this mass. Two clarifiers can have the same endpoints and different internal storage. Whole-plant solids retention time therefore requires the inventory as well as the two outlet solids flows. The benchmark formulation of Alex et al. (2008) includes reactor and settler inventories in its sludge-age calculation.

The mechanistic acceptance envelope requires

$$X_E \leq s_{\ell} \leq X_U, \quad \ell = 2, \dots, L-1. \quad (6.51)$$

This is an endpoint envelope. It does not require monotonic concentration from one layer to the next. With positive layer volumes, minimizing the inventory assigns every internal layer X_E . Maximizing it assigns every internal layer X_U .

For four layers, the envelope bounds each of the two unknown terms separately,

$$V_2 X_E + V_3 X_E \leq V_2 s_2 + V_3 s_3 \quad (6.52)$$

$$\leq V_2 X_U + V_3 X_U. \quad (6.53)$$

Adding the known endpoint terms gives

$$(V_1 + V_2 + V_3) X_E + V_4 X_U \leq M, \quad (6.54)$$

$$M \leq V_1 X_E + (V_2 + V_3 + V_4) X_U. \quad (6.55)$$

For the hypothetical four equal layers above, these values are 103,000 and 301,000 g. The 181,000-g inventory lies inside the interval. If both internal concentrations were instead 400 g m^{-3} , the mass would still be 181,000 g. Thus matching endpoints and total inventory does not identify the internal distribution.

Figure 6.2 shows the same four-layer arithmetic as concentration bars. Changing the two internal layers changes stored mass while the surface and hopper concentrations remain fixed. The left and right panels attain the lower and upper interval endpoints derived from the envelope.

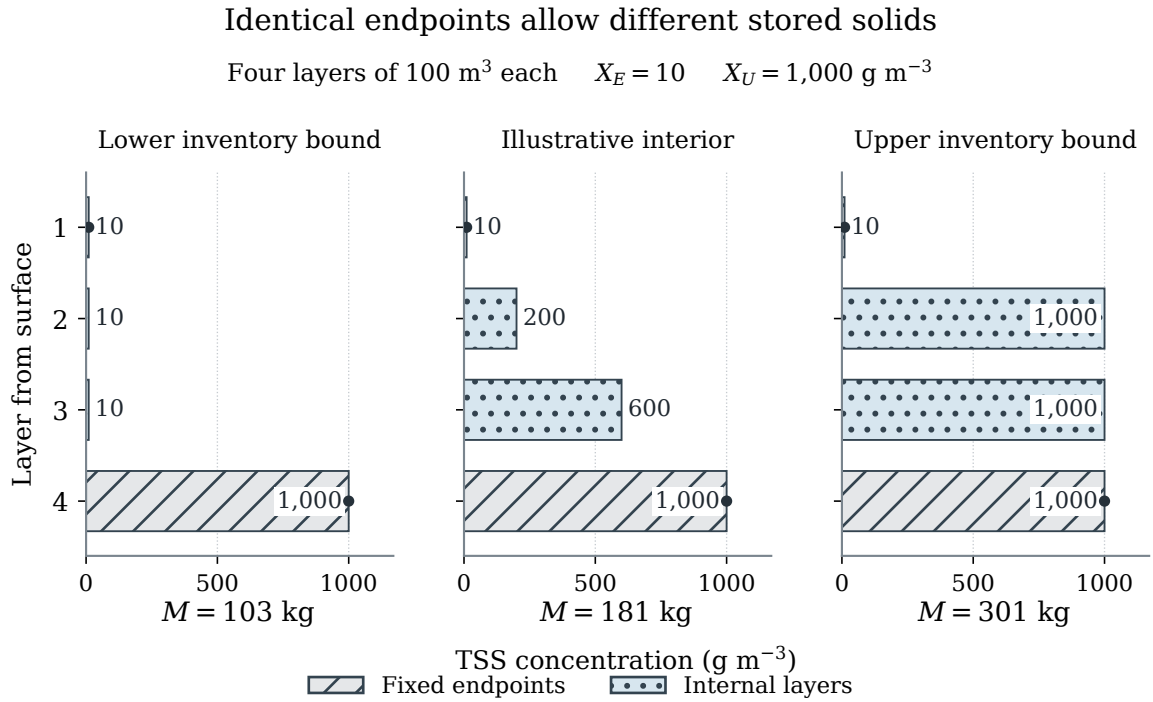


Figure 6.2: Hypothetical four-layer profiles with identical overflow and underflow concentrations. Each layer has volume 100 m^3 . The lower-bound, illustrative interior, and upper-bound profiles store 103, 181, and 301 kg of solids. Hatching identifies the fixed endpoints, while dotted fills identify internal layers. The diagram represents the adopted endpoint envelope.

The internal layers in the middle panel need not resemble those of another profile with the same inventory. Their total mass is the retained scalar information. The nonlinear layer equations supply the additional information needed to determine a mechanistic profile.

With L layers, the identical argument replaces $V_2 + V_3$ by $\sum_{\ell=2}^{L-1} V_{\text{cl},\ell}$. That internal-volume sum is $V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L}$. Collecting coefficients of each endpoint gives the general

interval,

$$M_{\text{cl}}^- = (V_{\text{cl}} - V_{\text{cl},L})X_E + V_{\text{cl},L}X_U, \quad (6.56)$$

$$M_{\text{cl}}^+ = V_{\text{cl},1}X_E + (V_{\text{cl}} - V_{\text{cl},1})X_U, \quad (6.57)$$

$$M_{\text{cl}}^- \leq M_{\text{cl}} \leq M_{\text{cl}}^+. \quad (6.58)$$

These bounds are necessary and sufficient for a profile with the prescribed endpoints and envelope to have the stated inventory. To see sufficiency, give every internal layer a common concentration s_* between X_E and X_U . The inventory is then

$$V_{\text{cl},1}X_E + V_{\text{cl},L}X_U + (V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L})s_*. \quad (6.59)$$

This expression moves continuously through the full interval as s_* varies. For equal-volume layers, it also gives a simple constructive profile for any admitted inventory.

As an illustrative calculation, take ten layers of 600 m^3 , $X_E = 10 \text{ g m}^{-3}$, and $X_U = 5,000 \text{ g m}^{-3}$. The allowable inventory interval is 3,054,000–27,006,000 g. Each inventory within this interval has an envelope-compatible profile. Joint projection enforces the interval, and layer-resolved mechanistic evaluation checks the flux equations.

For these ten equal layers, the calculation can be written out as

$$M_{\text{cl}} = 600(X_E + s_2 + s_3 + \cdots + s_9 + X_U), \quad (6.60)$$

$$M_{\text{cl}}^- = 600(9X_E + X_U), \quad (6.61)$$

$$M_{\text{cl}}^+ = 600(X_E + 9X_U). \quad (6.62)$$

The lower endpoint has nine layers at X_E and one at X_U . The upper endpoint has one layer at X_E and nine at X_U . This explains both coefficients in the compact interval rather than treating them as fitted constants.

6.2.3 Statistical prediction of the connected response

A response that preserves network and storage information

For a teaching example with one reactor and two components, denote a soluble component by S and a particulate component by X . The stored response is the explicit nine-entry column

$$\chi_{\text{toy}} = (m_S, m_X, c_{1,S}, c_{1,X}, g_{E,S}, g_{E,X}, g_{U,S}, g_{U,X}, M_{\text{cl}})^\top. \quad (6.63)$$

Adding a second reactor inserts $c_{2,S}$ and $c_{2,X}$ before the outlet coordinates and increases the length to eleven. Every reactor contributes F entries. The mixer and two outlets each contribute another F , while the clarifier inventory contributes one. The connected surrogate therefore predicts

$$\chi = \text{col}(m, c_1, \dots, c_{n_r}, g_E, g_U, M_{\text{cl}}) \in \mathbb{R}^{n_\chi}, \quad n_\chi = (n_r + 3)F + 1. \quad (6.64)$$

Here `col` stacks its arguments into one column. The response preserves concentrations needed for stage inventories, outlet flows needed for material balances, and clarifier storage needed for solids retention time. It omits the internal layer profile. The full mechanistic state is

$$y = \text{col}(c_1, \dots, c_{n_r}, s_1, \dots, s_L) \in \mathbb{R}_+^{n_y}, \quad n_y = n_r F + L. \quad (6.65)$$

The mapping from y to χ reconstructs the mixer, both outlets, and inventory deterministically. This mapping lets two different response dimensions support the same engineering calculations.

For the five-reactor case, the response count is $20 + 5(20) + 20 + 20 + 1 = 161$. The full-state count is $5(20) + 10 = 110$. Table 6.1 makes the response ordering explicit. A position identifies a coordinate and does not imply that neighboring entries have the same physical unit.

Table 6.1: Blocks in the 161-coordinate connected response.

Block	Positions	Coordinates
Mixer	1–20	$m_1, \dots, m_{20}$
Reactor 1	21–40	$c_{1,1}, \dots, c_{1,20}$
Reactor 2	41–60	$c_{2,1}, \dots, c_{2,20}$
Reactor 3	61–80	$c_{3,1}, \dots, c_{3,20}$
Reactor 4	81–100	$c_{4,1}, \dots, c_{4,20}$
Reactor 5	101–120	$c_{5,1}, \dots, c_{5,20}$
Overflow flows	121–140	$g_{E,1}, \dots, g_{E,20}$
Underflow flows	141–160	$g_{U,1}, \dots, g_{U,20}$
Clarifier inventory	161	M_{cl}

Complete second-order features and ridge estimation

The statistical input includes every free control and every fresh-influent component. Standardization first subtracts a fitting-set mean and then divides by a fitting-set standard deviation. For one operating coordinate and one influent coordinate,

$$z_1 = \frac{\vartheta_1 - \bar{\vartheta}_1}{s_{\vartheta_1}}, \quad z_2 = \frac{c_{in,1} - \bar{c}_{in,1}}{s_{in,1}}. \quad (6.66)$$

These are dimensionless coordinates. For hypothetical aeration 0.8, fitting mean 0.5, and fitting scale 0.1, the standardized value is 3. For hypothetical ammonium concentration 40 g m^{-3} with mean 30 and scale 10 g m^{-3} , the value is 1. The different native units no longer determine the regression penalty. For fitting-set centers and positive diagonal scales, define

$$z = \text{col} \left[D_{\vartheta, \text{fit}}^{-1}(\vartheta - \bar{\vartheta}), D_{in, \text{fit}}^{-1}(c_{in} - \bar{c}_{in}) \right] \in \mathbb{R}^{d_z}, \quad d_z = d_\vartheta + F. \quad (6.67)$$

For two standardized inputs, their symmetric product matrix is

$$zz^\top = \begin{bmatrix} z_1^2 & z_1 z_2 \\ z_2 z_1 & z_2^2 \end{bmatrix}. \quad (6.68)$$

Because multiplication commutes, $z_1 z_2$ and $z_2 z_1$ are the same feature. One triangular copy therefore gives the unique quadratic entries $(z_1^2, z_1 z_2, z_2^2)$. Including the linear entries and intercept produces six terms,

$$(1, z_1, z_2, z_1^2, z_1 z_2, z_2^2)^\top. \quad (6.69)$$

Three inputs would give an intercept, three linear terms, three squares, and three distinct cross-products, for ten features. In general there are d_z squares and $d_z(d_z - 1)/2$ cross-products. Their sum is $d_z(d_z + 1)/2$. The complete quadratic feature family is

$$r(z) = \text{col} \left[z, \text{vech}(zz^\top) \right], \quad (6.70)$$

$$\phi(\vartheta, c_{\text{in}}) = \text{col} \left[1, D_{r, \text{fit}}^{-1}(r(z) - \bar{r}) \right], \quad (6.71)$$

$$n_\phi = 1 + d_z + \frac{d_z(d_z + 1)}{2}. \quad (6.72)$$

The operator vech retains one triangular copy of the symmetric matrix. It includes every square and every pairwise product once. A product between ammonium loading and aeration allows their joint effect to differ from the sum of their separate effects. A square of the recycle coordinate permits local curvature (Nazlabadi et al., 2021; Selerio, 2026). These terms approximate the input-response relation.

Quadratic features undergo their own fitting-set centering and scaling after they are constructed. For the hypothetical standardized inputs $(3, 1)$, the unscaled nonconstant terms are $(3, 1, 9, 3, 1)$. If the first squared feature has fitting mean 1 and standard deviation 2, its final value is $(9 - 1)/2 = 4$. If the cross-product has fitting mean 0.2 and standard deviation 0.5, its final value is $(3 - 0.2)/0.5 = 5.6$. This second standardization prevents squares with large variance from receiving a weaker effective ridge penalty merely because they are numerically larger (Hastie et al., 2009).

A monomial is a single polynomial term, such as z_1^2 or $z_1 z_2$. The feature set containing each distinct term once spans the same quadratic functions as an ordered feature set that retains both $z_1 z_2$ and $z_2 z_1$. Their two coefficients enter prediction only through their sum. For an effective cross-product coefficient b , an ordered feature set can assign $b/2$ to each duplicate. The squared coefficient penalty is then $b^2/2$, compared with b^2 for one unique coefficient. Thus removing duplicates preserves the polynomial family but changes the regularization convention unless the penalty is adjusted. The connected fit selects its penalty for the stated feature set containing each distinct term once. For 27 inputs, the count is $1 + 27 + 378 = 406$.

For response coordinate k , write its standardized target as $\tilde{\chi}_{ik} = (\chi_{ik} - \bar{\chi}_k)/s_{\chi, k}$.

Its fitted value is $C_{k0} + \sum_{j=1}^{n_\phi-1} C_{kj}\phi_{ij}$. The scalar ridge objective for all responses is

$$\frac{1}{nn_\chi} \sum_{i=1}^n \sum_{k=1}^{n_\chi} \left(\tilde{\chi}_{ik} - \sum_{j=0}^{n_\phi-1} C_{kj}\phi_{ij} \right)^2 + \frac{\gamma}{n_\chi} \sum_{k=1}^{n_\chi} \sum_{j=1}^{n_\phi-1} C_{kj}^2. \quad (6.73)$$

The index $j = 0$ is the intercept, with feature value one. It appears in prediction and does not appear in the penalty sum. Dividing a physical prediction error by $s_{\chi,k}$ is equivalent to weighting its squared error by $1/s_{\chi,k}^2$. Response scaling therefore gives each coordinate comparable importance in the joint statistical score.

For example, an illustrative concentration error of 10 g m^{-3} with a response scale of 20 g m^{-3} contributes $0.5^2 = 0.25$ before averaging. An inventory error of 10,000 g with a scale of 200,000 g contributes $0.05^2 = 0.0025$. Comparing the unscaled numbers would give the opposite impression. The scales define what counts as a large error.

Let $\Phi_{\mathcal{I}}$ contain the feature rows for fitting indices $\mathcal{I}$ and let $\tilde{Y}_{\mathcal{I}}$ contain standardized joint responses. With $R_\phi = \text{diag}(0, 1, \dots, 1)$, the coefficient matrix solves

$$\hat{C}_\gamma = \arg \min_C \left\{ \frac{\|\tilde{Y}_{\mathcal{I}} - \Phi_{\mathcal{I}} C^\top\|_F^2}{|\mathcal{I}|n_\chi} + \frac{\gamma}{n_\chi} \|C R_\phi\|_F^2 \right\}. \quad (6.74)$$

The intercept is unpenalized. A positive γ regularizes correlated quadratic features. Hastie et al. (2009) explain this stabilization and the corresponding trade-off between model flexibility and coefficient shrinkage. The raw prediction is

$$\chi_{\text{raw}} = \bar{\chi} + D_\chi \hat{C}_\gamma \phi(\vartheta, c_{\text{in}}). \quad (6.75)$$

The response scale D_χ is fixed from fitting data. It places the inventory, component concentrations, and outlet flows on comparable statistical scales. Joint projection then applies the physical constraints to the raw prediction.

Differentiating the scalar ridge objective for one response column with coefficient vector β gives its normal equations,

$$(\Phi^\top \Phi + n\gamma R_\phi) \hat{\beta} = \Phi^\top \tilde{\chi}. \quad (6.76)$$

The term $n\gamma R_\phi$ adds positive curvature to the nonintercept coefficient directions. In a hypothetical one-feature teaching problem, take three feature and target values both equal to $(-\sqrt{3/2}, 0, \sqrt{3/2})$. Their means are zero and their population standard deviations are one. Symmetry gives a zero intercept. With slope b , the objective becomes $(1-b)^2 + \gamma b^2$. Differentiation gives $2(b-1) + 2\gamma b = 0$, so $\hat{b} = 1/(1+\gamma)$. For $\gamma = 1$, the fitted slope is 0.5 rather than the unpenalized value of one. This hypothetical calculation shows coefficient shrinkage.

Finally, the physical prediction is recovered coordinate by coordinate as $\chi_{\text{raw},k} = \bar{\chi}_k + s_{\chi,k} \sum_j \hat{C}_{kj} \phi_j$. A hypothetical mean of 100 g m^{-3} , scale of 20 g m^{-3} , and standardized prediction of 0.5 give 110 g m^{-3} . This return to physical units is required before imposing component balances.

The quadratic feature family is retained from ICSOR (Selerio, 2026). Its coefficients describe the raw statistical relation. If projection is active, a raw coefficient alone does not give the derivative of the final projected response. That derivative also depends on the constraints, their control dependence, and the active inequalities.

A separate positive model for overflow solids

Low overflow TSS calls for a different error scale from large reactor solids concentrations. An absolute error of 1 g m^{-3} is 1% of 100 g m^{-3} and 100% of 1 g m^{-3} . A joint standardized response score may not adequately describe this distinction at the discharge point. The relative-error argument of Tofallis (2015) motivates a separate logarithmic model.

For positive mechanistic targets, define

$$\ell_E = \log(X_E/X_\star), \quad X_\star = 1 \text{ mg L}^{-1}. \quad (6.77)$$

The ratio inside the logarithm is dimensionless. The scalar ridge fit uses the same complete quadratic features,

$$\hat{\beta}_{E,\gamma_E} = \arg \min_{\beta} \left\{ \frac{\|\ell_{E,\mathcal{I}} - \Phi_{\mathcal{I}}\beta\|_2^2}{|\mathcal{I}|} + \gamma_E \|R_{\phi}\beta\|_2^2 \right\}. \quad (6.78)$$

Its concentration-scale prediction is

$$\widehat{X}_{E,\log} = X_\star \exp(\hat{\beta}_{E,\gamma_E}^\top \phi) > 0. \quad (6.79)$$

A twofold concentration error has the same logarithmic discrepancy at low and high concentrations. Exponentiation gives a positive statistical estimate that enters the joint mass conservation projection.

For hypothetical targets of 2 and 8 mg L^{-1} , the dimensionless logarithms are $\log 2$ and $\log 8$. If a single fitted log value represents both equally, squared log error is minimized at

$$\frac{\log 2 + \log 8}{2} = \log 4. \quad (6.80)$$

Back-transformation gives $1 \times \exp(\log 4) = 4 \text{ mg L}^{-1}$. The arithmetic mean is 5 mg L^{-1} , which explains why exponentiation is not an unbiased concentration-scale mean adjustment. A fitted log value of -2 instead gives 0.1353 mg L^{-1} . It remains positive even though it is below the reference concentration $X_\star$. The reference establishes units and does not impose a lower bound.

Under squared logarithmic error, direct exponentiation represents a conditional geometric center rather than an unbiased concentration-scale mean (Tofallis, 2015). No concentration-scale bias adjustment is applied. The separate model also avoids imposing a constant solids-removal fraction across all influents and controls. Such a fraction would disregard variation in feed composition, hydraulics, recycle, and biological state.

Its fitted endpoint supplies the empirical closure introduced in Chapter 2. The physical balances leave the division of solids between the two outlets undetermined. The fitted overflow solids concentration supplies the additional relation used in joint projection. It remains a statistical estimate rather than a mass conservation law.

6.2.4 Joint projection onto the declared plant constraints

Equalities and inequalities at fixed controls

The projection adjusts every connected response block together. The required matrix rows can first be assembled for a hypothetical one-reactor, two-component plant. Use the nine-coordinate order already given for χ_{toy} . Let $r_I = 1$, $r_R = 0.5$, and $w = 0.1$, so $q_P = 2.5$, $q_C = 1.5$, $q_U = 0.6$, and $q_E = 0.9$. These hypothetical values illustrate assembly of the constraint rows. The soluble component is S and the particulate component X has unit TSS weight. Suppose their sum is fixed by mass conservation in the reactor and the fresh concentrations are $(30, 20)$ g m⁻³.

The two mixer rows, one inventory row, two clarifier rows, soluble pass-through, and fitted endpoint are then

$$2.5m_S - c_{1,S} - \frac{5}{6}g_{U,S} = 30, \quad (6.81)$$

$$2.5m_X - c_{1,X} - \frac{5}{6}g_{U,X} = 20, \quad (6.82)$$

$$c_{1,S} + c_{1,X} - m_S - m_X = 0, \quad (6.83)$$

$$g_{E,S} + g_{U,S} - 1.5c_{1,S} = 0, \quad (6.84)$$

$$g_{E,X} + g_{U,X} - 1.5c_{1,X} = 0, \quad (6.85)$$

$$g_{U,S} - 0.6c_{1,S} = 0, \quad (6.86)$$

$$g_{E,X} = 0.9(10) = 9. \quad (6.87)$$

The last row illustrates a fitted overflow concentration of 10 g m⁻³. The inventory row is written without its harmless unit-length factor to keep the numbers clear. Multiplying a zero equality by a positive factor does not change its solutions. In the same coordinate order, the explicit equality matrix is

$$\begin{bmatrix} 2.5 & 0 & -1 & 0 & 0 & 0 & -5/6 & 0 & 0 \\ 0 & 2.5 & 0 & -1 & 0 & 0 & 0 & -5/6 & 0 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1.5 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1.5 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -0.6 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \chi_{\text{toy}} = \begin{bmatrix} 30 \\ 20 \\ 0 \\ 0 \\ 0 \\ 0 \\ 9 \end{bmatrix}. \quad (6.88)$$

Every row expresses one previously derived relation. A zero entry means that the coordinate is absent from that relation. The inventory coordinate has a zero column because these equalities do not determine stored solids.

The illustrative response

$$\chi_{\text{toy}} = (30, 50, 30, 50, 27, 9, 18, 66, 18, 000)^\top \quad (6.89)$$

satisfies all seven rows. For example, its particulate mixer row is $2.5(50) - 50 - (5/6)(66) = 20$. Its clarifier particulate row is $9 + 66 = 1.5(50)$. Underflow concentration is $66/0.6 = 110 \text{ g m}^{-3}$. These arithmetic checks verify the linear constraints.

The full physical equality operator $\mathcal{H}(\vartheta)\chi = b(\vartheta, c_{\text{in}})$ collects

$$\text{col} \begin{bmatrix} q_P m - r_I c_{n_r} - (r_R/q_U)g_U - c_{\text{in}} \\ \{A(c_i - c_{i-1}) - d_i^{-1} A \xi_i\}_{i=1}^{n_r} \\ g_E + g_U - q_C c_{n_r} \\ E_S g_U - q_U E_S c_{n_r} \end{bmatrix} = 0. \quad (6.90)$$

The empirical overflow relation is appended as

$$t_X^\top g_E = q_E \widehat{X}_{E, \log}(\vartheta, c_{\text{in}}). \quad (6.91)$$

The complete operator is denoted $\mathcal{H}_{\log}$ with right-hand side $b_{\log}$. It has $2F + n_r K + F_S + 1$ rows. The last row fixes total overflow TSS. It does not prescribe its particulate composition.

The remaining teaching-plant requirements are also individual matrix rows. With three 100-m^3 layers, $V_{\text{cl}} = 300 \text{ m}^3$ and $V_1 = V_3 = 100 \text{ m}^3$. Densification is $0.6c_{1,X} - g_{U,X} \leq 0$. The inventory inequalities follow by substituting $X_E = g_{E,X}/0.9$ and $X_U = g_{U,X}/0.6$ into the endpoint bounds and multiplying by $0.9(0.6) = 0.54$,

$$120g_{E,X} + 90g_{U,X} - 0.54M_{\text{cl}} \leq 0, \quad (6.92)$$

$$-60g_{E,X} - 180g_{U,X} + 0.54M_{\text{cl}} \leq 0. \quad (6.93)$$

The lower row has coefficient $120 = 0.6(300 - 100)$ and coefficient $90 = 0.9(100)$. These are hydraulic and volume factors. They are not regression coefficients. At the illustrative response above, densification gives $30 - 66 = -36$, while both inventory rows give $-2,700$. The corresponding inventory interval is $13,000\text{--}23,000 \text{ g}$, which contains $18,000 \text{ g}$.

The three inequality rows can also be displayed in the response order,

$$\begin{bmatrix} 0 & 0 & 0 & 0.6 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 120 & 0 & 90 & -0.54 \\ 0 & 0 & 0 & 0 & 0 & -60 & 0 & -180 & 0.54 \end{bmatrix} \chi_{\text{toy}} \leq 0. \quad (6.94)$$

Non-negativity would add one row $-\chi_{\text{toy},k} \leq 0$ for each of the nine coordinates. The inequality matrix has the same column order as the equality matrix. Changing that order in one operator would constrain the wrong physical quantities.

For any plant size, collecting the same row patterns gives the physical inequality operator,

$$\mathcal{G}(\vartheta)\chi = \text{col} \begin{bmatrix} q_U E_X c_{n_r} - E_X g_U \\ q_U (V_{\text{cl}} - V_{\text{cl},L}) t_X^\top g_E + q_E V_{\text{cl},L} t_X^\top g_U - q_E q_U M_{\text{cl}} \\ q_E q_U M_{\text{cl}} - q_U V_{\text{cl},1} t_X^\top g_E - q_E (V_{\text{cl}} - V_{\text{cl},1}) t_X^\top g_U \end{bmatrix} \leq 0. \quad (6.95)$$

The first block enforces particulate densification. The last two rows are the inventory bounds multiplied by the positive product $q_E q_U$. This multiplication removes divisions without changing feasibility. Non-negativity requires $\chi \geq 0$ separately.

At fixed $(\vartheta, c_{\text{in}})$, every constraint is affine in χ . This remains true for the overflow closure because its fitted right-hand side is then a known scalar. The endpoint-envelope inequalities also remain affine even though their coefficients depend on the controls. The distinction between fixed-control linearity and control dependence is central to the numerical formulation.

The scaled nearest-response problem

Projection size is first defined for an individual response coordinate. If the raw value is $\chi_{\text{raw},k}$ and its positive scale is s_k , then

$$u_k = \frac{\chi_k - \chi_{\text{raw},k}}{s_k}. \quad (6.96)$$

The scalar projection objective is $\frac{1}{2} \sum_{k=1}^{n_\chi} u_k^2$. Thus a change of two scale units in one component costs $\frac{1}{2}(2)^2 = 2$, while a change of half a scale unit costs $\frac{1}{2}(0.5)^2 = 0.125$. With $D_\chi = \text{diag}(s_1, \dots, s_{n_\chi})$, the response relation becomes $\chi = \chi_{\text{raw}} + D_\chi u$.

For a mixer component row, the substitution is explicit,

$$q_P(m_{\text{raw},f} + s_{m,f}u_{m,f}) - r_I(c_{\text{raw},n_r,f} + s_{n_r,f}u_{n_r,f}) \quad (6.97)$$

$$- \frac{r_R}{q_U}(g_{\text{raw},U,f} + s_{U,f}u_{U,f}) = c_{\text{in},f}. \quad (6.98)$$

The scales multiply unknown projections, while all raw values are known at the current inputs. Moving the raw terms to the right gives one linear equation in u . The other balances are transformed in the same way. Let D_b and D_g be positive diagonal row scales. The projected standardized displacement solves

$$\hat{u} = \arg \min_{u \in \mathbb{R}^{n_\chi}} \frac{1}{2} u^\top u, \quad (6.99)$$

$$\text{subject to } D_b^{-1} \{ \mathcal{H}_{\log}(\vartheta)(\chi_{\text{raw}} + D_\chi u) - b_{\log}(\vartheta, c_{\text{in}}) \} = 0,$$

$$\chi_{\text{raw}} + D_\chi u \geq 0,$$

$$D_g^{-1} \mathcal{G}(\vartheta)(\chi_{\text{raw}} + D_\chi u) \leq 0,$$

$$\hat{\chi} = \chi_{\text{raw}} + D_\chi \hat{u}. \quad (6.100)$$

The identity Hessian defines a strictly convex quadratic program. In physical coordinates, its metric is $D_x^{-\top} D_x^{-1}$. Response scaling determines which physical changes count as large projections. Positive constraint-row scaling changes numerical conditioning while leaving the feasible set unchanged. Sturm and Silva (2025) motivate species-sensitive weighting for projection. Boyd and Vandenberghe (2004) supply the convex optimization principles used for feasibility, uniqueness, and optimality.

The physically constrained surrogate combines the original ICSOR separation of prediction and projection with a connected plant response (Selerio, 2026). The manifold projection developed by Valente et al. (2025) provides related output-projection context. Analytic neural-output constraints in Beucler et al. (2021), flux learning for mass conservation in Sturm and Wexler (2020), and constrained training in Ruckstuhl et al. (2021) provide alternative locations for physical enforcement. Li et al. (2024) include governing-equation residuals in physics-informed unit-operation models, including an activated sludge reactor. A residual penalty makes disagreement costly during fitting. It differs from an equality that each admitted prediction must satisfy. The present plant balances and inventory interval follow from the declared process model.

For a nonempty feasible set, closedness, coercivity, and strict convexity give a unique projected response. Feasibility is checked jointly for the fitted endpoint, mass conservation, non-negativity, and densification. For instance, demanding overflow solids above every feed solids concentration allowed by the other balances contradicts $X_E \leq X_N$. The empirical closure is therefore capable of making an otherwise feasible physical polyhedron empty. An infeasible fitted endpoint is recorded as a failed projection.

Figure 6.3 distinguishes the three inputs to the joint projection. The raw response supplies the starting point for projection. Derived physical relations define balance and envelope requirements. The separate logarithmic prediction supplies a fitted endpoint equality. Its empirical origin remains relevant even though it enters the same equality system as the physical balances.

The projected response minimizes standardized adjustment within the stated response scales and feasible set. The physical rows and fitted endpoint jointly define that set.

An independently auditable optimality certificate

Consider a separate hypothetical two-component projection. The raw prediction is $(-2, 9)$, its coordinate scales are $(1, 2)$, and the required total fixed by mass conservation is 10. The physical-coordinate problem is

$$\min_{c_1, c_2} \frac{1}{2}(c_1 + 2)^2 + \frac{1}{8}(c_2 - 9)^2, \quad c_1 + c_2 = 10, \quad c_1 \geq 0, \quad c_2 \geq 0. \quad (6.101)$$

The second coefficient is $1/(2 \times 2^2)$. A change of one physical unit in component 2 costs less than the same change in component 1 because its scale is larger.

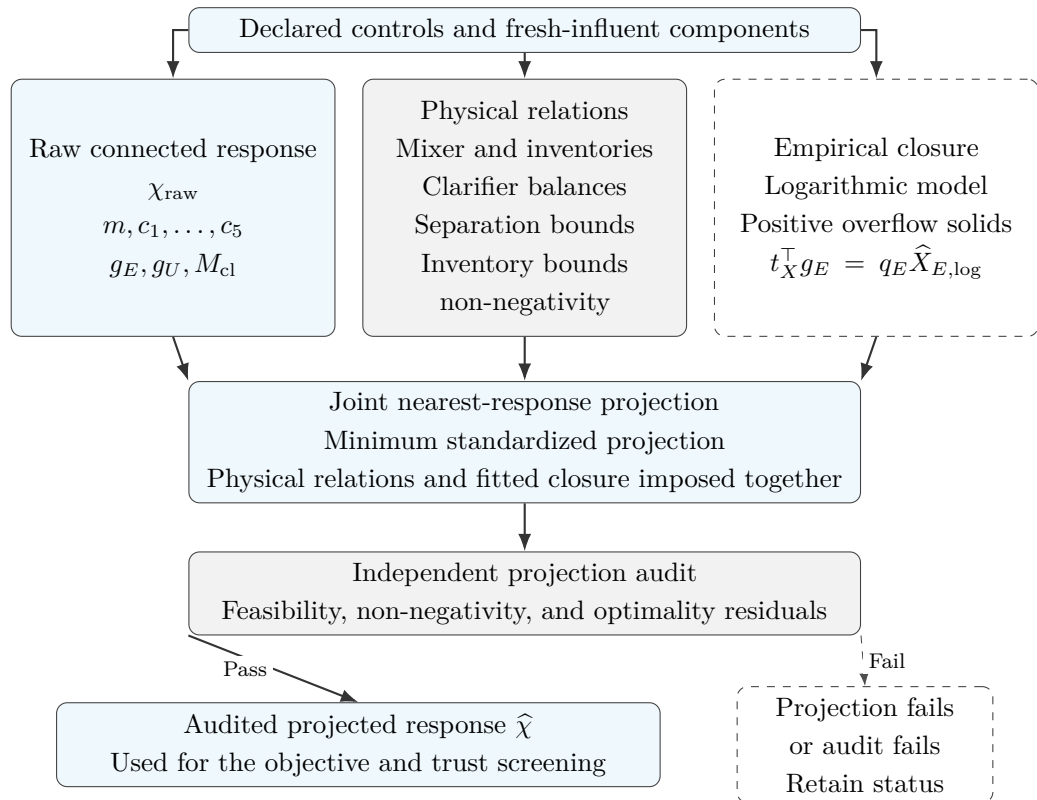


Figure 6.3: Conceptual information flow in the connected projection. The dashed closure block denotes an empirically fitted relation rather than a physical identity. Projection corrects all response blocks jointly. The audit checks the imposed relations at their numerical tolerances.

First omit non-negativity and introduce an equality multiplier λ . Stationarity and the total fixed by mass conservation give

$$c_1 + 2 + \lambda = 0, \quad (6.102)$$

$$\frac{1}{4}(c_2 - 9) + \lambda = 0, \quad (6.103)$$

$$c_1 + c_2 = 10. \quad (6.104)$$

The first two equations imply $c_1 = -2 - \lambda$ and $c_2 = 9 - 4\lambda$. Their sum gives $7 - 5\lambda = 10$, so $\lambda = -0.6$ and $(c_1, c_2) = (-1.4, 11.4)$. Mass conservation alone leaves a negative component.

Now impose the active boundary $c_1 = 0$, which forces $c_2 = 10$. Use multiplier $\mu_1 \geq 0$ for $-c_1 \leq 0$ and $\mu_2 \geq 0$ for $-c_2 \leq 0$. The Lagrangian is the objective plus $\lambda(c_1 + c_2 - 10) - \mu_1 c_1 - \mu_2 c_2$. Since component 2 is positive, its inequality is inactive and $\mu_2 = 0$. Its stationarity equation gives $\lambda = -0.25$. Component 1 then gives $2 - 0.25 - \mu_1 = 0$, so $\mu_1 = 1.75$. Both inequality multipliers are non-negative, and both products $\mu_j c_j$ vanish. These checks certify the boundary solution rather than merely declaring it plausible.

In standardized coordinates, $u = (2, 0.5)^\top$. The same problem has

$$E = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad e = 3, \quad G_u = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad g = \begin{bmatrix} -2 \\ 9 \end{bmatrix}. \quad (6.105)$$

The equality is $u_1 + 2u_2 = 3$. The inequality rows are the two physical non-negativity requirements after substitution. This example explains the standardized form before its use for the larger plant.

At fixed inputs, write the standardized problem as

$$\min_u \frac{1}{2} u^\top u, \quad Eu = e, \quad G_u u \leq g. \quad (6.106)$$

Let $\lambda^=$ be unrestricted equality multipliers and $\lambda^\leq \geq 0$ inequality multipliers. The optimality conditions are primal feasibility, dual feasibility, stationarity, and complementarity,

$$u + E^\top \lambda^= + G_u^\top \lambda^\leq = 0, \quad (6.107)$$

$$\lambda_j^\leq (g_j - (G_u u)_j) = 0 \quad \text{for every } j. \quad (6.108)$$

To derive the dual value, collect the linear terms in the Lagrangian. Define $v = E^\top \lambda^= + G_u^\top \lambda^\leq$. Completing the square gives

$$\frac{1}{2} u^\top u + v^\top u - e^\top \lambda^= - g^\top \lambda^\leq = \frac{1}{2} \|u + v\|_2^2 - \frac{1}{2} \|v\|_2^2 \quad (6.109)$$

$$- e^\top \lambda^= - g^\top \lambda^\leq. \quad (6.110)$$

The first term has minimum zero at $u = -v$. Minimizing over u therefore gives the dual value,

$$d(\lambda^=, \lambda^\leq) = -\frac{1}{2} \|E^\top \lambda^= + G_u^\top \lambda^\leq\|_2^2 - e^\top \lambda^= - g^\top \lambda^\leq. \quad (6.111)$$

For primal- and dual-feasible quantities, the gap is

$$\Gamma = \frac{1}{2}\|u + E^\top \lambda^\leq + G_u^\top \lambda^\leq\|_2^2 + (\lambda^\leq)^\top (g - G_u u) \geq 0. \quad (6.112)$$

It vanishes exactly when stationarity and complementarity hold. This gives a mathematical certificate of the unique projection (Boyd & Vandenberghe, 2004). Numerical acceptance checks the individual residuals independently. A small reported gap does not replace a non-negativity or equality check.

The hypothetical two-component example permits every certificate number to be checked. With $\lambda^\leq = -0.25$ and $\lambda^\leq = (1.75, 0)^\top$,

$$E^\top \lambda^\leq + G_u^\top \lambda^\leq = (-2, -0.5)^\top = -u, \quad (6.113)$$

$$g - G_u u = (0, 10)^\top, \quad (6.114)$$

$$\frac{1}{2}u^\top u = \frac{1}{2}(4 + 0.25) = 2.125, \quad (6.115)$$

$$d = -2.125 - 3(-0.25) - [-2(1.75)] = 2.125. \quad (6.116)$$

The primal and dual values agree. Stationarity is zero coordinate by coordinate, and the complementarity sum is $1.75(0) + 0(10) = 0$. The gap is consequently zero. A multiplier with the wrong sign would fail dual feasibility even if some other residual happened to be small.

Every distinct trial receives a projection solve from an independent initialization. An operator-splitting quadratic solver of the type studied by Stellato et al. (2020) supplies a numerical candidate. Multipliers are reconstructed independently through bounded-variable least squares for the audit. Numerical retries change the solver settings while preserving the mathematical problem.

Error bounds in the projection metric

Let $\mathcal{C}$ be the fixed-input feasible set and use the norm $\|v\|_{D_\chi^{-1}} = \|D_\chi^{-1}v\|_2$. This norm has the scalar meaning

$$\|v\|_{D_\chi^{-1}}^2 = \sum_{k=1}^{n_\chi} \frac{v_k^2}{s_k^2}. \quad (6.117)$$

To see the error argument, temporarily work in standardized coordinates. Write $a_k = \chi_{\text{raw},k}/s_k$, $q_k = \hat{\chi}_k/s_k$, and $t_k = \chi_{\text{ref},k}/s_k$. The transformed feasible set is convex because the scales define an invertible linear change of coordinates.

For any feasible comparison point z , the segment $q + \tau(z - q)$ remains feasible for $0 \leq \tau \leq 1$. The projection objective cannot decrease when leaving its minimizer along that segment (Boyd & Vandenberghe, 2004). Its derivative at zero must therefore satisfy

$$\sum_k (q_k - a_k)(z_k - q_k) \geq 0. \quad (6.118)$$

When the reference itself is feasible, take $z = t$. Reversing both signs gives $\sum_k (a_k - q_k)(q_k - t_k) \geq 0$. The componentwise identity

$$(a_k - t_k)^2 = (a_k - q_k)^2 + (q_k - t_k)^2 \quad (6.119)$$

$$+ 2(a_k - q_k)(q_k - t_k) \quad (6.120)$$

then implies, after summing over coordinates,

$$\sum_k (q_k - t_k)^2 \leq \sum_k (a_k - t_k)^2 - \sum_k (a_k - q_k)^2. \quad (6.121)$$

The last sum is exactly the squared standardized projection. Returning to the physical-coordinate notation, if a mechanistic reference χ_{ref} belongs exactly to $\mathcal{C}$, the projection variational inequality gives

$$\|\hat{\chi} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 - \|\hat{u}\|_2^2. \quad (6.122)$$

Thus the projection cannot increase aggregate squared error in this particular metric to an exactly feasible target. Component errors, physical-unit scores, and operating objectives are assessed separately.

In the hypothetical weighted two-component problem, take the feasible reference $(3, 7)$. Raw squared error is $(-2 - 3)^2 + (9 - 7)^2/4 = 26$. Projected squared error is $(0 - 3)^2 + (10 - 7)^2/4 = 11.25$. Projection size squared is $2^2 + 0.5^2 = 4.25$. The bound reads $11.25 \leq 26 - 4.25 = 21.75$. The second component's absolute error increases from 2 to 3 even though the aggregate weighted error decreases. The weighted aggregate and individual component errors therefore describe different effects.

The fitted overflow equality usually differs from the mechanistic endpoint. A physically consistent mechanistic target can therefore lie outside $\mathcal{C}$. Define

$$\delta_{\text{feas}} = \|P_{\mathcal{C}}(\chi_{\text{ref}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}. \quad (6.123)$$

Let p be the closest feasible point to t in standardized coordinates. The previous segment argument now applies to $z = p$. Split the remaining inner product into a feasible-point part and a reference-distance part,

$$\sum_k (a_k - q_k)(q_k - t_k) = \sum_k (a_k - q_k)(q_k - p_k) \quad (6.124)$$

$$+ \sum_k (a_k - q_k)(p_k - t_k). \quad (6.125)$$

The first term is non-negative. The magnitude of the second is at most

$$\left[\sum_k (a_k - q_k)^2 \right]^{1/2} \left[\sum_k (p_k - t_k)^2 \right]^{1/2} = r_{\text{corr}} \delta_{\text{feas}}. \quad (6.126)$$

This is the scalar Cauchy-Schwarz inequality. It bounds how far an infeasible reference can weaken the projection argument. Let $r_{\text{corr}} = \|\chi_{\text{raw}} - P_{\mathcal{C}}(\chi_{\text{raw}})\|_{D_{\chi}^{-1}}$. Apply the

projection variational inequality with the feasible comparison point $P_C(\chi_{\text{ref}})$. Expanding the squared error and bounding the remaining inner product gives

$$\|P_C(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 - r_{\text{corr}}^2 + 2r_{\text{corr}}\delta_{\text{feas}} \quad (6.127)$$

$$\leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 + \delta_{\text{feas}}^2. \quad (6.128)$$

Taking square roots therefore gives the more easily interpreted bound

$$\|P_C(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_\chi^{-1}} \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_\chi^{-1}} + \delta_{\text{feas}}. \quad (6.129)$$

The distance includes any finite numerical infeasibility of the reference. The reference distance is evaluated against both the physical identities and empirical equality. The weighted-projection discussion in Sturm and Silva (2025) likewise separates mass conservation from predictive improvement. The original-model evaluation checks biological kinetics, nonlinear settling, and local dynamic stability.

For a hypothetical illustration of the empirical-closure issue, let the total fixed by mass conservation still be 10 but also require the first component to equal a fitted value of 4. The feasible set then contains only $(4, 6)$. Suppose the mechanistic reference and raw prediction are both $(2, 8)$ with scales $(1, 2)$. They satisfy the physical total and non-negativity but do not satisfy the fitted equality. Raw error is zero. Projection introduces weighted error $\sqrt{(4-2)^2 + (6-8)^2/4} = \sqrt{5}$. The reference distance to the set is also $\sqrt{5}$, so the general bound is met exactly. Physical identities alone would have given the wrong premise for the stronger guarantee.

Control-dependent sensitivities

With fixed constraint matrices and an affine parameter-dependent right-hand side, a parametric quadratic projection can have a piecewise affine solution (Tøndel et al., 2003). Here the projection matrices and endpoint also vary with the controls. Hydraulic coefficients, inverse flow ratios, the input features, and the exponential endpoint all depend on ϑ . The projected plant response is generally piecewise smooth where regularity holds. Active-set transitions define boundaries between its smooth regions.

The differentiation can first be seen in a scalar hypothetical projection,

$$\min_u \frac{1}{2}u^2, \quad b(\theta)u = e(\theta), \quad (6.130)$$

where θ is one control and $b(\theta) \neq 0$. Stationarity and feasibility are $u + b\lambda = 0$ and $bu = e$. Written together, they form the two-by-two system

$$\begin{bmatrix} 1 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ e \end{bmatrix}. \quad (6.131)$$

Differentiate the two scalar equations with respect to θ . The product rule gives $u' + b'\lambda + b\lambda' = 0$ and $b'u + bu' = e'$. Moving known products to the right leaves

$$\begin{bmatrix} 1 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} u' \\ \lambda' \end{bmatrix} = \begin{bmatrix} -b'\lambda \\ e' - b'u \end{bmatrix}. \quad (6.132)$$

The matrix is unchanged. Its right-hand side now contains derivatives of both coefficients and targets.

For $b = 1 + \theta$ and $e = \theta$, the solution is $u = \theta/(1 + \theta)$ and $\lambda = -\theta/(1 + \theta)^2$. At the hypothetical control $\theta = 1$, the values are $u = 0.5$ and $\lambda = -0.25$. The derivative system becomes

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} u' \\ \lambda' \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.5 \end{bmatrix}. \quad (6.133)$$

Its second row gives $u' = 0.25$ and its first gives $\lambda' = 0$. Direct differentiation of $u = \theta/(1 + \theta)$ gives the same $u' = 1/(1 + \theta)^2$. Ignoring b' would instead differentiate a problem whose constraint coefficient did not vary.

For a stable active set, stack equality rows and active inequality rows into $B(\vartheta)$ and their right-hand sides into $b_a(\vartheta)$. The stationary system is

$$\begin{bmatrix} I & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda_a \end{bmatrix} = \begin{bmatrix} 0 \\ b_a \end{bmatrix}. \quad (6.134)$$

The componentwise product rule is now applied to $u + B^\top \lambda_a = 0$ and $Bu = b_a$,

$$\partial_j u + B^\top \partial_j \lambda_a = -(\partial_j B)^\top \lambda_a, \quad (6.135)$$

$$B \partial_j u = \partial_j b_a - (\partial_j B)u. \quad (6.136)$$

For a control ϑ_j , putting these derivative equations into one block system gives

$$\begin{bmatrix} I & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} \partial_j u \\ \partial_j \lambda_a \end{bmatrix} = \begin{bmatrix} -(\partial_j B)^\top \lambda_a \\ \partial_j b_a - (\partial_j B)u \end{bmatrix}. \quad (6.137)$$

Since D_χ is fixed after fitting, the physical response derivative is $\partial_j \hat{\chi} = \partial_j \chi_{\text{raw}} + D_\chi \partial_j \hat{u}$. For the endpoint, first differentiate the scalar fitted logarithm, which gives $\partial_j (\hat{\beta}^\top \phi) = \hat{\beta}^\top \partial_j \phi$. Exponential differentiation then gives $\partial_j \hat{X}_{E,\log} = \hat{X}_{E,\log} \hat{\beta}^\top \partial_j \phi$. Applying the product rule to its flow-weighted value gives the endpoint derivative,

$$\partial_j (q_E \hat{X}_{E,\log}) = (\partial_j q_E) \hat{X}_{E,\log} + q_E \hat{X}_{E,\log} \hat{\beta}_{E,\gamma_E}^\top \partial_j \phi. \quad (6.138)$$

Ignoring this term would differentiate a different optimization problem.

These derivatives are used only when active rows are independent, the stationary matrix is well conditioned, active inequality multipliers are strictly positive, and independent perturbation checks pass. Near degeneracy, different active descriptions can correspond to the same unique response. Uniqueness of the projection therefore does not ensure a reliable classical derivative. A value-only outer method remains available when the derivative chain cannot be validated.

6.2.5 Treatment priorities, resource measures, and operating safeguards

A common engineering objective

The effluent-quality map in Table 3.1 gives

$$z_E = I_{\text{comp}} c_E = \text{col}(\text{COD}, \text{TN}, \text{TP}, \text{TSS}). \quad (6.139)$$

Let the four diagonal entries of D_T be $T_{\text{COD}}, T_{\text{TN}}, T_{\text{TP}}, T_{\text{TSS}}$. Dividing each effluent quantity by its own scale gives a dimensionless quality contribution. The scalar weighted sum is

$$\begin{aligned} j_Q = \varpi_{Q,\text{COD}} \frac{z_{E,\text{COD}}}{T_{\text{COD}}} &+ \varpi_{Q,\text{TN}} \frac{z_{E,\text{TN}}}{T_{\text{TN}}} \\ &+ \varpi_{Q,\text{TP}} \frac{z_{E,\text{TP}}}{T_{\text{TP}}} + \varpi_{Q,\text{TSS}} \frac{z_{E,\text{TSS}}}{T_{\text{TSS}}}. \end{aligned} \quad (6.140)$$

For positive fixed normalization D_T and quality priorities $\varpi_Q \geq 0$ with $\mathbf{1}^\top \varpi_Q = 1$, this sum is written compactly as

$$j_Q = \varpi_Q^\top D_T^{-1} z_E. \quad (6.141)$$

The four totals describe different treatment concerns. They do not identify which individual nutrient transformations occur. A nearly constant TN across a reactor can accompany conversion of ammonium to nitrite. A decrease in overflow TP can arise from particulate separation rather than an additional clarifier reaction. The nutrient-control guidance of U.S. Environmental Protection Agency (2009) and component-resolved treatment framework of X. Zhang et al. (2026) support inspecting species and solids flows alongside aggregate quality measures.

Normalizing a bounded setting subtracts its lower bound and divides by the full range. This gives zero at the lower bound and one at the upper bound. For example, $H = 18$ h within 6–36 h gives $(18 - 6)/(36 - 6) = 0.4$. Aeration instead sums physical transfer capacities. Each product $V_i k_L a_i$ has units $\text{m}^3 \text{d}^{-1}$, so its fixed reference has the same units. Wasted solids are proportional to $Q_0 w X_U$. Dividing this rate by its fixed reference $Q_0 w^U X_U^{\text{ref}}$ cancels Q_0 . The resource terms are

$$j_H = \frac{H - H^L}{H^U - H^L}, \quad j_A = \frac{\sum_{i \in \mathcal{A}} V_i k_L a_i}{E_A^{\text{ref}}}, \quad (6.142)$$

$$j_I = \frac{r_I - r_I^L}{r_I^U - r_I^L}, \quad j_R = \frac{r_R - r_R^L}{r_R^U - r_R^L}, \quad (6.143)$$

$$j_W = \frac{w X_U}{w^U X_U^{\text{ref}}}. \quad (6.144)$$

The set $\mathcal{A}$ contains stages that can be aerated. The aeration reference $E_A^{\text{ref}} > 0$ is fixed before optimization. It is the maximum transfer-capacity numerator over the declared operating box. A fixed reference prevents an increase in selected volume from being canceled by a changing denominator.

The aeration proxy measures transfer capacity. Actual oxygen transfer also depends on the dissolved-oxygen deficit in Equation (6.15). The energy-management model of Miederer et al. (2025) distinguishes aeration and other energy demands. The treatment model of X. Zhang et al. (2026) makes the oxygen-transfer driving force explicit. The dimensionless terms are engineering measures of capacity and resource use.

Write the overall coefficients in the same order as the six terms. The complete scalar objective is

$$J = \varpi_1 j_Q + \varpi_2 j_H + \varpi_3 j_A + \varpi_4 j_I + \varpi_5 j_R + \varpi_6 j_W. \quad (6.145)$$

Its compact vector form is

$$J(\vartheta, \chi, \varpi) = \varpi^\top \text{col}(j_Q, j_H, j_A, j_I, j_R, j_W), \quad \varpi \geq 0, \quad \mathbf{1}^\top \varpi = 1. \quad (6.146)$$

The overall vector ϖ separates quality from capacity and resource priorities. The vector ϖ_Q separates priorities among the quality measures. These coefficients are choices about trade-offs. They are not percentages of the objective attained at an operating point.

For the equal-volume case,

$$j_A = \frac{H}{36 \text{ h}} \frac{a_3 + a_4 + a_5}{3}. \quad (6.147)$$

This follows from $V_i = Q_0 H / (24 n_r)$ and $k_L a_i = 47 a_i$. For hypothetical settings with all three aeration coordinates equal to one, reducing H from 36 h to 12 h reduces this proxy from one to one-third. The normalized aeration settings have stayed the same, but the total transfer capacity has changed with volume. A comparison based only on the aeration coordinates would miss this effect.

The baseline priority vectors are

$$\varpi = (0.50, 0.15, 0.20, 0.05, 0.05, 0.05)^\top, \quad \varpi_Q = \frac{1}{4} \mathbf{1}_4. \quad (6.148)$$

Quality has the largest individual coefficient. Aeration has the largest resource coefficient. The importance of oxygen supply and solids age to treatment and energy use is discussed by Leu et al. (2012). The exact coefficients remain case-study preferences. Changes in those preferences define separate operating questions and require separately reported decisions.

A hypothetical objective calculation can now be followed through all six terms. Use effluent values (40, 5, 1, 4) mg L⁻¹ and illustrative fixed quality scales (100, 10, 2, 20) mg L⁻¹. With equal quality priorities, $j_Q = (0.4 + 0.5 + 0.5 + 0.2)/4 = 0.4$. These illustrative scales are not estimated study quantities. Let $H = 18$ h, $(a_3, a_4, a_5) = (0.8, 0.6, 0.4)$, $r_I = 1$, $r_R = 0.5$, $w = 0.02$, and $X_U = 6,000$ g m⁻³.

The total biological volume is $Q_0 H / 24 = 7,500$ m³, so each equal tank has volume 1,500 m³. Aeration capacity is $1,500(47)(0.8 + 0.6 + 0.4) = 126,900$ m³ d⁻¹. Table 6.2 gives every normalized term and weighted contribution. Their sum is 0.353. This hypothetical arithmetic illustrates the declared objective.

Table 6.2: Hypothetical calculation of dimensionless objective contributions.

Term	Calculation	Value	Weighted value
j_Q	$(0.4 + 0.5 + 0.5 + 0.2)/4$	0.4	0.200
j_H	$(18 - 6)/(36 - 6)$	0.4	0.060
j_A	$126,900/423,000$	0.3	0.060
j_I	$(1 - 0)/(4 - 0)$	0.25	0.0125
j_R	$(0.5 - 0.25)/(1.25 - 0.25)$	0.25	0.0125
j_W	$(0.02)(6,000)/[(0.05)(15,000)]$	0.16	0.008

The overall priority of 0.50 for quality multiplies j_Q after its four internal quality weights have been applied. A quality priority of one-quarter is therefore not an overall objective coefficient of one-quarter. Also, a resource reference need not be a hard maximum. In particular, j_W can exceed one if its numerator exceeds the declared reference. A denominator used for normalization does not itself impose feasibility.

Solids retention time and clarifier loading

Stored solids and lost solids have different units. For two reactors, the stored mass would be $V_1X_1 + V_2X_2 + M_{cl}$ in grams. Its loss rate would be $Q_EX_E + Q_0wX_U$ in grams per day. Dividing mass by mass per day gives days. The same accounting applies to all five reactors.

External solids leave in the overflow and waste-sludge streams,

$$\dot{M}_X = Q_0(q_EX_E + wX_U). \quad (6.149)$$

Return sludge does not count as an external loss. With $X_i = t_X^\top c_i$, the whole-plant solids retention time is

$$\theta_X = \frac{\sum_{i=1}^{n_r} V_i X_i + M_{cl}}{\dot{M}_X}, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.150)$$

Its numerator includes the clarifier inventory. Omitting that mass would underestimate solids age for the same external loss rate. A wasting ratio alone also does not determine solids age because wX_U and the stored mass both matter (Alex et al., 2008).

For a hypothetical accounting example, use $Q_0 = 10,000 \text{ m}^3 \text{ d}^{-1}$, $w = 0.02$, $X_E = 10$, and $X_U = 6,000 \text{ g m}^{-3}$. The two external loss contributions are

$$Q_0q_EX_E = 10,000(0.98)(10) = 98,000 \text{ g d}^{-1}, \quad (6.151)$$

$$Q_0wX_U = 10,000(0.02)(6,000) = 1,200,000 \text{ g d}^{-1}. \quad (6.152)$$

Their sum is $1,298,000 \text{ g d}^{-1}$. At a total reactor volume of $7,500 \text{ m}^3$, assigning every reactor $3,000 \text{ g TSS m}^{-3}$ gives $22,500,000 \text{ g}$ in the reactors. If the clarifier inventory is $7,500,000 \text{ g}$, the total inventory is $30,000,000 \text{ g}$. The illustrative solids retention time is

$30,000,000/1,298,000 = 23.11$ d. This hypothetical ratio illustrates the inventory and external-loss accounting.

Clarifier surface loading divides the overflow volume rate by area. Clarifier solids loading divides the feed solids mass rate by area (Henze et al., 2008). Since one gram is 10^{-3} kilograms, its conversion appears before the area division. The descriptive clarifier loading measures are

$$\ell_{\text{surface}} = \frac{Q_0 q_E}{A_{\text{cl}}}, \quad \ell_{\text{solids}} = \frac{10^{-3} Q_0 q_C X_N}{A_{\text{cl}}}. \quad (6.153)$$

The factor 10^{-3} converts grams to kilograms. The surface quantity is in m d^{-1} and the solids quantity is in $\text{kg m}^{-2} \text{ d}^{-1}$. These quantities and θ_X are descriptive measures in the baseline comparison.

For the same hypothetical flow with $r_R = 0.5$, $q_C = 1.5$, $A_{\text{cl}} = 1,500 \text{ m}^2$, and $X_N = 3,000 \text{ g m}^{-3}$, the surface loading is $9,800/1,500 = 6.53 \text{ m d}^{-1}$. The feed solids rate is $10,000(1.5)(3,000) = 45,000,000 \text{ g d}^{-1}$, or $45,000 \text{ kg d}^{-1}$. Dividing by area gives $30 \text{ kg m}^{-2} \text{ d}^{-1}$. Using underflow instead of clarifier-feed solids in this formula would calculate a different loading quantity.

The common engineering safeguards are

$$X_U \leq X_U^{\max}, \quad X_N \geq X_F^{\min} > 0, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.154)$$

The positive feed-solids limit keeps particulate outlet reconstruction defined. The positive external-loss limit keeps solids retention time defined. The underflow bound represents the admitted thickening or solids-handling range. Their numerical values are plant-specific problem inputs selected before search. An underflow limit should be based on accepted sludge-handling conditions (Henze et al., 2008). The feed and loss floors should prevent undefined calculations without excluding physically relevant low-solids operation. These engineering inputs are distinct from the fixed $X_U^{\text{ref}} = 15,000 \text{ g m}^{-3}$ used only to normalize the wasted-solids objective.

Trust diagnostics for operating search

The trust diagnostics are numerical checks used to restrict the operating search to specified levels of data support and model consistency. They assess input coverage, projection size, fitted clarifier behavior, and reactor balance residuals. They are safeguards for selecting candidate aeration, recycle, and wasting settings. Their fixed thresholds define the accepted search region. The need to manage approximation error near candidate controls follows the model-management approaches of Alexandrov et al. (1998) and Hameed et al. (2026).

The trust diagnostics examine model consistency within the region admitted by projection. The projection diagnostic first squares each standardized movement, averages these squares over the response coordinates, and takes the square root. For

the hypothetical two-component projection $u = (2, 0.5)$, it is $\sqrt{(4 + 0.25)/2} = 1.458$. Dividing by the number of coordinates makes its interpretation comparable across response dimensions.

Feature leverage uses the feature vector and the regularized fitting geometry. In a hypothetical intercept-plus-one-feature example with three standardized feature values $(-\sqrt{3/2}, 0, \sqrt{3/2})$ and $\gamma = 1$,

$$\Phi^\top \Phi + 3\gamma R_\phi = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}. \quad (6.155)$$

Its inverse is $\text{diag}(1/3, 1/6)$. At query feature $(1, z)$, leverage is $1/3 + z^2/6$. It is $1/3$ at the center, $1/2$ at $z = 1$, and $11/6$ at $z = 3$. The largest development value in this teaching example is $7/12$. These hypothetical numbers illustrate weaker joint feature support away from the observed range.

The split diagnostic compares each particulate underflow flow with the flow expected from a common recovery factor. In a hypothetical two-particulate example with unit TSS weights, let the feed component concentrations be $(40, 60)$, so $X_N = 100$. An underflow component-flow vector $(20, 30)$ has total 50 and gives numerators $100(20) - 50(40) = 0$ and $100(30) - 50(60) = 0$. Changing the outlet allocation to $(25, 25)$ preserves total underflow solids but gives numerators 500 and -500 . With illustrative fixed split scales of 500, the standardized values are 1 and -1 , and their root mean square is one. Total solids consistency therefore need not establish composition consistency.

The reactor diagnostic first evaluates the kinetic balances, then divides each residual by its fixed balance scale, and finally computes their root mean square. In the hypothetical two-component reactor, a projected state $(9, 6)$ preserves the upstream total of $(12, 3)$. With $d = 2$ and the reaction rate held at 8 for this illustration, its two residuals are $2(12 - 9) - 8 = -2$ and $2(3 - 6) + 8 = 2$. A balance scale of 10 for each component gives standardized residuals $(-0.2, 0.2)$ and a diagnostic of 0.2. The invariant is satisfied even though the kinetic equations are not.

These four fixed diagnostics are written compactly as

$$\kappa_{\text{corr}} = \frac{\|D_X^{-1}(\hat{\chi} - \chi_{\text{raw}})\|_2}{\sqrt{n_X}}, \quad (6.156)$$

$$\kappa_{\text{lev}} = \phi^\top (\Phi^\top \Phi + n_{\text{dev}} \gamma R_\phi)^{-1} \phi, \quad (6.157)$$

$$\kappa_{\text{split}} = \left[\frac{1}{F_X} \sum_{f \in X} \left\{ \frac{X_N g_{U,f} - (t_X^\top g_U) c_{n_r,f}}{d_{\text{split},f}} \right\}^2 \right]^{1/2}, \quad (6.158)$$

$$\kappa_{\text{reac}} = \frac{\|D_{\text{bal}}^{-1} f_{\text{sm,reactor}}(\vartheta, \hat{\chi})\|_2}{\sqrt{n_r F}}. \quad (6.159)$$

The first measures standardized projection size. A large value identifies a raw response that requires a large movement to satisfy the imposed constraints. The projection is constrained and reported separately from the engineering objective.

The second diagnostic is regularized feature leverage. It measures input support within the fitted quadratic feature map. Sahigara et al. (2012) compare applicability-domain assessments and motivate checking where a statistical model is being used. A control vector inside its coordinate bounds may still have weak support in the joint feature space.

The third diagnostic checks the common particulate-recovery assumption in Equation (6.48). If every particulate component has the same recovery, its numerator vanishes. This diagnostic supplements component mass conservation and densification with a check on the reduced mechanistic composition relation.

The fourth diagnostic checks smooth reactor kinetics at the projected concentrations. It includes hydraulic transport, net biological conversion, and oxygen transfer on separate scales. Clarifier-layer residuals are evaluated separately with the layer-resolved mechanistic model. Its smoothing width is 10^{-8} under the definitions given below.

A row scale is based on the sizes of its separate physical terms before cancellation. Suppose a hypothetical row has terms $(100, -100)$ in one development state and $(200, -200)$ in another. Its net balance is zero in both states, but its terms have substantial magnitude. Their mean squared magnitude is $(100^2 + 100^2 + 200^2 + 200^2)/4 = 25,000$, giving a root mean square of 158.11. Scaling only the canceled net balance would lose this information. For a constraint row with p terms b_{ij} over n_{dev} development states, its fixed scale is

$$d_b = \max \left\{ 1, \left[\frac{1}{n_{\text{dev}}p} \sum_{i=1}^{n_{\text{dev}}} \sum_{j=1}^p b_{ij}^2 \right]^{1/2} \right\}. \quad (6.160)$$

The same convention defines inequality and split scales. The overflow row uses the mechanistic overflow flow and the out-of-fold fitted endpoint flow as its two terms. The floor of one is in the corresponding numerical row units. It is not a concentration floor.

Projection, split, and reactor limits use nearest-rank 95th percentiles of grouped out-of-fold development diagnostics. If sorted values are $v_{(1)}, \dots, v_{(n_{\text{dev}})}$, the limit is $v_{(\lceil 0.95n_{\text{dev}} \rceil)}$. The leverage limit is the maximum development leverage under the final fitted feature map. These quantities are fixed before optimization. They define empirical search thresholds. A diagnostic with zero limit remains constrained to zero and uses one only as a numerical row scale.

The surrogate decision problem is consequently

$$\begin{aligned} \min_{\vartheta} \quad & J(\vartheta, \hat{\chi}(\vartheta, c_{\text{in}}), \varpi), \\ \text{subject to} \quad & \vartheta \in \mathcal{V}, \\ & g_{\text{eng}}(\vartheta, \hat{\chi}) \leq 0, \\ & g_{\text{trust}}(\vartheta, \chi_{\text{raw}}, \hat{\chi}) \leq 0, \\ & \hat{\chi} \text{ satisfies Equation (6.100).} \end{aligned} \quad (6.161)$$

The lower problem is a convex quadratic projection. The upper problem is generally nonconvex. This distinction follows the nested-problem perspective described by Colson et al. (2007). The trust-region analysis of Alexandrov et al. (1998) explains why reliable approximation management needs more than a small average prediction error. The four fixed diagnostics supply the admission checks for this operating search.

6.2.6 Bounded products and convex relaxation

Flow and concentration products help explain why process optimization remains nonlinear even when a surrogate is linear in its fitted coefficients. A bounded product provides a simple example. Let $x_L \leq x \leq x_U$, $y_L \leq y \leq y_U$, and let z represent the product xy . The four products

$$(x - x_L)(y - y_L), \quad (x_U - x)(y_U - y), \quad (x_U - x)(y - y_L), \quad (x - x_L)(y_U - y)$$

are non-negative. Expanding the first gives $xy - x_Ly - y_Lx + x_Ly_L \geq 0$. Replacing xy by z gives one lower bound on z . Expanding the other three products gives the remaining bounds. Together they form the McCormick relaxation (McCormick, 1976)

$$\begin{aligned} z &\geq x_Ly + y_Lx - x_Ly_L, \\ z &\geq x_Uy + y_Ux - x_Uy_U, \\ z &\leq x_Uy + y_Lx - x_Uy_L, \\ z &\leq x_Ly + y_Ux - x_Ly_U. \end{aligned} \tag{6.162}$$

The four inequalities are linear in the variables x , y , and z . Every exact product within the stated bounds satisfies them. Their intersection nevertheless permits points that do not satisfy $z = xy$.

For a hypothetical pair with $0 \leq x, y \leq 1$, the bounds become $z \geq 0$, $z \geq x + y - 1$, $z \leq x$, and $z \leq y$. At $x = y = 0.5$, they permit $0 \leq z \leq 0.5$. The exact product is only $z = 0.25$. A favorable objective obtained at another permitted value of z is therefore not, by itself, an attainable process response. Tighter variable bounds can improve the relaxation, but they do not generally replace the exact product relation. Process-network optimization uses this distinction when constructing bounds and assessing solutions (Rubio-Castro et al., 2013; L. Yang et al., 2014).

This example separates a convex relaxation from the physical projection used here. The relaxation enlarges a nonlinear feasible set to obtain an optimization bound. The projection selects a response satisfying the declared mass conservation and non-negativity conditions near a predicted response. The present fixed-configuration plant retains the surrogate-assisted and direct mechanistic searches described below. Its plant connections remain fixed throughout these searches.

6.2.7 A direct mechanistic comparator

The full-state constrained optimization problem

The direct route retains the controls and full reduced mechanistic state. It re-constructs $\chi(\vartheta, c_{\text{in}}, y)$ and evaluates the same engineering objective. Its finite-volume clarifier equations require a positive feed-solids denominator at solver iterates. An auxiliary variable $\widetilde{X}_F \geq X_F^{\min}$ supplies that denominator, with equality $\widetilde{X}_F = t_X^\top c_{n_r}$ imposed at feasibility.

In a hypothetical one-reactor, two-component, three-layer illustration, the full state is $(c_S, c_X, s_1, s_2, s_3)$. It has five entries rather than the nine entries of the reduced statistical response. A simple conversion rate kc_S gives two reactor equations,

$$0 = d(m_S - c_S) - kc_S, \quad (6.163)$$

$$0 = d(m_X - c_X) + kc_S. \quad (6.164)$$

The three layer equations are those obtained from the interface-flow balances above, divided by their own volumes. The direct problem imposes all five equations rather than replacing the two reactor equations by only their sum fixed by mass conservation. Its three-layer envelope contains two scalar inequalities, $s_1 - s_2 \leq 0$ and $s_2 - s_3 \leq 0$. Four layers would require $(s_1 - s_2, s_1 - s_3, s_2 - s_4, s_3 - s_4) \leq 0$. This ordering explains how the envelope is assembled for ten layers.

The auxiliary denominator also has a concrete role. An infeasible trial can have zero feed solids while a nonzero layer concentration is being adjusted. Division by the actual feed value would then be undefined. An independently bounded $\widetilde{X}_F$ keeps the trial calculation defined. The equality $\widetilde{X}_F = X_N$ must subsequently hold, so this device does not allow a zero-feed trial to be accepted as a feasible physical state. It provides a usable calculation during the search, not an alternative physical feed concentration.

Let f_{sm} collect all $n_r F$ reactor equations and all L smoothed layer equations. Let

$$g_{\text{env}}(y) = \text{col}(s_1 \mathbf{1}_{L-2} - s_{2:L-1}, s_{2:L-1} - s_L \mathbf{1}_{L-2}). \quad (6.165)$$

The direct nonlinear program is

$$\begin{aligned} & \min_{\vartheta, y, \widetilde{X}_F} J(\vartheta, \chi(\vartheta, c_{\text{in}}, y, \widetilde{X}_F), \varpi), \\ & \text{subject to } f_{\text{sm}}(\vartheta, c_{\text{in}}, y, \widetilde{X}_F) = 0, \\ & \quad \widetilde{X}_F = t_X^\top c_{n_r}, \quad \widetilde{X}_F \geq X_F^{\min}, \\ & \quad \vartheta \in \mathcal{V}, \quad y \geq 0, \quad g_{\text{env}}(y) \leq 0, \\ & \quad g_{\text{eng,red}}(\vartheta, \chi) \leq 0. \end{aligned} \quad (6.166)$$

The reduced engineering vector removes the feed-solids row already imposed by the auxiliary equality and lower bound. The auxiliary does not alter a feasible physical state. It keeps the outlet mapping defined while a nonlinear solver moves through infeasible trial states.

Smooth switches and continuation

The exact kinetic and settling functions contain minima, maxima, positive parts, possible zero denominators, and a receiver switch. Derivative-based optimization benefits from smooth approximations to these operations. The chemical-process optimization treatment of Buzzi-Ferraris and Manenti (2013) provides the general derivative and conditioning context. The exact identities

$$\max(a, b) = \frac{a + b + |a - b|}{2}, \quad \min(a, b) = \frac{a + b - |a - b|}{2} \quad (6.167)$$

show where the nondifferentiable absolute value enters. Replacing $|r|$ by $\sqrt{r^2 + \delta^2}$ removes its corner for every $\delta > 0$. The positive part uses the same replacement in $(v + |v|)/2$. A denominator regularization instead replaces $1/b$ by $b/(b^2 + \delta^2)$. These are different operations chosen for different numerical difficulties. For positive width $\delta = \epsilon S$, define

$$\max_{\delta}(a, b) = \frac{1}{2} \left[a + b + \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.168)$$

$$\min_{\delta}(a, b) = \frac{1}{2} \left[a + b - \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.169)$$

$$[v]_{+, \delta} = \frac{1}{2} \left[v + \sqrt{v^2 + \delta^2} \right], \quad (6.170)$$

$$\left(\frac{a}{b} \right)_{\delta} = \frac{ab}{b^2 + \delta^2}. \quad (6.171)$$

For negative v , the positive-part expression is evaluated as $\delta^2/[2(\sqrt{v^2 + \delta^2} - v)]$ to avoid subtraction of nearly equal values. These approximations change the equations near a switch. Independent evaluation uses the exact equations at the selected controls.

For hypothetical values $a = 2$, $b = 1$, and $\delta = 0.1$, the smooth maximum is $(3 + \sqrt{1.01})/2 = 2.00249$ and the smooth minimum is 0.99751. The exact values are 2 and 1. For $v = -0.2$, the smooth positive part is $(-0.2 + \sqrt{0.05})/2 = 0.01180$, while the exact positive part is zero. At $b = 0$, the regularized quotient $ab/(b^2 + \delta^2)$ is defined and equals zero. At $a = 1$ and $b = 0.1$, it is 5 for $\delta = 0.1$, compared with the exact quotient 10. These differences illustrate why smaller widths and an independent exact-model evaluation are needed.

The stable positive-part expression follows by rationalization,

$$\frac{v + \sqrt{v^2 + \delta^2}}{2} = \frac{(v + \sqrt{v^2 + \delta^2})(\sqrt{v^2 + \delta^2} - v)}{2(\sqrt{v^2 + \delta^2} - v)} \quad (6.172)$$

$$= \frac{\delta^2}{2(\sqrt{v^2 + \delta^2} - v)}. \quad (6.173)$$

The numerator becomes $v^2 + \delta^2 - v^2 = \delta^2$. This form avoids canceling a negative v against a nearly equal square root.

For the receiver switch, first measure where the receiver concentration lies across its transition band. Subtracting the lower edge $X_t - h$ and dividing by the band width $2h$

maps that band to $[0, 1]$. For receiver concentration s , the smooth switch is zero below $X_t - h$, one above $X_t + h$, and

$$6\zeta^5 - 15\zeta^4 + 10\zeta^3, \quad \zeta = \frac{s - X_t + h}{2h} \quad (6.174)$$

within the transition band. This polynomial matches the values and first two derivatives at the band endpoints. It gives smooth interpolation between the two gravitational-flux rules.

For the illustrative band $X_t = 3,000$ and $h = 10 \text{ g m}^{-3}$, concentrations 2,990, 3,000, and 3,010 correspond to $\zeta = 0, 0.5, 1$. The switch values are 0, 0.5, and 1. At 2,995, $\zeta = 0.25$ and the switch is 0.10352. At 3,005, it is 0.89648. If the unrestricted gravitational flux is J_1 and the receiver-limited flux is J_2 , interpolation gives $(1 - \sigma)J_1 + \sigma J_2$ for switch value σ . The switch therefore changes a flux rule gradually rather than only replacing a reported concentration.

Each argument scale S is its development-set root mean square with a floor of one. Reactor arguments are pooled across stages. They include total oxidized nitrogen, fermentable substrate plus acetate, the hydrolysis denominator, the polyphosphate and stored-carbon denominators, and remaining storage capacity. Clarifier arguments include $s - f_{\text{ns}}X_N$ and gravitational solids flux. The velocity scale is 250 m d^{-1} . These scales stay fixed throughout continuation.

The direct route reduces smoothing through three stages,

$$(\epsilon, h) = (10^{-6}, 10), (10^{-7}, 3), (10^{-8}, 1), \quad (6.175)$$

where h is in g TSS m^{-3} . Each stage initializes from the preceding stage. Continuation helps a solver approach sharper switches while preserving a usable starting state. The final smooth candidate still requires an exact-model evaluation.

Jacobian structure and globalization

The Jacobian entry J_{jk} is the partial derivative of residual f_j with respect to state coordinate y_k . It describes how one balance changes when one coordinate changes while the others are held fixed. For a hypothetical two-reactor soluble conversion with rate $kc_{i,S}$, soluble pass-through in the recycle gives

$$m_S = \frac{c_{\text{in},S} + (r_I + r_R)c_{2,S}}{q_P}. \quad (6.176)$$

The two soluble residuals are $f_{1,S} = d_1(m_S - c_{1,S}) - kc_{1,S}$ and $f_{2,S} = d_2(c_{1,S} - c_{2,S}) - kc_{2,S}$. Their two-by-two state block is

$$\frac{\partial(f_{1,S}, f_{2,S})}{\partial(c_{1,S}, c_{2,S})} = \begin{bmatrix} -(d_1 + k) & d_1(r_I + r_R)/q_P \\ d_2 & -(d_2 + k) \end{bmatrix}. \quad (6.177)$$

The off-diagonal upper-right entry is created by recycle. Dropping it would treat the mixer as independent of the final reactor. For hypothetical values $d_1 = d_2 = 2$, $k = 1$,

$r_I = 1$, $r_R = 0.5$, and $q_P = 2.5$, the block is $\begin{bmatrix} -3 & 1.2 \\ 2 & -3 \end{bmatrix}$. Its entries have units d^{-1} because the residual has concentration-per-day units and its derivative is with respect to concentration.

Finite differences permit an independent check of those entries. Increasing $c_{1,S}$ by 0.01 g m^{-3} changes $f_{1,S}$ by $-0.03 \text{ g m}^{-3} \text{ d}^{-1}$, giving $-0.03/0.01 = -3$. Increasing $c_{2,S}$ by the same amount changes m_S by 0.006 and $f_{1,S}$ by 0.012 , giving 1.2 . These values are exact for the hypothetical linear teaching equations. In a nonlinear residual, the check instead uses a small step h ,

$$J_{jk} \approx \frac{f_j(y + he_k) - f_j(y - he_k)}{2h}. \quad (6.178)$$

Here e_k changes only coordinate k . Near a non-negative boundary, a central step can leave the allowed state region. A second-order forward alternative is

$$J_{jk} \approx \frac{-3f_j(y) + 4f_j(y + he_k) - f_j(y + 2he_k)}{2h}. \quad (6.179)$$

The step has the physical unit of y_k . Scaling coordinates makes its relative size comparable across different component bases.

For three layers at fixed feed conditions, only neighboring concentrations enter each interface flux. Their layer Jacobian therefore has the pattern

$$J_{\text{layer}} = \begin{bmatrix} \star & \star & 0 \\ \star & \star & \star \\ 0 & \star & \star \end{bmatrix}, \quad (6.180)$$

where a star denotes an entry that may be nonzero. Feed solids and outlet reconstruction add connections to reactor coordinates. The zeros describe the declared dependency structure and are not approximations made to reduce computation.

The direct residual is sparse in its mechanistic state. A reactor balance depends on its own component vector and the preceding stage. Recycle reconstruction adds dependencies on final-reactor and underflow quantities. A layer balance depends on nearby layer concentrations, feed solids, and hydraulic controls. The inventory and reporting maps are linear aggregations. These dependencies provide a structured Jacobian rather than a dense arbitrary response map.

Scaling is applied before testing Jacobian conditioning or comparing residual sizes. Different component bases include oxygen, COD, nitrogen, phosphorus, solids, and charge equivalents. An unscaled norm could be dominated by a large solids balance and conceal a nutrient-balance defect. Analytic derivatives or differentiated smooth equations supply the optimization Jacobian. Independent finite differences are retained for derivative checks. Sparse linear algebra exploits the local reactor and layer connections without omitting recycle terms.

The direct search uses the interior-point filter method studied by Wächter and Biegler (2006). Its line search manages progress in objective and constraint violation. This numerical globalization controls acceptance of local steps. The surrogate route similarly separates numerical search progress from the final feasibility and stationarity audits.

6.3 Methodology

6.3.1 Case-study domain and fixed parameters

Topology and free controls

The case study has five equal-volume reactors and ten clarifier layers. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls. The free-control vector is

$$\vartheta = (H, a_3, a_4, a_5, r_I, r_R, w)^\top \in \mathbb{R}^7. \quad (6.181)$$

For every stage, $h_i = H/5$. The unaerated settings are $a_1 = a_2 = 0$, the aerated transfer coefficients are $k_L a_i = 47a_i \text{ d}^{-1}$, and $\xi_i = 0$. These are case-study conditions rather than structural restrictions on the connected formulation.

Table 6.3: Plant geometry and operating domain.

Quantity	Value or interval
Fresh-influent flow Q_0	10,000 m ³ d ⁻¹
Reactors n_r , layers L , feed layer ℓ_F	5, 10, 5
Reactor volume fraction f_i^V	1/5 for every stage
Clarifier area and total depth	1,500 m ² and 4.0 m
Clarifier total and individual layer volumes	6,000 m ³ and 600 m ³
Nominal total HRT H	6–36 h
Each of a_3, a_4, a_5	0–1 independently
Mixed-liquor recycle r_I	0–4
Return-sludge ratio r_R	0.25–1.25
Waste-sludge ratio w	0.001–0.05
Aeration reference E_A^{ref}	423,000 m ³ d ⁻¹
Wasted-solids reference X_U^{ref}	15,000 g TSS m ⁻³

The equal volumes provide a compact unaerated-to-aerated train. The layer depth is 0.4 m. The operating intervals preserve both positive clarifier outlet flow ratios. Together with 20 influent coordinates, the seven controls give 27 statistical inputs, 406 features, and 161 reduced-response coordinates. The mechanistic state has 110 coordinates. The projection has 76 equality rows and 12 physical inequality rows. Adding response non-negativity gives 173 inequality rows.

Changing the number of reactors or free controls changes the statistical dimensions. Changing the layer count changes the mechanistic dimension and inventory mapping but does not add coordinates to the aggregate surrogate response. In either case, physical and predictive validation must be repeated. The layer discretization enters through the mechanistic targets and inventory bounds.

Fresh-influent intervals

Table 6.4 defines independent intervals for the plant experiment. The nominal influent is their componentwise midpoint. The concentration bases and quality-map weights are those in Table 3.1. The plant alkalinity interval is 1.6–5.2 mol charge equivalents m^{-3} . This interval defines the connected-plant influent design.

Table 6.4: Independent fresh-influent intervals for the connected plant.

Component	Lower	Upper	Constituent basis
S_O	0	0.5	oxygen
S_F	20	180	COD
S_A	5	80	COD
S_{NH4}	12	55	nitrogen
S_{NO2}	0	3	nitrogen
S_{NO3}	0	8	nitrogen
S_{N2}	0	2	nitrogen
S_{PO4}	2	18	phosphorus
S_I	10	90	COD
S_{ALK}	1.6	5.2	charge equivalents
X_I	20	120	COD
X_S	60	280	COD
X_H	15	100	COD
X_{PAO}	5	60	COD
X_{PP}	2	20	phosphorus
X_{PHA}	1	30	COD
X_{AOB}	0.5	8	COD
X_{NOB}	0.5	8	COD
X_{MeP}	0	12	solids
X_{MeOH}	0	12	solids

All concentration values are in grams of the stated constituent per m^3 , apart from alkalinity. Numerically, g m^{-3} equals mg L^{-1} . Independent interval sampling distributes candidates across the declared experimental domain.

Stoichiometric, kinetic, and settling parameters

The constants in Tables 6.5–6.8 define the plant case. They are fixed parameters rather than fitted kinetic estimates. A parameter's concentration basis follows the associated component. Symbols with several process subscripts on the same row have equal values. The two nitrifier populations remain separate, with different growth rates and oxygen half-saturation constants.

Table 6.5: Fixed yields, constituent contents, and conversion factors.

Parameter	Value	Unit or interpretation
Y_H, Y_{PAO}	0.625	g biomass COD per g substrate COD
Y_{PHA}	0.20	g stored COD per g phosphorus
Y_{PO4}	0.40	g phosphorus per g stored COD
Y_{AOB}	0.18	g biomass COD per g nitrogen
Y_{NOB}	0.08	g biomass COD per g nitrogen
f_{SI}	0	soluble inert fraction
f_{XI}	0.10	particulate inert fraction
$i_{N,SI}$	0.01	g nitrogen per g COD
$i_{N,SF}, i_{N,XS}$	0	g nitrogen per g COD
$i_{N,XI}$	0.02	g nitrogen per g COD
$i_{N,BM}$	0.07	g nitrogen per g biomass COD
$i_{P,SI}, i_{P,SF}, i_{P,XS}$	0	g phosphorus per g COD
$i_{P,XI}$	0.01	g phosphorus per g COD
$i_{P,BM}$	0.02	g phosphorus per g biomass COD
$i_{P,MeP}$	1/4.87	g phosphorus per g metal-phosphate solids
γ_{OH}	3.45	g metal-hydroxide solids per g phosphorus

6.3.2 Mechanistic sampling, fitting, and prediction assessment

Simulation groups and accepted cases

The joint domain has seven control coordinates and twenty influent coordinates. Strength-one Latin-hypercube sampling places one point in each stratum of every coordinate. Independent random stratum orderings prevent a shared ordering from forcing artificial coordinate correlations. Open-unit jitter is a random fractional position strictly between zero and one. It places a candidate influent or control value inside each selected stratum, excluding its endpoints. The comparison of sampling designs by McKay et al. (1979) provides the basis for this coverage-oriented design.

Table 6.6: Fixed hydrolysis and heterotrophic rate and saturation constants.

Parameter	Value	Unit
K_H	3.0	d^{-1}
$\eta_{\text{hyd},\text{NO}_3}, \eta_{\text{hyd},\text{NO}_2}$	0.6	dimensionless
$\eta_{\text{hyd},\text{fe}}$	0.4	dimensionless
$K_{\text{O},\text{hyd}}, K_{\text{O},\text{H}}, K_{\text{O},\text{PAO}}$	0.20	g oxygen m^{-3}
$K_{\text{NO}_3,\text{hyd}}, K_{\text{NO}_2,\text{hyd}}, K_{\text{NO}_x,\text{hyd}}$	0.5	g nitrogen m^{-3}
K_X	0.10	g substrate COD per g biomass COD
μ_H	6.0	d^{-1}
q_{fe}	3.0	d^{-1}
b_H	0.40	d^{-1}
$\eta_{\text{H},\text{NO}_3}, \eta_{\text{H},\text{NO}_2}$	0.9	dimensionless
K_F, K_{fe}, K_A	4.0	g COD m^{-3}
$K_{\text{NO}_3,\text{H}}, K_{\text{NO}_2,\text{H}}, K_{\text{NO}_x,\text{H}}$	0.5	g nitrogen m^{-3}
$K_{\text{NH}_4,\text{H}}, K_{\text{NH}_4,\text{PAO}}, K_{\text{NH}_4,\text{NOB}}$	0.05	g nitrogen m^{-3}
$K_{\text{PO}_4,\text{H}}, K_{\text{PO}_4,\text{PAO}}, K_{\text{PO}_4,\text{nit}}$	0.01	g phosphorus m^{-3}
$K_{\text{ALK},\text{H}}, K_{\text{ALK},\text{PAO}}$	0.10	mol equivalents m^{-3}

The study design attempts 8,000 development candidates and 2,000 holdout candidates from separate seeded streams. Rejected candidates are retained in the accounting and are not replaced. The two-start physical acceptance procedure determines which candidates enter a statistical partition. Acceptance counts are recorded for both partitions.

The sampling arithmetic can be understood with four strata in one hypothetical coordinate. Each stratum has width $1/4$ in the unit interval. Number the strata from zero to three. Selecting stratum two and a within-stratum fraction of 0.6 gives $(2 + 0.6)/4 = 0.65$. For an illustrative physical interval from 10 to 20, this becomes $10 + 0.65(20 - 10) = 16.5$. Applying independent permutations to the other coordinates retains one observation in each marginal stratum without forcing the same combinations in every coordinate.

The exact jitter expression uses the same idea at a much finer resolution. The floor operation $\lfloor v \rfloor$ means the largest integer no greater than v . Dividing U by 2^{11} and taking its floor yields an integer from zero to $2^{53} - 1$. Adding 0.5 selects the midpoint of the corresponding fine stratum. Division by 2^{53} places that midpoint inside the unit interval. In exact arithmetic, the extreme values are 2^{-54} and $1 - 2^{-54}$. Both are strictly inside the interval. Endpoint checks remain necessary when values are represented with finite numerical precision.

For an exact numerical randomization convention, each coordinate uses a descending Fisher–Yates permutation of its strata. The jitter from an unsigned 64-bit random

Table 6.7: Fixed storage, nitrifier, and chemical rate and saturation constants.

Parameter	Value	Unit
q_{PHA}	5.0	g stored COD per g biomass COD per d
q_{PP}	0.60	g phosphorus per g biomass COD per d
μ_{PAO}	0.56	d ⁻¹
η_{PAO,NO_3}	0.07	dimensionless
η_{PAO,NO_2}	0.90	dimensionless
$b_{PAO}, b_{PP}, b_{PHA}, b_{AOB}$	0.20	d ⁻¹
$K_{NO_3,PAO}, K_{NO_2,PAO}, K_{NO_x,PAO}$	0.5	g nitrogen m ⁻³
K_{PS}	0.20	g phosphorus m ⁻³
K_{PP}	0.01	g phosphorus per g biomass COD
K_{max}	0.34	g phosphorus per g biomass COD
K_{IPP}	0.02	g phosphorus per g biomass COD
K_{PHA}	0.01	g stored COD per g biomass COD
μ_{AOB}	1.81	d ⁻¹
μ_{NOB}	1.52	d ⁻¹
b_{NOB}	0.17	d ⁻¹
$K_{O,AOB}$	0.74	g oxygen m ⁻³
$K_{O,NOB}$	1.75	g oxygen m ⁻³
$K_{NH_4,AOB}, K_{NO_2,NOB}$	0.5	g nitrogen m ⁻³
$K_{ALK,nit}, K_{ALK,PRE}, K_{ALK,chem}$	0.50	mol equivalents m ⁻³
k_{PRE}	1.0	m ³ per g solids per d
k_{RED}	0.60	d ⁻¹

Table 6.8: Fixed settling and oxygen-transfer parameters.

Quantity	Value
Maximum settling velocity v_{max}	250 m d ⁻¹
Theoretical settling velocity v_0	474 m d ⁻¹
Hindered-settling coefficient r_h	0.000576 m ³ per g TSS
Low-concentration coefficient r_p	0.00286 m ³ per g TSS
Nonsettleable fraction f_{ns}	0.00228
Receiver threshold X_t	3,000 g TSS m ⁻³
Oxygen saturation S_O^*	8.5 g oxygen m ⁻³
Maximum aerated-stage k_La	47 d ⁻¹

Table 6.9: Randomization and fitting settings specified before assessment.

Setting	Value
Development Latin-hypercube seed	100042
Holdout Latin-hypercube seed	100043
Influent-scenario Latin-hypercube seed	314159
Grouped-fold permutation seed	271828
Number of fitting folds	5
Raw-response and logarithmic ridge grids	$\{10^{-8}, 10^{-7}, \dots, 10^2\}$
Standard-deviation divisor	fitting-set size
Near-constant coordinate cutoff	$10^{-12} \max(1, \max_i v_i)$
Low-overflow group	at or below development 25th percentile
High-overflow group	at or above development 90th percentile

integer U is

$$\zeta_U = \frac{\lfloor U/2^{11} \rfloor + 0.5}{2^{53}} \in (0, 1). \quad (6.182)$$

This definition excludes exact stratum boundaries. Accepted states retain their candidate ordering before the grouped-fold permutation is applied.

The fitting and prediction assessments use the states admitted by the integration, balance, branch, and stability checks. Rejection counts retain the corresponding outcomes from the attempted design.

Two initial states and the nonsmoothed steady solve

The first initial state assigns c_{in} to every reactor and fresh-influent TSS to every clarifier layer. The second retains the influent soluble concentrations and multiplies the particulate concentrations in reactors 1–5 by (1.5, 2, 2.5, 3, 3.5). Its final-reactor TSS sets a layer profile with multipliers

$$(0.002, 0.005, 0.01, 0.03, 0.10, 0.50, 1.25, 2.25, 3.25, 4.00) \quad (6.183)$$

from surface to hopper. The different solids distributions probe sensitivity to the initial biomass and storage state. Two-start state agreement is one of the numerical acceptance checks.

The positive-state transformation follows from the scalar chain rule. Treat y_f as the numerical concentration in its assigned native unit and introduce $\eta_f = \log y_f$ for $y_f > 0$. The logarithm therefore acts on the numerical coordinate, with one native unit as its

implicit reference. Differentiating its inverse gives

$$y_f = \exp(\eta_f),$$

$$\frac{dy_f}{dt} = \exp(\eta_f) \frac{d\eta_f}{dt} = y_f \frac{d\eta_f}{dt}.$$

The original component balance requires $dy_f/dt = f_f(y)$. Equating the two expressions and dividing by positive y_f gives $d\eta_f/dt = f_f(y)/y_f$. The balance is transformed rather than replaced by another physical process.

For a hypothetical concentration of two native units and a balance rate of -0.4 native units per day, the transformed rate is $-0.4/2 = -0.2$ per day. A simple half-day illustrative step in logarithmic coordinates gives

$$\eta_{new} = \log 2 + 0.5(-0.2), \quad y_{new} = \exp(\eta_{new}) = 2 \exp(-0.1) \approx 1.8097.$$

This example explains the algebra of a positive coordinate. The prescribed numerical integration retains the backward differentiation method and its error controls. The integration error controls and steady-state acceptance checks are applied separately.

Time integration uses the original nonsmoothed equations. A logarithmic state transformation keeps integrated coordinates positive. Before this transformation alone, an initial coordinate smaller than 10^{-8} is raised to 10^{-8} in its own physical unit. For $y_f = \exp(\eta_f)$, the transformed balance is $\dot{\eta}_f = f_f(y)/y_f$. This initialization safeguard is distinct from clipping a final predicted concentration.

The integration horizon contains two nested choices. First, the wasting coordinate is bounded below by 0.001 for this horizon calculation. Second, its reciprocal term is compared with a minimum horizon of 400 days. At a hypothetical $w = 0.01$, the inner maximum is 0.01, so the reciprocal term is $50/0.01 = 5000$ and the horizon is 5000 days. At $w = 0.20$, that term is 250, so the horizon is 400 days. At $w = 0$, the inner floor gives $50/0.001 = 50,000$ days. These arithmetic examples illustrate the declared rule and are not claims about actual convergence durations.

The values zero and 0.20 illustrate the branches of the horizon rule and lie outside the case-study wasting interval. They are not admitted operating candidates.

The backward differentiation method provides an integration route for stiff equations (Curtiss & Hirschfelder, 1952). The horizon is

$$T = \max\{400, 50/\max(w, 0.001)\} \text{ d.} \quad (6.184)$$

The calculation can stop early when its scaled residual falls below 10^{-9} . Dynamic relaxation supplies an initial state for algebraic steady calculation, consistent with the continuation rationale of Kelley and Keyes (1998) and its activated sludge use in Selerio (2026). A bounded least-squares polishing calculation is applied only if the integrated endpoint fails the physical audit. Solver convergence and physical acceptance use separate tolerances.

Table 6.10: Steady-state generation and physical acceptance settings.

Setting	Value
Integration relative tolerance	10^{-7}
Transformed integration absolute tolerance	10^{-9}
Maximum integration step	$T/100$
Polishing evaluation limit	5,000
Polishing step, function, and gradient tolerances	10^{-9} each
Term-scaled balance tolerance	10^{-6}
State non-negativity tolerance	10^{-10}
Reaction-rate non-negativity tolerance	10^{-12}
Two-start scaled state agreement	10^{-6}
Rightmost reduced-state Jacobian eigenvalue	at most -10^{-8} d^{-1}
Eigenvalue agreement between two step sizes	10^{-6} d^{-1}
Matrix mass conservation and rank audit tolerance	10^{-10}

A residual first adds the signed physical terms in a balance. For hypothetical incoming, consumption, and outgoing contributions of 100, -60 , and -39.99999 in one consistent component-rate unit, the remaining imbalance is

$$100 - 60 - 39.99999 = 0.00001.$$

The largest absolute term is 100. Dividing the absolute imbalance by that scale gives $0.00001/100 = 10^{-7}$. The floor of one in the scale prevents division by a very small reference when all terms are near zero. The resulting dimensionless ratio assesses cancellation relative to the terms in that component equation. It is evaluated separately for each balance rather than obtained by adding balances with different constituent units.

For a balance with separately evaluated terms b_j , the acceptance residual is

$$r_{\text{bal}} = \frac{|\sum_j b_j|}{\max(1, \max_j |b_j|)}. \quad (6.185)$$

Reactor generation scales repeat $\max(1, c_{\text{in},f}^U)$ for each component. Layer scales use $\max(1, X_N)$. State agreement is tested in these scaled coordinates.

Local stability asks what happens to a small perturbation about an equilibrium. A first-order expansion of the dynamic equations has the form $\dot{\delta y} = J_{\text{dyn}} \delta y$, where entry (f, j) of J_{dyn} is the derivative of balance f with respect to state j . In a hypothetical independent two-coordinate example,

$$\begin{bmatrix} \dot{\delta y}_1 \\ \dot{\delta y}_2 \end{bmatrix} = \begin{bmatrix} -0.2 & 0 \\ 0 & 0.05 \end{bmatrix} \begin{bmatrix} \delta y_1 \\ \delta y_2 \end{bmatrix}.$$

The scalar equations are $\dot{\delta y}_1 = -0.2\delta y_1$ and $\dot{\delta y}_2 = 0.05\delta y_2$. Their perturbations evolve as $\delta y_1(0) \exp(-0.2t)$ and $\delta y_2(0) \exp(0.05t)$ when time is in days. One decays and the

other grows. The two diagonal eigenvalues are -0.2 and 0.05 per day, so the rightmost eigenvalue is positive and this hypothetical equilibrium fails the stability condition.

For a coupled system, an eigenvector gives a combination of coordinates that behaves as one local mode. The real part of its eigenvalue determines growth or decay. The rightmost eigenvalue means the largest real part among all modes. This is a local test because the linear expansion describes small perturbations. Checking derivatives with two finite-difference steps tests numerical sensitivity of the estimated modes, while the stated negative threshold tests their local decay.

Reduced-state local stability is assessed by the rightmost eigenvalue of the dynamic Jacobian. Finite differences use normalized steps $\epsilon_{\text{mach}}^{1/3}$ and $2\epsilon_{\text{mach}}^{1/3}$. Central differences are used away from the non-negative boundary. Second-order forward differences are used near it. Both eigenvalue estimates must pass the upper-bound and agreement checks. These checks assess local stability of the reactor-and-aggregate-layer reference model.

Both starts must provide finite states, satisfy non-negativity and rate checks, pass component and invariant mass conservation checks, obey the operating domain and layer envelope, and meet the layer-flux residual tolerance. Generation also requires identical branch-classification tuples between the two starts. These tuples record the active kinetic and settling branches. A state that meets a residual test but fails another physical check is not admitted as a training target.

Fold-local fitting and two separately selected penalties

Every component and location from one accepted plant state stays in the same fold. This grouping prevents the same simulated plant from contributing training information and an assessment target within a fold. Fold sizes differ by at most one. Centers, input scales, quadratic-feature scales, and response scales are recomputed inside each fitting fold. The holdout contributes none of these quantities. The information-leakage analysis of Kaufman et al. (2012) explains why assessment information must be kept outside fitting choices.

The one-standard-error rule can be followed with a hypothetical five-fold comparison. Suppose the candidate with the smallest mean score has fold scores 0.40 , 0.42 , 0.38 , 0.41 , and 0.39 . Their mean is 0.40 . The sum of squared deviations from that mean is

$$0^2 + 0.02^2 + (-0.02)^2 + 0.01^2 + (-0.01)^2 = 0.001.$$

The sample standard deviation is $\sqrt{0.001/4} \approx 0.01581$. The standard-error summary is $0.01581/\sqrt{5} \approx 0.00707$. The admissible mean-score threshold is consequently $0.40 + 0.00707 = 0.40707$.

If a stronger hypothetical penalty has mean score 0.406 , it lies within the threshold. A still stronger candidate with mean score 0.430 lies outside. The rule retains the

largest candidate penalty within the threshold. It does not select the numerical score 0.40707 as a penalty. This hypothetical example explains the regularization choice.

Each candidate penalty is assessed by the standardized root mean square error of the complete raw response. The largest penalty within one standard error of the smallest mean score is selected. The rule favors stronger regularization when the score does not clearly distinguish the smaller penalty (Hastie et al., 2009). Projected errors do not choose the raw-response penalty. The logarithmic model uses the same folds, grid, and one-standard-error rule but is scored on its logarithmic target. Its penalty is selected separately.

The augmented ridge equations are solved with column-pivoted orthogonal-triangular factorization and audited using singular-value decomposition. A coordinate whose standard deviation meets the near-constant cutoff in Table 6.9 is rejected rather than assigned an arbitrary scale. Positive ridge penalties and the unpenalized intercept give a unique coefficient solution under the stated fitting design. The fitted penalties and coefficient matrices are retained as estimation quantities. The candidate grid defines the eligible penalties, and cross-validation selects their fitted values.

After selection, both models are refitted once on all accepted development states. Grouped out-of-fold predictions supply trust calibration and projection assessment. The overflow equality for an out-of-fold state uses the corresponding out-of-fold logarithmic model, not the final fit that has seen that state. The final models supply holdout predictions and optimization evaluations.

The overflow target must be strictly positive before logarithmic fitting. Predicted logarithms and their back-transformations must be finite. The development-set low-overflow and high-overflow strata are fixed by the percentiles in Table 6.9. Separate stratum assessment can reveal an error pattern concealed by a central concentration range. The imbalanced-regression analysis of Shahbazi et al. (2026) motivates attention to sparse or extreme response ranges. Separate concentration-stratum scores describe these response ranges.

Scientific admission and projection audits

Scientific admission is the decision to accept the fitted plant surrogate for operating assessment under the specified numerical criteria. Out-of-fold predictions are calculated for development cases excluded from the corresponding fit. They provide checks on new-case prediction within that development design before the surrogate is used to select controls.

Admission of the connected surrogate requires finite and independently audited out-of-fold projections for all accepted development states. The complete raw-response normalized root mean square error and the raw clarifier-inventory normalized error must each be below one. Because errors use development response scales, the threshold requires aggregate error below one scale unit. This threshold defines the case-study

admission criterion.

Every projection is assessed for equality feasibility, physical inequalities, non-negativity, dual feasibility, stationarity, and complementarity. The overflow-closure residual is also reported separately. A failed admission criterion remains a failure rather than triggering an unreported refit or changed penalty selection rule.

Table 6.11: Projection solve and independent audit settings.

Setting	Value
Solver absolute and relative tolerances	10^{-12} each
Initial iteration limit	100,000
Polishing	enabled
Retry values of algorithm parameter ρ_{alg}	0.01 and 10
Retry iteration limit	200,000 each
Adaptive ρ_{alg} during retries	disabled
Scaled equality and inequality tolerances	10^{-8} each
Dual, stationarity, and complementarity tolerances	10^{-8} each
Physical non-negativity tolerance	10^{-10}
Active-set identification tolerance	10^{-7}

Prediction metrics and their denominators

The error calculations can be expanded with two hypothetical observations of one concentration coordinate. Let the reference values be 2 and 6 and the predictions be 3 and 4 in the same native unit. Subtracting reference from prediction gives residuals $3 - 2 = 1$ and $4 - 6 = -2$. Their different summaries are

$$\begin{aligned}
 \text{mean squared error} &= \frac{1^2 + (-2)^2}{2} = 2.5, \\
 \text{root mean square error} &= \sqrt{2.5} \approx 1.5811, \\
 \text{mean absolute error} &= \frac{|1| + |-2|}{2} = 1.5, \\
 \text{bias} &= \frac{1 + (-2)}{2} = -0.5.
 \end{aligned}$$

The mean squared error has squared concentration units. The other three quantities have concentration units. Bias retains the signs and can be small when positive and negative errors cancel.

For the same illustrative values, the reference mean is $(2 + 6)/2 = 4$. The squared error of the two predictions is $1^2 + (-2)^2 = 5$. Always predicting the reference mean instead would give squared error $(2 - 4)^2 + (6 - 4)^2 = 8$. Their ratio is $5/8$, so the coefficient of determination is $1 - 5/8 = 0.375$. The denominator describes variation of

the assessment reference. It is not a training scale or a concentration limit. The general formulas record these same operations over all assessed observations.

For a response coordinate f and assessment state i , define the signed error $a_{if} = p_{if} - t_{if}$, where p is predicted and t is mechanistic. Physical-unit root mean square error, mean absolute error, and bias are

$$\left(\text{mean}_i a_{if}^2\right)^{1/2}, \quad \text{mean}_i |a_{if}|, \quad \text{mean}_i a_{if}. \quad (6.186)$$

Normalized joint-response metrics replace a_{if} by $a_{if}/(D_\chi)_{ff}$ and state whether they pool errors across coordinates. Coordinatewise coefficients of determination use

$$R_f^2 = 1 - \frac{\sum_i (p_{if} - t_{if})^2}{\sum_i (t_{if} - \bar{t}_f)^2}. \quad (6.187)$$

A constant assessment target has an undefined denominator and is recorded accordingly. An arithmetic mean of coordinate scores is distinct from a score calculated on pooled observations.

Normalization changes the numerical comparison by assigning a denominator to each coordinate. In the preceding hypothetical error example, suppose the training response scale is two native units. Dividing residuals 1 and -2 by two gives 0.5 and -1 . Their standardized root mean square error is

$$\sqrt{\frac{0.5^2 + (-1)^2}{2}} = \sqrt{0.625} \approx 0.7906.$$

If a descriptive holdout report instead uses the observed range $6 - 2 = 4$, its normalized residuals are 0.25 and -0.5 . Their root mean square summary is approximately 0.3953. The predictions have not changed. The denominator has changed. This explains why a training-scale admission score and a holdout-range reporting score must be identified separately.

Location-quality summaries cover the mixer, five reactors, overflow, and underflow. Joint constraints link their errors, so the assessment retains each location alongside the plant aggregate. These reporting quantities connect prediction assessment to the treatment and resource measures used in operating decisions (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019). Outlet flows are divided by their own positive flow ratios before applying the quality map. For location ℓ and composite j , define the holdout reporting range and normalized error as

$$r_{\ell j} = \max_i t_{i\ell j} - \min_i t_{i\ell j}, \quad e_{i\ell j} = \frac{p_{i\ell j} - t_{i\ell j}}{r_{\ell j}}. \quad (6.188)$$

For positive ranges, an aggregate score pools the 8×4 location-composite coordinates. A location score pools four composites and a composite score pools eight locations. The displayed R^2 summaries average the corresponding coordinatewise scores. The holdout ranges serve only reporting. They do not enter fitting, admission, objective normalization, or trust calibration.

The overflow model receives concentration-scale and logarithmic assessment separately. The concentration report includes root mean square error, mean absolute error, bias, normalized error, and R^2 . The logarithmic report includes root mean square error and bias. Strict positivity of Equation (6.79) does not replace these assessments.

Exploratory simulation informs the specified response, closure, and acceptance rules. Development data supply input transformations, response scales, selected penalties, overflow fitting, and trust calibration. The separate holdout stream evaluates the resulting fitted procedure. The operating evaluation uses independently solved connected responses at the selected controls. Fitted coefficients, preprocessing, priorities, trust thresholds, engineering safeguards, and numerical settings remain fixed throughout these assessments (Kaufman et al., 2012).

6.3.3 Operating searches and convergence evidence

A common baseline and an explicit basin-sensitivity experiment

A normalized control measures a fraction of its permitted interval. Consider an illustrative interval from 10 to 30 physical units. A control value of 15 lies five units above its lower bound in an interval of width twenty, so its normalized value is $5/20 = 0.25$. The midpoint 20 maps to 0.5, and the upper bound 30 maps to one. Conversely, a normalized value of 0.75 maps back to $10 + 0.75(30 - 10) = 25$. The general forward and inverse relations therefore use the same interval width,

$$\vartheta_j = \vartheta_j^L + (\vartheta_j^U - \vartheta_j^L)\eta_j.$$

This transformation allows operating variables with different units to use comparable numerical step sizes. It does not assign them equal importance in the engineering objective.

The baseline search for each route starts at the center of the normalized operating box. For each coordinate,

$$\eta_j = \frac{\vartheta_j - \vartheta_j^L}{\vartheta_j^U - \vartheta_j^L} \in [0, 1]. \quad (6.189)$$

The nominal influent is the midpoint of the fresh-influent intervals. The baseline explores one initial basin. The baseline records a local search from this common starting point.

For seven controls, the normalized center has seven entries equal to 0.5. Moving only the first coordinate upward by one quarter of its interval makes that coordinate 0.75. Moving it downward makes that coordinate 0.25. The six other entries remain 0.5 in both cases. Writing out the entries gives

$$\begin{aligned} \eta_{\text{first},+} &= (0.75, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top, \\ \eta_{\text{first},-} &= (0.25, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5)^\top, \\ e_1 &= (1, 0, 0, 0, 0, 0, 0)^\top. \end{aligned}$$

The unit coordinate vector e_1 records which coordinate moves. Two signs for each of seven coordinates give fourteen displaced starts. Adding the center gives fifteen. The compact expression below writes this same collection without listing every vector.

A planned multistart sensitivity experiment evaluates basin dependence separately from this baseline. It uses the same 15 starting control vectors for both routes,

$$\eta^{(0)} = \frac{1}{2}\mathbf{1}_7, \quad \eta^{(j,\pm)} = \frac{1}{2}\mathbf{1}_7 \pm \frac{1}{4}e_j, \quad j = 1, \dots, 7. \quad (6.190)$$

Each start receives the same route-specific budget and audit rules. A direct start obtains its mechanistic initial state from an independently accepted solve at those fixed controls. Failed initial states remain failures. The best independently validated feasible candidate is retained, with all starting-point outcomes and total search work reported. The reported baseline uses the single center start. The fifteen-start design defines a separate sensitivity analysis.

The distinction matters when interpreting a faster search. An algorithm that visits fewer basins can finish quickly while missing a better candidate. The expensive-function benchmarking study of Bliet et al. (2023) motivates keeping objective quality, evaluation work, and computational time as separate comparison dimensions.

Surrogate search and derivative fallback

The initial surrogate outer method is sequential least squares programming, based on the constrained optimization approach of Kraft (1988). Each trial obtains a fully audited joint projection. The active-set derivative chain is used only after the rank, conditioning, multiplier, and independent derivative checks pass. An active-set change requires reconstruction of the projection and its derivatives.

When that chain cannot be validated, a deterministic value-only quadratic-approximation method continues within the normalized bounds (Ragoneau, 2022). It uses the same audited projection and does not substitute an approximate lower solve. Finite differences are used for independent derivative audits rather than as an unqualified derivative replacement. The best feasible visited point is solved again from an independent projection initialization before retention.

First-order evidence and finite-resolution polling

A candidate receives first-order stationarity evidence only when the derivative chain, lower projection certificate, and upper constrained-stationarity audit pass. This qualification concerns stationarity of the declared surrogate optimization problem at that candidate.

Coordinate and coupled directions can have different feasibility near an active constraint. Consider a hypothetical two-control problem with $x + y = 1$, non-negative controls, and objective $J = x$. At $(0.5, 0.5)$, any nonzero move in x alone or y alone

Table 6.12: Surrogate search, sensitivity, and finite-poll settings.

Setting	Value
Initial outer iteration limit	250
Initial outer function tolerance	10^{-10}
Active slack tolerance	10^{-7}
Required active inequality multiplier	greater than 10^{-8}
Domain-aware active-set perturbation	10^{-6}
Stationary-system condition number times machine precision	at most 10^{-8}
Sensitivity residual and scaled state reproduction	10^{-8} each
Independent derivative steps	10^{-6} and 5×10^{-7}
Derivative relative-error tolerance	10^{-5}
Upper feasibility and stationarity tolerance	10^{-6} each
Value-only fallback iteration and evaluation limits	250 each
Initial and final fallback trust radii	0.25 and 10^{-6}
Fallback feasibility tolerance	10^{-6}
Coarse and fine poll radii	10^{-3} and 10^{-4}
Coarse and fine direction counts	14 and 106
Absolute and relative sufficient-decrease tolerances	10^{-8} each
Direction-rank tolerance	10^{-10}
Certification evaluation limit	10,000
Ray growth factor and maximum probes	2 and 16
Objective relative tie tolerance	10^{-10}

violates the equality. A diagonal move that lowers x and raises y by the same amount preserves it. For a unit direction $(-1, 1)^\top/\sqrt{2}$ and illustrative step radius 0.1, the trial is

$$\begin{aligned}x_{trial} &= 0.5 - 0.1/\sqrt{2} \approx 0.4293, \\y_{trial} &= 0.5 + 0.1/\sqrt{2} \approx 0.5707, \\x_{trial} + y_{trial} &= 1.\end{aligned}$$

The objective decreases by approximately 0.0707. An axis-only poll would contain no feasible nonzero trial in this example, even though a feasible improving direction exists.

In seven coordinates, each of the seven axes has a positive and a negative direction, giving fourteen. There are $7(6)/2 = 21$ unordered coordinate pairs. Each pair admits four sign combinations, giving $4(21) = 84$ pairwise directions. Eight regular-simplex directions supplement them. A regular simplex is the higher-dimensional counterpart of an equilateral triangle. Its directions distribute around a center rather than following only the coordinate axes. The fine direction count is $14 + 84 + 8 = 106$. These finite collections test coordinate and coupled moves.

If the derivative evidence is unavailable, the convergence procedure performs complete feasible-direction polls at normalized radii 10^{-3} and 10^{-4} . The coarse poll contains the 14 signed coordinate directions. The fine poll adds 84 signed pairwise diagonals and eight regular-simplex directions from a Helmert construction. The pairwise count is $4\binom{7}{2}$. Coupled directions can reveal feasible improvements that an axis-only test misses near a constraint.

Each complete poll must contain a feasible nonzero displacement. A rank calculation is diagnostic because active constraints can reduce the dimension of the feasible cone. An accepted improvement triggers a new complete poll. A fine-radius move also triggers revalidation at the coarse radius. After an improving step, up to 16 search-only probes grow along the same ray by successive factors of two. These probes may accelerate search but supply no convergence certificate.

Absence of sufficient decrease in both complete revalidated polls gives evidence only for the tested directions and radii. An incomplete poll, a failed calculation, or an exhausted certification budget leaves stationarity unresolved. The direction design follows the role of feasible polling discussed in the direct-search analysis of Audet and Dennis (2006).

Direct continuation, recovery, and endpoint checks

The direct route uses the interior-point filter algorithm of Wächter and Biegler (2006) through the three smoothing stages. Its endpoint audit independently checks all smooth reactor and layer equations, inequalities, non-negativity, multipliers, and stationarity. The full-state audit is separate from the final solver status.

If the initial direct search fails, one recovery may start from the same influent's certified surrogate controls and repeat the same smoothing sequence. Recovery is not

attempted after an initially successful search or from an uncertified surrogate candidate. Because this recovery uses a surrogate decision, it is a dependent rescue calculation rather than independent basin exploration. Its status and computational cost are reported separately.

Table 6.13: Direct continuation and exact-evaluation qualification settings.

Setting	Value
Interior-point iteration limit per smoothing stage	2,500
Convergence and constraint tolerances	10^{-8} each
Dual and complementarity tolerances	10^{-6} each
Endpoint equality acceptance	10^{-8}
Endpoint inequality and stationarity acceptance	10^{-6} each
Endpoint active-set tolerance	10^{-7}
Receiver-switch ambiguity distance from X_t	1 g TSS m^{-3}
Other branch ambiguity distances	$10^{-6}S$
Two-start agreement under both prescribed state scales	10^{-6}
Engineering feasibility tolerance after exact evaluation	10^{-6}

A validated feasible candidate can be retained when its search budget is exhausted or stationarity remains unresolved. Its convergence qualification remains visible. A case with no validated feasible candidate remains a search failure. A favorable success flag alone is insufficient for either route.

6.3.4 Independent mechanistic verification and fair comparison

The two searches use different response calculations but must be assessed on the same original-model basis. Figure 6.4 separates the search, native-candidate audit, independent physical evaluation, and comparison eligibility. Each stage retains its own outcome.

The common evaluation stage supplies treatment responses at fixed controls rather than returning information to the fitted models. A valid physical response can still violate a retained engineering safeguard and remain useful as a diagnostic. Separating these outcomes prevents numerical convergence, physical validity, and decision eligibility from being treated as the same property.

Evaluation at fixed selected controls

Every available selected decision is evaluated with its controls and influent held fixed. Both initial states are solved on the original nonsmoothed reactor and ten-layer clarifier equations. The independently solved state supplies the reference treatment

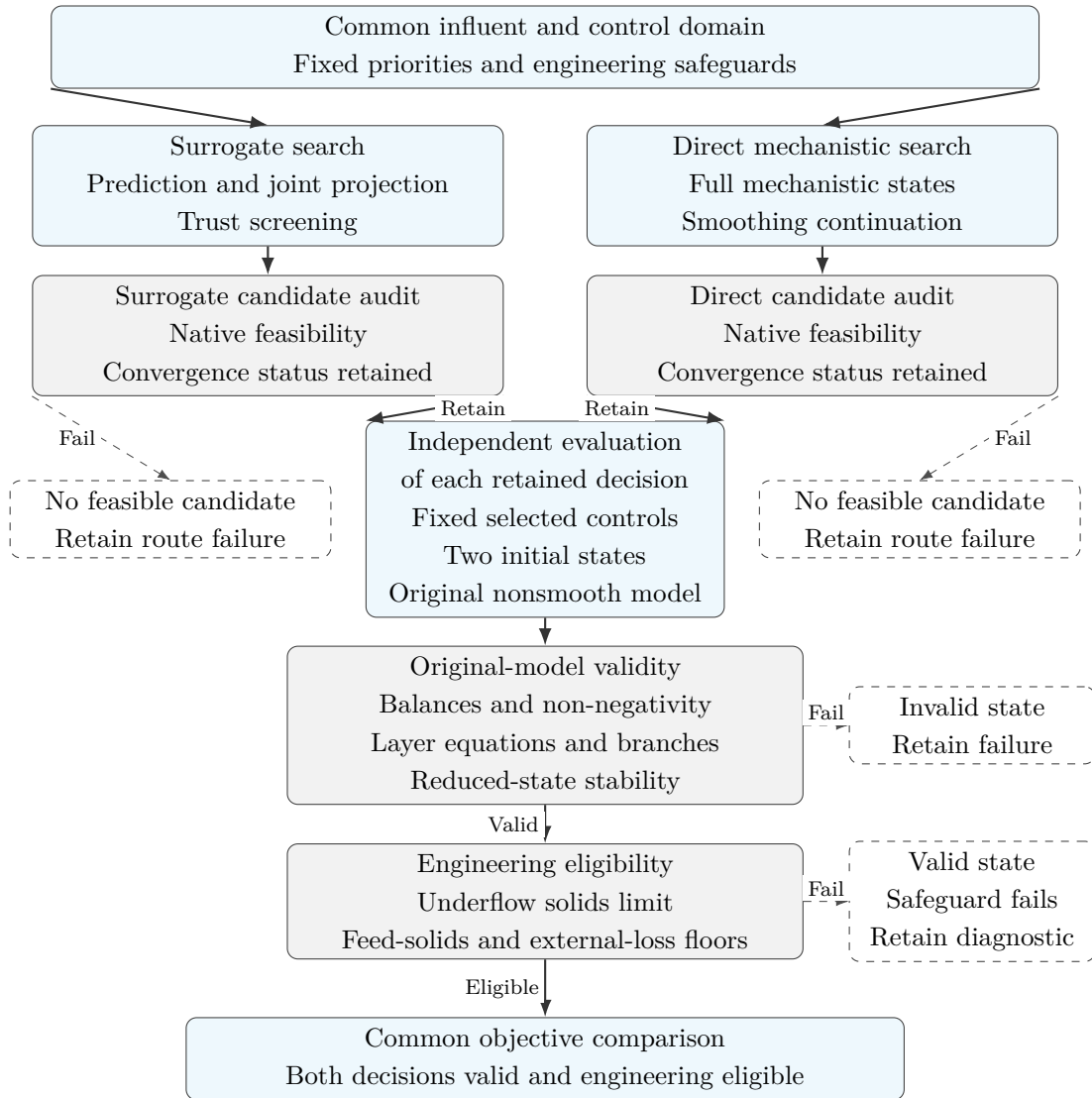


Figure 6.4: Conceptual paired-search and verification workflow. Each retained control vector receives its own independent original-model evaluation. The surrogate and smooth-direct native responses do not supply the reference treatment outcome. Original-model validity and engineering eligibility are distinct checks. Only eligible pairs enter the common-objective comparison, while failed or qualified cases remain in the accounting.

outcome for either route. The role of an original-model check follows the feasibility and validation requirements discussed by Bhosekar and Ierapetritou (2018).

Exact evaluation uses the physical checks in Table 6.10. Two-start state agreement is tested under both generation scales and fixed state scales. The fixed scales are development-set coordinatewise median absolute deviations with a floor of one. A valid evaluation must pass mass conservation, non-negativity, layer residuals, the domain, the endpoint envelope, and the reduced-state stability test.

Branch tuples must agree unless an evaluation start is boundary-ambiguous. Ambiguity bands cover the receiver threshold, zero storage capacity, zero settling offset, zero or maximum settling velocity, and equal adjacent gravitational fluxes. An ambiguity flag qualifies the state-level branch comparison. It does not prove that every mismatching branch falls within its own ambiguity band. Agreement of the physical states and all other audits is still required. Smooth stationarity or smooth-reference equivalence near such a boundary requires that qualification.

This verification is especially relevant near a prescribed nutrient limit. The aeration-scheduling framework of Ren et al. (2026) explicitly considers influent disturbances and effluent ammonium safeguards. This approach supports evaluating the particular effluent quantity and treatment requirement that governs an operating decision.

Native and common-reference objectives

Let route S denote the connected surrogate and route M the direct smooth mechanistic search. Their native responses are $\chi_{\text{nat},S} = \hat{\chi}$ and $\chi_{\text{nat},M} = \chi_{\text{sm},M}$. The independently solved responses are $\chi_{\text{ref},S}$ and $\chi_{\text{ref},M}$. Raw and projected surrogate predictions are also evaluated, where defined, at both retained control vectors.

Route-specific normalization can change an objective even when the physical response is identical. Consider an illustrative two-quality case with positive development standard deviations 0.5 and 2 in the corresponding native units. A scale with a floor of one gives denominators 1 and 2. The unfloored scale retains 0.5 and 2. For a hypothetical response of 0.8 and 4, their standardized coordinates are

$$\begin{aligned}\text{floored scale} &= (0.8/1, 4/2) = (0.8, 2), \\ \text{unfloored scale} &= (0.8/0.5, 4/2) = (1.6, 2).\end{aligned}$$

With equal quality weights, the means are 1.4 and 1.8. This difference arises entirely from the normalization. Writing the diagonal matrices makes the componentwise denominators explicit,

$$D_{T,S}^{\text{toy}} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad D_{T,M}^{\text{toy}} = \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix}.$$

These matrices illustrate the two general definitions. The comparison of selected decisions uses one common reference normalization so that such a numerical choice is not mistaken for better treatment.

For positive development-set effluent standard deviations σ_j , native quality normalization is

$$D_{T,S} = \text{diag}(\max(1, \sigma_j)), \quad D_{T,M} = \text{diag}(\sigma_j). \quad (6.191)$$

The unit floor is in the physical unit of each quality coordinate. The common reference objective uses $D_{T,M}$ for both decisions. Actual fitted standard deviations are estimation quantities and are not specified as theoretical constants.

A hypothetical comparison first assigns native objective values of 0.90 and 0.95 to two operating choices. Prediction favors the first choice by $0.95 - 0.90 = 0.05$. Independent mechanistic evaluation gives 1.10 and 0.98 at the same choices, reversing their ordering. Two distinct subtractions describe this result. For the first choice, the native-minus-reference difference is $0.90 - 1.10 = -0.20$. Its negative sign means that the native calculation presents a smaller objective than the independent evaluation.

For an illustrative direct-route decision with native objective 0.95 and reference objective 0.98, its native-minus-reference difference is -0.03 . Comparing the independently evaluated decisions instead gives $1.10 - 0.98 = 0.12$. On the shared objective basis, this positive difference means the surrogate-selected decision has the larger objective among these two choices. Both projected responses can satisfy the imposed balances while the independently evaluated objectives have this ordering. The arithmetic compares these two retained choices on a common reference basis (Alexandrov et al., 1998; Pedrozo et al., 2025). The compact expressions distinguish within-route prediction differences from between-route decision differences.

The native-to-reference and route-to-route differences are

$$\Delta J_{\text{nat-ref},k} = J_{\text{nat},k}(\vartheta_k, \chi_{\text{nat},k}) - J_{\text{ref}}(\vartheta_k, \chi_{\text{ref},k}), \quad k \in \{S, M\}, \quad (6.192)$$

$$\Delta J_{S-M} = J_{\text{ref}}(\vartheta_S, \chi_{\text{ref},S}) - J_{\text{ref}}(\vartheta_M, \chi_{\text{ref},M}). \quad (6.193)$$

The first can reflect both response approximation and route normalization. The second compares independently evaluated decisions using a common objective basis (Bhosekar & Ierapetritou, 2018; Pedrozo et al., 2025). The route-to-route difference is reported as a paired objective comparison.

An objective comparison and a control comparison also have different denominators. A hypothetical objective difference of 0.12 with reference direct objective 0.98 has direct-relative percentage $100(0.12)/0.98 \approx 12.24\%$. The symmetric scaled difference uses the denominator $\max(1, 1.10, 0.98) = 1.10$ and gives $0.12/1.10 \approx 0.1091$. The first uses one route as its reference. The second uses a symmetric magnitude scale with a floor.

For an illustrative control interval from 10 to 30, two selected settings of 25 and 20 differ by five physical units. Their normalized difference is $5/(30 - 10) = 0.25$. If this is the only differing coordinate among seven, the root mean square difference is $\sqrt{0.25^2/7} \approx 0.0945$, while the maximum absolute difference is 0.25. The average spreads one change over all seven coordinates. The maximum keeps the largest change visible.

Standardized response errors at each decision retain the raw, projected, and native quantities separately. A symmetric relative objective difference divides ΔJ_{S-M} by $\max(1, |J_S|, |J_M|)$. A direct-relative percentage divides $100\Delta J_{S-M}$ by J_M only when that denominator is nonzero. Normalized control differences are

$$d_j = \frac{\vartheta_{S,j} - \vartheta_{M,j}}{\vartheta_j^U - \vartheta_j^L}, \quad d_{\text{rms}} = \left(\frac{1}{7} \sum_{j=1}^7 d_j^2 \right)^{1/2}, \quad d_{\text{max}} = \max_j |d_j|. \quad (6.194)$$

These summaries distinguish similar objective values from similar controls.

Removal follows from the material entering and leaving on the stated concentration basis. For a hypothetical fresh-influent concentration of 200 mg L^{-1} and an effluent concentration of 2 mg L^{-1} , the concentration reduction is $200 - 2 = 198 \text{ mg L}^{-1}$. Dividing by 200 gives a fraction of 0.99, and multiplying by one hundred gives 99%. Algebraically,

$$100 \frac{z_{\text{in},j} - z_{E,j}}{z_{\text{in},j}} = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right).$$

This step requires a positive influent denominator. It compares concentrations on one reporting basis. A component inventory balance additionally uses the appropriate flow rates and represented outlets.

Selected-decision removal for composite j is

$$\mathcal{R}_j = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right), \quad (6.195)$$

when the fresh-influent denominator is positive. Predicted-to-reference removal differences use percentage points. Concentration percentage errors instead divide by the mechanistic effluent concentration. For a hypothetical influent of 200 mg L^{-1} and effluent values of 2 and 4 mg L^{-1} , the effluent prediction error is 100% relative to 2. The removal difference is only one percentage point. Reporting both makes their different denominators clear.

Validity, engineering eligibility, and failures

A paired objective comparison requires both routes to supply natively feasible candidates, valid exact evaluations, and satisfaction of the common retained engineering safeguards after evaluation. A valid physical evaluation that violates an underflow, feed-solids, or external-loss safeguard still retains its diagnostic objective and treatment response. It does not enter the eligible paired comparison. Solids retention time and loading measures are reported for every finite evaluation and do not add eligibility thresholds.

These status fields are not collapsed into one success label. Missing or nonfinite quantities, an undefined feature scale, a failed factorization, or an inconsistent numerical audit remain computational failures. A finite state that fails a scientific threshold

Table 6.14: Distinct questions retained in failure and qualification accounting.

Assessment	Question answered
Generation acceptance	Did both original-model starts pass the required physical and numerical checks?
Projection feasibility	Did the fitted closure and physical constraints admit an audited joint response?
Native engineering feasibility	Did the search candidate satisfy its declared optimization constraints?
Convergence qualification	Did derivative audits or complete finite polls supply their stated evidence?
Exact-model validity	Did independent original-model evaluation converge to an accepted state?
Retained engineering feasibility	Did the exact response pass the common safeguards?
Paired eligibility	Were both decisions available and admissible on the common reference basis?

remains a diagnostic state with the failed criterion identified. Generation rejection, optimization failure, projection infeasibility, and branch qualification have different causes and denominators.

Influent scenarios and time accounting

Ten influent vectors from a separate twenty-dimensional Latin hypercube define the scenario experiment. They use the seed in Table 6.9 and do not enter fitting, calibration, or tuning. Both routes are attempted for the nominal influent and every scenario. Available selected decisions receive exact evaluation even when the other route fails. These cases examine sensitivity to the specified influent vectors.

Optimization time is the duration of the route’s initial search. For the surrogate, it includes regression, projection, and any value-only fallback inside that search. For the direct route, it includes the initial three-stage smoothing sequence. Development sampling, fitting, convergence certification, conditional recovery, and exact-model evaluation are outside this interval. Failed initial attempts remain in the time accounting. Scenario aggregates use the ten influent cases and keep the nominal case separate. Both routes use the same processor allocation and thread count.

Search time is accompanied by evaluated points, iterations, projection retries, fallback use, completed smoothing stages, recovery status, and convergence qualification. The benchmark concerns identified by Blik et al. (2023) make these workload quantities useful when comparing expensive-function methods. Objective values and search qualification accompany the timing comparison.

A complete workflow account additionally includes development, certification, recovery, and verification. These tasks have separate cost terms. Repeated use can amortize development effort, but the relevant number of later decisions depends on the full per-decision cost and on acceptable mechanistic decision quality (Bhosekar & Ierapetritou, 2018; Blik et al., 2023). The complete-cost calculation uses all of these terms.

The effect of repeated use can be shown with hypothetical total costs. Suppose development requires 100 seconds, and each later surrogate-assisted decision requires three seconds in total for search, qualification, and verification. Suppose a directly evaluated decision requires six seconds for the corresponding complete task. For twenty decisions, the costs are $100 + 20(3) = 160$ seconds and $20(6) = 120$ seconds. For fifty decisions, they are 250 and 300 seconds. These hypothetical numbers illustrate amortization.

More generally, let n_{dec} be the number of later decisions, T_{dev} the development cost, and T_S and T_M the complete per-decision costs. Expanding the comparison gives

$$\begin{aligned} T_{dev} + n_{dec}T_S &< n_{dec}T_M, \\ T_{dev} &< n_{dec}(T_M - T_S), \\ n_{dec} &> \frac{T_{dev}}{T_M - T_S} \quad \text{when } T_M > T_S. \end{aligned}$$

The illustrative threshold is $100/(6 - 3) = 33\frac{1}{3}$, so at least thirty-four complete decisions are needed to cross it. If the complete per-decision cost is not smaller, a positive development cost cannot be recovered by this timing argument. The calculation also assumes that the compared decisions meet the required quality and feasibility conditions. Quality and feasibility are assessed through the common mechanistic evaluation.

The resulting evaluation keeps four types of evidence separate. Statistical assessment concerns approximation of accepted mechanistic states. Projection audits concern the imposed network and inventory constraints. Search audits concern local numerical evidence for the declared optimization problem. Original-model verification concerns treatment and feasibility at the selected controls. Together these assessments evaluate the connected surrogate as an operating decision aid (Bhosekar & Ierapetritou, 2018; Pedrozo et al., 2025).

6.4 Results

Treatment outcomes for Extended ICSOR and Smooth NLP are compared through independent evaluation with the original mechanistic equations. The nominal influent is denoted N, and the ten influent scenarios are denoted S1–S10. Optimization time is assessed separately from prediction and decision quality.

6.4.1 Accepted development and holdout states

The development design retained 7386 of 8000 attempted states. The holdout design retained 1845 of 2000 attempted states. Their acceptance fractions were 92.33% and 92.25%. Table 6.15 gives the corresponding rejection counts. Rejected states were not replaced. The accepted states supplied the fitting and holdout assessments.

Table 6.15: Realized mechanistic samples for the connected-plant assessment. The separate development and holdout designs retained accepted states without replacing rejected candidates.

Design	Attempted	Accepted	Rejected	Accepted (%)
Development	8000	7386	614	92.33
Holdout	2000	1845	155	92.25

The raw-response selection retained a ridge penalty of 0.1. Its mean grouped cross-validation normalized root error was 0.659. This score used the fitting response scales across the complete joint response. The holdout composite scores use the location-specific reporting ranges.

6.4.2 Aggregate and local holdout accuracy

Projection reduced the holdout nRMSE from 0.04690 to 0.04306. The relative reduction calculated before rounding was 8.19%. The nMAE decreased from 0.03044 to 0.02568, a reduction of 15.65%. Mean location-level R^2 increased from 0.925 to 0.938. Figure 6.5 separates the aggregate, location, and composite summaries. All eight location summaries and all four composite summaries had lower nRMSE after projection.

The composite nRMSE values decreased from 0.04741 to 0.04586 for COD, from 0.03724 to 0.03454 for TN, from 0.04170 to 0.03349 for TP, and from 0.05852 to 0.05476 for TSS. TP showed the largest relative improvement among these composite summaries. TSS retained the largest projected error. The location-specific holdout ranges define the weighting of these composite scores.

Projection reduced nRMSE in 30 of the 32 location-composite combinations. Figure 6.6 identifies the two exceptions. COD error increased by 0.47% in the final reactor and by 2.36% in the overflow. Overflow TP error decreased from 0.02948 to 0.01795, a reduction of 39.10%. The location-specific assessment identifies both the reductions and increases in prediction error.

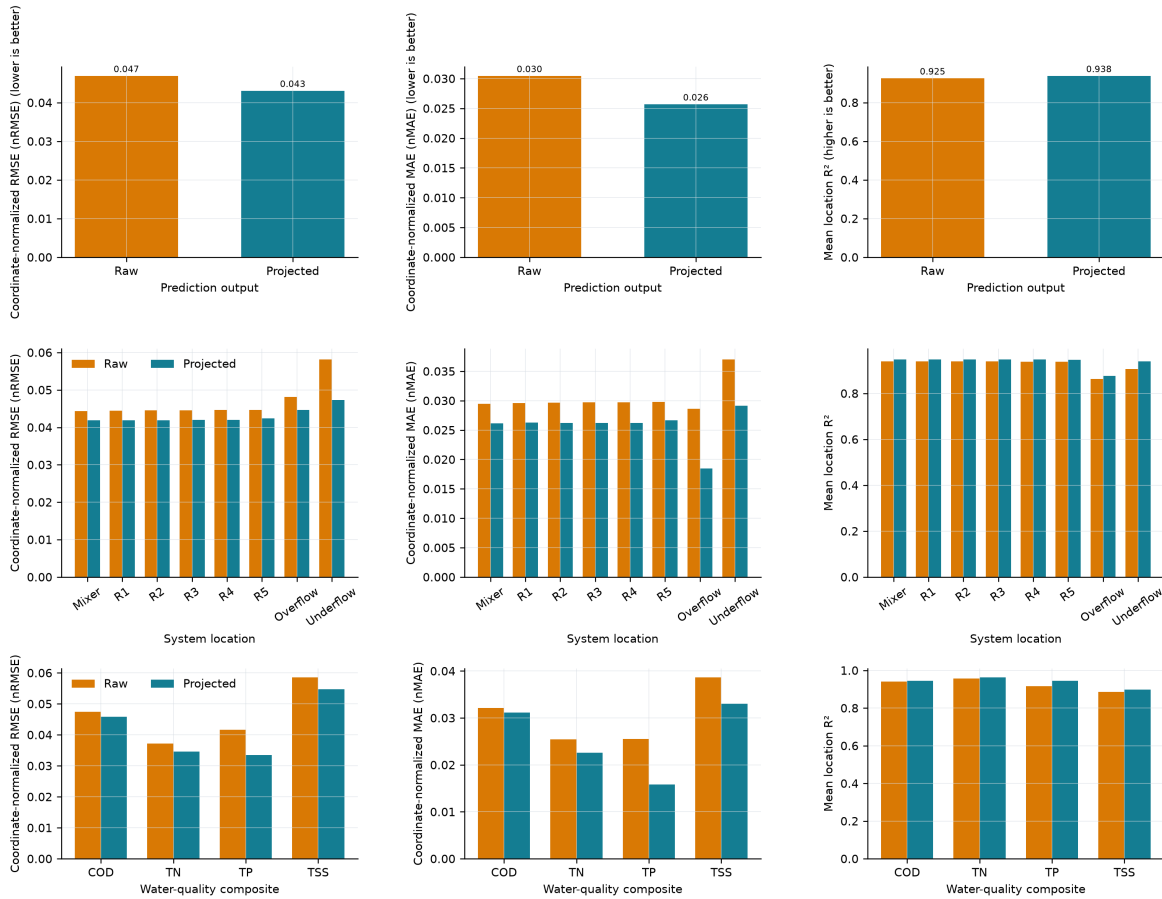


Figure 6.5: Raw and projected composite accuracy on 1845 holdout plant states. Rows show aggregate, location, and composite summaries. Columns show nRMSE, nMAE, and mean location-level R^2 . Errors are normalized by the mechanistic holdout range of each location and composite. The R^2 summaries average coordinate-specific scores rather than pooling locations into one score.

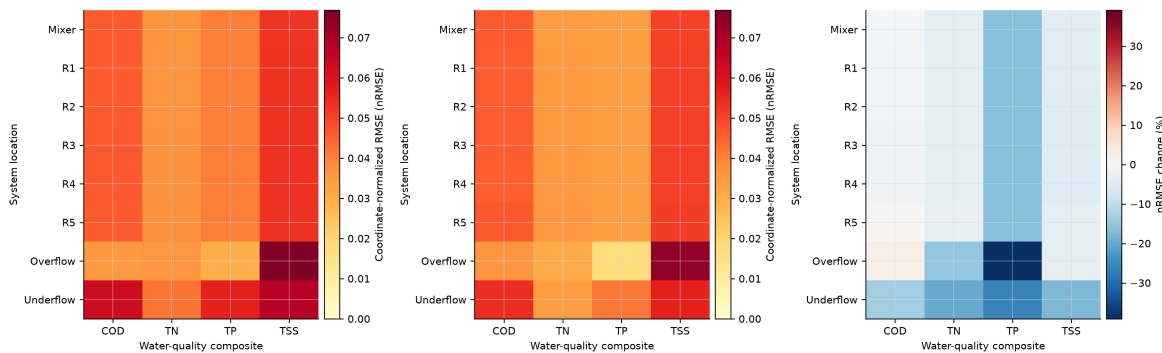


Figure 6.6: Holdout nRMSE by treatment location and composite. Raw and projected heatmaps share their color scale. The third panel shows percentage change relative to raw error. Negative values indicate improvement. The mixer, five reactors, overflow, and underflow are assessed separately.

6.4.3 Parity across the connected treatment system

Figure 6.7 pools the 14,760 location-observations for each composite while retaining location medians. Much of the density lay near the one-to-one line. The projected mean location-level R^2 values were 0.944 for COD, 0.963 for TN, 0.945 for TP, and 0.899 for TSS. These values average the location-specific scores.

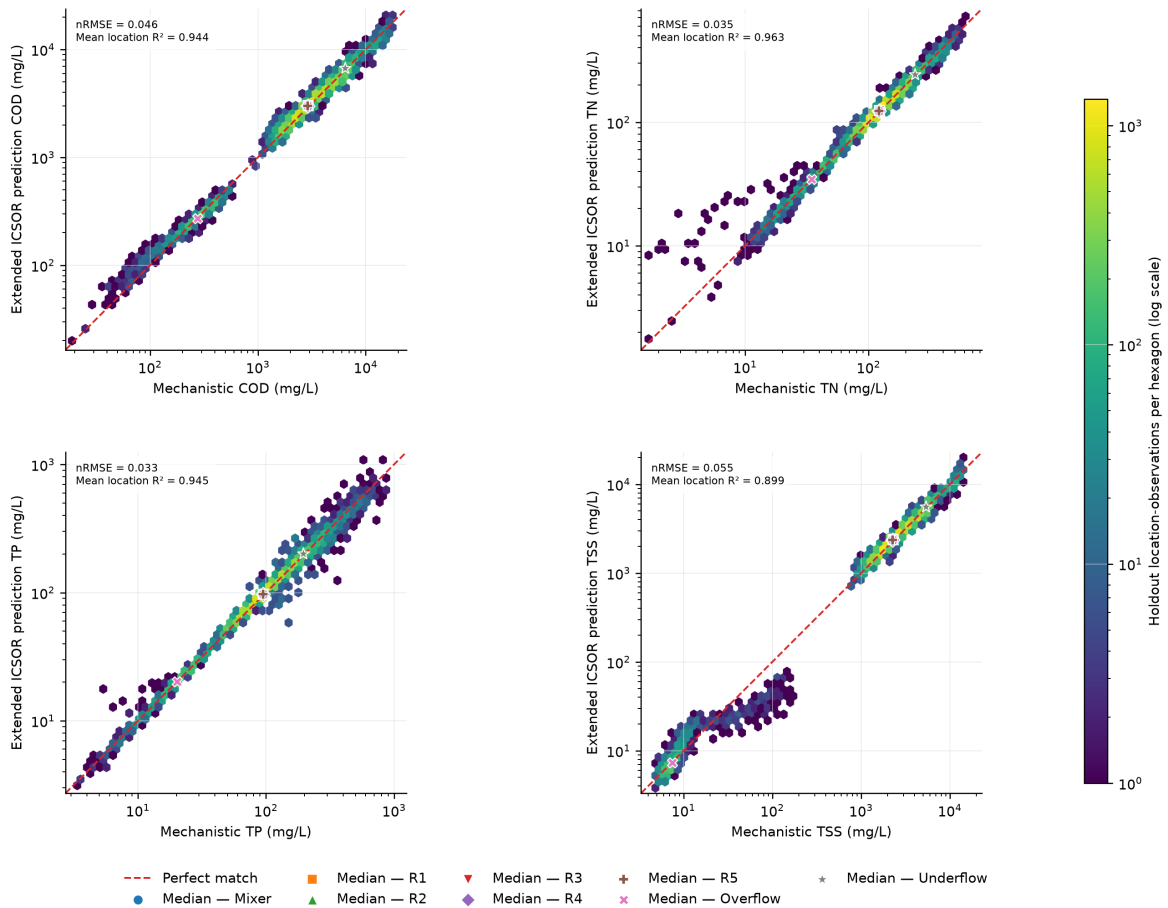


Figure 6.7: Projected composite parity across eight locations and 1845 holdout states. Both concentration axes are logarithmic. Hexagon color gives the number of location-observations, and labelled markers identify each location’s median. The one-to-one line indicates exact agreement. Metric annotations retain location-specific normalization and average location-level R^2 .

The overflow remained an important source of error. Its projected location-average R^2 was 0.876, below the reactor averages of approximately 0.947. The projected overflow TSS nRMSE was 0.07392. TSS was underpredicted in 1008 of the 1845 holdout states, or 54.6%. The remaining states had exact or overpredicted TSS values.

6.4.4 Physical audits and search qualifications

Every raw holdout response had at least one mass conservation violation in the independent physical audit. Negative entries occurred in 1765 of the 1845 raw responses. The corresponding projected summaries recorded no mass conservation, non-negativity, or network-inequality violations under their stated audit definitions. These counts summarize the imposed numerical balance and inequality checks.

The surrogate route retained an initially feasible candidate for the nominal influent and all ten scenarios. The initial direct searches retained candidates for the nominal influent and nine scenarios. The S1 direct attempt did not retain a validated feasible candidate, and its permitted recovery did not resolve that search outcome. The two-route figures nevertheless include a separately evaluated valid direct decision for S1. This decision is included normally in the treatment and objective comparisons. It is not counted as a successful initial direct search.

Table 6.16 distinguishes these availability results from convergence evidence. Positive first-order qualification flags were recorded for the surrogate decisions at N, S6, and S10. Seven other surrogate decisions had finite-poll qualification. S4 remained unresolved. No retained initial direct candidate had a positive first-order certificate. The independent mechanistic responses supplied the reference treatment outcomes for these retained decisions.

Table 6.16: Qualification of the initial center-start searches. Each route was attempted in eleven cases. Finite-poll qualification uses the specified directions and radii. The separately evaluated S1 direct decision appears in the paired figures but does not alter these initial-search counts.

Outcome	Extended ICSOR	Smooth NLP
Initial searches	11	11
Retained feasible initial candidates	11	10
First-order qualification	3	0
Additional finite-poll qualification	7	0
Retained candidates with unresolved qualification	1	10
Cases without a retained initial candidate	0	1

6.4.5 Prediction fidelity at selected controls

Figure 6.8 compares Extended ICSOR predictions with independent mechanistic outcomes at its selected controls. Across the 44 case-composite comparisons, median absolute concentration error was 12.75% of the mechanistic effluent value. Mean absolute error on that percentage basis was 18.43%. Median and mean absolute removal errors

were 3.26 and 5.23 percentage points. These selected-decision summaries had different denominators and a different input distribution from the holdout scores.

In S6, predicted COD was 151.10 mg L⁻¹ and the mechanistic value was 99.21 mg L⁻¹. In S10, the corresponding values were 232.23 and 167.78 mg L⁻¹. The predictions underestimated COD removal by 8.51 and 10.44 percentage points. These two COD predictions were conservative about treatment removal.

The nutrient errors in S4 had the opposite implication. Predicted TN was 2.67 mg L⁻¹ against a mechanistic value of 7.47 mg L⁻¹. Predicted TP was 2.08 mg L⁻¹ against 4.68 mg L⁻¹. TN and TP removals were overstated by 16.97 and 27.33 percentage points. These errors remained after projection and trust screening.

TSS was overpredicted in ten of the eleven selected surrogate decisions. Predicted values ranged from 5.53 to 8.05 mg L⁻¹, while mechanistic values ranged from 5.28 to 6.41 mg L⁻¹. In S1, values of 7.78 and 5.77 mg L⁻¹ differed by only 0.61 percentage points in removal. A small removal error could therefore coexist with a substantial relative effluent-concentration error.

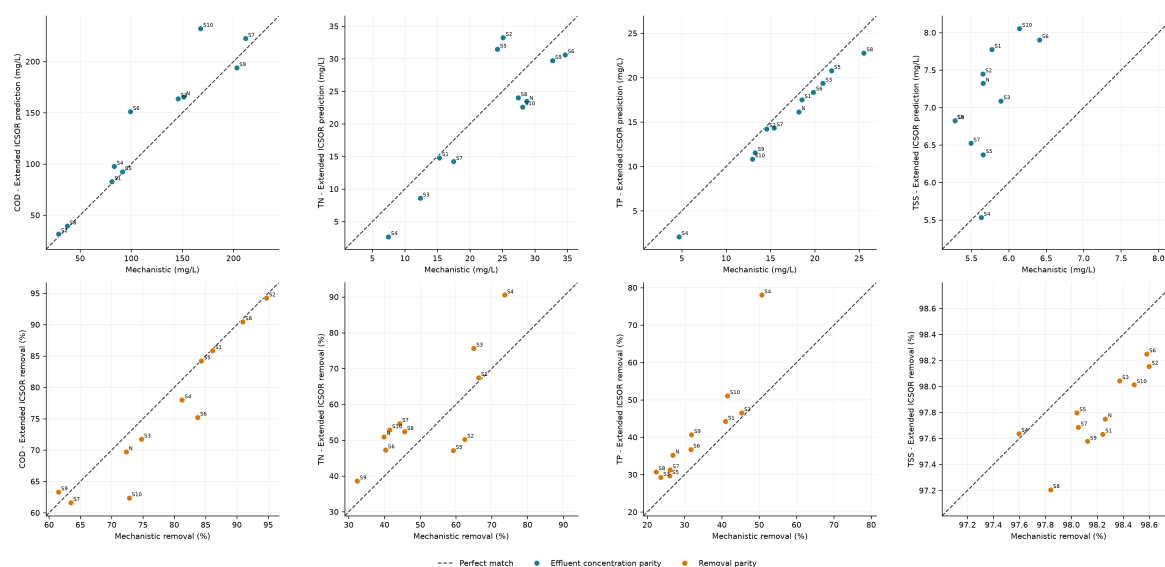


Figure 6.8: Extended ICSOR predictions at selected controls compared with independent original-model evaluation. The upper row shows effluent concentration parity in mg L⁻¹. The lower row shows removal parity in percent. Columns correspond to COD, TN, TP, and TSS. Teal identifies concentrations and orange identifies removals. Labels identify N and S1–S10. Dashed lines indicate exact agreement.

6.4.6 Treatment outcomes and operating choices

Figure 6.9 compares both routes on the mechanistic reference basis. Smooth NLP produced lower COD, TN, and TP in the nominal case and nine of the ten scenarios. Nominal COD was 151.26 mg L⁻¹ at the surrogate decision and 57.60 mg L⁻¹ at the

direct decision. In S9, values were 203.24 and 19.08 mg L⁻¹. Smooth NLP had lower TSS in six of the eleven comparisons. The TSS ordering therefore varied among cases.

Extended ICSOR selected the 6-h HRT lower bound at N, S6, S7, S9, and S10. Smooth NLP selected the 36-h upper bound in S5 and S10 within numerical tolerance. Since fresh flow was fixed, these HRT choices changed biological reactor volume. The comparison therefore combined reactor capacity selection with operating control.

Both routes selected the upper aeration bound in reactor 3 for every case within numerical tolerance. Aeration in reactors 4 and 5 also reached the upper bound in many cases. Both routes selected the return-sludge lower bound of 0.25. Smooth NLP selected internal recycle ratios of 4.00 in S2 and S10 and 3.44 in S9. Its other values were effectively zero. Extended ICSOR selected ratios between zero and 0.804.

The distinction between nominal HRT and residence time per passage was visible in S2. Smooth NLP selected $H = 31.40$ h with $r_I = 4.00$ and $r_R = 0.25$. The reactor-train flow ratio was 5.25, giving a residence time per passage of approximately 5.98 h. Recycle flow reduced the residence time for each passage through the train.

Wasting frequently reached its upper bound. Its resource contribution nevertheless depended on the selected underflow concentration through wX_U . Neither the wasting ratio nor an aeration setting alone described the corresponding material or resource measure.

6.4.7 Mechanistic treatment-train profiles

Figure 6.10 follows the original-model concentrations at each route's selected controls. All four composites increased from fresh influent to the mixer for both routes in all eleven cases. Recycle mixing brought concentrated material back into the biological train (Alex et al., 2008). The mixer increase reflected internal circulation.

Changes across the reactors differed from the concentration decrease across the clarifier. The former reflected the combined soluble and particulate state after biological conversion. The latter reflected particulate separation under the nonreactive clarifier assumption. A nearly constant composite concentration can accompany conversion among its component forms (Henze et al., 2000; Jeppsson & Diehl, 1996).

The large TSS decrease between reactor 5 and the final effluent showed the contribution of solids separation to treated-water quality. Species concentrations, outlet solids flows, and the reactor and clarifier inventories support interpretation of nitrification, denitrification, phosphorus uptake, and solids age (Henze et al., 2008; Wentzel et al., 1992).

6.4.8 Quality and resource trade-offs

Smooth NLP had the lower common-reference objective in the nominal case and all ten scenarios. Figure 6.11 separates the total objective from quality and weighted

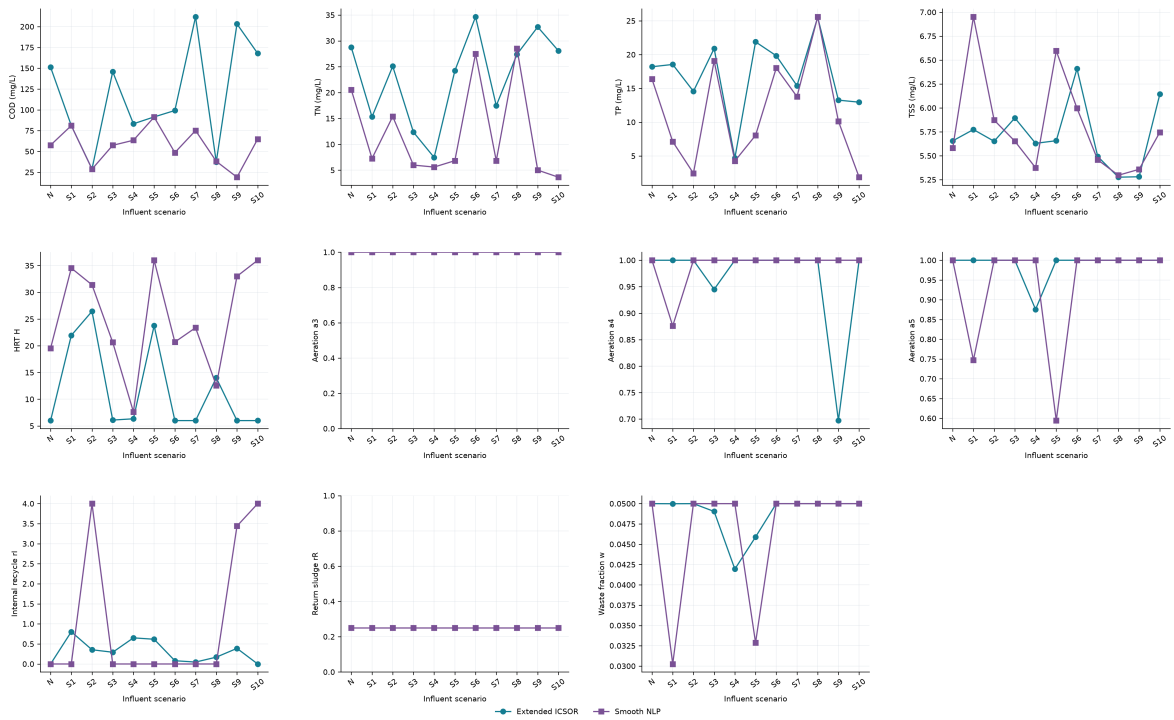


Figure 6.9: Mechanistic effluent outcomes and selected controls for both optimization routes. The top row shows COD, TN, TP, and TSS in mg L^{-1} . The remaining panels show HRT, aeration in reactors 3–5, internal recycle, return sludge, and wasting. The unused panel is omitted. N and S1–S10 appear in every comparison. The constant aeration and return-sludge panels use plain zero-to-one axes.

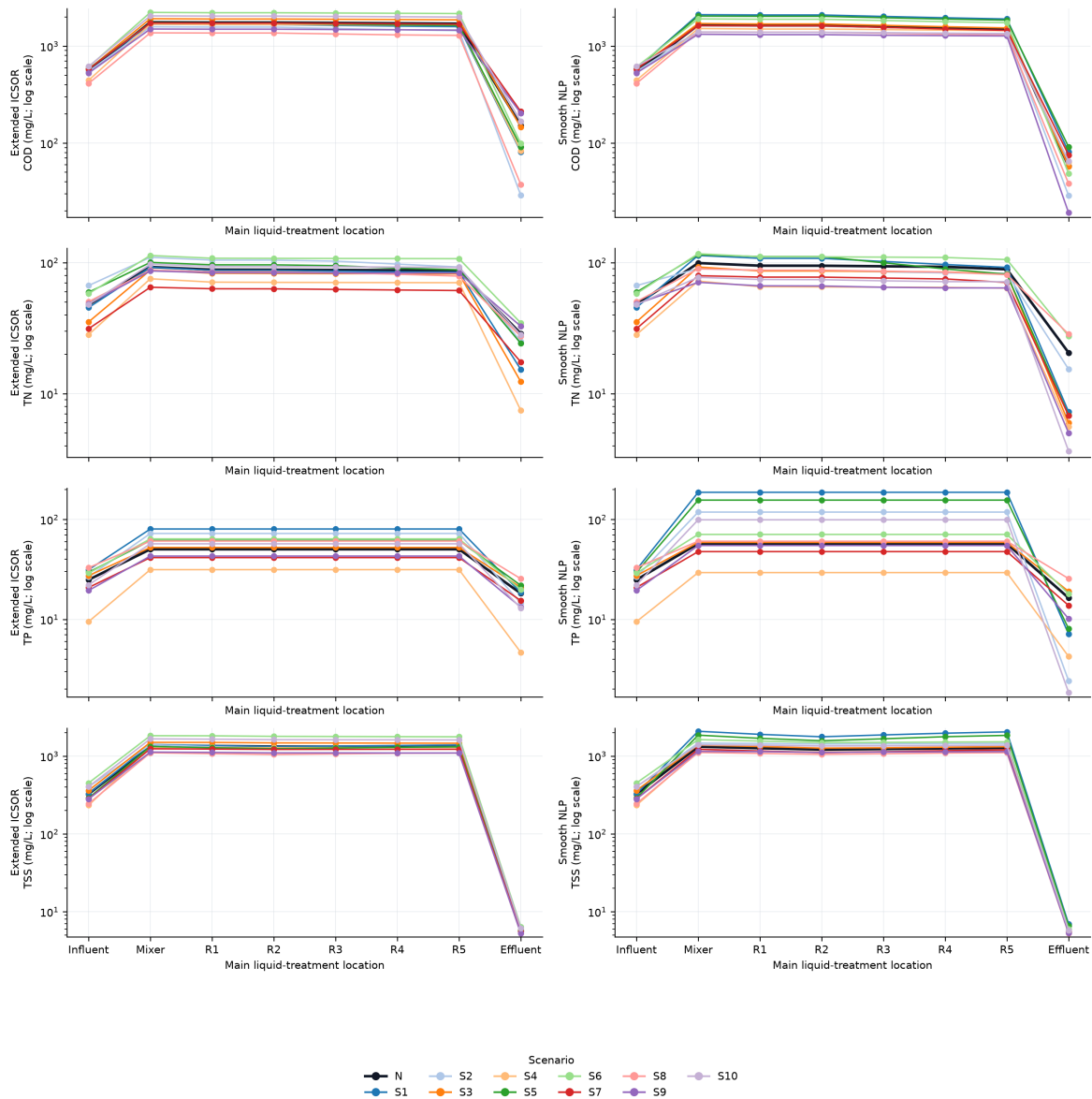


Figure 6.10: Mechanistic concentration profiles at controls selected by Extended ICSOR and Smooth NLP. Rows show COD, TN, TP, and TSS from fresh influent through the mixer and reactors 1–5 to clarifier effluent. Each composite has identical logarithmic concentration limits for both routes. Colors identify N and S1–S10. Underflow is not placed in the main liquid-treatment path.

resource contributions. Nominal objectives were 0.8647 for Extended ICSOR and 0.7862 for Smooth NLP. The surrogate value was 9.99% higher relative to the direct value. S5 had the largest absolute difference, with values of 1.0101 and 0.7022. S8 had the smallest difference, with values of 0.9279 and 0.9234.

Smooth NLP had lower normalized quality cost in ten of the eleven cases. Its weighted resource contribution was higher in every case except S8. In S10, quality cost decreased from 1.4748 to 0.3699. Its weighted contribution decreased by approximately 0.5525. Weighted resource cost increased from 0.0557 to 0.4191. The resulting total objective decreased from 0.7931 to 0.6040. Better weighted treatment was obtained while accepting a larger capacity and resource contribution.

S8 illustrated the opposite trade-off. Quality cost increased from 1.5856 to 1.6110, but weighted resource cost decreased from 0.1351 to 0.1179. Smooth NLP therefore had a lower total objective even though every evaluated effluent composite was slightly higher. The resource reduction determined the lower total objective in this case.

The development quality scales were 106.61, 13.07, 7.11, and 17.38 mg L⁻¹ for COD, TN, TP, and TSS. All exceeded one. The two native normalization matrices in Equation (6.191) consequently coincided in this experiment. Both route objectives therefore used the same quality normalization.

The volume dependence of the aeration proxy was also important. When $H = 6$ h and all three aeration settings equal one, Equation (6.147) gives $j_A = 1/6$. Frequent low-HRT surrogate decisions thus received relatively small capacity and aeration contributions despite maximum aeration settings. The contributions measured normalized capacity and resource proxies.

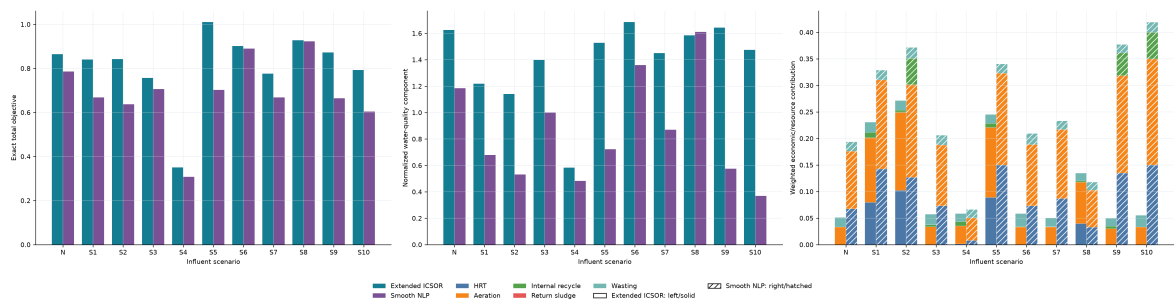


Figure 6.11: Common-reference total objective, normalized quality cost, and weighted resource contributions. Both routes are evaluated with the original mechanistic model and common quality scales. Resource stacks show HRT, aeration, internal recycle, return sludge, and wasting. Solid bars identify Extended ICSOR, and hatched bars identify Smooth NLP. The quality panel shows j_Q , whose contribution to the total objective is $0.50j_Q$.

6.4.9 Optimization time

Extended ICSOR had shorter recorded optimization time for N and every scenario. Figure 6.12 shows all eleven comparisons. Across S1–S10, times ranged from 0.758 to 2.702 s for Extended ICSOR and from 3.932 to 8.253 s for Smooth NLP. Means were 1.726 and 5.236 s. The ratio of these means was approximately 3.03. Nominal times were 0.985 and 5.062 s.

Scenario-specific direct-to-surrogate time ratios ranged from 1.90 to 7.99. The largest ratio occurred in S10. The comparison included initial failed attempts, including the S1 direct attempt. Candidate retention is recorded separately in Table 6.16.

The recorded times cover the initial optimization intervals defined in the methodology. Extended ICSOR had both a shorter search interval and a higher mechanistic objective in the paired comparisons.

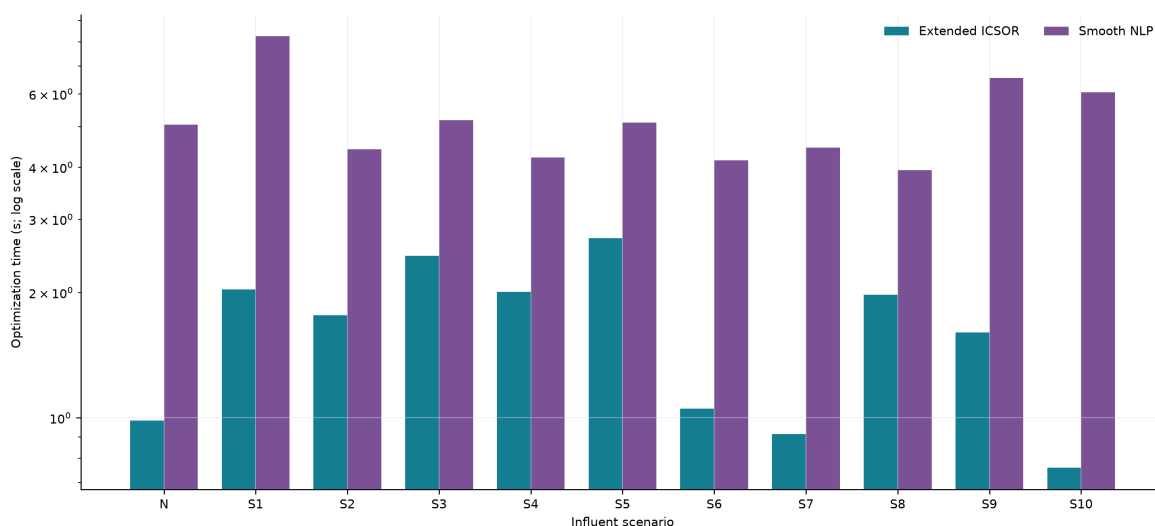


Figure 6.12: Recorded optimization time for N and S1–S10 on a logarithmic seconds axis. The interval includes the initial surrogate search with projection and fallback or the initial direct smoothing sequence. It excludes generation, fitting, certification, conditional recovery, and independent mechanistic evaluation. Initial failed attempts remain in the timing account. Scenario mean times use S1–S10 only.

6.5 Discussion

6.5.1 Connected enforcement and its statistical benefit

The joint response made the mixer, reactor train, clarifier outlets, and stored solids available in one prediction. Joint projection linked the material returned to the mixer with the material leaving the reactors and clarifier. It also reconciled reactor inventories with external effluent and waste-sludge flows. The observed reduction in holdout error

supported this connected formulation. Improvement across all eight location summaries and all four composite summaries showed that the statistical benefit extended across the treatment system.

The separate logarithmic model supplied a positive overflow solids endpoint. Joint projection combined that fitted endpoint with component balances, soluble continuity, particulate densification, and storage bounds. The two local COD increases and the remaining overflow errors describe how these simultaneous adjustments affected individual reporting quantities. Equation (6.122) relates weighted error reduction to feasibility of the reference, and Equation (6.129) includes its distance from the complete projection set. The location-specific results provide the corresponding assessment in the reported composite quantities. This connection between constraint choice, projection weighting, and prediction error follows the analysis of Sturm and Silva (2025) and Kircher and Votsmeier (2025).

6.5.2 Holdout accuracy and selected-decision fidelity

The operating search concentrates evaluations around preferred settings, so prediction errors at selected controls have a different distribution from errors across the holdout design (Bhosekar & Ierapetritou, 2018; Pedrozo et al., 2025). S4 illustrated the engineering importance of this assessment. Its projected TP prediction was 2.08 mg L^{-1} , whereas the independently solved value was 4.68 mg L^{-1} . For an illustrative treatment requirement of 3 mg L^{-1} , the predicted and mechanistic values fall on opposite sides of the threshold. The corresponding TN discrepancy also changed the estimated nutrient removal substantially.

Independent mechanistic evaluation supplied treatment outcomes from the biological kinetics and nonlinear clarifier equations at each selected control vector. Projection supplied network consistency, and trust screening supplied fixed checks on data support and model residuals. Their combination connected rapid candidate selection with a common process reference for judging the resulting decisions. The approximation-management work of Hameed et al. (2026) and the treatment-safeguard setting of Ren et al. (2026) support this emphasis on the effluent response that governs the operating choice.

6.5.3 Decision quality under declared priorities

Smooth NLP had lower common-reference objectives in all eleven paired comparisons. Extended ICSOR had shorter optimization times. These findings identify a trade-off between search computation and the quality of the retained decisions under the fixed priorities. Response approximation, trust restrictions, and the local search paths provide possible explanations for the selected controls. Each route is assessed through the same mechanistic response and quality scales.

The S8 outcome showed how resource priorities affect the preferred decision. Its lower direct objective accompanied slightly higher effluent concentrations because the weighted resource contribution decreased sufficiently. In S10, the direct decision accepted a larger resource contribution to obtain much lower quality cost. Separate presentation of quality and resource terms made both choices understandable under the declared weights (Aboagye et al., 2021; Padrón-Páez et al., 2020).

The free HRT coordinate connected reactor capacity with operating choices. At fixed influent flow, increasing HRT increased biological reactor volume. The aeration contribution then changed with both volume and the selected transfer coefficient. For application to an installed plant, fixed reactor volumes and specified treatment constraints define the corresponding operating problem. The component-resolved analysis of X. Zhang et al. (2026) and energy treatment of Miederer et al. (2025) support explicit accounting of hydraulic capacity, oxygen-transfer capacity, and resource demand.

6.5.4 Computational performance and operating assessment

The surrogate search varied seven controls and evaluated responses through second-order regression and joint projection. The direct search optimized controls together with reactor and clarifier states through a smoothing sequence. These computational structures were consistent with the shorter recorded surrogate search times. Development, fitting, qualification, recovery, and physical verification form the other terms in the complete-cost accounting described in the methodology. Retaining their costs alongside decision quality follows the benchmarking principles discussed by Blik et al. (2023).

First-order qualification and finite polling supplied distinct numerical evidence at the retained candidates. The finite polls tested specified coordinate and coupled directions at two resolutions, following the role of directional exploration in Audet and Dennis (2006). Their qualification results accompany the independently evaluated treatment outcomes and objective values. The resulting decision aid combines connected material accounting, rapid candidate evaluation, and a mechanistic assessment of the selected aeration, recycle, wasting, and capacity settings.

6.5.5 Data Availability

The simulation data and numerical results supporting this study are publicly available at <https://github.com/egbertoselerio1997/surrogate-optimization>.

Chapter 7

Conclusion and Outlook

7.1 Synthesis of the research findings

The central research problem was how to use rapid activated sludge predictions for engineering analysis when prediction accuracy alone does not establish a usable treatment state or a good operating decision. The dissertation addressed this problem through complete component prediction, physical enforcement, explicit regression, and independent assessment of selected controls. Its general answer is that these capabilities can be connected through a common component-state representation. They still require separate evidence. A physically consistent state can be inaccurate, an inspectable regression can have higher error than a flexible learner, and a faster search can select a worse decision on the original mechanistic basis.

The adopted component state retained organic substrate, nitrogen species, phosphorus storage, biomass, and solids. The reporting map then produced COD, TN, TP, and TSS (Henze et al., 2000). This order allowed the same response to support material accounting, treatment reporting, and operating assessment without treating a reported total as a unique component state. The findings below answer the four research questions in Section 1.3. Their numerical evidence is reported in Sections 4.4, 5.4, and 6.4.

7.1.1 Complete reactor prediction and data availability

The first research question concerned the accuracy of different learners as training data changed and inputs moved beyond the sampled domain. The common reactor benchmark showed that the preferred learner depended on the simulation budget. At 500 cases, XGBoost and LightGBM had the lowest observed raw component nRMSE values of 0.435 and 0.439. At 10,000 cases, the multilayer perceptron had the lowest raw and projected values of 0.150 and 0.149. This answers the comparative gap with evidence from a shared component target rather than separate prediction tasks. It also shows why selection at one training size cannot establish the preferred model at every

size (Borisov et al., 2024; Viering & Loog, 2023).

The exterior assessment supplied a different answer about the usable input range. The multilayer perceptron again had the lowest error, but its projected nRMSE was 0.251 under mild exterior inputs and 0.521 under severe exterior inputs. Every projected exterior state still passed the physical checks. The larger severe-domain error therefore did not arise from failure of those checks. For practical use, an unfamiliar loading or operating condition needs an accuracy assessment even when projection returns a valid balance. The benchmark supports model selection within declared data and domain conditions, not unrestricted extrapolation or one universal learner ranking.

7.1.2 Physical enforcement and its consequences

The second research question concerned what projection changes in the same raw prediction. Every raw assessment state failed the specified mass conservation tolerance. Some predictors returned only non-negative raw concentrations, which showed that positivity alone was insufficient. Joint enforcement returned states that passed both requirements in all 766,350 assessed projection instances. These instances included repeated assessment across nested design sizes rather than that many distinct simulated reactor states. The maximum invariant residuals were 2.06×10^{-13} in the learning domain and 1.72×10^{-13} in the exterior assessment. The result demonstrates numerical enforcement of the adopted balances and bounds across different learners. It applies the existing projection of Kircher and Votsmeier (2025) to a common activated sludge task rather than introducing a new projection principle.

The accuracy effect was mixed. At the largest design size, projection improved 188 of 260 predictor–component comparisons, but aggregate component nRMSE decreased for only five of the thirteen predictors. Every predictor had lower TN and TP root error and higher COD and TSS root error. Thus, physical enforcement closed the compliance gap without eliminating the approximation gap. A treatment analysis concerned mainly with nutrients can reach a different assessment of the adjustment from one concerned with organic matter, biomass, or solids. The result supports separate component and composite reporting and confirms the importance of the adjustment metric discussed by Sturm and Silva (2025).

Displacement and computational cost completed this answer. They described how much a raw prediction changed and the effort needed to return the checked state. Projection dominated the repeated prediction cost of the fastest raw linear and neural models under the stated batch protocol. A rapid regression evaluation is therefore not the same as a rapid complete prediction. The paired assessment provides a basis for selecting an enforcement procedure together with a learner while retaining accuracy, physical compliance, and complete prediction cost as distinct criteria (Bhosekar & Ierapetritou, 2018).

7.1.3 Interpretation of the complete regression response

The third research question concerned how an explicit regression could support interpretable prediction with mass conservation and non-negativity. ICSOR supplied named operating, loading, curvature, and interaction terms, followed by learned component coupling and checked projection (Selerio, 2026). These stages made it possible to trace an input contribution to the returned component state. In one fitted model, 44 of 210 unique COD influent-curvature terms exceeded their reporting threshold, while 371 of 380 shared off-diagonal coupling entries exceeded theirs. The representation was selective in influent curvature but dense in coupling. Its contribution to interpretation was an explicit response structure, not a uniformly sparse model or uniquely identified biological parameters.

The accuracy evidence supported a conditional practical use for this structure. At 500 cases, ICSOR had the lowest mean held-out pooled composite root error of 10.79, compared with 11.26 for XGBoost and 11.45 for LightGBM. It also had the lowest reported root-error learning-curve area of 6.33 after normalization by the design-size interval. At 10,000 cases, repeated-split errors were 5.94 for ICSOR and 4.62 for the multilayer perceptron. LightGBM had the best mean rank across the broader assessment. These outcomes favor considering ICSOR when an explicit equation and limited simulation data matter, while retaining flexible learners when the main priority is lower full-size error. Accuracy scoring used checked ICSOR states and affine-aligned comparator states, so these comparisons do not isolate the effect of regression structure from output enforcement.

The physical assessment showed that interpretation must include projection. Affine projection sufficed for 436 of the 2000 fixed-partition cases. The remaining 1564 cases required constrained linear projection, and every final state passed both adopted physical checks. The analytical raw and affine sensitivities and the hypothetical example of an active concentration bound showed why a raw coefficient is not necessarily the final local slope. An engineer examining an aeration response must follow coupling and the active projection constraints at the operating point of interest. The model therefore addresses the interpretation gap through a traceable calculation, while coefficient stability and causal process identification remain separate questions (Selerio, 2026; Shen et al., 2023).

7.1.4 Connected prediction and operating decisions

The fourth research question concerned the support that joint prediction and projection could provide for connected plant decisions. The response retained the mixer, five biological reactors, clarifier overflow, clarifier underflow, and stored solids. Joint enforcement linked shared component flows and inventories rather than checking each unit in isolation (Alex et al., 2008; Jeppsson & Diehl, 1996). The development and holdout designs retained 7386 and 1845 mechanistic states. On the holdout, projection

reduced aggregate nRMSE from 0.04690 to 0.04306, an improvement of 8.19%, and reduced aggregate nMAE by 15.65%. It improved 30 of 32 location-composite nRMSE values. The two increases were COD in the final reactor and overflow. This evidence addresses the network-consistency gap and shows a broad, but not uniform, statistical benefit from the complete connected procedure.

The decision assessment identified a limit that the holdout improvement could not resolve. In S4, Extended ICSOR predicted TN and TP concentrations of 2.67 and 2.08 mg L⁻¹, while independent mechanistic evaluation gave 7.47 and 4.68 mg L⁻¹. These errors remained after projection and trust screening. A nutrient requirement between the predicted and reference concentrations would lead to different judgments about the selected setting. The result answers the decision-quality gap by showing why treatment outcomes must be calculated at the controls actually selected, not inferred from average holdout performance (Alexandrov et al., 1998; Pedrozo et al., 2025).

Smooth NLP had a lower common-reference objective in the nominal case and all ten influent scenarios. Nominal values were 0.8647 for Extended ICSOR and 0.7862 for Smooth NLP. Extended ICSOR had shorter recorded search times but did not match the direct route's objective values. The answer to the fourth question is therefore qualified. Joint prediction and projection can support rapid exploration with a consistent material account, but they do not replace mechanistic decision evaluation. The comparisons concern available local decisions under the declared priorities and do not establish a global optimum or superiority of one route in every plant setting.

7.2 Contributions and gaps addressed

The dissertation contributes a common complete-component reactor benchmark. This addresses the lack of directly comparable evidence about learner behavior under changing data availability and input conditions. It retains both small component concentrations and reported treatment quantities. The contribution is the defined activated sludge comparison and its evidence, not a claim that one learner is always preferable.

A second contribution is the paired assessment of a common physical projection across learners. It makes the distinction between physical enforcement and statistical improvement measurable. The adopted projection and stoichiometric foundations come from existing work (Kircher & Votsmeier, 2025; Sturm & Wexler, 2022). The dissertation's contribution lies in applying them to complete activated sludge predictions and evaluating compliance, error redistribution, displacement, and cost together. This also establishes the limits of interpreting a feasible component state as a fully accurate mechanistic response.

A third contribution is the structured regression and analysis of its complete returned response. Named terms expose operating and loading relationships, while component

coupling represents shared output dependence. Stage-specific physical diagnostics and sensitivities distinguish what the regression learns from what projection imposes. This addresses the gap between having visible coefficients and explaining the final treatment prediction. The structure supports inspection without presenting fitted associations as causal kinetic parameters (Selerio, 2026).

The fourth contribution is a connected component-state formulation with joint projection and independent decision verification. It includes the shared streams and solids inventories that a reactor-only output cannot represent. Common mechanistic evaluation then connects those predictions to treatment and resource choices. The contribution includes the observed limitation of the surrogate decisions. Demonstrating lower search time alongside higher reference objectives supplies useful decision evidence rather than assuming that computational speed establishes engineering superiority.

Table 7.1 links the gaps to their answers and practical uses. The integration establishes a consistent assessment logic, not one numerical ranking across all three studies. Reactor component errors use training-based scales, the structured comparison pools native composite coordinates, and plant holdout errors use location-specific reference ranges. Their validation designs and enforcement procedures also differ. These scores should therefore be interpreted within their own assessments rather than combined into one cross-study measure.

Table 7.1: Research gaps, observed answers, and practical uses of the integrated findings.

Research gap	Observed answer	Practical use
Uncertain complete-state learner performance	The leading learner changed with data budget, and exterior accuracy deteriorated despite physical compliance	Select for the available data and intended input range
Training does not establish physical compliance	All 766,350 projected instances passed both checks, but aggregate error improved for only five of thirteen learners at the largest size	Check balances and bounds separately from treatment accuracy
Visible terms do not explain the final response	ICSOR exposed operating and loading terms, while most fixed-partition cases required constrained projection	Interpret the coupled and projected response at the operating point
Consistent unit predictions do not establish good plant decisions	Joint prediction improved 30 of 32 local scores, but faster surrogate searches selected higher reference objectives	Verify selected controls with the original plant equations

Together, these contributions connect process knowledge with statistical approxi-

mation and decision evidence. Reaction stoichiometry supplies the invariant relations, while unit boundaries supply transport and inventory relations (Alex et al., 2008; Henze et al., 2000). Regression approximates the response within the sampled conditions. Projection imposes the declared physical requirements. Interpretation examines the complete mapping, and independent mechanistic evaluation establishes the reference outcome at selected controls. The integrated contribution is this defined connection and the evidence needed to assess each part. It does not claim invention of activated sludge stoichiometry, general projection, or surrogate optimization.

7.3 Practical implications

7.3.1 Treatment and resource priorities

The operating results help engineers compare treatment gains with the resources required to obtain them. In S10, the direct decision reduced normalized quality cost from 1.4748 to 0.3699 while increasing the weighted resource contribution from 0.0557 to 0.4191. The total objective decreased from 0.7931 to 0.6040. This choice accepted greater capacity and resource contributions for better weighted treatment. In S8, a lower resource contribution produced a slightly lower total objective while all four effluent composites were slightly higher. Reporting the separate contributions helps engineers understand the selected setting and relate it to their treatment priorities (Aboagye et al., 2021; Padrón-Páez et al., 2020).

The selected HRT values also support capacity planning. Varying HRT at fixed fresh flow changed the biological reactor volume. The direct route reached the 36-h upper bound in S5 and S10, while the surrogate reached the 6-h lower bound in several cases. Comparing these settings connects reactor capacity with the treatment obtained. Aeration can be assessed alongside reactor volume because both influence oxygen-transfer capacity. Wasted-solids production can likewise be assessed from wasting flow and underflow concentration. For operating analysis at an installed plant, the existing reactor volumes and equipment capacities provide the basis for comparing control settings (Henze et al., 2008; Miederer et al., 2025).

The separation of quality and resource terms provides a practical basis for developing facility-specific objectives. Effluent concentrations and removal values can express treatment targets. Electricity demand, recycle pumping, and sludge-handling costs can express resource priorities. Their weights specify how each quantity contributes to the preferred decision. Engineers can use the resulting contributions to explain a choice and compare alternatives under different priorities (Miederer et al., 2025; Padrón-Páez et al., 2020).

7.3.2 A verified decision workflow

The surrogate supports screening and candidate selection. The engineer first defines the plant boundary, influent composition, available controls, and treatment targets. The fitted model then provides component responses for comparing alternatives. Joint projection reconciles the predicted streams and inventories before treatment and resource quantities are compared. Physical residuals and projection displacement provide diagnostic information for selecting candidates for detailed evaluation.

Independent evaluation with the original reaction and settling equations supplies final-effluent concentrations, shared solids flows, inventories, and the common-reference objective at the selected controls. These quantities provide a consistent basis for choosing among candidate settings. The evaluations can also guide additional mechanistic cases near promising operating conditions. This connects further surrogate fitting with the decisions that the model is intended to support (Alexandrov et al., 1998; Hameed et al., 2026).

Component and location information also supports process diagnosis. The increase from fresh-influent to mixer concentrations shows the material circulated through recycle. The TSS decrease across the clarifier shows the contribution of solids separation. Separate ammonium and oxidized-nitrogen concentrations reveal changes among nitrogen forms when TN changes little. Examining these quantities alongside returned solids and inventories helps engineers relate aeration, recycle, and wasting choices to the treatment response (Alex et al., 2008; Henze et al., 2000; Jeppsson & Diehl, 1996).

7.3.3 Computational value

The measured scenario mean initial-search times were 1.726 s for Extended ICSOR and 5.236 s for Smooth NLP. Their ratio was approximately 3.03. The shorter surrogate search interval provides computational value for repeated exploration of operating alternatives. Rapid screening is useful when an engineer needs to compare several influent conditions or combinations of aeration, recycle, and wasting settings.

Repeated application distributes model-development effort across multiple operating comparisons. Engineers can use a fitted surrogate to explore alternatives and reserve detailed mechanistic calculations for selected candidates. This workflow focuses process calculations on the operating decisions of interest while using rapid prediction for broader comparison (Bhosekar & Ierapetritou, 2018; Bliet et al., 2023).

7.4 Directions for future research

7.4.1 Adaptive sampling around operating decisions

Future studies can generate new mechanistic cases where the surrogate is most influential in the operating search. A first surrogate can screen the declared domain and identify candidate controls. Mechanistic calculations at those candidates can then supply additional training states. Sampling can concentrate on large nutrient discrepancies, strong projection displacements, and regions close to treatment requirements. Repeating fitting and operating assessment would connect the simulation budget to the decisions it is intended to support. This proposal develops the reference-informed model management of Alexandrov et al. (1998) and the gray-box approximation management discussed by Hameed et al. (2026).

An informative comparison would hold the mechanistic evaluation budget fixed and compare uniform sampling, error-based sampling, and decision-based sampling. The outcomes should include effluent errors at selected controls, the common-reference objective, the number of accepted decisions, and the complete time required to deliver them. The S4 nutrient discrepancies provide a concrete target for such a study. Additional cases near those selected operating conditions could test whether focused sampling improves TN and TP prediction where the operating choice depends on them.

7.4.2 Projection aligned with treatment priorities

The reactor results motivate further study of projection weights and the quantities they emphasize. A projection that minimizes changes in component coordinates can redistribute error differently from an assessment based on reported nutrients or solids. Future formulations can compare component scaling, composite-sensitive weighting, and weights based on prediction uncertainty. Each formulation should retain the same mass conservation and non-negativity requirements so that changes in treatment accuracy can be attributed to the adjustment metric. The species-weighted correction of Sturm and Silva (2025) provides a precedent for choosing adjustment weights from component magnitude and uncertainty.

The connected plant offers an additional test of this idea. Weighting can distinguish internal state accuracy from final-effluent accuracy while preserving the unit connections. Studies can assess whether nutrient-focused weights improve selected-control predictions without producing large errors in biomass, solids inventory, or other treatment locations. This would make the relation between physical adjustment and the intended engineering decision more explicit.

7.4.3 Uncertainty-aware prediction and operating safeguards

Uncertainty can be developed alongside the projected point prediction. Repeated fits, model ensembles, or probabilistic component predictions can describe variation in the estimated treatment response. Each sampled state can be projected onto the same physical set. The resulting distributions can then be evaluated for their coverage of mechanistic reference concentrations, particularly for TN, TP, and overflow TSS. The conservation-aware probabilistic framework of Hansen et al. (2024) and the differentiable probabilistic projection of Utkarsh et al. (2025) provide starting points for this extension.

Operating studies can use these distributions to formulate treatment safeguards. For example, a candidate setting can be selected to keep an upper predictive bound for TP below a specified treatment requirement. Mechanistic and plant-based evaluations can test the frequency with which those safeguards are met. The effluent ammonium safeguards considered by Ren et al. (2026) illustrate how treatment requirements can guide aeration decisions. Separating uncertainty associated with surrogate fitting, influent measurements, and mechanistic parameters would also show which information most improves the decision (Machado et al., 2014; Petersen et al., 2003).

7.4.4 Stable interpretation of the complete response

Further interpretation work can examine the stability of coefficient groups across repeated training samples and operating domains. The present structure provides natural groups for operating terms, loading terms, curvature, and coupling. Resampling can show which groups remain prominent and which individual coefficients vary. Comparisons based on the complete predicted response can help identify different coefficient combinations that describe similar treatment behavior. This work would address the coefficient-stability question identified by Selerio (2026).

Sensitivity studies should also follow the final projection. Derivatives can be evaluated at representative operating points and checked against direct perturbations of the returned concentrations. Mapping the active constraints would show where a nutrient response changes smoothly and where a projection boundary alters its slope. Comparing these projected sensitivities with mechanistic sensitivities would connect model inspection more closely to aeration, recycle, and wasting decisions (Selerio, 2026; Shen et al., 2023).

7.4.5 Dynamic prediction and control

A dynamic extension can replace steady-state targets with trajectories of component concentrations and stored solids. The governing balances would include the time derivatives of reactor and clarifier inventories (Alex et al., 2008; Takács et al., 1991). Training data can represent changing influent loads, aeration schedules, recycle settings,

and sludge wasting. Physical enforcement can then account for the material transported and transformed during each prediction interval.

This extension would support studies of predictive control under daily influent variation and load disturbances. The evaluation can include transient nutrient peaks, recovery time, inventory changes, oxygen demand, and the computation required at each control update. The field-tested aeration controller of Bernardelli et al. (2020) and the wastewater control review of Croll et al. (2023) provide examples for developing and evaluating such dynamic strategies. The connected component representation provides a useful starting point because it already retains the locations and inventories that influence delayed treatment responses.

7.4.6 Plant validation, topology, and resource models

Validation with measured plant data is a major next step. Studies can combine influent fractionation, calibrated mechanistic parameters, unit concentrations, clarifier solids measurements, and recorded controls (Machado et al., 2014; Newhart et al., 2019; Petersen et al., 2003). A prospective assessment can fix the fitting and operating procedure before collecting the final evaluation period (Kaufman et al., 2012). Testing across seasons and loading conditions would provide evidence about how the component surrogate transfers to observed treatment behavior.

Operating objectives can also be developed for specific facilities. An installed plant has defined reactor volumes, aeration capacity, recycle capacity, and discharge requirements. Electricity demand, chemical use, sludge handling, and operating costs can replace or supplement the engineering proxies used here (Henze et al., 2008; Miederer et al., 2025). Multiobjective studies can present the attainable treatment and resource combinations before a preference is selected. Extensions to other reactor arrangements, attached-growth units, and resource-recovery connections can retain the same principle of defining component states and boundary streams before physical enforcement is imposed (Aboagye et al., 2021; Durkin et al., 2024; Henze et al., 2008).

More extensive search comparisons can evaluate several initial controls, hybrid surrogate and mechanistic searches, and the fifteen-start basin-sensitivity design described in Chapter 6. Recording state generation, fitting, prediction, projection, search, and verification costs would establish when surrogate development becomes worthwhile for repeated decisions (Bhosekar & Ierapetritou, 2018; Bliet et al., 2023). The central assessment would remain the treatment outcome and common-reference objective obtained for a given computational budget.

7.5 Concluding perspective

The dissertation shows how complete component prediction, physical enforcement, interpretation, and decision verification can form one activated sludge analysis framework. Its findings answer the research questions without treating one capability as proof of another. Learner choice depended on the available data and input range. Projection enforced the adopted physical requirements but did not improve every error. Structured regression made the response inspectable but did not have the lowest error at the largest data budget. Connected projection improved plant prediction, while mechanistic evaluation exposed higher objectives at the faster surrogate-selected decisions.

The practical contribution is a defined basis for rapid exploration with explicit checks on what is predicted and what is selected. The surrogate can help screen alternatives, while the original mechanistic model supplies the reference treatment outcome at candidate controls. Plant measurements and facility-specific priorities are needed to extend that basis to actual operation. Computational efficiency is useful when it helps deliver an acceptable treatment decision, not when search time alone is reduced.

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List of Publications and Presentations

Publication

Selerio, E. F. (2026). An interpretable statistical surrogate of activated sludge systems that preserves mass conservation and component non-negativity. *Digital Chemical Engineering*, 20, Article 100329. doi: 10.1016/j.dche.2026.100329

Selerio, E. (2026, October 5). Plant-wide optimization of activated sludge systems using a physically constrained statistical surrogate. *SSRN*. doi: 10.2139/ssrn.7567818

Presentation

Selerio, E. F. (2023, April 21–22). *Mass transport, biofilm accumulation, and biotransformation in recycling down-flow hanging sponge systems* [Poster presentation]. 4th International Conference on Sustainable Energy Ecosystems (SEECON 2023), University of San Carlos–Talamban Campus, Cebu City, Philippines.